# Companion to *On the equation $n'' = n$ and a problem of Erdős and Barbeau*: explorations, closures and the square sieve

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**Abstract**

This companion collects the exploratory and negative material of the note *On the equation $n'' = n$ and a problem of Erdős and Barbeau on products of prime–reciprocal sums*, cited below as the core note. Nothing here is needed for the results stated there, with one exception: the minus–layer density theorem, that both Pythagorean layers have density zero on the squarefree stratum, is proved here. What is otherwise here is the record of which routes were tried and where each fails. It contains the local heuristic model, the function–field lens locating the obstruction as archimedean, the breeder form, the deviation ladder and the plus layer, two closures on the existence route, and the square sieve at the mass–two wall — both the range at which its analytic input does not close, and the far smaller one at which it does.

## 1 The certificate programme and its anatomy

The barrier of Section 4 bounds a solution; it does not exclude one. This section records the programme that attempted the exclusion level by level, its completeness, and the point at which it provably stops. Nothing here is used by the results of the core note.

**Remark 1.1** (Verification tier, and why the ladder stops here)**.** Theorem 4.3 is machine–checked in Lean 4, and so is Proposition 4.5: the theorem `erdos307_sixty` formalises the entire argument; the forced–primes and element bounds, the Pythagorean plus–square, and a pruned depth–first search over the pool whose *completeness is proved, not assumed* (`dfs_sound`), the search itself running under the kernel; on top of the two independent unverified programs. The level cannot be climbed further by finite means: already at $|U| = 60$ the family $U = \{p_1, \dots, p_{59}\} \cup \{q\}$ qualifies for *every* prime $q > 277$, and on it the two square conditions read $x^2 = Aq + \Pi_{59}$, $y^2 = Bq + \Pi_{59}$ with $A, B = N_{59} \pm 2\Pi_{59}$, i.e. the Pell–type system $Bx^2 - Ay^2 = -4\Pi_{59}^2$ with 113–digit coefficients. We verified that no accessible modulus obstructs it (mod 32, and all odd $\ell < 4000$: the conditions pin $q \equiv 5 \pmod 8$ and thin each $\ell \mid \Pi_{59}$ to about half of its residue classes and each $\ell \nmid \Pi_{59}$ to about a quarter, but never to zero), consistent with the local–solubility theme of Section 2. The obstruction, however, lives at the primes *of* $A$ and $B$, and it is detectable *without factoring*: Proposition 1.2 below closes this family for *every* $q$ by a single Jacobi–symbol computation. (An exact sweep to $q \le 10^{12}$, `hunt/tail_sweep.rs`, $1.25 \times 10^{11}$ candidates, none with $Aq + \Pi_{59}$ even a single square, which corroborates it independently.) The tail family also exposes the self–similarity of the ladder:

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writing $a = qm$, so that $m\,b = \Pi_{59}$ partitions the first 59 primes, its two closing conditions collapse to the exact equation

$$D(m)\,D(b) + m^2 \;=\; \Pi_{59}$$

together with the divisibility $m \mid D(b)$ (then $q = D(b)/m$, required prime); verified exhaustively on a 12–prime toy universe. Equivalently, the deficit $1 - \sigma(m)\sigma(b)$ must equal $m^2/\Pi_{59}$ *exactly*: the circularity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$ of Section 5 in its purest form, with the deviation forced to be the perfect square $m^2$. Closing even this one family is thus a $2^{58}$–fold exact–coincidence search of the same type as #307 itself (expected count $\sim 2^{58}/\Pi_{59} \approx 10^{-95}$): each rung of the ladder past 60 contains the whole problem. The one case with *no* free large prime (a 60–set drawn entirely from small primes) is instead a finite search, and empty in the minimal range: a direct both–squares enumeration of all 60–prime sets with $T > 2$ supported in the first 70 primes (4,011,289 sets; `hunt/close60.rs`) finds none passing even the plus–square, so no level–60 solution has all its primes $\leq 349$. (The first 68 primes give 1,258,448 sets, also empty, and reproduce independently.) By Corollary 5.5 this verdict is decisive rather than provisional: at these masses a support passing the plus– and minus–squares, with the parity of Proposition 5.1(2), *is* a two–cycle, so a survivor would have been a solution and not a candidate. The enumeration extends with compute.

**Proposition 1.2** (The canonical tail family is empty: a reciprocity kill)**.** *No solution of #307 has support $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$, for any prime $q$.*

*Proof.* A solution supported on $U$ forces $x^2 = Aq + \Pi_{59}$ with $x = a + b$ and $A = N_{59} + 2\Pi_{59}$ (Remark 1.1). Every prime $\ell \mid A$ exceeds 277: for $\ell \leq 277$ one has $\ell \mid \Pi_{59}$, so $\ell \mid A$ would force $\ell \mid N_{59}$, contradicting $\gcd(N_{59}, \Pi_{59}) = 1$ (rigidity). Hence $\ell \nmid \Pi_{59}$, and reduction mod $\ell$ gives $x^2 \equiv \Pi_{59} \pmod{\ell}$, requiring the Legendre symbol $(\Pi_{59} \mid \ell) \neq -1$. But the Jacobi symbol satisfies

$$\left(\tfrac{\Pi_{59}}{A}\right) \;=\; -1$$

(a finite gcd–speed computation, script `code/tailkill.py`; no factorisation of the 114–digit $A$ is needed), so some $\ell \mid A$ of odd multiplicity has $(\Pi_{59} \mid \ell) = -1$. The family is empty, for every $q$.

The companion symbol at $B = N_{59} - 2\Pi_{59}$ equals $-1$ as well; necessarily: for any admissible base $S$ (writing $D = D_S$, $N = N_S$, $A_S, B_S = N \pm 2D$) one has $A_S \equiv B_S \pmod 8$ (as $4D \equiv 8 \bmod 16$) and $A_S \equiv B_S \equiv N \pmod p$ for every odd $p \mid D$, while the quadratic–reciprocity sign difference carries the exponent $\frac{p-1}{2} \cdot \frac{A_S - B_S}{2}$ with $\frac{A_S - B_S}{2} = 2D$ even; hence $(D \mid A_S) = (D \mid B_S)$ always; one binary certificate per tail family. Running it over all 49,961 admissible bases of Proposition 4.5 proves 31,219 of the one–new–prime families at level 60 empty (62.5%, curiously near 5/8); the remaining 18,742 are Jacobi–inconclusive: their $-1$–Legendre primes, if any, occur to an even count, visible only through a factorisation of $A_S$. A factoring attack on those (trial division and Brent–Pollard $\rho$ over a Montgomery core, `hunt/survivor_kill.rs`; the found factors are kept as a pairwise–coprime basis, so a kill certificate (an odd–multiplicity basis piece $P$ with $(D_S \mid P) = -1$) needs no primality testing at all), pushed to $\rho$–budget $10^6$ with a Pollard $p-1$ stage, closes a further 15,264: in total 46,483 of the 49,961 one–new–prime families at level 60 are proven empty (93.0%). The 3,478 still open resist all of this for a structural reason, not a want of effort: each is Baillie–PSW *composite* yet has no prime factor below $\sim 10^{12}$, so its $A_S$ (or $B_S$) is a *balanced* semiprime of two $\sim$57–digit primes, splitting one is factoring at cryptographic scale. Two cheap tests confirm no shortcut survives: a product–tree batch–GCD over all 37,484 values $A_S, B_S$ finds only shared primes below $10^9$ (already found by $\rho$, so *no* free kills), and Baillie–PSW leaves exactly 34 families in which $A_S$ and $B_S$ are both *probable* primes; these are immune *conditionally on that primality* (a prime $A_S$ with $(D_S \mid A_S) = +1$ has no inert factor, hence no congruence obstruction for any $q$).

The conditionality is not pedantry and not removable by more of the same computation: Baillie–PSW is rigorous only in its *composite* verdicts, so the 46,483 kills are unconditional while the 34 immunities are not. Upgrading them would require primality *certificates* for thirty-four 114–digit integers (ECPP or a Pocklington factorisation of $A_S - 1$). The elementary route does not reach: immunity needs *both* $A_S$ and $B_S$ prime, and partial factorisation of $n-1$ (trial division to $3 \times 10^6$ together with the fully split part) leaves $\log F/\log n$ below $1/3$ on at least one side of every one of the 34, so neither Pocklington ($F > \sqrt{n}$) nor Brillhart–Lehmer–Selfridge ($F > n^{1/3}$) applies to any family: 0 of 34 are certifiable this way (`code/immune_certify.py`). This is now settled. Running the Adleman–Pomerance–Rumely –Cohen–Lenstra test on all 68 integers proves every one of them prime in 8 seconds of wall clock, so the 34 immunities upgrade from [C] to [P] and are *unconditional* (`code/immune_prove.gp`, PARI/GP 2.17.4, whose `isprime` is a proof and not a probable–prime test). The run reproduces the whole chain independently: 49,961 admissible bases, 31,219 Jacobi kills, and, among the 81 bases whose $A_S$ and $B_S$ are both prime, exactly the 34 with $(D_S \mid A_S) = +1$; the other 47 carry $(D_S \mid A_S) = -1$ and were already killed. The primality is moreover backed by *independently checkable* evidence rather than a verdict: `code/immune_certify.gp` emits Atkin–Morain ECPP certificates for all 68 integers, archived at `certs/immune_ecpp.txt` (512 KB) so a third party can check them without trusting the prover. That file is a `primecertexport` dump and is not machine–readable: PARI's `read` parses its header as a polynomial, so `primecertisvalid` cannot consume it, and earlier versions of this paper and of the repository documented that command in error. The working check is `code/immune_cert_verify.gp`, which extracts the 68 asserted subjects and re–proves each by ECPP in 1.3 seconds: 68/68 prime, 0 failures. The structural half of the campaign is now formal: the identity $(D_S \mid A_S) = (D_S \mid B_S)$, which is what makes one binary test decide a family rather than two, is proved in Lean 4 as `Erdos307.jacobiSym_A_eq_B` (`lean/Erdos307/Campaign.lean`), on the three standard axioms and with no `native_decide`; it follows from periodicity, $A_S - B_S = 4D_S$ being exactly the period of $J(a \mid \cdot)$. The primality itself is not formalised and cannot presently be: mathlib's only route to a large prime is `lucas_primality`, which needs a complete factorisation of $n-1$, and that is precisely what the 0 of 34 above rules out. Note the two counts answer different questions and both are correct: 81 is the number of prime pairs, 34 the number of *immune* families among them. A complete kill/immune split of the residual is in any case beyond feasible computation, and the honest state of the single–tail sector is 46,483 *empty* [P], 34 *immune and now unconditionally* [P] *by the* APRCL *run above,* 3,478 *factoring–hard.* □

**Proposition 1.3** (Sector decomposition of level 60)**.** *Let $U$ be a 60–element prime support with $T(U) > 2$. Then exactly one of the following holds:* (i) $T(U) - 2 \geq 1/\max U$*, and* $U = S \cup \{q\}$ *for an admissible base $S$ (one of the* 49,961 *of Proposition 4.5); the* single–tail sector*, or* (ii) $T(U) - 2 < 1/\max U$*, and, writing $p > q$ for the two largest elements and $W$ for the rest,* $T(W \cup \{p\}) < 2$ *and* $T(W \cup \{q\}) < 2$*; the* pair sector*, whose members escape every single–tail family.*

*Proof.* $U$ lies in the tail family of $U \setminus \{r\}$ iff $T(U) - 1/r \geq 2$, which is easiest for $r = \max U$; (i) is that case ($T = 2$ being impossible by rigidity, the base is admissible). In case (ii) no removal reaches 2, and a fortiori the two stated 59–subsets stay below 2. □

**Remark 1.4** (The campaign; completeness of the certificate; the arity barrier)**.** The reciprocity certificate captures *every* $q$–independent congruence obstruction of a tail family: at a prime $\ell \mid A_S B_S$ the system degenerates to the rank–one congruences $x^2 \equiv D_S$ or $y^2 \equiv D_S \pmod{\ell}$, while at $\ell \nmid 2A_S B_S$ the conditions define conics in $(x, q)$ or $(y, q)$ with $\sim \ell$ points, deleting at most half of the residue classes of $q$ and never all (the 2–adic layer pins $q \equiv 5 \bmod 8$ without emptiness); by CRT and Dirichlet, admissible $q$ survive any finite set of such conditions. Hence a tail family admits a congruence kill iff $(D_S \mid \ell) = -1$ for some $\ell \mid A_S B_S$, which the Jacobi symbol detects for 31,219 bases, and partial factorisation (`hunt/survivor_kill.rs`) for a further 15,264: in total 46,483 of the 49,961 single–tail families at level 60 are proven empty (93.0%), 3,478 remaining open (each a balanced ∼57–digit semiprime, factoring–hard). The weapon is intrinsically *arity–one*, and provably fails on the pair sector. With $s = p + q$, $t = pq$, a pair–sector member requires the two simultaneous squares $x^2 = A_W t + D_W s$ and $y^2 = B_W t + D_W s$ (the would–be third condition $s^2 - 4t = \square$ is automatic, being $(p-q)^2$), together with the reciprocity–linked constraints $(D_W p \mid q) = (D_W q \mid p) = 1$ (mod $q$, a solution with $q \mid a$ has $N_U \equiv D_W p$, forcing $D_W p$ to be a square). At a prime $\ell \mid A_W$ the condition $x^2 \equiv D_W s \pmod{\ell}$ constrains only $s$, which the free second prime absorbs; the pair system is in fact locally soluble at *every* modulus (a theorem, Proposition 1.5 below, corroborated by a direct scan over all $\ell \le 600$) so, unlike the single–tail family, the pair sector admits *no* congruence obstruction at all. It is a genuine two–parameter surface per base, #307 in miniature (Remark 1.7); reciprocity cannot touch it, and only a direct both–squares search or a descent argument can.

The sector is nevertheless an explicit finite object, and small. A level–60 support $U$ is covered by the one–new–prime campaign exactly when $T(U \setminus \{\max U\}) > 2$, the bases of Proposition 4.5 being the 59–prime sets of mass exceeding 2; the complement is $U = R \cup \{m\}$ with $|R| = 59$, $m = \max U$ and $T(R) \le 2 < T(R) + 1/m$. The mass ladder at $k = 60$ puts every element of $R$ below $793.67 \times 2 = 1587.4$, so the complement is a search over 59–subsets of the 250 primes under 1588 with $T(R) \in (2 - 1/\max R,\, 2]$. There are exactly 18,234,653 of them, the largest $\max R$ being 1583 (`code/pairsector_count.rs`, corroborated by an independent binned counting DP in `code/pairsector_countdp.rs`). Each is an arity–*one* tail family in $m$: Proposition 1.5 denies a congruence kill to the family with both new primes free, not to these, so the $q$–independent certificate does apply to each of the 18.2 million bases. Whether it kills them is not settled here; the point is that the pair sector is a finite target of measured size against which the existing weapon is admissible.

**Proposition 1.5** (The pair sector admits no congruence kill)**.** *Let $W$ be any pair–sector base and $A = A_W$, $B = B_W$, $D = D_W$, so that a pair–sector solution requires $x^2 = A\,\alpha\beta + D(\alpha + \beta)$ and $y^2 = B\,\alpha\beta + D(\alpha + \beta)$ with $\alpha, \beta$ the two new primes. For every prime power $\ell^j$ this system has a solution in units $\alpha, \beta$ and integers $x, y$ at a nonsingular point. Consequently, by CRT and Dirichlet, no finite system of congruence conditions empties any pair–sector family: the arity barrier of Remark 1.4 is a theorem, and the certificate programme provably ends at arity one.*

Formalised, in part. *`Erdos307.no_pinning_isUnit` is the step that makes the tail–kill mechanism unavailable, `collapse_identity` is regime* (2)*'s observation that the two square conditions are one, `regime_one_first` and `regime_one_second` are the explicit unit of regime* (1)*, and `targets_agree_mod_eight` is the mod–8 collapse of regime* (3)*. The lift from $\ell$ to $\ell^j$ that regime* (1) *needs is `square_lifts_to_prime_powers`, proved from `hensel_all_powers`: if $\ell$ is an odd prime not dividing $c$ and $c$ is a square modulo $\ell$, it is a square modulo every power of $\ell$. Regime* (2) *carried a citation to Weil's bound in earlier revisions; it does not need one. Its two steps are `Erdos307.pairlocal_inner_sum`, the evaluation of the inner sum of $S_{fg}$ as $-\chi\big((A\alpha{+}D)(B\alpha{+}D)\big)$ by the quadratic complete sum of Proposition 1.9, with its side condition discharged as $4D^2\alpha^2 \ne 0$; and `pairlocal_count_pos`, the arithmetic turning the resulting $O(\ell)$ bounds into a positive count from $\ell \ge 17$. 0 `sorry`* (`lean/Erdos307/PairLocal.lean`, `lean/Erdos307/PairLocalCount.lean`)*. Checked directly: over* 872 *random bases with $A - B = 4D$ and every odd $\ell \le 400$, the count of admissible $(\alpha, \beta)$ was positive in every case and tracks $(\ell-1)^2/4$; the crude bounds $|S_f|, |S_g| \le 2\ell$, $|S_{fg}| \le 3\ell$ held in all* 409 *cases tested* (`code/pairlocal_count.gp`)*.*

*Proof.* First, no $\ell$ pins either target to a constant: that needs $\ell \mid A$ (or $B$) *and* $\ell \mid D$, whence $\ell \mid N$, contradicting $\gcd(N, D) = 1$ (rigidity); so the tail–kill mechanism is unavailable, and solubility is decided in three regimes.

*(1) $\ell$ odd, $\ell \mid D$.* This covers every odd $\ell \leq 103$: at level 60, omitting a prime $p$ caps $T(U)$ at $T_{61} - 1/p$, and $1/p > T_{61} - 2 = 0.00944\ldots$ exactly for $p \leq 103$, so all such primes lie in $U$, hence in $W$ (the two adjoined primes are the largest). Take $\beta = 1$, $x = 1$, $\alpha = (1 - D)/(A + D) \bmod \ell^j$, a unit since $A + D \equiv N \not\equiv 0 \pmod{\ell}$: the first equation holds by construction, and the second target $(B + D)\alpha + D \equiv N \cdot N^{-1} = 1 \pmod{\ell}$ is a unit congruent to a square, so $y$ exists modulo $\ell^j$.

*(2) $\ell$ odd, $\ell \nmid 2D$* (in particular every $\ell \geq 107$ outside $W$). The two conditions are secretly *one*: setting $m = (x^2 - y^2)/4D$ and $u = (x^2 - Am)/D$, the relation $A - B = 4D$ gives the polynomial identity $Bm + Du = y^2$. Only *existence* is wanted, and counting the pairs $(\alpha, \beta)$ directly settles it with no curve and no Weil bound. Write $\chi$ for the quadratic character, $f = A\alpha\beta + D(\alpha + \beta)$ and $g = B\alpha\beta + D(\alpha + \beta)$. Summing $\frac{1}{4}(1 + \chi(f))(1 + \chi(g))$ over $\alpha, \beta \in \mathbb{F}_\ell^\times$ and discarding the at most $2(\ell - 1)$ pairs with $fg = 0$,

$$4\,\#\{\text{both nonzero squares}\} \;\geq\; (\ell - 1)^2 + S_f + S_g + S_{fg} - 8(\ell - 1).$$

For fixed $\alpha$ each target is *linear* in $\beta$, so the inner sums of $S_f$ and $S_g$ vanish by the linear evaluation above, leaving $|S_f|, |S_g| \leq 2\ell$. And $fg$ is a *product of two linear forms* in $\beta$, so its inner sum is the quadratic evaluation, whose side condition $ad \neq bc$ is here $D\alpha(A - B)\alpha = 4D^2\alpha^2 \neq 0$ — exactly the hypotheses $\ell \nmid 2D$ and $\alpha \neq 0$; summing the resulting $-\chi\big((A\alpha + D)(B\alpha + D)\big)$ over $\alpha$ is that same evaluation again, so $|S_{fg}| \leq 3\ell$. Hence the count is positive as soon as $(\ell - 1)^2 - 7\ell - 8(\ell - 1) > 0$, i.e. from $\ell \geq 17$, far below the $\ell \geq 107$ in play. The point is nonsingular ($\alpha\beta \neq 0$; the Jacobian contains $\mathrm{diag}(2x, 2y)$), so it lifts to every $\ell^j$.

*(3) $\ell = 2$.* Both targets are odd and agree modulo 8 (as $\alpha\beta$ is odd, $D(\alpha + \beta) \equiv 0 \bmod 4$, and $(A - B)\alpha\beta = 4D\alpha\beta \equiv 0 \bmod 8$), and a finite table over ($A \bmod 8$, $D \bmod 8$) shows that for every base some achievable pair $(\alpha\beta, \alpha + \beta) \bmod 8$ makes the common target $\equiv 1 \pmod{8}$; a 2–adic square, soluble modulo $2^j$ for every $j$.

All three regimes, the collapse identity, the threshold $1/103 > T_{61} - 2 > 1/107$, and the mod–32 layer were verified computationally on three bases and every $\ell \leq 600$ (`code/pairlocal.py`; the measured density of admissible $(x, y)$ in regime (2) is 0.497–0.503 against the predicted $\frac{1}{2}$). $\square$

**Remark 1.6** (Anatomy of the certificate; the empirical 5/8 law)**.** Writing $D = 2D_1$, the reciprocity signs in the certificate cancel ($\frac{A-1}{2} + \frac{N-1}{2} = N - 1 + 2D_1$ is even), leaving the clean product

$$\left(\tfrac{D}{A}\right) \;=\; \left(\tfrac{2}{A}\right)\left(\tfrac{D_1}{N}\right), \qquad \left(\tfrac{2}{A}\right) = +1 \iff N \equiv 3, 5 \pmod{8},$$

and one layer down $(N \mid D_1) = \prod_{p \mid D_1}\left(\frac{D/p}{p}\right)$, since $N \equiv D/p \pmod{p}$: the cofactor structure of rigidity recurs inside its own certificate. This product evaluates in closed form: with $D/p = 2D_1/p$ and quadratic reciprocity,

$$(N \mid D_1) = (2 \mid D_1) \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{s + \binom{t}{2}}, \quad s = \#\{p \mid D_1 : p \equiv 3, 5\ (8)\},\ t = \#\{p \mid D_1 : p \equiv 3\ (4)\},$$

a Rédei genus character of $D_1$, *independent of $N$*. This is forced, not merely observed: $(N \mid D_1) = \prod_{p \mid D_1}(N \mid p)$, and cofactor survival gives $N \equiv D/p = 2D_1/p \pmod{p}$, so $(N \mid p) = (2 \mid p)\,(D_1/p \mid p)$. The first factors multiply to $(-1)^s$ since $(2 \mid p) = -1$ exactly for $p \equiv 3, 5 \pmod{8}$; and, each unordered pair $\{p, q\}$ occurring once in each of the two cofactors,

$$\prod_{p \mid D_1}\left(\frac{D_1/p}{p}\right) = \prod_{\{p,q\} \subset D_1}\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{\binom{t}{2}},$$

by quadratic reciprocity, since $\frac{p-1}{2}$ is odd exactly for $p \equiv 3 \pmod 4$. (Checked independently, prime by prime, on 20,000 bases: 0 violations.) The inner symbol is thus not a coin but a genus invariant; its empirical fairness (49.9%, 50.0% over the 49,961 bases) is that character's equidistribution; and $(D_1 \mid N)$ follows by the reciprocity sign $(-1)^{\frac{D_1-1}{2}\frac{N-1}{2}}$, hence depends on $N$ only through $N \bmod 4$. In fact the entire certificate closes into a finite functional. Let $v(S) = (n_1, n_3, n_5, n_7) \bmod 4$ count the odd primes of $S$ in the residue classes $1, 3, 5, 7$ modulo 8. Every factor above is a function of $v$ alone: $s \equiv n_3 + n_5$, $t = n_3 + n_7$ (with $\binom{t}{2}$ needing only $t \bmod 4$), $D_1 \equiv 3^{n_3}5^{n_5}7^{n_7} \pmod 8$, $S(D_1) \equiv (n_1+n_5) - (n_3+n_7) \pmod 4$, and (by the mod–8 law of Proposition 5.14, $D_1' \equiv D_1 S(D_1) \pmod 8$) also $N = D_1 + 2D_1' \equiv D_1 + 2\big(D_1 S(D_1) \bmod 4\big) \pmod 8$. Hence

$$(D \mid A) \;=\; K\big(v(S)\big) :$$

*the kill is decided by sixteen residue counts modulo* 4. The realized vectors are exactly the coset $n_1 + n_3 + n_5 + n_7 \equiv 58 \equiv 2 \pmod 4$ (all 64 of its classes occur), $K$ is constant on each, and the functional reproduces the direct 113–digit Jacobi computation on every one of the 49,961 bases (`code/kill_classvector.py`). The five–eighths law is then an exact finite statement, and its mechanism is the *reciprocity switch*: for $N \equiv 1 \pmod 4$ the reciprocity sign is inert and the kill reduces to the genus coin; exact conditional rate $12{,}699/24{,}975 = 0.5085$; while for $N \equiv 3 \pmod 4$ the sign couples the parity of $t$ into the product and the rate jumps to $18{,}520/24{,}986 = 0.7412$ (by class modulo 8: 0.509, 0.741, 0.508, 0.741 at $N \equiv 1, 3, 5, 7$; the 3/4 classes are $N \equiv 3 \pmod 4$, not the $(2 \mid A)$ classes). The blend $\frac{1}{2} \cdot \frac{24975}{49961} + \frac{3}{4} \cdot \frac{24986}{49961} = 0.62503$ against the exact $31{,}219/49{,}961 = 0.62487$ is the whole of the "coincidence". The correlation is an ensemble property, a random–subset proxy gives ≈0.45/0.55; and the functional yields no second certificate: it recomputes the same symbol.

**Remark 1.7** (The kill–program is one–sided)**.** Every kill is a proof that no solution hides in that family, but if #307 has a solution, the solution's own family is unkillable, so no accumulation of kills can overshoot the truth: the program's asymptote *is* the existence question. Together with the self–similarity of Remark 1.1 (each rung past 60 contains the whole problem), this locates the campaign precisely: it corners where a solution could live, one certified region at a time, and can terminate only if #307 has answer NO, which the heuristics disfavour.

**Remark 1.8** (The reciprocity kills are the local factors of the expected count)**.** The divergent naive count of Section 2 and the reciprocity sieve of Proposition 1.2 are the *same* computation. Decompose the expected number of solutions over tail families $S \cup \{q\}$: the summand for a base $S$ is $\kappa_S \left(D_S\sqrt{\text{mass}^2 - 4}\right)^{-1} \sum_q q^{-1}$, where $\kappa_S = \prod_\ell(\text{local solubility density at } \ell)$ is exactly the reciprocity data. A reciprocity–killed base carries an inert $\ell \mid A_S$ with $(D_S \mid \ell) = -1$, so $N_S+2D_S \equiv D_S \pmod \ell$ is a non–residue for *every* $q$, forcing $\kappa_S = 0$; the family drops from the count entirely. An immune base ($A_S, B_S$ both prime, both residues) has no inert prime, so $\kappa_S > 0$ and its summand is a divergent $\sum_q q^{-1}$ (verified on an explicit 114–digit immune base: local density 1 at every tested prime, partial sums growing without bound). Hence the campaign of Remark 1.4 is a Selberg–sieve evaluation of the singular series: the kills remove ∼93% of the count's mass, and the residual log–divergence (the entire "weakly favours YES" signal) is carried by the 34 conditionally immune families together with the pair sector. (Conditionally, because their immunity rests on the Baillie–PSW primality of $A_S, B_S$; the pair sector's contribution, by contrast, is unconditional, being a theorem, Proposition 1.5.) This does not touch the reliability of the count (which ignores the $N(r) \le 1$ rigidity), but it pins down *where* the heuristic's YES lives.

**Proposition 1.9** (Local completeness of the reciprocity sieve)**.** *For an admissible base $S$ and an odd prime $\ell$, the admissible tail residues $Q_\ell = \{q \bmod \ell : (A_S q + D_S \mid \ell) \neq -1$ and $(B_S q + D_S \mid \ell) \neq -1\}$*

*fall into exactly three regimes.* (i) $\ell \mid D_S$, *which covers every* $\ell \le 167$ *uniformly over the* 49,961 *admissible bases, since* $1/p \ge (T_{59} - 2) + \frac{1}{281}$ *forces every prime* $p \le 167$ *into every base: then* $A_S \equiv B_S \equiv N_S \not\equiv 0 \pmod{\ell}$*, the two conditions collapse to the single condition* $(N_S q \mid \ell) \ne -1$*, and* $|Q_\ell| = (\ell+1)/2 \ni 0$*.* (ii) $\ell \mid A_S B_S$*: one condition is the constant* $(D_S \mid \ell)$*, so* $Q_\ell = \emptyset$ *iff* $(D_S \mid \ell) = -1$ *(exactly the certificate of Proposition 1.2) and otherwise* $|Q_\ell| \ge (\ell-1)/2$*.* (iii) $\ell \nmid A_S B_S D_S$ *(hence* $\ell \ge 173$*): the complete character sums give* $|Q_\ell| \ge (\ell-3)/4 - 2 \ge 40$ *(the linear sums vanish, the quadratic sum is* $-\chi(A_S B_S)$*, boundary terms* $\le 2$*). (Regimes* (i) *and* (ii) *are formalised:* `Erdos307.regime_one_collapse`*,* `regime_one_same_argument` *and* `regime_two_constant` *(*`lean/Erdos307/LocalComplete.lean`*). Regime* (iii)*'s counting step is* `regime_three_count`*, and its two complete character–sum hypotheses are now theorems rather than citations:* $\sum_q \chi(aq+b) = 0$ *for* $a \ne 0$ *is* `Erdos307.sum_quadraticChar_affine`*, the reindexing* $q \mapsto aq+b$ *followed by* `quadraticChar_sum_zero`*; and* $\sum_q \chi((aq+b)(cq+d)) = -\chi(ac)$ *for* $ad \ne bc$ *is* `Erdos307.sum_quadraticChar_quad` *proved by factoring out* $\chi(ac)$*, translating to* $\sum_u \chi\big(u(u+e)\big)$ *with* $e \ne 0$*, writing* $u(u+e) = u^2(1+e/u)$ *for* $u \ne 0$ *so that the character value is* $\chi(1+e/u)$*, and reindexing by the involution* $u \mapsto e/u$ *of the nonzero elements to reach* $\sum_{t \ne 0} \chi(1+t) = 0 - \chi(1) = -1$*. Both hold over any finite field of odd characteristic.* 0 *sorry* (`lean/Erdos307/CompleteSums.lean`)*. Checked against direct evaluation for all odd primes* $\ell \le 59$*: the linear sum over every* $(a,b)$ *with* $a \ne 0$*, and the quadratic sum over all* 33,008,952 *quadruples with* $ad \ne bc$ *— 0 violations* (`code/complete_sums_check.gp`).) *Consequently* $Q_\ell \ne \emptyset$ *forces* $|Q_\ell| \ge 2$ *at every modulus, the surviving local density is positive (e.g.* $\prod_{\ell \le 300} |Q_\ell|/\ell \approx 2.2 \times 10^{-19} > 0$ *on the canonical base), and by CRT* no finite system of congruences (no covering) can annihilate an unkilled family*: the binary certificate of Proposition 1.2 is the complete local theory of the tail sector, there is no "second kill layer," and the one–sidedness of Remark 1.7 is structural rather than an artefact of method. (All three regimes verified exhaustively on the canonical base: collapse and* $|Q_\ell| = (\ell+1)/2$ *at* $\ell \in \{3,5,7,101,277\}$*; the Weil floor at every prime* $281 \le \ell \le 600$*;* $A_S, B_S$ *carry no factor below* $10^6$*, consistent with the balanced–semiprime wall of Remark 1.4.)*

**Proposition 1.10** (What a barrier improvement must supply)**.** *Let* $(a,b)$ *be a solution with* $a < b$ *and put* $t = b/a$*, so that* $t = \sigma(a)$ *and, by* $\sigma(a)\sigma(b) = 1$*,*

$$\sigma(U) \;=\; \sigma(a) + \sigma(b) \;=\; t + \tfrac{1}{t}.$$

*For* $n$ *with* $T_n > 2$ *let* $t_n > 1$ *be the root of* $t + 1/t = T_n$*. Since* $\sigma(U) \le T_{|U|}$*,*

$$\sigma(a) > t_n \implies |U| \;\ge\; n+1,$$

*and this is the only route by which a mass argument can raise the barrier. Numerically*

| $n$ | 59 | 60 | 61 | 62 | 63 | 64 |
|---|---|---|---|---|---|---|
| $T_n$ | 2.0023502 | 2.0059089 | 2.0094424 | 2.0128554 | 2.0161127 | 2.0193282 |
| $t_n$ | 1.0496677 | 1.0798804 | 1.1020081 | 1.1199914 | 1.1352477 | 1.1490254 |

So the barrier level is decided entirely by how close $\sigma(a)$ may lie to 1, and each further level demands a definite gap: $|U| \ge 61$ needs $\sigma(a) > 1.0799$, a full 8% above 1.

That is why the mass mechanism is exhausted here rather than merely difficult. The only unconditional lower bound on the gap is $\sigma(a) - 1 = (a' - a)/a \ge 1/a$, which is worthless at $a > 10^{56}$; and $\sigma$ genuinely comes close to 1 from above on very few primes, the Sylvester-type set $\{2,3,7,41\}$ giving $\sigma = 1.00058\ldots$ with four. Indeed no bound in $\omega(a)$ alone can exist: the greedy Sylvester construction gives, for each $k$, squarefree $a$ with $\omega(a) = k$ and $0 < |\sigma(a) - 1|$ doubly exponentially small in $k$ — from below $1 - \sigma = 7.8 \times 10^{-592}$ at $k = 12$, and from above

$\sigma - 1 = 4.6 \times 10^{-591}$ at the same $k$ (`code/sigma_gap.gp`). So $|\sigma(a) - 1| \geq f(\omega(a))$ is false for every positive $f$, and the unconditional $1/a$ is of the right shape. Nothing in the cycle equations forbids $a' - a = 1$. A barrier past 60 must therefore come from a mechanism that bounds $\sigma(a)$ away from 1 without appealing to the size of $U$ — which is what Theorem 4.3 does in the opposite direction — and we know of none.

The same reduction locates the extreme case. Writing $c = b - a$, the identity $\sigma(U) - 2 = (t-1)^2/t = c^2/D$ is the minus layer $D' - 2D = c^2$, and $\gcd(c, D) = \gcd(b - a, ab) = 1$ because $\gcd(a, b) = 1$. Since an admissible support contains every prime $\leq 167$ (Remark 1.1), $c$ is divisible by none of them, so $c = 1$ or $c \geq 173$. The branch $c = 1$ is $D' = 2D + 1$, equivalently $a' = a + 1$ and $(a+1)' = a$; it is the closest a solution could come to $\sigma(a) = 1$, and it is exactly the configuration a mass argument cannot reach. Below $3 \times 10^6$ the equation $a' = a + 1$ has only the solutions 30, 858, 1722 and 66,198, and none extends: for the first three $a + 1$ is prime, so $(a + 1)' = 1$, and for the last $a + 1$ is not squarefree. The branch is not excluded, only unsearched, since the barrier puts it above $10^{56}$. The level–59 computation that gives $|U| \geq 60$ is, in these terms, an exhaustive verification that $\sigma(a) > t_{59}$, and Proposition 4.8 says the same verification is unavailable one level up.

**Remark 1.11** (The Lehmer reading of the tail is withdrawn)**.** Remark 1.16 below reads the surviving question as the primality of a term $q_n$ of the *exponential* Pell orbit, and Remark 6.26 names that orbit as the only live branch. Proposition 4.7 withdraws the reading. The orbit is genuinely infinite and both square conditions do hold along all of it; what fails is that no term past $N$ can be prime. The single identity responsible is $A - B = 4D$, which the earlier treatment used only to form the Pell system and never to factor $4Dq$.

**Remark 1.12** (The threshold in largest–prime coordinates, and which sectors it leaves alive)**.** Remark 1.1 already writes a cycle as $a = qm$ with $q$ the largest prime; Bado [27] makes the same reduction exact and independent of the level. With $q = P^+(ab)$, $q \mid a$, write $a = qd$, $b = e$: then $e = d + qd'$, $e' = qd$, and $q = (e - d)/d'$. Here $\sigma(a) = 1/q + \sigma(d)$, so $t$ — or $1/t$, according to which side carries $q$; $t + 1/t$ is symmetric and the reading below is unaffected — is $\sigma(d)$ up to $1/q$, and Proposition 1.10 becomes a statement about the *sector* $d$. It is very uneven. (The rows below take $q$ astronomically large, which Proposition 4.5 supplies once $d$ is small: at $q = 281$ the first row would read 1295.)

| $d$ | $\sigma(d)$ | $\sigma + 1/\sigma$ | $\lvert U\rvert \geq$ |
|---|---|---|---|
| $2$ | 0.500000 | 2.500000 | 1413 |
| $2 \cdot 3$ | 0.833333 | 2.033333 | 69 |
| $2 \cdot 3 \cdot 7$ | 0.976190 | 2.000581 | 59 |
| $2 \cdot 3 \cdot 7 \cdot 43$ | 0.999446 | 2.000000 | 59 |
| $2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$ | 0.999979 | 2.000000 | 59 |
| $3$ | 0.333333 | 3.333333 | $\sim 1.1 \times 10^8$ |

The last row is a Mertens estimate, the exact $K$ being past $\pi(10^6)$. Inverting $t + 1/t \leq T_K$ in the same convention gives the windows: $|U| \leq 60$ forces $\sigma(d) \in [0.9260, 1.0799]$ — the $t_{60}$ of the table above — and $|U| \leq 68$ forces $\sigma(d) \in [0.8372, 1.1945]$ (`code/sector_window.gp`). So a sector can be cheap only if $\sigma(d)$ sits within a few per cent of 1. The primary pseudoperfect numbers, having $\sigma(n) = 1 - 1/n$ exactly, are among those extremal sectors — they are not the only ones, the Sylvester–type $\{2, 3, 7, 41\}$ above being another — and by Corollary 8.7 they are the $u = 1$ solutions of #307's coprime relaxation, so Erdős #313 sits at the floor of this barrier as well.

The reduction is not new to the tail theory: it *is* the parametrisation of Proposition 4.7, with $\alpha = d$ and $\beta = e$. Substituting $e = d + qd'$ and $e' = qd$ into $\alpha^2 q^2 - Nq + (\beta^2 - D)$ at $\alpha = d$ gives $e(e - d - qd') = 0$ identically (`code/alpha_is_d.gp`; the discriminant identity is $4d^2$ times the same vanishing, not a second fact). Two things follow. Everything proved about tail families is a statement about sectors: the reciprocity kill of Proposition 1.2, and the factoring wall behind the families it leaves open. And the sector $d = 2$ is $\alpha = 2$, which the table puts at $|U| \geq 1413$, so the level–60 programme never meets it. Negatively, the reduction does not shorten that residual: the unknown is the divisor $\alpha \mid D$, of which there are $2^{59}$, and the only $\alpha$–independent certificate is the Jacobi one already used.

**Remark 1.13** (The sector barrier sharpened by $d'$, and a second derivation of 60)**.** The sector reading above uses only the mass. It can be sharpened, and the sharpening is what Bado [6] exploits at $d = 42$. Since $e = d + qd'$, one has $\gcd(e, d) = \gcd(qd', d) = 1$ and $\gcd(e, d') = \gcd(d, d') = 1$, so

$$\operatorname{supp}(e) \text{ avoids } \operatorname{supp}(d) \cup \operatorname{supp}(d') \qquad \text{in every sector,}$$

and the mass $\sigma(e)$, which is $1/\sigma(d)$ to many places, must be carried by primes drawn from a thinned set. Two further identities come with it, both immediate from $e' = qd$: $de - d'e' = d^2$ and $\sum_{r|e} 1/r = d/d' - d^2/(d'e)$; and the associated defect $\Delta_d(R) = dR - d'R'$ satisfies $\Delta_d(Rr) = r\Delta_d(R) - d'R$ with terminal value $\Delta_d(e) = d^2$ (`code/sector_general.gp`, symbolic in $d, d', q, R, R'$). When $d$ is primary pseudoperfect, $d' = d - 1$; that is the only reason those sectors look distinguished.

Running the resulting bound sector by sector — mass $d/d'$ from primes avoiding $\operatorname{supp}(d) \cup \operatorname{supp}(d')$, together with the parity constraint that $d$ even forces $e$ odd and $\omega(e)$ even — gives

$$\min_{d} \big(1 + \omega(d) + \omega(e)\big) \;=\; 60,$$

over all sectors $d$ built from primes up to 47, attained at $d = 15470 = 2 \cdot 5 \cdot 7 \cdot 13 \cdot 17$ and at many others, among them the primary pseudoperfect $47058 = 2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$ (`code/sector_dprime.gp`). This is Theorem 4.3's constant recovered by a route sharing nothing with its proof: that one is an exhaustive computation over the 49,961 admissible 59–prime supports, this one a largest–prime descent and a reciprocal–mass count. It does not improve the barrier — it bottoms out at exactly 60 — but two independent derivations of the same constant are worth recording, and the sectors where it is tight are where any improvement must act.

At $d = 42$ the mass bound alone gives $\omega(e) \geq 62$ and $|U| \geq 66$; [6] obtains $\omega(e) \geq 64$ and $|U| \geq 68$ by adjoining the residue conditions $\sum_{r|e} r^{-1} \equiv 0 \pmod 3$ and $\pmod 7$ and $\prod_{r|e} r \equiv 1 \pmod{41}$ and maximising over admissible supports. We have re-run that certificate independently, obtaining the same maximum and the same 51,281 reachable states. The residue sieve cannot be pushed further as a sieve: the complete system of local conditions is the sector equation itself. For each $r \mid e$, reducing $e' = qd$ modulo $r$ gives $\prod_{s|e,\, s \neq r} s \equiv -d^2/d' \pmod r$, one condition per prime of $e$; conversely these, with $e \equiv d \pmod{d'}$ and $1 < \sigma(e) < 2$, force $e' = qd$ exactly, both sides lying in $(e, 2e)$. Bado's conditions modulo 3 and 7 are the sums of these; the maximiser certifying his bound violates them at five of its six smallest primes, and adjoining the three at $5, 11, 13$ to the dynamic programme lowers the 64–prime maximum from 1.03032 to 1.02853, still above 42/41 (`code/sector42_znam_violations.gp`, `code/sector42_znam_dp.rs`). The next step is therefore not a sieve but an exact enumeration, and it goes through.

**Proposition 1.14** (The $d = 42$ sector at 64 primes)**.** *No solution of #307 has $a = 42q$ with $q$ prime and $\omega(b) = 64$. Hence in that sector $\omega(b) \geq 66$ and $|P \cup Q| \geq 70$.*

*Proof (finite verification).* Write $e = b$, $T = 42/41$, and let $A_1 < A_2 < \cdots$ be the primes other than $2, 3, 7, 41$, $S_j = \sum_{i \le j} 1/A_i$. By the sector identity, $\sigma(e) = T - 1764/(41e)$, and $e > 3.5 \times 10^{57}$ by Theorem 4.3, so $T - 10^{-55} < \sigma(e) < T$. Exactly, $A_{64} = 337$, $A_{197} = 1229$, $A_{198} = 1231$, and $S_{61} + 3/1231 < T$ (`code/sector42_k64.gp`). If $m$ of the 64 primes of $e$ exceed 1229 then $\sigma(e) \le S_{64-m} + m/1231$, which for $m \ge 3$ is below $S_{61} + 3/1231 < T - 10^{-55}$; so $m \le 2$. Next, if $\mathrm{supp}(e) = S \cup \{\ell\}$ with $R = \prod S$, the identity $42e - 41e' = 1764$ reads $\ell\,\Delta_{42}(R) = 41R + 1764$, so $\Delta_{42}(R) > 0$ and $\ell$ is the integer $(41R + 1764)/\Delta_{42}(R)$: the last prime is determined by the others.

*Case $m \le 1$.* Take $\ell$ the prime exceeding 1229 if $m = 1$, else the largest prime of $e$, so $\ell \ge A_{64}$. The other 63 primes $S$ lie among $A_1, \ldots, A_{197}$ with $\sigma(S) = \sigma(e) - 1/\ell \in [T - 1/A_{64} - 10^{-55}, T)$. The enumeration tests 36,901,596,763 sets, every such $S$ among them; for none is $(41R+1764)/\Delta_{42}(R)$ an integer.

*Case $m = 2$.* Let $\ell_2 \le \ell_1$ be the two primes exceeding 1229, $S$ the other 62, and $\delta = T - \sigma(S)$. Then $1/\ell_1 + 1/\ell_2 \in (\delta - 10^{-55}, \delta)$, so $\delta < 2/1231$ and $\ell_2 \in (1/\delta,\, 2/\delta + 1)$. The enumeration tests 166,213 sets, every such $S$ among them; the least $\delta$ among them is $1.539 \times 10^{-6}$, so every $\ell_2$ range is finite (largest upper bound 1,299,387). For each $S$ and each prime $\ell_2$ in its range, $S \cup \{\ell_2\}$ determines $\ell_1$ as above; it is never an integer.

The enumeration prunes on the mass in floating point with a margin of $10^{-9}$ on both ends of each window, so no set in a window is lost; the integrality test is a double–double prefilter with a conservative band followed by exact integer arithmetic (7,454,365 and 2,495 exact tests respectively), the integer layer checked against PARI to the digit (`code/sector42_k64.rs`, two runs with independent big–integer implementations). The non–enumerative half of the argument — the forced–prime identity and its converse, the sign condition, the case–split inequality and the parity step — is formalised in `lean/Erdos307/Sector42.lean`, with the two finite computations appearing as explicit hypotheses of the concluding statement. So $\omega(e) \ne 64$. By [6], $\omega(e) \ge 64$ and $\omega(e)$ is even ($e$ is odd and $e' = 42q$ is even); hence $\omega(e) \ge 66$, and $|P \cup Q| = |\{2, 3, 7, q\}| + \omega(e) \ge 70$. □

**Remark 1.15.** The same argument decides $\omega(e) = 66$ in principle, but with up to four primes beyond the truncation the nested ranges multiply, and the enumeration is no longer a twenty–minute job; we have not run it. What the proposition shows is where the sector programme actually stops: not at the residue sieve, whose limit is 68, but at the cost of an exact enumeration in which the last one or two primes are forced by the rest. The two–large–prime case, a factoring problem in general ($(\Delta\ell_1 - 41R)(\Delta\ell_2 - 41R) = 1681R^2 + 1764\Delta$), is finite here only because the smaller of the two is trapped in $(1/\delta, 2/\delta)$ and $\delta$ is bounded below over a finite list.

Three consequences, in increasing order of what they cost.

*The gate is finite, not Lehmer.* The tail primes of a base lie in the explicit finite set $\{2(N \pm k)/m^2 : m \mid 4D,\ k^2 = AB + Dm^2\}$. So the second of the three gates is a search over the divisors of $4D$, not a primality question along an exponential sequence, and the classification of the residual difficulty as Lehmer–class rather than Schinzel–class does not survive. Checked sound *and complete* against exhaustive search on 1200 bases with zero mismatches, and stepped out along a real orbit, in `code/tailbound.gp`: on the base $D = 1$, $N = 7$ (bound $N = 7$) the orbit gives $q_1 = 7$ prime, then $q_2 = 741895$, $q_3 = 76921173511$, and every later term composite, with both square conditions holding throughout.

*The count reverses sign, and the search has now been run.* The expected number of admissible $m$ is $\sum_{m|4D}(AB + Dm^2)^{-1/2} \approx D^{-1/2}\sum_{m|4D} m^{-1}$, which on the first immune base is $2.4 \times 10^{-56}$ and below $10^{-38}$ summed over all 34 immune families, the opposite of the "weakly favours YES" of Remark 1.8 on exactly the families that reading was about. That prediction is now checked. The full divisor space is $4 \cdot 2^{58} \approx 1.15 \times 10^{18}$ per family and is not enumerable, but because the hit probability falls like $1/(m\sqrt{D})$ the expected count is carried by small $m$: restricting to $\omega(m) \le 8$ leaves 99.9998% of the expected count. Over all 34 families that slice is $3.0777 \times 10^{11}$ candidates, and

$$\text{it contains no } m \text{ with } AB + Dm^2 \text{ a perfect square}$$

(4,933 survive the modular cascade, exact `issquare` kills all of them; the earlier $\omega(m) \le 7$ pass, $4.7081 \times 10^{10}$ candidates and 771 survivors, agrees).

The same search has been run over *every* admissible base, which matters because Proposition 4.7 requires no primality of $A_S$ or $B_S$, only the algebra. The 34 immune families are the ones left after the Jacobi kill and the factoring attack, and their immunity is conditional on a Baillie–PSW verdict; the other 3,444 of the 3,478 open families are open precisely because $A_S$ or $B_S$ is a balanced semiprime nobody can split. The tail search does not care. Over all 49,961 admissible bases at $\omega(m) \le 4$, which is 99.2% of the expected count per base, the run is $9.1296 \times 10^{10}$ candidates, 1,469 survive the modular cascade, and exact `issquare` kills all 1,469. Every survivor was additionally checked to satisfy $m \mid 4D$, the hypothesis the proposition needs. So

$$\text{no admissible base admits a Pythagorean tail prime with } \omega(m) \le 4,$$

unconditionally and with no primality input. Where Proposition 1.2 leaves 3,478 of the one–new–prime families at level 60 open, of which 34 only conditionally immune, this leaves none open on the stated slice. It is a slice result and not a proof: the untested remainder is 0.8% of the expected count per base, and $\omega(m) \ge 5$ is untested. The run is `code/tailsearch.rs` (a cascade of quadratic–residue filters modulo primes coprime to $2D$, 771 survivors) with every survivor then tested exactly by `issquare` in `code/tailsearch_verify.gp` (0 squares). A positive control recovers the known solutions of the toy base $D = 1$, $N = 7$, and an exhaustive exact run at $\omega(m) \le 2$ agrees with the filter, so the cascade is not silently over–rejecting.

It is worth recording what the slice covers in the other measure, since the two are thirteen orders apart. The parametrisation runs over $m \mid 4D$ with $m$ even, so $m = 2\alpha$ with $\alpha \mid D$, and at level 59 that is $2^{59} = 5.7646 \times 10^{17}$ divisors. Splitting on the parity of $\alpha$, the divisors with $\omega(m) \le 4$ number $2\sum_{j\le 3}\binom{58}{j} = 65{,}136$, and those with $\omega(m) \le 8$ number $2\sum_{j\le 7}\binom{58}{j} = 692{,}376{,}800$. As fractions of the space the bound $q \le N$ actually proves finite, these are

$$1.130 \times 10^{-13} \qquad \text{and} \qquad 1.201 \times 10^{-9},$$

against the 99.2% and 99.9998% of the *expected count* quoted above. Both figures are correct and they measure different things: the expected count concentrates on small $\omega$, the hit probability falling like $1/(m\sqrt{D})$, so a slice can carry almost all of the expectation while carrying almost none of the space. Only the second pair of numbers is a statement about what has been checked. This is VERIFIED on the stated slice, not on the full divisor set: the residual 0.0018% of the expected count is untested.

The filter primes must be coprime to $2D$, which is itself a structural fact rather than a convenience. Since $D$ contains every prime up to 167, for any $\ell \mid D$

$$AB + Dm^2 - N^2 = D\big(m^2 - 4D\big) \equiv 0 \pmod{\ell},$$

so the value is the square $N^2$ modulo $\ell$ for *every* $m$, and as $\gcd(D, N) = 1$ Hensel lifts that root to every power of $\ell$. The tail search therefore has no local obstruction at any prime dividing $D$, the analogue of Proposition 1.9 for this formulation; the classical 64/63/65/11 pre–filter for squares rejects nothing here, measured at a 75% pass rate modulo 64 and 100% modulo 63. Formalised as `tail_value_mod_dvd` and `hensel_lift_identity`; the induction from the step to all powers is routine and is not formalised.

*The paper's own height estimate now says the same thing.* Remark 1.16 puts the least point of a nonempty fiber at height $\exp(10^{112})$ on regulator grounds, so its $q \approx x^2/A$ carries some $10^{112}$ digits, against a bound of 115. Two independent estimates, one from divisor counting and one from the class–number formula, agree that no immune family admits a Pythagorean tail prime. Neither is a proof: the bound $q \le N$ is proved, the emptiness is HEURISTIC.

Formalised in `lean/Erdos307/TailBound.lean` (`tail_factor`, `tail_quadratic`, `tail_discriminant`, `tail_bound`, `tail_finite`) on the three standard axioms, with a non–vacuity witness at the base above.

**Remark 1.16** (The immune signal, and where its Lehmer reading broke)**.** Superseded by Proposition 4.7 and Remark 1.11: the orbit described here is real, but the inference that primality along it is the residual question is not, because the tail prime is bounded by $N$. What follows is the orbit, the three gates, and the structural points that survive.

For a reciprocity–immune base the Pell system of Remark 1.1, $B_S x^2 - A_S y^2 = -4D_S^2$, is locally soluble everywhere ($\kappa_S > 0$), and a Pythagorean pair in the tail family $S \cup \{q\}$ is *exactly* a solution with $q = (x^2 - D_S)/A_S = (y^2 - D_S)/B_S$ a positive prime. Its solutions grow like $\varepsilon^n$ in the fundamental unit of the order of discriminant $4A_SB_S$ (224 digits, $A_SB_S$ a nonsquare, on an explicit immune base). Writing $x_{n+1} = tx_n - x_{n-1}$ with $x_n = (\alpha^n + \beta^n)/2$ and $\alpha\beta = 1$, the squares $x_n^2$ have characteristic roots $\alpha^2, \beta^2, 1$ and the constant shift is absorbed by the root 1, so

$$q_n = \frac{x_n^2 - D_S}{A_S} \qquad \text{satisfies} \qquad q_{n+3} = (t^2-1)q_{n+2} - (t^2-1)q_{n+1} + q_n,$$

a non–degenerate ternary linear recurrence (verified on four Pell parameters). A reciprocity–killed base is the complementary degeneration: there the same system is locally *insoluble* ($\kappa_S = 0$), so no candidate exists at all, and the dichotomy of Proposition 1.2 is Pell–solubility versus Pell–insolubility.

The return is milder than "$k = 0$" suggests: on the Pythagorean locus the deviation $k = (a' - b)/a$ is an *integer* (the identity $v(D(u) - v) = u(u - D(v))$ with $\gcd(u, v) = 1$), forced into $\{0, -1\}$ by the near–balanced immune ratio, so $k = 0$ is a sign choice rather than a measure–zero coincidence, and the naive Wieferich shortcut that would pin $q$ is vacuous, since $x^2 \equiv D_S \pmod q$ holds identically from $x^2 = A_S q + D_S$.

Neither square condition carries the difficulty alone. Imposing only the minus–square, $d^2 = B_S q + D_S$, the admissible $d$ satisfy $d^2 \equiv D_S \pmod{B_S}$, and along such a class $d = d_0 + tB_S$ one has

$$q(t) \;=\; \frac{d_0^2 - D_S}{B_S} \;+\; 2d_0\, t \;+\; B_S\, t^2,$$

a quadratic in $t$ (toy bases: $3 + 54t + 173t^2$, $7 + 238t + 1693t^2$, $2 + 446t + 17357t^2$): a Bunyakovsky question, of the same class as the $p = 5$ family of Proposition 5.15(iii), and conjecturally routine. It is the *conjunction* of the two squares that replaces the quadratic parameter by a fundamental unit and turns the orbit exponential.

*Where the reading broke.* That exponential growth was taken to place the question in the Lehmer class, on two grounds: no algebraic factorisation is available, $D_S$ being squarefree so that $x^2 - D_S$ is irreducible, and no covering congruence can kill the family, by Proposition 1.9. Both observations are correct, and neither is the relevant one. The factorisation that does exist is not of $x^2 - D_S$ but of $x^2 - y^2 = 4D_S q$, available because $A_S - B_S = 4D_S$ *exactly*: the identity was used here only to build the Pell system, never multiplicatively. Proposition 4.7 draws it, and the ternary recurrence above then supplies no prime term past $q \le N$.

Three nested gates separate an immune family from a solution: fiber nonemptiness, a tail prime, and the $k = 0$ return. The first is a class–group condition, since Hasse supplies only rational points, and it is quantitative: immunity makes $A_S B_S$ a product of two distinct primes, hence squarefree, so the conic lives in the near–maximal order of $\mathbb{Q}(\sqrt{A_S B_S})$ of discriminant $\sim A_S B_S \approx 10^{224}$, and the class–number formula puts the expected regulator at $R = \sqrt{\Delta}\, L(1,\chi)/(2h) \approx 10^{112}$; barring an anomalously large class number, the *least* point of a nonempty fiber sits at height $\exp\bigl(10^{112}\bigr)$. That estimate is used in Remark 1.11 as independent corroboration that the second gate is empty. A direct sweep to $q \le 10^{12}$ was structurally incapable of reaching even the first gate.

**Remark 1.17** (The affine sieve does not reach a rank–one orbit)**.** The modern instrument for primes in orbits is the affine sieve of Bourgain, Gamburd and Sarnak [45], which produces $r$–almost–primes in orbits of thin groups. It is the one method designed for exactly the shape of question Remark 1.16 isolates, so its failure here is worth recording precisely rather than left implicit.

The affine sieve needs a spectral gap: the congruence quotients of the acting group must form an expander family, and that is what supplies the level of distribution the sieve consumes. Here the $q_n$ run along a Pell orbit, so the acting group is the cyclic group generated by the fundamental unit of the order of discriminant $4A_S B_S$. Cyclic groups are abelian, hence amenable, and have no spectral gap; expansion is precisely the property an abelian group lacks. The sieve therefore returns nothing, and it returns nothing on $2^n - 1$ for the same reason. The obstruction is rank, not thinness as such: non–elementary thin orbits, Apollonian packings among them, do yield almost–primes.

The natural repair fails for a reason worth stating too. One may embed the rank–one orbit in a larger thin group whose orbit does expand, and sieve there. What the affine sieve then delivers is infinitely many almost–primes in the *larger* orbit, inside which our sequence has relative density zero; the sieve does not localise to a prescribed sub–orbit. Expansion is bought at the cost of losing the sequence.

This meets the unit–rank trichotomy of Remark 3.14 from a second direction. There, rank zero gave a classical gate and denied a continuous family, positive rank the reverse. Here, rank one is exactly the cell in which the affine sieve has no traction. Stated as a single missing property: what is wanted is a substitute for expansion at rank one, and that is the Mersenne problem itself, not a lemma about #307.

**Remark 1.18** (The gap to the record, measured)**.** It is worth stating how far the required statement is from the best that is known, because the distance is not one of degree.

Write a sequence's *density exponent* $\theta$ for $\#\{n \le X\} \asymp X^{\theta + o(1)}$. The sparsest sets currently proved to contain infinitely many primes are the values of binary forms: $a^2 + b^4$ at $\theta = 3/4$, by Friedlander and Iwaniec [40], and $a^3 + 2b^3$ at $\theta = 2/3$, by Heath-Brown [41], the latter being the sparsest polynomial form known to capture its primes. Our $q_n$ grow like $\varepsilon^{2n}$, so $\#\{n : q_n \le X\} \asymp \log X$ and

$$\theta = 0.$$

The distance from $2/3$ to $0$ is categorical rather than quantitative. Both records are bilinear arguments: they run a sieve of dimension one and close it with a Type II estimate obtained by averaging over a set of size a positive power of $X$. Every gain in such an argument is a gain of a power of $X$, and against $\log X$ terms a power of $X$ gains nothing; below $\theta = 1/2$ the standard sieve axioms already fail, and at $\theta = 0$ there is no set left to average over. This is the same wall as Remark 1.17 approached from the analytic rather than the spectral side, and it is why every sequence of exponent zero, Mersenne, Fibonacci, Lucas and ours alike, is open together.

The structure our sequence does carry is the standard one and confers nothing. From $A_S q_n = x_n^2 - D_S$ one gets $x_n^2 \equiv D_S$ for every prime $p \mid q_n$ not dividing $D_S$, so all prime factors of $q_n$ lie in the density-$\frac{1}{2}$ set where $D_S$ is a quadratic residue. That constrains the factorisations without excluding any, which is exactly the position the primitive divisor theorems already left us in: divisors, never primality.

**Remark 1.19** (The residue, atomised, and where its difficulty comes from)**.** Decomposing the question into independently known statements locates the obstruction on a single atom and identifies its source.

Two atoms carry theorems. Every $q_n$ beyond the thirtieth has a primitive prime divisor [48]; and the greatest prime factor satisfies $P^+(q_n) > n \exp\bigl(\log n/(104 \log\log n)\bigr)$, by Stewart [42], which settled questions of Schinzel and Erdős and was the first general advance on Bang and Carmichael. Every other atom, that $q_n$ be squarefree infinitely often, that $\omega(q_n)$ be bounded infinitely often, that $q_n$ be a $P_k$, and primality itself, is open.

The chain that would close the crux is $P^+$ large, hence $\omega$ small, hence prime. It breaks at the first link, and by a measurable amount. Stewart's bound certifies $\log P^+ \geq (1 + o(1)) \log n$ against $\log q_n \approx 2n \log \varepsilon$, so the fraction of the term's size that is certified is of order $\log n/n$: 10.6% at $n = 10$, 0.32% at $n = 10^3$, $4 \times 10^{-4}$% at $n = 10^6$. Primality is the fraction 1. No sharpening of the constant 104 changes this, since $n^{1+o(1)}$ is the shape linear forms in logarithms produce. The Subspace Theorem is the other available instrument and does not help: it is stronger under weaker hypotheses but ineffective [43], and an ineffective statement cannot exhibit the single prime term a construction needs.

The source is visible in the comparison with the polynomial case. For $n^2 + 1$ the greatest prime factor exceeds $n^{1.3}$ infinitely often [44], that is $q^{0.65}$, a *positive proportion* of the term's size, and on such sequences almost-primality is available too. For an exponentially parametrised sequence only $(\log q)^{1+o(1)}$ is available, a vanishing proportion. The gap between the two regimes is exactly the gap between polynomial and exponential parametrisation, and Proposition 5.5 proves #307 admits no polynomial one. So the obstruction recorded in Remark 1.17 as a question of rank, in Remark 1.18 as one of density exponent, and here as one of $P^+$, is a single obstruction with three faces, and its source is Proposition 5.5: forced exponentiality is what places the residue in the class where nothing of the required strength is known.

Two things sharpen that attribution, and both make the obstruction less incidental rather than more. First, the boundary is exactly where Proposition 5.5 sits. The fraction of a term's size that Stewart's bound certifies is $\log P^+/\log q_n \approx \log n/\log q_n$, so it stays bounded below precisely when $\log q_n \ll \log n$, that is when $q_n$ grows polynomially: degree $d$ gives the constant $1/d$. Every super-polynomial rate sends it to 0, and not only the exponential one: $\exp(\sqrt{n})$ and $n^{\log n}$ do as well. So polynomial growth is not one convenient case among several, it is the whole of the favourable region, and it is exactly what Proposition 5.5 excludes.

Second, the hypotheses of Proposition 5.5 are worth testing, since a gap there would dissolve everything above. It assumes a *fixed* number of polynomials, and its proof uses that finiteness when it splits $\sum_i 1/p_i$ into constants plus a remainder tending to 0. A family whose support grows with the parameter is therefore not covered. That gap is real and useless: a growing support means a growing number of simultaneous primality conditions, which is a harder demand than one, not an easier one, and it does not return the family to the polynomial regime. The proposition forbids the case that would help and leaves uncovered only cases that would not.

**Remark 1.20** (The squarefree atom, and what it reduces to)**.** Among the four open atoms one is structurally easier, and it is worth saying why and how far it gets. Squarefreeness of $q_n$ asks only for an *upper* bound on a count of bad events, never for the production of a prime, so it is the one atom not subject to the parity obstruction that blocks bounded $\omega$, $P_k$, and primality.

The argument it wants is a union bound. For a prime $p \nmid 2A_S D_S$, $p^2 \mid q_n$ forces $x_n^2 \equiv D_S \pmod{p^2}$; the Pell orbit is purely periodic modulo $p^2$ with period $\pi(p^2)$, there are at most two square roots of $D_S$, and each is met a bounded number of times per period, so the density of bad $n$ attached to $p$ is at most $C/\pi(p^2)$. Where $\pi(p^2) = p\,\pi(p)$, which is the generic behaviour, and $\pi(p) \asymp p$, this is $\asymp C/p^2$, and the union bound closes as soon as $C\sum_p p^{-2} < 1$. The constant is comfortable: $\sum_p p^{-2} = 0.4522474\ldots$, so any $C < 2.21$ suffices, and a positive proportion of the $q_n$ would be squarefree, hence infinitely many.

What is missing is the hypothesis $\pi(p^2) = p\,\pi(p)$. Its failure is precisely a Wall–Sun–Sun prime for the sequence, and there the density is $\asymp 1/p$ rather than $\asymp 1/p^2$, whose sum diverges and destroys the bound. No such prime is known for any classical sequence and their finiteness is unproved, so the atom does not close: it *reduces* to a Wall–Sun–Sun statement. That is a sharper position than “open”, since it names the obstruction and shows it is a single hypothesis rather than a programme, but it is a reduction and not a proof.

The unconditional pieces are formalised. `Erdos307.primeSquareSum_certificate` gives $\sum_p p^{-2} < \frac{1}{2}$ in the finite form used above, the first six primes together with the bound $\frac{1}{2}\sum_{m\ge 16} m^{-2} < \frac{1}{30}$ for the odd primes beyond; `good_density_pos` and `squarefree_positive_density_of_bound` are the union bound and the conditional implication. All carry 0 `sorry` and the three standard axioms (`lean/Erdos307/Squarefree.lean`).

Two further points fix the standing of this atom, and both are negative. First, on the immune bases the argument is blocked at its *lowest* rung as well as its highest. The density attached to a single small $p$ is $\#\{n \bmod \pi(p^2) : p^2 \mid q_n\}/\pi(p^2)$, read off the Pell orbit modulo $p^2$, and that needs the fundamental solution, the object Proposition 6.4 places at $10^{24.7}$ operations. For a Lucas sequence with a known fundamental unit the small-prime rung is a finite computation; here it is not. The ladder is therefore blocked at both ends, by non-computability below and by Wall–Sun–Sun above. Second, and more to the point, squarefreeness is not what the construction needs: Remark 1.16 requires $q$ to be *prime*, and a squarefree $q$ with several prime factors is useless there, so even the dream lemma of this atom would leave #307 exactly where it stands. Of the four atoms this is the easiest, and it is also the only one whose resolution would not advance the problem.

## 2 Local anatomy and a heuristic model

**Lemma 2.1** (Local anatomy)**.** *Let $a$ be squarefree and $q$ a prime with $q \nmid a$. Then $q \mid a'$ if and only if $\sum_{p\mid a} p^{-1} \equiv 0 \pmod q$, where $p^{-1}$ is the inverse of $p$ modulo $q$. If $q \mid a$, then $q \nmid a'$.*

Formalised. *`Erdos307.anatomy_iff`, with the factorisation $a' \equiv a\sum_{p\mid a} p^{-1}$ proved as `csum_eq_mul_recipSum` rather than assumed; the second half is `anatomy_on_support` and is Rigidity* (`lean/Erdos307/Frame.lean`)*.*

*Proof.* $a' = a\sum_{p\mid a} 1/p$. If $q \nmid a$ then $q \mid a'$ iff $q \mid a\sum_{p\mid a} p^{-1}$ iff $\sum_{p\mid a} p^{-1} \equiv 0 \pmod q$. If $q \mid a$, then by Lemma 2.1 applied to the prime set of $a$, $q \nmid a'$. □

Thus the cycle equation $a' = b$, $b' = a$ factors, prime by prime, into $\omega(a)+\omega(b)$ local congruences of exactly the shape that governs amicable–pair heuristics: for each $q \mid b$ one needs $\sum_{p\mid a} p^{-1} \equiv 0 \pmod q$, and symmetrically. These local conditions, however, are individually *empty of obstruction*:

**Proposition 2.2** (No single–modulus congruence obstruction)**.** *For every modulus $m$ with $2 \le m \le 210$, the reduced cross–match system $N_P \equiv D_Q$, $N_Q \equiv D_P \pmod m$ is realisable by disjoint squarefree prime sets $P, Q$, in both structural branches:* $\gcd(m, D_P D_Q) = 1$, *and the branch in* *which a prime divisor of $m$ lies in $P \cup Q$. Hence no individual modulus in this range can rule out a solution of #307.*

*Construction.* Write $\Sigma_S \equiv \sum_{p \in S} p^{-1} \pmod{m}$, so $N_S \equiv D_S \Sigma_S$. The reduced system is $\Sigma_P \equiv D_Q D_P^{-1}$, $\Sigma_Q \equiv D_P D_Q^{-1} \pmod{m}$; its only coupling is $\Sigma_P \Sigma_Q \equiv 1$, automatic, with $D_P, D_Q$ otherwise free. By Dirichlet each unit class mod $m$ contains primes, fixing $D_P, D_Q$; appending a prime $r \equiv 1 \pmod{m}$ leaves $D_S$ fixed while sending $N_S \mapsto r N_S + D_S \equiv N_S + D_S$, i.e. incrementing $\Sigma_S$ by 1, so (when $\gcd(D_S, m) = 1$) every target class is reached with disjoint sets. In the branch $\ell \mid m$, $\ell \in P$, reduction mod $\ell$ collapses the system to $N_Q \equiv D_P \equiv 0 \pmod{\ell}$, met by inverse–paired residues in $Q$; composite $m$ needs no separate gluing, the increment acting directly mod $m$. We verified both branches for all $2 \le m \le 210$ (for prime and prime–power moduli the inside branch requires the directed construction above; seeding a prime divisor of $m$ into $P$, a blind search missing these as a search–power artefact, not an arithmetic obstruction). □

**Remark 2.3** (Precise scope)**.** This excludes only a *single fixed–modulus* congruence disproof. The witnesses are $m$–dependent (the modulus–6 witness already fails modulo 210), and a single $(P, Q)$ solving the system for *all* $m$ simultaneously is, for bounded integers, equivalent to the integer identities $N_P = D_Q$, $N_Q = D_P$ (that is, to #307 itself) so per–modulus solvability is *no* evidence for existence. Nor does it exclude other finite or local negative arguments (size/Mertens bounds, density, magnitude–congruence hybrids, or a Hasse–type failure across infinitely many moduli); indeed our own $|U| \ge 59$ and $2.09 \times 10^{56}$ barriers are finite–data results fully consistent with it. Together with Theorem 3.1 (every non–archimedean shadow trivially negative for degree reasons) it brackets the problem: no single–modulus obstruction over $\mathbb{Z}$, fatal non–archimedean analogues over $\mathbb{F}_q[t]$, the difficulty living at the archimedean place.

**A first–order count.** Model a candidate squarefree $a$ as "$P$–abundant" ($\sum_{p|a} 1/p > 1$) and ask for $b = a'$ to itself be $Q$–abundant with $\sum_{q|b} 1/q = 1/s$. Two empirical inputs (Section 6) calibrate this:

- the density of $P$–abundant $n$ is $\delta = 0.0420$ (and $\approx 0.0176$ among squarefree $n$); a theorem, not merely an observation (Proposition 2.6);
- for a qualifying squarefree $a$, $b = a'$ is squarefree about 95% of the time, but its mass $\sum_{q|b} 1/q$ has mean $\approx 0.047$ against the required $\approx 0.934$; the empirical tail probability $\Pr[\sum_{q|b} 1/q \ge 1/s]$ is 0 in 40,920 samples at these scales (Section 6).

The gap is structural: at small scale a $P$–abundant $a$ hogs the cheap mass–carrying small primes, and $b = a'$, being coprime to $a$, is both too small and too coprime to assemble reciprocal mass $1/s$. The extremal case is stark: for the primorial $a = \prod_{i \le k} p_i$, though $\sum_{p|a} 1/p$ rises through 2 at $k = 59$, the cofactor sum $a'$ is essentially prime, with $\sum_{q|a'} 1/q < 0.2$ for every $k < 100$ (and only 0.0014 at $k = 59$): the family that *maximises* the forward mass is the most starved on the return. Equivalently, for squarefree $n$ with $n'$ squarefree, $n'' = n\,\sigma(n)\,\sigma(n')$, so the two–step return ratio $n''/n = \sigma(n)\sigma(n')$ is the #307 product itself; over *all* such $n < 10^6$ it stays strictly below 1, never exceeding 0.54 (maximal at $n = 969969 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$), so $D$ is two–step contracting throughout the tested range and must reverse to unit ratio for a solution to exist. The first scale at which $b$ could plausibly carry the required mass is of the order of the barrier ($\sim 10^{56}$); this dynamical mass–gap is *consistent with* the proved barrier rather than an independent derivation of it (the empirical tail probability 0 at small scale is equally consistent with NO and with YES–beyond–the–barrier). A naive "expected number of solutions" built from the local congruences (one factor $\sim 1/q$ per condition) diverges like a constant times $\log X$; this is a divergent naive count of exactly the type known to be unreliable, so we treat it only as very weak, non–rigorous suggestive evidence that solutions could exist at or beyond the barrier.

**Remark 2.4** (Calibration)**.** We present the count of the previous paragraph as a heuristic, not a theorem. Heuristics of amicable–pair type are known to be delicate; we report it as weak evidence, consistent with the divergence of the refined expected count, that the answer to #307 is plausibly YES (and hence that the period–one form of the Ufnarovski–Åhlander conjecture is plausibly false).

**Remark 2.5** (A cautionary refinement)**.** One can try to sharpen this into a quantitative rate using a Kubilius/Erdős–Wintner model, stratified over which small primes lie in the support of $a$ versus remain available to $b$. Two lessons emerge, both methodological. First, a plain Monte–Carlo estimate *undercounts* the rate by some $10^{61}$: the expectation is dominated by rare, balanced splits of the small primes that sampling almost never sees, so a naive estimate of this kind is misleadingly small and only an explicit stratified sum corrects it. Second, and this is why we record no numerical prediction, the stratified estimate is itself unstable: refining the stratification from the first 8 to the first 14 primes drops the rate by some 17 orders of magnitude and pushes the implied first occurrence from $\sim 10^{10^{38}}$ out past $\sim 10^{10^{55}}$, with no sign of convergence. The only robust conclusion is negative: every version places a hypothetical first solution *doubly exponentially* beyond any conceivable search, consistent with the constructive deficit of Section 4; the model does not reliably decide existence in either direction.

**Proposition 2.6** (The abundance density is a theorem)**.** *The additive function $f(n) = \sum_{p|n} 1/p$ satisfies the Erdős–Wintner three–series criterion [18] (here $\sum 1/p^2$ and $\sum 1/p^3$ converge), so $f$ has a limiting distribution and the density $\delta = \lim_{x\to\infty} \frac{1}{x}\#\{n \le x : f(n) > 1\}$ exists, equal to $\Pr\big[\sum_p X_p/p > 1\big]$ for independent $X_p \sim \mathrm{Bernoulli}(1/p)$. Moreover the constant carries a* certified *enclosure:*

$$0.04202 \;\le\; \delta \;\le\; 0.04223.$$

*Writing $S_1 = \sum_{p\le P_1} X_p/p$, $P_1 = 3\times 10^4$, and $T \ge 0$ for the tail, monotone upper and lower cdf envelopes of $S_1$ are propagated through the Bernoulli convolution on a grid of step $10^{-7}$ with the shift $1/p$ rounded down for the upper and up for the lower envelope (which preserves them, cdfs being nondecreasing) so $\delta \ge 1 - U(1)$ costs nothing (the tail is nonnegative), while $\delta \le 1 - L(1 - a) + \Pr[T \ge a]$ with the explicit Chernoff bound $\Pr[T \ge a] \le \exp(1.36 - P_1 a)$, $a = 10^{-3}$ (from $\log \mathbb{E}\, e^{\lambda T} \le \lambda s_2 + 0.72\lambda^2 s_3$ at $\lambda = P_1$, using $s_2 < 1/P_1$, $s_3 < 1/(2P_1^2)$ and $e^x - 1 \le x + 0.72x^2$ on $[0,1]$); float slack $10^{-9}$ exceeds the accumulated rounding by nine orders, and the machinery reproduces an exact four–prime reference to $2\times 10^{-16}$ (`code/delta_cert.py`). The point value $0.0420$, at the bottom of the enclosure, agrees to four places with the sieve to $2\times 10^7$; the upper edge is conservative, carried almost entirely by the $a$–window. As for the density of abundant integers, no closed form is expected.*

**Remark 2.7** (Two dynamical probes: no second obstruction)**.** Two measurements sharpen the picture beyond the first–order mass count, and both cut towards rather than against a solution. *(i) Image density.* A two–cycle needs *both* members in the image $D(\mathbb{N})$ (each is the other's derivative), so one might fear a second obstruction; that the high–mass numbers a cycle requires are seldom derivatives. The opposite holds: of the integers below $10^7$, 68.3% lie in $D(\mathbb{N})$, and the fraction *rises* with mass, from 61.7% at $\sum_{p|n} 1/p < 0.2$ to 78.2% at $\sum_{p|n} 1/p > 1$ and 80% beyond. The high–mass numbers are thus over–represented among derivatives; the second cycle condition is comfortably met. *(ii) The barrier is where the balanced mass–product reaches* 1. Set $r(n) = \big(\sum_{p|n} 1/p\big)\big(\sum_{q|n'} 1/q\big) = D^2(n)/n$, so $r = 1$ is exactly the cycle condition. Exhaustively over $n \le X$ the maximum of $r$ climbs

$0.50 \to 0.62$ as $X$ runs $10^3 \to 10^9$ (with no exact cycle, as the barrier demands), with no ceiling; the climb follows the balanced split $r \approx (S(k)/2)^2$, $S(k) = \sum_{i \le k} 1/p_i$, which reaches 1 precisely when $S(k) = 2$, i.e. $k \approx 59$. The dynamical trajectory thus reproduces the $|U| \ge 59$ barrier from the opposite direction. Neither probe decides existence; both *remove* hypothetical obstructions, leaving the exact archimedean closure as the sole gate.

**Remark 2.8** (Why the analytic method is starved: $N(r) \le 1$)**.** The rank–one rigidity (Section 9) has a sharp consequence for the circle–method machinery that resolves the *unrestricted* Egyptian–fraction problem. For a rational $r$ let $N(r) = \#\{\, P : \sum_{p \in P} 1/p = r \,\}$ count the finite prime sets with reciprocal sum $r$. Then

$$N(r) \le 1 :$$

by the Rigidity Lemma (Theorem 2.1) the sum $\sum_{p\in P} 1/p = N_P/D_P$ is already in lowest terms, so the reduced denominator of $r$ equals $D_P = \prod P$ and $P$ is exactly its set of prime divisors; uniquely determined ($N(r) = 1$ precisely when that denominator is squarefree and the numerator of $r$ is its cofactor sum). The circle method needs the opposite: a target hit by a positive proportion of subsets, so a main term dominates. This is exactly what Bloom's shift to $\{p-1\}$ manufactures, $p-1$ being composite and divisor–rich, a rational is the reciprocal sum of *many* subsets of $\{1/(p-1)\}$ (already among the first fourteen shifted primes the multiplicity reaches 3 and grows), whence [7] represents every positive rational there. Over the primes themselves $N(r) \le 1$ leaves nothing to average: the analytic engine that reaches $\{p - h\}$ is provably starved on $\{p\}$, and this is the method–level reason #307 (the rank–one, exact–prime case) lies beyond it.

# 3 A function–field lens: the obstruction is archimedean

Define an arithmetic derivative on $\mathbb{F}_q[t]$ by $P' = 1$ for monic irreducibles and the Leibniz rule, so $f' = \sum_i e_i\, f/P_i$ for $f = \prod_i P_i^{e_i}$ (the exponents $e_i$ read modulo the characteristic).

**Theorem 3.1.** *Over $\mathbb{F}_q[t]$ this derivative has no two–cycles and no nonzero fixed points. Indeed $\deg f' \le \deg f - 1$ for every nonconstant $f$. The analogues of the integer fixed points $p^p$ also vanish: $(P^p)' = p\,P^{p-1}P' = 0$ in characteristic $p$. The same holds verbatim for every twisted variant ($P' = u_P \in \mathbb{F}_q^\times$, Leibniz).*

*Proof.* Each term $e_i\, f/P_i$ has degree $\deg f - \deg P_i \le \deg f - 1$, so $\deg f' \le \deg f - 1$. A two–cycle $f' = g$, $g' = f$ would force $\deg g \le \deg f - 1$ and $\deg f \le \deg g - 1$, hence $\deg f \le \deg f - 2$; a fixed point $f' = f$ is excluded likewise. (Verified exhaustively over $\mathbb{F}_2$ to degree 7, $\mathbb{F}_3$ to degree 5, and $\mathbb{F}_5$ to degree 4; twisted variants over $\mathbb{F}_2$ and $\mathbb{F}_5$ on small supports through $2.3 \times 10^6$ twist configurations, none closing.) □

Formalised. `Erdos307.ff_no_cycle` is the theorem, on the single hypothesis $\deg f' + 1 \le \deg f$; `deg_drop_of_terms` is that hypothesis from the term bound, and `no_fixed_point_of_drop` excludes fixed points. The argument is stated on the degrees alone, so it holds verbatim for every twisted variant, the twist multiplying each term by a unit (`lean/Erdos307/FunctionField.lean`).

**Corollary 3.2** (The function–field Erdős–Barbeau equation is insoluble)**.** *For every finite field $\mathbb{F}_q$, all nonempty finite sets $P, Q$ of monic irreducibles, and all unit weights $u_\pi, v_\rho \in \mathbb{F}_q^\times$,*

$$\Big(\sum_{\pi\in P} \frac{u_\pi}{\pi}\Big)\Big(\sum_{\rho\in Q} \frac{v_\rho}{\rho}\Big) \ne 1.$$

*Proof.* Write $f = \prod P$, $g = \prod Q$, $D_u(f) = \sum_\pi u_\pi f/\pi$; the equation reads $D_u(f)\, D_v(g) = fg$. If some $\pi \in P \cap Q$, then $\pi^2 \mid fg$ while $D_u(f) \equiv u_\pi f/\pi \not\equiv 0$ and $D_v(g) \equiv v_\pi g/\pi \not\equiv 0 \pmod{\pi}$, so $\pi \nmid D_u(f) D_v(g)$; impossible; hence $\gcd(f, g) = 1$. The same congruence gives $\gcd(f, D_u(f)) = \gcd(g, D_v(g)) = 1$, so $g \mid D_u(f)$ and $f \mid D_v(g)$, forcing $D_u(f) = c\,g$, $D_v(g) = c^{-1} f$ with $c \in \mathbb{F}_q^\times$: a twisted two–cycle, excluded by the degree drop. The weighted product equation is thus insoluble over every $\mathbb{F}_q(t)$; it *is* soluble over the unit–rich number rings of Theorem 3.11, since any twisted cycle gives $(D_u(a)/a)(D_v(b)/b) = (b/a)(a/b) = 1$, and over $\mathbb{Q}$, where the admissible weights are $\pm 1$, it is #307 itself with the signed barrier of Remark 3.4, open beyond $2.09 \times 10^{56}$. The three regimes (non–archimedean monotonicity (closed, negative), unit–broken monotonicity (closed, positive), ordered–archimedean (open, barred)) now have theorems on two of three doors. $\square$

Over $\mathbb{Z}$, by contrast, $n' = n \sum_i e_i/p_i$ can exceed $n$ precisely because the archimedean absolute value lets a sum of strictly smaller terms outgrow the original; the "mass $> 1$" phenomenon $\sum 1/p > 1$, which has no degree–theoretic analogue. Theorem 3.1 thus localises the entire existence question to the archimedean place: every non–archimedean shadow of #307 is trivially negative. This is consistent with (and partly explains) both the census finding of Section 6 and Proposition 2.2 (no single modulus obstructs the cross–match system in the tested range); the difficulty is not local. Any eventual proof must therefore distinguish $\mathbb{Z}$ from $\mathbb{F}_q[t]$; over the latter the decisive tools are the degree and the Mason–Stothers/Wronskian inequality [19], neither of which has a $\mathbb{Z}$–analogue. The conjectural analogue, *abc*, does not fill the gap, and quantifiably so. Every cycle inherits, from $N' = a^2 + b^2$ with $N = ab$, the relation $(a - b)^2 + 4ab = (a + b)^2$ on coprime squarefree members, but this triple is *abc*–unexceptional, of quality $q = \log(a + b)^2 / \log \operatorname{rad}\big((a - b) \cdot ab \cdot (a + b)\big) \approx \frac{1}{2}$ (median 0.543, never above 0.74 across $6 \times 10^4$ coprime squarefree pairs; the only $q > 1$ cases are degenerate smooth–gap pairs such as $(5, 3)$, vanishingly far below the $10^{56}$ scale a solution must occupy). Granting *abc* in its strong explicit form would therefore yield no bound here: the function–field collapse rests on an inequality genuinely *absent* over $\mathbb{Z}$, not merely unproven, and the size mismatch is by a factor of two in the exponent. #307 lives in the *abc*–comfortable zone.

**Remark 3.3** (The free grading, and what evaluation destroys)**.** Theorem 3.1 has a structural reading. In the free Leibniz world (the polynomial ring $\mathbb{Z}[x_p]$ with $x_p' = 1$) the derivative of a squarefree monomial of degree $n$ is the elementary symmetric form $e_{n-1}$, homogeneous of degree $n - 1$: the formal derivative strictly lowers the grading, cycles are impossible, and over $\mathbb{F}_q[t]$ the degree realises this grading faithfully. Over $\mathbb{Z}$ the evaluation $x_p \mapsto p$ destroys homogeneity, and the data show it destroys it completely: for random squarefree $a$ of fixed size, $\omega(a')$ is governed by the size of $a'$ alone (Erdős–Kac), flat in $\omega(a)$ within each parity class (measured: mean $\omega(a') =$ 2.7–3.0 as $\omega(a)$ runs over $3, 5, 7$, and 3.6–4.0 over $4, 6, 8$, the offset being the parity anatomy of Theorem 4.18). A two–cycle is therefore a doubly anti–typical event: an additively assembled integer must carry multiplicative rank far in the Erdős–Kac tail *and* on an exactly prescribed support; the two independent prices whose product is the heuristic rarity of Section 2. The deeper additive–multiplicative content collapses, on inspection, into the $S$–unit form of Proposition 3.7: dividing the cycle equation by $D_Q$ exhibits it as an $n$–term unit equation; one more coordinate system in which the same exactness reappears unchanged.

**Remark 3.4** (The obstruction is *ordered*–archimedean)**.** The barrier is finer than "archimedean": its engine is AM–GM on real, *positive* masses, i.e. the ordering of $\mathbb{R}$. Over $\mathbb{Z}[i]$ (a PID, so the Rigidity Lemma and cross–matching survive verbatim, with cycles defined up to units), the masses $\sum 1/\pi$ are *complex*: phases can align so that $\left|\sum 1/\pi\right| > 1$ with as few as three Gaussian primes (e.g. $\frac{1}{1+i} + \frac{1}{2+i} + \frac{1}{2-i} = 1.3 - 0.5i$), and the 59–prime bound has no analogue. The ordered–archimedean thesis thus predicts that small derivative two–cycles *may* exist over $\mathbb{Z}[i]$ where they are barred over

$\mathbb{Z}$, and the prediction is confirmed. Two facts make the contrast precise. *First*, phases do not help over $\mathbb{Z}$: for any *signed* derivative ($p' = \pm 1$, Leibniz) a two–cycle still satisfies $|s||t| = 1$ with $|s| \le \sigma(a)$, $|t| \le \sigma(b)$ and disjoint supports (the rigidity proof is unchanged), so $\sigma(a) + \sigma(b) \ge 2$ and the full 59–prime, $2.09 \times 10^{56}$ barrier applies to every choice of signs: real phases cannot cancel. *Second*, fourth roots of unity suffice instantly: Leibniz derivatives on $\mathbb{Z}[i]$ with unit prime–values ($\pi' \in \{\pm 1, \pm i\}$) possess genuine two–cycles at norm $\sim 10^4$. The smallest:

$$a = (1+i)(2+7i)(6+11i) = -129 - i, \qquad b = 3\,(2+i)(1+2i)(6+5i) = -75 + 90i,$$

with $a' = b$ under the *standard* values $\pi' = 1$ on the primes of $a$, and $b' = a$ exactly under $\pi' = i$ on $3$, $2+i$, $1+2i$ and $\pi' = 1$ on $6+5i$; seven prime classes ($N(a) = 16{,}642$, $N(b) = 13{,}725$) against the fifty–nine that $\mathbb{Z}$ demands. Exhaustive search finds exactly sixteen such cycles up to conjugation and direction having a member of norm $\le 2 \times 10^7$ with $\omega \le 5$ on the enumerated side ($\omega$–profiles $(3,4)$, $(4,4)$, $(5,4)$, and even $(4,6)$) and the population grows steadily (4 below $3 \times 10^5$, 16 below $2 \times 10^7$), consistent with an infinite, slowly growing family; in six of the sixteen one side requires no twist at all. All sixteen were verified exactly, by independent recomputation; no twisted *fixed* point ($a' = ua$) exists in the same range. The strict two–sided normalisation $\pi' = 1$ (first–quadrant representatives) has no cycle up to units (nor any fixed point, zero–derivative, or unit–derivative element) with $N(a) \le 2 \times 10^9$ ($9.4 \times 10^8$ candidates, partners by full factorisation); this is the expected order rather than a separate rigidity, a single fixed phase system being heuristically $4^{\omega-1}$–fold rarer than the union of all systems. The contrast with $\mathbb{Z}$ survives this accounting: over $\mathbb{Z}$ *every* one of the $2^{\omega-1}$ sign systems is barred by the theorem above, while over $\mathbb{Z}[i]$ the union of phase systems is demonstrably populated from norm $1.7 \times 10^4$ on. The barrier of #307 is thus localised precisely: it is the impossibility of phase cancellation in the ordered field $\mathbb{R}$, where the only available phases $\pm 1$ provably gain nothing; while the moment fourth roots of unity are admitted, the analogous existence problem closes at seven primes. The cycles found come in conjugate pairs (supports not conjugation–stable); a conjugation–*stable* Gaussian cycle would descend to rational data through the norm form $a^2 + b^2$, i.e. precisely the Pythagorean layer of Proposition 10.2; now a concrete target rather than a speculation.

**Remark 3.5** (The descended operator; the symmetric sector)**.** Conjugation–stable supports admit an exact descent to $\mathbb{Z}$. Such a squarefree Gaussian integer is an associate of an odd squarefree rational $m$ (split primes entering as whole conjugate pairs; the $(1+i)$–sector is empty by a short parity argument), and a conjugation–equivariant unit assignment makes the twisted derivative rational:

$$D(m) \;=\; \sum_{\substack{p \mid m \\ p \equiv 1\,(4)}} t_p \,\frac{m}{p} \;+\; \sum_{\substack{q \mid m \\ q \equiv 3\,(4)}} \varepsilon_q \,\frac{m}{q}, \qquad t_p \in \{\pm 2x_p, \pm 2y_p\}, \quad \varepsilon_q = \pm 1,$$

where $p = x_p^2 + y_p^2$; a Leibniz derivative on odd squarefree integers whose prime values have size $2\sqrt{p}$ rather than 1, and whose masses $\sim 2/\sqrt{p}$ reach 2 with four primes, so no 59–prime barrier exists for it. A two–cycle of $D$ is exactly a conjugation–stable Gaussian cycle and would plant the phenomenon on rational soil. Parity constrains the hunt: $D(m)$ is odd iff $m$ carries an odd number of primes $\equiv 3 \pmod 4$, so every member must contain such a prime. One level deeper, since $q^{-1} \equiv q \pmod 8$ for odd $q$, the descended operator obeys $D(m) \equiv m\,W(m) \pmod 8$ with $W(m) = \sum_{p \mid m} t_p\, p$, so a two–cycle forces $W(m) \equiv W(m') \equiv mm' \pmod 8$; a mod–8 constraint on the sign assignment that prunes the $4^{\#\mathrm{split}} 2^{\#\mathrm{inert}}$ phase enumeration severalfold in future searches (verified on $2 \times 10^5$ random assignments; `code/peeling_groupoid.py`). An exhaustive search over members $m \le 10^9$ (Gaussian norm $10^{18}$, $\omega \le 8$, $1.4 \times 10^9$ closure tests) found no cycle, no fixed point $|D(m)| = m$, and no vanishing $D(m) = 0$. The search has since been extended two decades further in $m$, on a box narrower in the other parameters: $\omega \in [4, 6]$ and odd primes below 600, with $m$ running over $[10^9, 10^{10}]$ and then $[10^{10}, 10^{11}]$. Both tiers are exhaustive over their 5,778 prime pairs and return nothing: 10,004,024 members and $6.94 \times 10^8$ closure tests on the first, 22,218,430 members and $1.82 \times 10^9$ closure tests on the second, so $3.22 \times 10^7$ members and $2.52 \times 10^9$ closure tests in all, with no two–cycle at any point (`hunt/descended_hunt2.rs`, `hunt/run_tier2_par.sh`). The tier boxes are not nested in the one above, being narrower in $\omega$ and in the prime range while wider in $m$, so the two counts complement rather than supersede one another; and the tiers test closure only, not the fixed–point and vanishing conditions recorded above. We claim no obstruction from this: equivariance prunes the available phases fourfold per split prime, and a phase count makes emptiness at this depth the expected order. The descent target remains open over $\mathbb{Z}[i]$, but not in the real–quadratic world, where the corresponding sector contains a fully rational pair (Remark 3.6). Finally, the descended coefficients are classical objects: at a split prime $p$ the four values $\{\pm 2x_p, \pm 2y_p\}$ are exactly the Frobenius traces of the congruent–number curve $y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$) and of its twists ($a_p \in \{\pm 2x_p, \pm 2y_p\}$ verified by point counting for all $p < 120$) so the conjugation–stable sector is the arithmetic derivative twisted by the Hecke eigensystem of $\mathbb{Q}(i)$, and its mass statistics are governed unconditionally by Hecke's equidistribution theorem: the first point of contact between #307 and automorphic forms. Concretely: over random squarefree $m$ the three–series conditions converge (the variance term is $\sum 2/p^2$), so the descended mass $D(m)/m$ has an Erdős–Wintner *limiting distribution* whose prime components are arcsine laws scaled by $1/\sqrt{p}$; the required input, equidistribution of the Hecke angles, is visible exactly (Kolmogorov–Smirnov distance 0.0041 from uniform over the 8,977 split primes below $2 \times 10^5$, well inside the 95% band 0.0144).

**Remark 3.6** (The wall is the orderability of $\mathbb{Q}$)**.** The cycles over $\mathbb{Z}[i]$ permit a dissection of the barrier that no amount of theory over $\mathbb{Z}$ could provide. The entire structural chain of this note (rigidity, cross–matching, the Pythagorean identity, the split discriminant) transfers verbatim to twisted derivatives over $\mathbb{Z}[i]$: Leibniz holds for disjoint supports, so a cycle $(a, b)$ has $D(N) = a^2 + b^2$ for $N = ab$, with $D(N) - 2N = (a - b)^2$ and $2N - D(N) = \big(i(a-b)\big)^2$ (all verified exactly on the cycles found). The last equation exposes the wall: over $\mathbb{Z}[i]$, where $-1 = i^2$ is a square, *both* discriminant factors are squares automatically, and the plus/minus distinction of Proposition 5.13 (which over $\mathbb{Z}$ carries the entire obstruction, via $(a - b)^2 \geq 0 \Rightarrow \sigma(N) \geq 2$) has no content. The unique non–transferring link is the sign–definiteness of squares: the barrier is, in the end, the *formal reality* of $\mathbb{Q}$ in the sense of Artin–Schreier. (The two Artin–Schreier theorems thus bracket the problem: their characteristic–2 map builds the tower of Proposition 5.8, their theory of orderable fields names the wall.) This classifies where the mechanism can operate at all: it needs an ordering making squares nonnegative (formal reality, hence a real embedding) *and* bounded twisting phases (finite unit group); by Dirichlet's unit theorem, unit rank $r_1 + r_2 - 1 = 0$ with $r_1 \geq 1$ forces degree one. $\mathbb{Q}$ *is the only number field over which the mass barrier can operate*: over imaginary quadratic fields squares lose sign–definiteness (over $\mathbb{Z}[i]$ even $-1$ is a square), and over every other field the unit rank is positive and twisted masses are unbounded. This is a classification of the wall's habitat, not an existence claim elsewhere, but over $\mathbb{Z}[i]$ existence is no longer a claim: it is a list of sixteen. The ladder of imaginary quadratic rings confirms the mechanism quantitatively. Over $\mathbb{Z}[\sqrt{-2}]$, whose only units are $\pm 1$, precisely the twisting freedom that provably gains nothing over $\mathbb{Z}$; sign–twisted cycles nevertheless exist from norm 11,462 on ($a = -90 - 41\sqrt{-2}$, $b = 75 - 45\sqrt{-2}$, seven prime classes): the complex embedding alone carries the phenomenon. Over the Eisenstein ring $\mathbb{Z}[\omega]$, with six units, cycles appear at norm 2,964 with *six* prime classes ($a = -40 + 22\omega$, $b = 56 - 21\omega$); the smallest derivative two–cycle exhibited in any ring, and the population at fixed height grows with the unit group: at norm $\leq 2 \times 10^7$ the censuses read 14, 56, and 278 twisted–cycle instances for unit groups of order 2, 4, 6 (a factor ∼4–5 per two extra units, as a phase count predicts) while the strict canonical systems of all three rings remain empty to norm $2 \times 10^9$ ($3 \times 10^9$ candidates in total, partners by full factorisation). Two further objects appeared over $\mathbb{Z}[\sqrt{-2}]$: a composite with *unit* derivative ($a' \sim 1$; impossible over $\mathbb{Z}$, where $n' = 1$ characterises the primes), and an *anti–holomorphic* cycle: $a = -3163-1474\sqrt{-2}$ satisfies $a' = -\bar{a}$ exactly, under a single conjugation–equivariant sign assignment closing both directions ($N(a) = N(b) = 14{,}349{,}921$, five prime classes each side). The equation $a' = u\bar{a}$ is vacuous over $\mathbb{Z}$ only in its *one–dimensional* reading, where it says $n' = \pm n$. In the reading that matches the problem it is not vacuous but central: by Proposition 5.2 the Pythagorean layer over $\mathbb{Z}$ *is* the anti–holomorphic equation $\mathcal{D}z = \mu\bar{z}$ for the coordinatewise derivative on $z = a + bi$, and #307 is exactly its unit–multiplier case. The ring ladder and the main problem are therefore instances of one equation rather than analogues; what singles out $\mathbb{Z}$ is the extra constraint $\operatorname{Im}\mu = 1$, which confines the multiplier to a line meeting $\mathbb{Z}[i]^\times$ in the single point $\mu = i$. That the anti–holomorphic sector is thin is uniform across the ladder: over $\mathbb{Z}[\sqrt{-2}]$ the cycle above is still the *only* one after widening the census simultaneously in all three directions; prime–norm $\leq 2 \times 10^4$, $\omega \leq 8$, product–norm $\leq 10^{10}$ (a factor 700 past the cycle itself), $8.09 \times 10^8$ supports, `hunt/antiholo_hunt.rs`, and over $\mathbb{Z}$ the sector is empty below $10^{112}$ by Corollary 10.3. We record the plateau as consistent with genuine finiteness, not as evidence for it: a random model for $a' = -\bar{a}$ predicts a count growing like $\log X$, which the tested range cannot separate from a constant. All cycles were verified exactly by independent recomputation. Finally, the second pillar was tested in isolation: over $\mathbb{Z}[\sqrt{2}]$ (*totally real*, so squares are sign–definite in both embeddings, but with infinite unit group $\pm(1+\sqrt{2})^{\mathbb{Z}}$) twisted cycles appear already at norm 882: with unit twists of height $\leq 2$, $a = 21\sqrt{2}$ (classes $\sqrt{2}$, $-1+2\sqrt{2}$, $1+2\sqrt{2}$, 3) cycles exactly with $b = 51 + 13\sqrt{2}$, of norm 2263 and only *two* prime classes (against the fifty–nine that $\mathbb{Z}$ demands) and the composite $369 - 15\sqrt{2}$ has derivative exactly $\varepsilon^6 = 99 + 70\sqrt{2}$, a unit. Both pillars of the classification are thus load–bearing, and each is now confirmed by explicit counterexample on the soil where it alone fails: drop formal reality and cycles appear (imaginary quadratic rings, norms $10^3$–$10^4$); keep formal reality but drop unit finiteness and cycles appear (real quadratic, norm 882, the smallest in any ring); over $\mathbb{Q}$, where both hold, the barrier stands at $10^{112}$ by theorem. The bracketing of $\mathbb{Q}$ is complete from both axes. The deep census sharpens it: at norm $\leq 2 \times 10^7$ the real–quadratic ring carries 30,204 twisted–cycle instances against 14, 56, 278 for the imaginary rings with $2, 4, 6$ units infinite unit rank dominates by two orders of magnitude, and contains both the *minimal* architecture, $a = \sqrt{2}\,(5 - \sqrt{2})$ cycling with $b = 7$ at norms $(46, 49)$, two prime classes per member (the rational prime 7 being composite in the ring), the smallest derivative cycle found in any ring; and a *fully rational* pair:

$$D(3707) = 1547, \qquad D(1547) = 3707 \qquad (3707 = 11 \cdot 337, \quad 1547 = 7 \cdot 13 \cdot 17),$$

exactly, with unit heights $\leq 2$ on both sides, verified independently: two ordinary integers in a derivative two–cycle. The descent phenomenon that is empty over $\mathbb{Z}[i]$ to norm $10^{18}$ is realised over $\mathbb{Z}[\sqrt{2}]$ at four digits. The canonical system of $\mathbb{Z}[\sqrt{2}]$ remains cycle–free to norm $2 \times 10^9$ ($7.7 \times 10^8$ candidates), now with seven unit–derivative composites.

The cycle phenomenon over rings with units is governed by the theory of unit equations, which supplies its rigidity theorem.

**Proposition 3.7** ($S$–unit rigidity of twisted cycles)**.** *Let $\mathcal{O}$ be the ring of integers of a number field and $S$ a finite set of primes of $\mathcal{O}$. If $(a, b)$ is a twisted two–cycle with* $\operatorname{supp}(ab) \subseteq S$*, then dividing $D(a) = b$ by $b$ exhibits a solution of*

$$x_1 + \cdots + x_{\omega(a)} = 1, \qquad x_i = u_i\,\frac{a/\pi_i}{b},$$

*in the $S$–unit group of the field, and likewise for $b$. By the theorem of Evertse–Schlickewei–Schmidt [46] (cf. [39]), such an equation has only finitely many nondegenerate solutions; hence* every fixed support carries at most finitely many twisted cycles free of vanishing subsums*, and the species can be infinite only through growing supports; precisely the behaviour of every census above. (Vanishing subsums correspond to partial zero–derivative patterns; none occurred in any search. The relation was verified exactly on the $\mathbb{Z}[\sqrt{2}]$ specimen: the four $S$–unit terms of $a = 21\sqrt{2}$ sum to 1.)*

**Remark 3.8** (Galois folding, and the thinness of symmetric sectors)**.** Over a quadratic field with automorphism $\sigma$, a $\sigma$–equivariant choice of unit values ($u_{\pi^\sigma} = u_\pi^\sigma$) makes $D$ commute with conjugation, so the *single* equation $D(a) = a^\sigma$ already implies that $(a, a^\sigma)$ is a twisted two–cycle (checked symbolically and on 1,988 random instances). In the two–prime case the partner is forced: $\rho = -\big(\bar{u}v\,\pi + \bar{v}\,(\pi^\sigma)^2\big)/\big(N(u) - N(\pi)\big)$, with integrality a congruence and the entire residue of the problem the primality of $|N(\rho)|$ along Pell–exponentially growing sequences; Lehmer territory rather than Bunyakovsky. Whether a folded sector is starved or rich is decided by its degrees of freedom. When the fold leaves no slack the sector is thin: the folded *two*–prime family over $\mathbb{Z}[\sqrt{2}]$ is empty in the tested box (36,000 parameter triples, none even integral), the conjugation–stable sector over $\mathbb{Z}[i]$ is empty to Gaussian norm $10^{18}$, and the anti–holomorphic cycle over $\mathbb{Z}[\sqrt{-2}]$ is unique in its census. But with *three* primes the folded equation $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ carries three unit unknowns against two real equations, and this underdetermined system is *populated*: an exhaustive box ($p_i \le 137$, unit heights $\le 4$) contains twenty–four solutions over eleven prime triples, the smallest being

$$a = (3+\sqrt{2})(5-2\sqrt{2})(5-\sqrt{2}) = 57 - 16\sqrt{2}, \qquad D(a) = a^\sigma = 57 + 16\sqrt{2},$$

with unit values $\varepsilon^2$, $\varepsilon$, $-\varepsilon^{-1}$: *the derivative conjugates the element*, at norm 2737 on both sides, the equivariant return being automatic (verified exactly). Existence of twisted cycles over the four rings tested is a theorem by exhibition; for *every* ring with infinite unit group it is now reduced to the abundance of lattice points on these underdetermined unit systems (measured plentiful already at $k = 3$) a geometry–of–numbers question rather than a primality one.

**Remark 3.9** (The forced third prime: a Schinzel–type reduction over $\mathbb{Z}[\sqrt{2}]$, unblocked)**.** Within that abundance the arithmetic is sharper than free lattice–counting. Fixing $\pi_1, \pi_2$ and the units $u_1, u_2, u_3$, the folded equation rearranges to $\pi_3\,(u_1\pi_2 + u_2\pi_1) - (\pi_1\pi_2)^\sigma \pi_3^\sigma = -u_3\pi_1\pi_2$, a $2\times 2$ integer linear system for $\pi_3 = (x, y)$ whose determinant is $N(u_1\pi_2 + u_2\pi_1) - N(\pi_1\pi_2)$ (an identity, verified on all 168 solutions of the box). So $\pi_3$ is *forced* by $(\pi_1, \pi_2, u_1, u_2, u_3)$, and infinitude of twisted cycles is exactly the primality of this forced $\pi_3$ for infinitely many such data a Schinzel/Bunyakovsky–type condition over $\mathbb{Z}[\sqrt{2}]$. Crucially it is *not* the regime of Proposition 5.5: the exponential unit freedom $u_i = \pm\varepsilon^{k_i}$ keeps the family non–polynomial yet populated, so the polynomial–rigidity that forbids reducing the *rational* #307 to such a hypothesis does not apply here. A Schinzel–type theorem over $\mathbb{Z}[\sqrt{2}]$ would render twisted infinitude unconditional, whereas none touches #307 itself; the twisted problem sits strictly closer to the reach of current sieve theory than its rational shadow. Concretely the twisted solution set is *abundant* at accessible scale; thousands of cycles at modest norm (Proposition 3.12), and the count of distinct prime–norm signatures grows roughly in proportion to the norm bound in our census (e.g. $6, 12, 20$ at bounds $40, 80, 160$, unit heights saturating); whereas the rational Pythagorean layer is *empty* below $10^{112}$ (Corollary 10.3). The twisted and rational problems thus differ not in the difficulty of the search but in whether solutions exist to be found at all: for the twist they are plentiful and the only question is their primality, exactly the Schinzel–shaped residue above. That residue is a *pair*, not a triple or a Lehmer sequence: fixing $\pi_1$ and bounding the units, the cycle count still grows with $N(\pi_2)$ (verified:

$33, 48, 60$ for $\pi_1 = 3+\sqrt{2}$ at $N(\pi_2) \le 60, 120, 240$), so $\pi_1$'s primality is free and only the *simultaneous* primality of $\pi_2$ and the forced $\pi_3$ remains. Setting up the sieve pins the difficulty precisely. Writing $C = u_1\pi_2 + u_2\pi_1$, $E' = (\pi_1\pi_2)^\sigma$, $F = -u_3\pi_1\pi_2$, the forced third prime is $\pi_3 = (\bar{C}F + E'\bar{F})/\det$ with $\det = N(C) - N(E')$ a binary quadratic form in $(c, d) = \pi_2$. The first condition is the clean form $N(\pi_2) = c^2 - 2d^2$; the second, however, is *not* a second form but the rational condition $N(\pi_3) = N(G)/\det^2$ ($G = \bar{C}F + E'\bar{F}$ quadratic, $N(G)$ quartic, $\gcd(N(G), \det) = 1$), evaluated on the integral locus $\det \mid G$, and $N(\pi_3)$ is unbounded, spiking as $\det \to 0$ (values to $\sim 10^{11}$ already at $N(\pi_2) \le 2 \times 10^4$). On this *frozen* ($\pi_1$, units) slice the question is primality along the folded conic bundle, and the slice is thin: its unimodular fibres $\det = \pm 1$ are *empty* by a genus obstruction (there $\det = \pm 1$ would need $x^2 - 2y^2 \in \{-55, -43\}$, both non–residues). But the thinness is an artefact of freezing exactly the parameters that carry the abundance. The *full* twisted variety $D(a) = a^\sigma$ ($\pi_1, \pi_2, \pi_3$ and their units all free) is a cubic threefold whose solution count grows with positive exponent; a positive–density attack must run there, not on a fixed fibre, so the fixed–fibre conic sieve is a closed dead end. The honest placement is then: the reachable object is the full threefold at the Heath–Brown $x^3 + 2y^3$ frontier, compounded by a *coupled* second primality ($N(\pi_2)$ and $N(\pi_3)$ simultaneously prime); the two–sparse–conditions/parity barrier where the circle method loses minor–arc cancellation. This is strictly below the exponential/Lehmer tier of the immune families (Remark 1.16, where no sieve reaches at all), but a generation beyond present technology; it is where genuinely new analytic input for the twist would first have to bite.

The natural attempt to *prove* that infinitude (forcing one prime to be the conjugate of another so that a single free prime remains, whence Dirichlet would suffice) is blocked, and the block is a clean structural law.

**Proposition 3.10** (No conjugate prime pair in a twisted cycle)**.** *Let $D$ be a Leibniz derivative with unit prime–values over a quadratic ring with automorphism $\sigma$, and $a$ a squarefree twisted cycle $D(a) = a^\sigma$. Then no two prime factors of $a$ are $\sigma$–conjugate. Equivalently a twisted cycle's support is disjoint from its conjugate; it is never conjugation–stable, which is exactly why the exhibited $\mathbb{Z}[i]$ cycles are asymmetric, coming in conjugate pairs of* distinct *supports rather than on one symmetric support. This* separates *two problems easily conflated: the twisted equality $D(a) = a^\sigma$ treated here, and the symmetric target of Remark 3.5, which is a genuine two–cycle $D(m) = m'$ ($m \ne m'$, both conjugation–fixed) of the* descended *rational operator; not a twisted equality. The present proposition bars the first from the symmetric locus; it does not bear on the second, which remains genuinely open (the exhaustive emptiness there, Remark 3.5, is unexplained by it).*

*Proof.* Suppose a split prime $\pi$ and its distinct conjugate $\pi^\sigma$ both divide $a$. Reducing $D(a) = \sum_i u_i\,(a/\pi_i)$ modulo $\pi$ kills every term but the one at $\pi_i = \pi$, so $D(a) \equiv u_\pi\,(a/\pi) \not\equiv 0 \pmod{\pi}$ ($u_\pi$ a unit and $\pi \nmid a/\pi$ by squarefreeness). Yet $a^\sigma = \prod_i \pi_i^\sigma$ carries the factor $(\pi^\sigma)^\sigma = \pi$, so $a^\sigma \equiv 0 \pmod{\pi}$; hence $D(a) \ne a^\sigma$. An exhaustive $\mathbb{Z}[\sqrt{2}]$ search confirms no cycle carries a conjugate pair. □

We codify what is theorem and what remains open in this circle.

**Theorem 3.11** (Existence of twisted derivative two–cycles)**.** *Leibniz derivatives with unit prime–values admit genuine two–cycles over $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$, and $\mathbb{Z}[\sqrt{2}]$. The proof is by exhibited certificate, each verified exactly by independent recomputation: the smallest instances are, in order of norm, $\big(\sqrt{2}(5-\sqrt{2}), 7\big)$ over $\mathbb{Z}[\sqrt{2}]$ at norms $(46, 49)$; the Eisenstein cycle at norm* 2,964*; the sign–twisted cycle over $\mathbb{Z}[\sqrt{-2}]$ at norm* 11,462*, and the cycle over $\mathbb{Z}[i]$ at norm* 16,642*; together with the rational pair $D(3707) = 1547$, $D(1547) = 3707$ over $\mathbb{Z}[\sqrt{2}]$.*

**Proposition 3.12** (Geometry of the solution set)**.** *Over* $\mathbb{Z}[\sqrt{2}]$*, consider the Galois–folded system* $u_1\pi_2\pi_3+u_2\pi_1\pi_3+u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ *of Remark* 3.8. (i) *For* every *triple of split primes the system is solvable with the* $u_i$ *on the real unit hyperbola* $\{|st| = 1\}$*: solving the second embedding for* $u_2$ *and letting the third coordinate tend to zero, the first–embedding residual sweeps* $\pm\infty$ *while the remaining terms stay bounded, so a root exists by continuity; the real solution set is a nonempty curve* (*verified numerically on* 200 *random triples, all solvable*)*. The volume of the solution variety never vanishes.* (ii) *By Proposition* 3.7 *each fixed support carries finitely many integral solutions, so infinitude can only come from growing supports.* (iii) *The measured integral hit rates are* 11 *of* 455 *prime triples in the smallest box, and* 30,204 *cycles of norm* $\leq 2\times10^7$ *at twist height* $\leq 2$. (iv) *What remains open is precisely whether the discrete unit lattice meets the real curve in integer points for infinitely many supports, inhomogeneous Diophantine approximation on exponentially parametrised curves. Two shortcuts are closed: bounded–unit polynomial families are obstructed (the rational component of the return forces unit coefficients to track growing prime coordinates, which bounded units cannot), and even the cleanest forward family (*$D\big(\sqrt{2}(x-\sqrt{2})\big) = x+2$ identically in $x$*, supplying a candidate cycle whenever* $x^2-2$ *and* $x+2$ *are prime) has sporadic returns: the closing two–term unit equation, effectively finite for each* $x$*, is solvable at* $x = 5$ *and provably insolvable at* $x = 15$ *and* $x = 21$*. Those failures are, however, exactly the rigid case: a* prime *return* $x\pm2$ *leaves the closing system no slack. The identity has four unit branches* $D\big(\sqrt{2}(x\pm\sqrt{2})\big) = x\pm2$ *(independent signs, twists* 1 *and* $\pm\varepsilon^{\pm1}$*), and when* $x\pm2$ *is composite squarefree the closing system gains one unit unknown per prime class against the same two equations, and does close:* $x = 107, 121, 387, 1693$ *yield cycles with rational partners* $b = 105, 119, 385, 1695$*, each verified exactly. Infinitude along this family is thereby reduced to Bunyakovsky for* $x^2-2$ *together with recurring composite–return closure: the softest form the gap takes anywhere in this paper. Even the primality half of that reduction can be removed. By the half–dimensional sieve [15, 16] (applied to* $(2y+1)^2-2 = 4y^2+4y-1$ *to keep* $x$ *odd),* $x^2-2$ *is a* $P_2$ *infinitely often; every prime factor of* $x^2-2$ *has* 2 *as a quadratic residue and therefore splits in* $\mathbb{Z}[\sqrt{2}]$*, so* $\sqrt{2}(x-\sqrt{2})$ *is infinitely often squarefree and all–split with* $\omega \in \{2,3\}$*. The candidate supports thus exist* unconditionally*; all that remains open on this route is whether the unit gates open along that supply infinitely often. Exactness migrates; it does not disappear.*

**Proposition 3.13** (No vanishing derivatives, no transport, and the price of exactness)**.** (i) *No squarefree* $z$ *with* $\omega(z) \geq 1$ *has* $D(z) = 0$*, for any twisted derivative over any of the rings considered (indeed over any UFD): modulo a prime* $\pi \mid z$ *every cofactor term but one vanishes, leaving* $D(z) \equiv u_\pi\, z/\pi \not\equiv 0 \pmod{\pi}$. *The zero–derivative counters of every census above are therefore theorems, not observations.* (ii) *Consequently cycles cannot be transported multiplicatively: for squarefree* $c$ *coprime to* $ab$*, Leibniz gives* $D(ca) = D(c)\,a + c\,D(a)$*, so* $(ca, cb)$ *is a cycle exactly when* $D(c) = 0$*; impossible. Cycles do not breed; each one requires fresh exactness.* (iii) *The two cycle equations can be bought by construction, at a conserved price. Buying* neither *leaves the generic stratum: two exact unit conditions (*30,204 *specimens). Buying* one *is free in two dual ways; forward,* $\pi := q-\varepsilon^m\sqrt{2}$ *makes* $D(\sqrt{2}\,\pi) = q$ *an identity (the four–branch family above is the case* $m = \pm1$*); return,* $\pi := (\rho+\varepsilon^{-1}\rho^\sigma)/\sqrt{2} = (c+2d)-d\sqrt{2}$ *for* $\rho = c+d\sqrt{2}$ *closes* $D(b) = \sqrt{2}\,\pi$ *identically for every associate* $b$ *of* $\rho\rho^\sigma$*, and the residue is one primality,* $|N(\pi)|$*, plus one exact unit gate, measured at* 2 *of* 50 *pairs with both primalities in the smallest box. The two hits re–derive, with no search, precisely the two census minimal–shape cycles with prime rational partner: norms* $(46, 49)$ *and* $(206, 49)$*. Buying* both *collapses the parameters to unit–indexed quadratics in* $q$ *with a square–discriminant condition; the resulting* $q = 7$ *line carries the known cycles at* $m = 1$ *and* $m = -2$ *(discriminants* 25 *and* 81*) together with one Galois image, and every other prime–norm rung out to* $|m| \leq 14$ *fails its return, verified exactly. Each purchased identity is repaid as a sparser arithmetic condition elsewhere: the hardness of exactness is conserved, never destroyed.*

**Remark 3.14** (The two faces of the gap: unit rank dichotomises the obstruction)**.** The invariant that governs abundance in Remark 3.6 (the unit rank $r = r_1 + r_2 - 1$ of Dirichlet) also governs the *type* of the residual open problem, and the two types are mutually exclusive. When $r = 0$ (the rings $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$) the unit group is finite, so a given pair of supports admits only $|\mathcal{O}^\times|$ twist assignments and hence finitely many candidate cycles: the solution locus is a set of *isolated points*, zero–dimensional, and infinitude can arise only from infinitely many *distinct* supports independently passing a closing test; a Bateman–Horn/Schinzel–type recurrence over prime tuples, with no continuous family to deform along. The constituent primalities are individually classical (each support prime is a value of a binary norm–form, so by Fermat–Dirichlet (and, for $\mathbb{Z}[i]$, by the Green–Sawhney theorem with *both* coordinates prime) there is no shortage of supports); it is their *simultaneous* closure that is not classical. This was tested directly: the minimal shape $a = (1 + i)\pi$ over $\mathbb{Z}[i]$ yields *no* cycle for any Gaussian prime $\pi$ of norm $\leq 4000$ under any of the sixteen twists, and the sixteen genuine cycles are irregular mixtures of inert and split classes ($\omega = 3, 4, 5$) with no one–parameter family among them; the signature of isolated points, not a curve. When $r \geq 1$ (every real quadratic ring, e.g. $\mathbb{Z}[\sqrt{2}]$), Proposition 3.12 makes the real solution locus a positive–dimensional curve in each support, but the unit lattice is an infinite rank–$r$ discrete set, and infinitude requires it to meet that curve in *integer* points infinitely often; inhomogeneous Diophantine approximation on an exponentially parametrised curve. No unit rank delivers both a classical gate and a continuous family: rank zero gives the former and denies the latter, positive rank the reverse. This is why no single ring lets the infinitude be *finished* by present methods; a structural account of the obstruction's universality, not a barrier theorem. It also locates $\mathbb{Z}$ exactly: the unique ring carrying *both* the rank–zero sporadicity *and*, by formal reality, the mass barrier; the worst cell of the trichotomy of Corollary 3.2, which is precisely why #307 is the hard representative of its family.

## 4 A breeder form and the feasibility of certificate search

Allowing *two* free primes on one side turns the divisor trick into a cleaner bilinear form, Euler's amicable–pair method, in the shape used by te Riele's breeding algorithms [13], and lets us state precisely the strongest honest conclusion: a *conditional* reduction, which nonetheless does not become a feasible finite search.

**Proposition 4.1** (Bilinear breeder form)**.** *Fix coprime squarefree $M_0, N$ and seek a two–cycle $a = M_0rp$, $b = Nq$ with $r, p, q$ distinct primes coprime to $M_0N$. Put $G := NM_0 - N'M_0'$ $(= \Delta_0)$, $c := M_0N'$, and $K := GN^2 + c^2$. Then the cycle equations $a' = b$, $b' = a$ are equivalent to*

$$(Gr - c)(Gp - c) = K, \qquad q = \frac{M_0rp - N}{N'}.$$

*Proof.* Expanding, $(Gr - c)(Gp - c) - K = -N'G\,(a' - b)$ (verified symbolically), so the bilinear equation is equivalent to $a' = b$; the second cycle equation $b' = a$ then determines $q$ as stated. $\square$

**Proposition 4.2** (The breeder is integral)**.** *Keep the notation of Proposition* 4.1 *and let $G \neq 0$. Then*

$$(Gr - c)(Gp - c) = K \iff G\,rp - c(r + p) = N^2,$$

*and every solution of that equation has*

$$N \mid a', \qquad q = \frac{M_0rp - N}{N'} = \frac{a'}{N} \in \mathbb{Z}.$$

*Proof.* Expanding, $(Gr - c)(Gp - c) - c^2 = G\big(G\,rp - c(r + p)\big)$ while $K - c^2 = GN^2$, so the two equations differ by the factor $G \neq 0$. Substituting $G = NM_0 - N'M_0'$, $c = M_0N'$, $a = M_0rp$ and $a' = M_0'rp + M_0(r + p)$ into $G\,rp - c(r + p) = N^2$ gives $NM_0rp - N'M_0'rp - M_0N'(r + p) = N^2$, that is

$$N\,(a - N) \;=\; N'\big(M_0'rp + M_0(r + p)\big) \;=\; N'a'.$$

Hence $N \mid N'a'$, and $N$ squarefree gives $\gcd(N, N') = 1$, so $N \mid a'$ and $q = a'/N$. □

This is worth isolating because the displayed $q = (M_0rp - N)/N'$ is a quotient by a quantity of the size of the barrier, and one might expect a candidate to be usable only when $N'$ happens to divide $M_0rp - N$ — a further demand of probability about $1/N'$, which would leave the construction hopeless for reasons having nothing to do with primality. There is no such demand. The expectation $\mathbb{E} \approx \big(\tau(K)/G\big)(\ln r \ln p \ln q)^{-1}$ is therefore the correct count: the breeder loses candidates only to primality, never to integrality. A scan of 51,132 frames (`code/breeder_integral.gp`) produced 9 admissible candidates, every one with integral $q$. Formalised in `lean/Erdos307/Breeder.lean`.

Each factorisation $K = d \cdot (K/d)$ with $d \equiv -c \pmod{G}$ proposes a candidate $(r, p) = \big((d + c)/G, (K/d + c)/G\big)$; a solution is any such candidate whose $r, p, q$ are all prime. A dual, partition–side form is occasionally useful:

**Proposition 4.3** (Subset form)**.** *For a support $U$ with $D = \prod_{p \in U} p$ and $\Sigma = \sum_{p \in U} D/p$, a partition $U = P \sqcup Q$ solves #307 iff $S = P$ satisfies $A(\Sigma - A) = D^2$, where $A := \sum_{p \in S} D/p$. Moreover $A(\Sigma - A) - D^2 = -(1 - s_P s_Q)\, D^2$.*

(Both identities verified symbolically and on $2 \times 10^4$ random supports.) The last identity says the "analytic" residual $1 - s_P s_Q$ and the "arithmetic" residual $A(\Sigma - A) - D^2$ are one equation up to the factor $D^2$: the two obvious steering handles carry no independent information.

**A conditional reduction.** Several ingredients of a construction are unconditional. Greedy Egyptian–fraction choice gives disjoint $P, Q$ with $0 < 1 - s_P s_Q = O(1/T)$ (solvability to any precision); a prime $r$ nearest $M_0N'/\Delta_0$ gives $G \leq \Delta_0^{0.475}(M_0N')^{0.525}$ unconditionally (Baker–Harman–Pintz [14]), and each small prime $\ell$ can be forced to divide $K$ by one congruence on a free prime slot of $N$ (Dirichlet then supplies the prime). Granting these, #307 reduces to a single prime–density statement,

(H) among the admissible candidates so produced, some triple $(r, p, q)$ is prime,

and (H) $\Rightarrow$ #307 is YES. This (H) is a Hardy–Littlewood/Bateman–Horn–type assertion for an explicit, computable, *non–polynomial* family; we know of no named conjecture implying it. That is not an accident of our construction:

**Proposition 4.4** (No fixed–shape family, of any growth rate)**.** *Let $P(t) = \{p_1(t), \ldots, p_k(t)\}$ and $Q(t) = \{q_1(t), \ldots, q_l(t)\}$ be families of integers of* any *shape — no polynomial, algebraic or analytic hypothesis is imposed — indexed by $t \in \mathbb{N}$, with $k, l$ fixed. Suppose each member takes positive prime values, each nonconstant member tends to $+\infty$, and $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for* infinitely many *$t$. Then every member is constant.*

*Proof.* Let $\varphi$ enumerate, in increasing order, the $t$ at which the relation holds; $\varphi$ is strictly monotone, so $\varphi(n) \to \infty$. Split each side into its constant members, contributing $A_P$ and $A_Q$, and its nonconstant ones, contributing $\varepsilon_P(t), \varepsilon_Q(t) > 0$ which tend to 0 because each nonconstant member tends to $+\infty$ and there are finitely many. Along $\varphi$ the relation holds at every index and the remainders still tend to 0, so Proposition 5.5's two steps apply verbatim: the limit gives $A_PA_Q = 1$, and subtracting leaves $A_P\varepsilon_Q + A_Q\varepsilon_P + \varepsilon_P\varepsilon_Q = 0$ with every term nonnegative and some term strictly positive if a nonconstant member exists. □

The proof of Proposition 5.5 uses polynomiality in exactly one place, to pass from "holding at infinitely many $t$" to "holding identically", and restricting to the subsequence removes that need. What the argument actually consumes is three hypotheses — finitely many members, positive values, nonconstant members growing — and none of them is about shape.

The strengthening is not cosmetic. Excluding polynomial families rules out a reduction of #307 to Bunyakovsky, Schinzel or Bateman–Horn, but it says nothing about *exponential* families, and those are what the classical technique for problems of this kind actually uses: Thābit's rule for amicable pairs runs on primes $3\cdot 2^n - 1$, $3\cdot 2^{n-1} - 1$, $9\cdot 2^{2n-1} - 1$, and Euler's generalisation is of the same shape. A member $3\cdot 2^n - 1$ is positive and tends to infinity, so it satisfies the hypotheses above and the family is excluded. Proposition 4.4 therefore closes the whole Thābit/Euler programme for #307, not merely its polynomial shadow: no fixed–shape rule of any growth rate can generate solutions, and the constructive calculus of Section 5 is not failing for want of a cleverer ansatz.

Two limits on the statement should be read with it. The hypothesis that the families have *fixed* cardinality is used, in the splitting into constants plus a remainder tending to 0, and a family whose support grows with $t$ is not covered; that gap is real, and harmless, since a growing support is a growing number of simultaneous primality demands rather than a weaker one. And positivity is a genuine hypothesis, not a consequence, for the reason recorded in Remark 4.5. Formalised as `Erdos307.no_family_of_any_shape` on three standard axioms, in the same shape as Proposition 5.5: the abstract core is proved, and the two facts the family supplies — that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to 0 — enter as hypotheses rather than being derived, so what is imported stays visible. The theorem's statement quantifies over arbitrary index types and arbitrary $f_P : \mathbb{N} \to \iota \to \mathbb{R}$; no polynomial occurs anywhere in it.

**Remark 4.5** (Erratum: positivity is a hypothesis, not a consequence)**.** Earlier versions asked only that each polynomial take prime values for infinitely many integers $t$, without requiring the values to be positive on a common branch. As printed that statement is *false*: take $P = (2, 2, t, -t)$ and $Q = (2, 2)$. Each member takes prime values for infinitely many integers, and

$$\left(\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{t} - \tfrac{1}{t}\right)\left(\tfrac{1}{2} + \tfrac{1}{2}\right) = 1$$

identically, with two nonconstant members present. The proof's step "every term on the left is nonnegative" silently assumes positivity, which is exactly the missing hypothesis; with it, as now stated, the proof is correct and the counterexample is excluded.

The formalisation carries the repaired statement, not the printed one: `Erdos307.nonconstant_empty` takes $0 < f_P(i)$ for every nonconstant member as an explicit hypothesis, so the Lean development was already stating the corrected proposition. This is the one place in the paper where formalisation caught a missing hypothesis before review did.

This upgrades the earlier computational observation (that the breeder family is not polynomially parametrised) to a theorem for *all* fixed–shape families: the prime–density hypothesis behind #307 is irreducibly non–polynomial, which is why no standard conjecture applies.

**Proposition 4.6** (Near–miss frames are extremal for $G$)**.** *Let $a$ be squarefree with $a'$ squarefree, and take the frame $(M_0, N) = (a, a')$, so that $M_0' = a'$ and $N' = a''$. Then*

$$G \;=\; NM_0 - N'M_0' \;=\; a'\,(a - a''),$$

*and since $a'' \neq a$ unless #307 is solved, $|a - a''| \geq 1$ and therefore $|G| \geq a'$.*

*Proof.* Direct substitution; the inequality is $|a - a''| \geq 1$ for the nonzero integer $a - a''$. $\square$

The point is not that the bound is large but that it runs the wrong way. Writing $r = a''/a$ for the near–miss ratio, $G = aa'(1 - r)$, so a frame built from a good near–miss has $1 - r$ small and $aa'$ correspondingly large, and the two cancel *exactly*, because $a(1 - r) = a - a''$ is an integer and a nonzero one. Improving the near–miss therefore does not shrink $G$; it drives $a'$ up and $G$ with it. The witness of Proposition 6.9, with $|1 - r| = 1.65 \times 10^{-13}$, gives $|G| \geq a' > 10^{112}$, against the $\log_{10} G \approx 13$ that the Baker–Harman–Pintz route reaches at the barrier. So the natural reading of "steer $\Delta_0$ small by steering $s_{M_0} s_N$ to 1" is not merely circular in the sense of Section 5: on near–miss frames it is inverted.

Small $|G|$ does occur, but only on small frames: over 1,143,888 coprime squarefree frames with $M_0, N \leq 2000$ the minimum is $|G| = 5$, at $(M_0, N) = (2, 3)$. That is the supply–demand conflict in raw form, and it can be measured. Binning by $|G|$ and taking the best frame in each bin (`code/breeder_supply.gp`) gives

| $\log_{10} \lvert G \rvert$ | 0.5 | 1.5 | 2.5 | 3.5 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|
| $\max \tau(K)/\lvert G \rvert$ | 0.600 | 0.313 | 0.214 | 0.053 | 0.0065 | 0.0030 |

The ratio is largest on the smallest frames, where it is already below 1, and decays monotonically from there. Since a productive frame needs $\tau(K)/|G|$ well above 1, no frame in the range supplies one, and the trend is against scale rather than with it.

**The descent, and why it cannot be iterated.** Adjoining a prime $w$ to $M_0$ acts on $G$ *linearly*: with $M_0 \mapsto M_0 w$ one has $M_0' \mapsto M_0' w + M_0$ and hence

$$G \longmapsto wG - M_0 N', \qquad \text{so} \qquad |G_{\text{new}}| = |G| \cdot |w - w^*|, \qquad w^* := \frac{M_0 N'}{G} = \frac{\sigma(N)}{\varepsilon},$$

writing $\varepsilon = 1 - \sigma(M_0)\sigma(N) = G/(M_0 N)$. Since $G = 0$ is #307, this is a descent towards a solution, and on the $\varepsilon$ scale it is quadratic: $\varepsilon_{\text{new}} = \varepsilon^2 \theta / \sigma(N)$ with $\theta = w - w^*$. It is a Newton iteration for #307.

Three facts close it. First, it cannot start below the barrier: $w^* = \sigma(N)/\varepsilon \geq 2$ needs $\varepsilon \leq \sigma(N)/2$, i.e. $\sigma(M_0)\sigma(N)$ near 1, which by AM–GM needs $\sigma(M_0 N) \geq 2$ and so $\omega(M_0 N) \geq 59$. Over 4,284,896 coprime squarefree frames with $M_0 \leq 5000$, $N \leq 3000$ the smallest $|\varepsilon|$ is 0.470, and $w^* < 2$ throughout.

Second, only $w = \lfloor w^* \rfloor$ or $\lceil w^* \rceil$ can shrink $|G|$, since $|G_{\text{new}}| < |G|$ demands $|w - w^*| < 1$. So a step descends only when one of those two integers is prime, with probability $\approx 2/\log w^*$.

Third, and decisively, a successful step squares the difficulty. From $w^* = M_0 N'/G$,

$$w^*_{\text{new}} = \frac{M_0 w N'}{G\theta} = w^* \cdot \frac{w}{\theta} \;\approx\; \frac{(w^*)^2}{\theta},$$

so $\log w^*$ more than doubles at every step while $\log |G|$ falls by only $|\log \theta| \approx 0.69$. The gain is linear in the number of steps $k$ and the cost is quadratic in the exponent: $|G|$ falls like $2^{-k}$, whereas the probability of surviving $k$ steps is $\prod_{j<k} 2/\log w_j^* \asymp 2^{-k^2/2}$. Starting from the greedy frame $N = 30$, $M_0 = \prod_{7 \leq p \leq 271} p$, where $\log |G| = 249$ and $w^* = 428.2$ (`code/breeder_descent.gp`):

| paths available | $10^2$ | $10^4$ | $10^6$ | $10^8$ | $10^{10}$ | $10^{12}$ |
|---|---|---|---|---|---|---|
| reachable depth $k$ | 3 | 5 | 6 | 7 | 8 | 8 |
| $\lvert G \rvert$ shrinks by | 8 | 32 | 64 | 128 | 256 | 256 |

against the factor $10^{108}$ required to reach $G = 0$. Even $10^{12}$ independent descent paths buy eight steps and a factor 256. This is the quantitative form of the circularity of Section 5: the Newton iteration for #307 exists, is exact, and converges quadratically, and each step costs a prime whose size is the square of the last one's.

**Why (H) is not a finite search.** It is tempting to read (H) as "run a certificate hunt that is expected to succeed." It is not, for a quantitative reason. The expected number of certificates from one frame is $\mathbb{E} \approx \big(\tau(K)/G\big) \cdot (\ln r \, \ln p \, \ln q)^{-1}$, and *both* factors are pinned by the geometry.

- *Demand.* $G = \Delta_0 = M_0 N(1 - s_{M_0} s_N)$ exactly, so $G$ is small only in the circular regime $s_{M_0} s_N \approx 1$; over $2 \times 10^4$ random coprime squarefree frames the ratio $G/(M_0 N)$ has median 1 and best value $\approx 10^{-0.23}$, and the honest Baker–Harman–Pintz reduction still leaves $\log_{10} G \approx 13$ at the barrier.

- *Supply.* $\tau(K)$ is bounded by the *size* of $K = GN^2 + c^2$ $(\geq c^2)$, which is an *output* of the frame, not a free dial. The maximal order of the divisor function [17, Thm. 317] caps $\log_2 \tau(K)$ at $\approx 36$ for $K \leq 10^{54}$ (the barrier geometry) and at $\approx 112$ even for $K \leq 10^{240}$; reaching $\tau(K) = 2^{120}$ needs $K \gtrsim 10^{263}$, i.e. frame scale far beyond the barrier.

Enlarging a frame grows $K \propto N^2$ but gains divisors only at rate $\log 2/\log\log K < 1$ per log–unit of $K$, while it grows $G$ at rate 1 per log–unit of $N$; so $\tau(K)/G$ cannot be driven up. Optimising $\mathbb{E}$ over all $(M_0, N)$ splits with geometry–consistent $K$ and the maximal–order $\tau$, and granting the most favourable demand and singular–series constants, yields $\log_{10} \mathbb{E} \leq -4$ at the barrier and *decreasing* with scale; empirically 385 factorable frames realise a median of 0 and a maximum of 1 admissible divisor, against the $\sim 10^5$ a single productive frame would need. A self–consistent accounting; capping the number $w$ of forced prime factors of $K$ by the frame's own residue freedom (counted generously), sharpens this to $\log_{10} \mathbb{E} \leq -8$ at $B = 56$, falling to $\approx -260$ at $B = 3000$: the deficit is *scale–invariant*, so no choice of scale, slot count, or budget split reaches expectation 1. (The construction is even self–similar: engineering $K$'s algebraic; e.g. Gaussian–integer; factorisation to supply divisors reduces to prescribing $N$ and $N'$ simultaneously, an arithmetic–antiderivative inverse problem of the same difficulty class as #307.) The breeder thus makes #307 conditional on (H) but does *not* render it a feasible finite computation: the supply–demand conflict through $K = GN^2 + c^2$ is exactly the circularity of Section 5 in quantitative form. There is also a structural reason the analogy with amicable pairs was bound to fall short: te Riele's breeding amplifies a *known seed* pair into new ones, whereas the barrier (Theorem 4.3) forbids any seed below $2.09 \times 10^{56}$ the method imports a precondition the problem cannot supply.

# 5 The deviation ladder, the plus layer, and a second route to the barrier

The material below accreted onto the "note added" above because it grew out of the imported union criterion, and it stayed there. None of it is prior work: it is the deviation ladder, the function–field places, the mod–8 law, the plus layer, the slot anatomy, and an independent derivation of the barrier constant, and it includes two results carrying Lean formalisations (Propositions 5.11 and 5.16). It is given its own section and subsections here so that it can be found.

### 5.1 The deviation ladder

**Proposition 5.1** (The deviation ladder)**.** *Let $(a, b)$ be a Pythagorean pair with $a < b$ and parameter $k$. Writing $\sigma(m) = \sum_{p|m} 1/p$, one has $k = \sigma(a) - b/a = a/b - \sigma(b) \in \mathbb{Z}$, and:*

(1) *$k \le 0$, with $k = 0$ exactly the (open) two–cycle case. Indeed $b' = a-bk > 0$ and $b > a \ge 1$ force $k \le 0$ (if $k \ge 1$ then $b' \le a - b < 0$). Moreover $|k| < \min(b/a, \sigma(b)) \le \sigma(b) \le \log\log b + O(1)$, so the ladder has finitely many rungs; and, since $\sigma(b)$ barely exceeds 2 at the relevant scales, in practice only one or two.*

(2) (Parity.) *For odd squarefree $m$, $m' \equiv \omega(m) \pmod 2$; for even squarefree $m$, $m'$ is odd. Hence if exactly one member is even, $k \equiv \omega(\text{odd member}) \pmod 2$; if both are odd, $\omega(a) \equiv \omega(b) \pmod 2$ and $k \equiv \omega(a) + 1 \pmod 2$. The case $k = 0$ recovers the parity dichotomy of Theorem 4.18.*

(3) *Each rung is the system $a' = b + ak$, $b' = a - bk$, equivalently $(\sigma(a) - k)(\sigma(b) + k) = 1$ (a shifted copy of the cross–match $st = 1$) so every rung is exactly as rigid as $k = 0$. Since $\sigma(a) + \sigma(b) = a/b + b/a$ is independent of $k$, Corollary 10.3 holds on* every *rung: the whole ladder is empty below $10^{112}$. (We make no quantitative claim that $k = 0$ is easier than the rest.)*

(4) *On the rung $k = -1$ the equations read $a' = b - a$, $b' = a + b$ and $a'^2 + b'^2 = 2(a^2 + b^2)$: the derivative acts on $z = a + bi$ as the orientation–reversing $\sqrt{2}$–similarity $z \mapsto (-1 + i)\,\bar{z}$.*

(5) *By uniqueness of the splitting (Bado, Thm. 7.5), each squarefree $N$ with $N'^2 - 4N^2 = \square$ carries, under the convention $a < b$, a well–defined invariant $k(N) \in \mathbb{Z}_{\le 0}$; #307 holds iff $k(N) = 0$ for some $N$, and by (3) the invariant lives on a domain that is empty below $10^{112}$.*

Formalised. *`Erdos307.k_nonpos` is the bound $k \le 0$ of* (1)*; the finiteness of the ladder, $|k| < \min(b/a, \sigma(b))$, is not formalised. Further, `rung_parity_mixed` and `rung_parity_odd` are* (2)*, `rung_cross` and `rung_mass_indep` are* (3) *(the second being the invariance that carries Corollary 10.3 onto every rung), and `rung_minus_one` is* (4)*. Corollary 5.5 is `k_determined`, with the confinement to $k \in \{0, -1\}$ entering as a hypothesis rather than proved, and Bado's uniqueness of the splitting in* (5) *cited* (`lean/Erdos307/Ladder.lean`)*.*

The whole ladder is one equation in $\mathbb{Z}[i]$, and the rung index is a Gaussian multiplier.

**Proposition 5.2** (Gaussian multiplier form of the ladder)**.** *For coprime squarefree $a, b$ put $z = a+bi \in \mathbb{Z}[i]$ and let $\mathcal{D}z := a'+b'i$ be the coordinatewise arithmetic derivative. Then, unconditionally,*

$$z \cdot \mathcal{D}z \;=\; (aa' - bb') \;+\; i\,(ab)'.$$

*Consequently $(a, b)$ is a Pythagorean pair if and only if $\mathcal{D}z = \mu\bar{z}$ for a Gaussian integer $\mu$ with $\operatorname{Im}\mu = 1$; that multiplier is $\mu = k + i$ with $k$ the deviation of Proposition 5.1, and*

$$k = \frac{aa' - bb'}{a^2 + b^2}, \qquad a'^2 + b'^2 = (k^2 + 1)(a^2 + b^2) = N(\mu)\,(a^2 + b^2), \qquad \gcd(a', b') \mid k^2 + 1.$$

*Hence #307 asks for a Pythagorean pair whose multiplier is a* unit*. Since $\{k + i : k \in \mathbb{Z}\} \cap \mathbb{Z}[i]^\times = \{i\}$, the two–cycle rung $k = 0$ is the unique rung on which $\mathcal{D}$ acts as an isometry ($a'^2+b'^2 = a^2+b^2$); every other rung is an expanding anti–similarity of ratio $\sqrt{k^2 + 1} > 1$, and so cannot be periodic. The condition $\operatorname{Im}\mu = 1$ is the Pythagorean equation and $\operatorname{Re}\mu = 0$ is #307: one coordinate of a single Gaussian integer each.*

*Proof.* Leibniz gives $\operatorname{Im}(z\,\mathcal{D}z) = a'b + ab' = (ab)'$ and $\operatorname{Re}(z\,\mathcal{D}z) = aa' - bb'$, which is the identity (verified symbolically and on 7,501 coprime squarefree pairs). If $(a,b)$ is Pythagorean then $(ab)' = a^2+b^2 = N(z)$, so $z\,\mathcal{D}z = (aa'-bb')+i\,N(z)$; dividing by $N(z) = z\bar{z}$ gives $\mathcal{D}z/\bar{z} = (aa'-bb')/N(z)+i$, which lies in $\mathbb{Z}[i]$ with real part the integer $k$ of Proposition 10.2. Conversely $\mathcal{D}z = \mu\bar{z}$ with $\operatorname{Im}\mu = 1$ forces $(ab)' = \operatorname{Im}(z\,\mathcal{D}z) = \operatorname{Im}(\mu)N(z) = a^2+b^2$. Substituting $a' = b + ak$, $b' = a - bk$ gives $\mathcal{D}z = (k+i)\bar{z}$ identically, whence the norm identity $|\mathcal{D}z|^2 = |\mu|^2|z|^2$. Finally $g = \gcd(a', b')$ divides both $k(b+ak) + (a-bk) = a(k^2+1)$ and $(b+ak) - k(a-bk) = b(k^2+1)$, and $\gcd(a,b) = 1$. $\square$

The ladder of Proposition 5.1 has a twin, visible only after extending the derivative to $\mathbb{Q}$ by the quotient rule $(a/b)' = (a'b - ab')/b^2$ [8].

**Proposition 5.3** (The co–ladder and the arithmetic Riccati equation)**.** *For coprime squarefree $a \neq b$ the following are equivalent:* (R) *$u = a/b$ satisfies the arithmetic Riccati equation*

$$u' \;=\; 1 - u^2;$$

(W) *$a'b - ab' = b^2 - a^2$;* (L) *there is $k \in \mathbb{Z}$ with* $a' = b + ak$ and $b' = a + bk$; *the ladder of Proposition 10.2 with the sign of one equation flipped. The deviation cancels in the quotient: on* (L)*, $a'b - ab' = (b+ak)b - a(a+bk) = b^2 - a^2$ identically, so the Riccati equation is the $k$–free form of this* second *ladder. On it:*

(i) (Positivity barrier.) *$k \geq 0$ always: with $a < b$, $k \leq -1$ would give $b' = a + bk \leq a - b < 0$. Hence $\sigma(a) + \sigma(b) = a/b + b/a + 2k > 2$, so every co–ladder pair has $\omega(ab) \geq 59$ and $ab > 7.9 \times 10^{112}$: the co–ladder is empty at every reachable scale* by positivity alone*; a second transversal relaxation of the two–cycle carrying the full barrier through a mechanism independent of the minus square. Positivity separates the two cases further. If $k \geq 1$ then $\sigma(a) = k + b/a > 2$, since $a < b$, so the* smaller *member alone carries mass exceeding 2: $\omega(a) \geq 59$ and $a \geq \Pi_{59} > 8.77 \times 10^{112}$, against* $\min(a,b) \geq 2.09 \times 10^{56}$ *for a two–cycle. A co–ladder point with $k \geq 1$ is therefore larger on its smaller member by a factor exceeding $4 \times 10^{56}$, and $k = 0$ is the only case the barrier permits below $\Pi_{59}$: the relaxation buys nothing in the range where anything could be found, which is the conclusion Proposition 7.25 reaches for the mass condition, here for the ladder.*

(ii) (The cycle is the intersection.) *A two–cycle is exactly a common point of the two ladders at $k_{\mathrm{sym}} = k_{\mathrm{anti}} = 0$: #307 asks for $u \in \mathbb{Q}_{>0} \setminus \{1\}$ satisfying the Riccati equation together with the Pythagorean equation of Proposition 10.2. In multiplier form the two ladders read $\mathcal{D}z = kz + i\bar{z}$ (symmetric; norm $(k^2+1)N(z) + 4kab$) and $\mathcal{D}z = (k+i)\bar{z}$ (antisymmetric; norm $(k^2+1)N(z)$), meeting at $\mathcal{D}z = i\bar{z}$.*

(iii) (Inversion symmetry.) *$(1/u)' = 1 - (1/u)^2$ is automatic from* (R)*; the Riccati layer is symmetric in $a \leftrightarrow b$, as* (L) *is.*

*Proof.* (W)$\Leftrightarrow$(L): the equation rearranges to $b(a' - b) = a(b' - a)$, and $\gcd(a,b) = 1$ forces $a \mid a' - b$, $b \mid b' - a$ with equal quotients; the converse is the displayed cancellation. (R)$\Leftrightarrow$(W) is the quotient rule. In (i), summing $\sigma(a) = b/a + k$ and $\sigma(b) = a/b + k$ gives the mass bound (strict, as $a \neq b$), and mass $> 2$ forces $\geq 59$ primes as in Theorem 4.3. In (iii), $(1/u)' = -u'/u^2 = (u^2-1)/u^2$. All identities were verified symbolically; exhaustively, no co–ladder pair with $ab \leq 10^7$ exists (3,918,188 candidate pairs through the $a \mid a' - b$ prefilter; the barrier's prediction), and a first small–box probe over $\mathbb{Z}[i]$ (canonical $\pi' = 1$) also finds none, though there the positivity mechanism is unavailable and the layer's status is genuinely open (`code/riccati_coladder.py`). $\square$

The two ladders embed in a plane, and the barrier becomes a half–plane law.

**Proposition 5.4** (The deviation half–plane)**.** *For coprime squarefree $a \neq b$ call the pair* integral *if both deviations $s = (a' - b)/a$ and $t = (b' - a)/b$ are integers, and place it at $(s,t) \in \mathbb{Z}^2$; the antisymmetric ladder is the line $t = -s$, the co–ladder is $t = s$, and a two–cycle is the origin. Then*

$$\sigma(ab) \;=\; \frac{a}{b} + \frac{b}{a} + s + t \;>\; 2 + (s+t),$$

*so* every *stratum in the closed half–plane $s + t \geq 0$ carries the full barrier: $\omega(ab) \geq 59$ and $ab > 7.9 \times 10^{112}$. Both ladder barriers are the two central lines of this one inequality, and #307 sits at the origin; on the boundary of the protected region, at the crossing of its two distinguished lines. The open half–plane $s + t \leq -1$ is unprotected, and is populated from $ab = 30$ on: over $ab \leq 10^7$ exactly* 21 *strata occur, all obeying the law, the nearest being $(0,-1)$ and $(-1,0)$ at $(a,b) = (6,5)$ and $(5,6)$; off–axis strata occur ($(1,-13)$ at $(690,53)$), so integrality does not force one exact side. The $(0,-1)$ stratum with $b = a'$ is the chain equation*

$$a'' + a' = a, \qquad \textit{equivalently} \qquad \sigma(a)\big(1 + \sigma(a')\big) = 1,$$

*a companion of the cross–match $\sigma(a)\sigma(b) = 1$ with the same shape and one shifted factor. Its solutions up to $10^7$ are $\{6,\ 42,\ 15590,\ 47058\}$ and they split in two. Every primary pseudoperfect number whose predecessor is* prime *solves it, trivially: then $a' = a - 1$ and $a'' = 1$ ($6, 42, 47058$; $1806$ fails, its predecessor $1805 = 5 \cdot 19^2$ being composite, and $2$ fails as $a'' = 0$). The remaining solution $15590 = 2 \cdot 5 \cdot 1559$ is not on the pseudoperfect line at all: $15590' = 10923 = 3 \cdot 11 \cdot 331$ and $10923' = 4667$, a genuine two–step closure. It is the first solution of the chain equation whose derivative is composite; the sequence $6, 42, 15590, 47058$ does not appear in the OEIS. All census claims verified exhaustively (`code/deviation_halfplane.py`).*

*Proof.* $\sigma(a) = a'/a = b/a + s$ and $\sigma(b) = a/b + t$ sum to the display, with strict inequality by AM–GM ($a \neq b$); mass $> 2$ forces $\geq 59$ primes as in Theorem 4.3. The ladder identifications are Propositions 10.2 and 5.3; the census is a finite computation. ☐

**Corollary 5.5** (Below mass 5/2 the rung is not a choice but a parity)**.** *Let $(a,b)$, $a < b$, be a Pythagorean pair with $\sigma(ab) < 5/2$. Then $k \in \{0,-1\}$, and $k$ is* determined *by the parity of $\omega$:*

$$\begin{array}{ll} \textit{exactly one member even:} & k = 0 \iff \omega(\textit{odd member})\ \textit{even}; \\ \textit{both members odd:} & k = 0 \iff \omega(a)\ \textit{odd}. \end{array}$$

*Hence, at these masses, the single–integer test of Proposition 10.1 together with one parity check is* equivalent *to #307, not merely necessary for it: a squarefree $N$ with $N'^2 - 4N^2 = \square$ whose splitting has the stated $\omega$–parity* is *a two–cycle. In particular the finite verifications of Proposition 4.5 and Remark 1.1 are decisive in both directions; a surviving candidate would have been a solution, not a candidate, and the whole level–59/60 regime lies in this range, its masses being confined to $(2,\ 2.00235]$, where $r = b/a \leq 1.0497$.*

*Proof.* $\sigma(ab) = r + 1/r$ with $r = b/a > 1$, and $r + 1/r < 5/2 \iff r < 2$. By Proposition 5.1(1), $k \leq 0$ and $|k| < b/a = r < 2$, so $k \in \{0,-1\}$; a set of two consecutive integers is separated by parity, and Proposition 5.1(2) supplies that parity from $\omega$. The stated equivalences are the two cases of that law at $k \equiv 0$. The mass range for $|U| \in \{59, 60\}$ is Theorem 4.3 and Proposition 4.5. ☐

**Corollary 5.6** (Diagonal stratification: the cost of the last lattice step)**.** *Let $m(x)$ be least with $\sum_{i\le m(x)} 1/p_i > x$. The diagonal $s+t=-j$ carries the barrier $\sigma(ab)>2-j$, hence $\omega(ab)\ge m(2-j)$ and $ab\ge \Pi_{m(2-j)}$:*

$$\begin{array}{llll} j\ge 2: & \textit{mass} > 0, & \textit{no barrier}, & ab\ge 2;\\ j=1: & \textit{mass} > 1, & \omega\ge 3, & ab\ge 2\cdot 3\cdot 5 = 30;\\ j=0: & \textit{mass} > 2, & \omega\ge 59, & ab\ge \Pi_{59} > 8.77\times 10^{112}. \end{array}$$

*The deviation plane is thus stratified by mass, one diagonal at a time, and #307 sits on the first diagonal carrying a nontrivial barrier. The $j=1$ floor is* attained*: the census minimum on that diagonal is exactly* 30*, at $(a,b)=(5,6)$ and $(6,5)$; so the barrier is sharp one step off the origin. The final lattice step therefore costs*

$$\Pi_{59}/30 \;\approx\; 2.9\times 10^{111},$$

*one hundred and eleven orders of magnitude for a single unit of $s+t$. This is the barrier of Theorem 4.3 in its most compressed form: not a bound on solutions, but the price of the last step of an integer lattice walk, paid because the diagonal index is a mass threshold and the mass thresholds* 1 *and* 2 *sit at* 3 *and* 59 *primes.*

**Remark 5.7** (The populated strata calibrate the heuristic model)**.** The unprotected strata are the first family of #307–type exact coincidences where the average–case model can be tested against exhaustive truth. Pricing the second exactness condition of a candidate pair at $1/b$ and re–running the census enumeration with identical filters gives expected counts 11.4, 18.3, 26.1, 34.2 at $ab\le 10^4,\dots,10^7$ against observed $6,12,18,23$: the ratio is stable near $2/3$ over four decades; the model has the right order of growth and a constant local correction, i.e. it is *calibrated* at the one–condition level. Composing two levels, however, already breaks the naive pricing: for the chain equation $a''+a'=a$ the crude spread price predicts $E=48.8$ against the observed 4; a twelvefold bias at reachable height, because the composition requires mass–window conditioning between the levels (the stratified–versus–naive lesson of Section 2, here measured rather than inferred). The reliability of averaging thus degrades measurably with each composed exact condition, and the two–cycle at the origin composes many. This is the quantitative face of the average–to–instance gap: the model earns trust stratum by stratum, and loses it condition by condition (`code/strata_calibration.py`).

The Pythagorean layer behaves very differently over function fields, and refines into two arithmetic sub–layers over $\mathbb{Z}$.

**Proposition 5.8** (The Pythagorean layer over function fields)**.** *Over $\mathbb{F}_q[t]$ with $q$ odd, $a^2+b^2=(ab)'$ has no coprime squarefree monic solutions: $\deg(a^2+b^2)=2\max(\deg a,\deg b)>\deg a+\deg b-1\ge\deg((ab)')$, the leading coefficient $1$ or $2\ne 0$ preventing cancellation. Over $\mathbb{F}_2[t]$ the layer is nonempty: up to degree* 8 *(exhaustively,* 23,548 *coprime squarefree pairs) its only solutions are the three pairs of the Artin–Schreier tower $x_{j+1}=x_j(x_j+1)$, namely $(t,t+1)$, $(t^2+t,t^2+t+1)$, $(t^4+t,t^4+t+1)$; each has $a'=(a+1)'=1$ and $(a(a+1))'=(a+1)+a=1$, and the climb halts at level* 4 *(there $(x_3+1)'=t^2+t\ne 1$). These pairs lie off the cycle stratum $k=0$, on the side the integer normalisation of Proposition 5.1(1) forbids. Cycles ($k=0$) remain impossible over every $\mathbb{F}_q[t]$ (Theorem 3.1): the function–field world separates the Pythagorean layer from the cycle layer, sharpening the integer obstruction to "the archimedean order forces $k\le 0$."*

The function–field immunity is more fragile than the characteristic: it is destroyed by inverting a single place.

## 5.2 Function–field places

**Proposition 5.9** (Inverting one place destroys the function–field obstruction)**.** *Over the $S$–integers $\mathbb{F}_q[t, t^{-1}]$, whose unit group $\mathbb{F}_q^\times \cdot t^{\mathbb{Z}}$ is infinite, unit–valued derivative two–cycles exist. Over $\mathbb{F}_3$ the smallest has degree* 3 *and only* two *prime classes:*

$$a = t^3 + t^2 + 1 = (t-1)(t^2 - t - 1), \qquad b = t^3 - t^2 - 1 = (t+1)(t^2 + t - 1),$$

*with $D(a) = b$ and $D(b) = a$ under the assignment $(u_1, u_2) = (t, 1)$ on each side, $a, b$ coprime squarefree and both coprime to $t$.*

*Proof.* $D(a) = t\,(t^2-t-1)+1\cdot(t-1) = t^3-t^2-1 = b$ and $D(b) = t\,(t^2+t-1)+1\cdot(t+1) = t^3+t^2+1 = a$, both exact in $\mathbb{F}_3[t]$; the factorisations are as displayed and share no factor. The values $t$ and 1 are units of $\mathbb{F}_q[t, t^{-1}]$, so this is a unit–valued Leibniz derivative in the sense used throughout Section 3. Verified independently by exhaustive search and by direct recomputation (`hunt/ff_twist.rs`). □

**Remark 5.10** (What the function–field theorem is really about)**.** Proposition 5.8 is a *degree* argument: with $\pi' = 1$ one has $\deg D(a) \le \deg a - 1$, so $D(a) = b$, $D(b) = a$ forces $\deg b \le \deg b - 2$. Sign twists cannot disturb it, because $\mathbb{F}_q^\times$ consists of constants and degree is blind to them, exactly as, over $\mathbb{Z}$, twisting by $\pm 1$ retains the full barrier (Remark 3.4). But allowing the values $t^k$, units once $t$ is inverted, makes $\deg D(a) \le \deg a - 1 + K$, and the two–cycle inequalities collapse to $0 \le 2K - 2$: the obstruction is vacuous from $K = 1$, and cycles duly appear at once. A census over $\mathbb{F}_3$ confirms both halves: at $K = 0$ the count is 0 at every degree tested, and at $K = 1, 2$ it is $4, 16$ / $4, 48$ / $12, 122$ for degrees $\le 4, 5, 6$. The moral sharpens Section 3's thesis rather than softening it. What immunises $\mathbb{F}_q[t]$ is not the characteristic, nor the finiteness of its unit group (cycles exist over $\mathbb{Z}[\sqrt{-2}]$, whose unit group has order 2) but the presence of a *single* distinguished place carrying a degree function that the derivative strictly lowers. Adjoining a second place destroys it. This is the exact structural analogue of the claim made for $\mathbb{Z}$: one archimedean place, one mass function the derivative must respect, and a barrier that survives every twist available in the ring.

**Proposition 5.11** (No place of $\mathbb{Q}$ carries the function–field descent)**.** *The mechanism isolated in Remark 5.10 (a place whose valuation the derivative strictly lowers) is unavailable over $\mathbb{Q}$, and this is a classification statement rather than a failure of ingenuity.*

(i) Archimedean place. $|D(n)| = n\,\sigma(n)$*, which exceeds $n$ for every $n$ of mass $> 1$; the derivative* expands*, first at $n = 30$, and on* 1.79% *of squarefree $n \le 10^6$. The absolute value is in any case not ultrametric, so the second half of the degree argument (*$\deg(x+y) \le \max$*) is unavailable too.*

(ii) $p$–adic places. *For squarefree $n$ with $p \mid n$, cofactor survival gives $v_p(D(n)) = 0 < 1 = v_p(n)$: the derivative lowers $v_p$ there. But it* raises *it elsewhere, there are squarefree $n$ with $p \nmid n$ and $p \mid D(n)$ (e.g. $p = 2$, $n = 15$; $p = 3$, $n = 14$; $p = 5$, $n = 6$), in comparable abundance to the lowering cases at every $p$ tested. So no $p$–adic place contracts uniformly.*

(iii) *By Ostrowski's theorem* (i) *and* (ii) *exhaust the places of $\mathbb{Q}$.*

*Hence no valuation–theoretic descent of the type that proves Proposition 5.8 can exist over $\mathbb{Q}$: the function–field argument fails to transfer for a structural reason internal to the classification of absolute values, not for want of a cleverer place. (Counts and witnesses: `code/no_place.rs`.)*

Formalised. *`Erdos307.not_descent_of_strictMono` gives* (i) *in a form stronger than stated,* killing *every* strictly *monotone valuation at once and not only the absolute value, on the witness*

$30 \mapsto 31$ (*der_thirty*); *padic_lowers_on_support is the lowering half of* (ii) *and is Rigidity, while* *raises_two*, *raises_three*, *raises_five* *check the three raising witnesses outright. Ostrowski enters* *no_place_descent* *as an explicit disjunction hypothesis rather than silently.* 0 *sorry* (`lean/Erdos307/NoPlace.lean`).

### 5.3 The mod–8 law and the plus layer

**Proposition 5.12** (Deviation calculus for $k$–cycles)**.** *For pairwise coprime squarefree $a_1, \dots, a_k$ with product $N$ and deviations $\kappa_i = (a_i' - a_{i+1})/a_i$ (indices mod $k$): the ratios $a_{i+1}/a_i$ have product 1 (telescoping), and $\sigma(N) = \sum_i a_{i+1}/a_i + \sum_i \kappa_i$. On the relaxed layer $\sum_i \kappa_i = 0$ (which contains the cycles $\kappa_1 = \cdots = \kappa_k = 0$) AM–GM gives $\sigma(N) \geq k$, so the distinct–prime barrier $m_k$ of Proposition 4.19 extends to the relaxed problem for every $k$, and to genuine $k$–cycles for $k \leq 3$; for $k \geq 4$ it inherits the pairwise–coprimality caveat of Proposition 4.19 (the bound then holds for the multiset reciprocal sum). At $k = 2$, $\kappa_1 = -\kappa_2$ is the deviation of Proposition 5.1 and $\sum \kappa_i = 0$ is $a^2 + b^2 = (ab)'$.*

**Proposition 5.13** (Split discriminant)**.** *For coprime squarefree $a, b$ with $N = ab$,*

$$N' + 2N - (a+b)^2 = N' - 2N - (a-b)^2 = (ab)' - a^2 - b^2.$$

*Hence $N$ is the product of a Pythagorean pair iff $N' - 2N$ and $N' + 2N$ are* both *perfect squares $d^2, s^2$ (their product is Bado's discriminant $N'^2 - 4N^2$), with $a, b = (s \pm d)/2$. The minus factor carries the obstruction: $N' - 2N = (a-b)^2 \geq 0$ forces $\sigma(N) \geq 2$, hence the 59–prime barrier, indeed* 0 *of* 103,596 *squarefree $N < 2 \times 10^5$ satisfy $N' - 2N = \square$ (Corollary 10.3 in one line). The plus factor is satisfiable at small scale (*114 *such $N < 2 \times 10^5$). In characteristic* 2 *the obstructing factor vanishes ($a - b = a + b$) and the criterion degenerates to $(a+b)^2 = N'$: the permissiveness of Proposition 5.8 and the barrier of Corollary 10.3 are the absence and presence of one and the same algebraic factor.*

Both squares of the split criterion are pinned modulo 8 by an elementary law that seems worth recording on its own.

**Proposition 5.14** (The mod–8 law of the derivative)**.** *For odd squarefree $m$ write $S(m) = \sum_{p|m} p$. Then*

$$m\,m' \equiv S(m) \pmod 8.$$

*Consequently:* (a) *modulo* 2 *this is the parity law $m' \equiv \omega(m)$ used in Theorem 4.18;* (b) *an all–odd two–cycle $a' = b$, $b' = a$ forces*

$$S(a) \equiv S(b) \equiv ab \pmod 8$$

*the two prime sums are congruent to each other and to the product; refining Theorem 4.18(ii);* (c) *in the mixed case $a = 2k$ ($k$ odd), the odd member obeys $b \equiv k\big(1 + 2S(k)\big) \pmod{16}$ and $S(b) \equiv 2kb \pmod 8$, and the even member the companion law $(2k)\,(2k)' \equiv 2 + 4S(k) \pmod{16}$;* (d) *for odd squarefree $N$ the plus quantity satisfies $N' + 2N \equiv N\big(S(N) + 2\big) \pmod 8$, so an odd plus–hit must have $N(S(N) + 2) \equiv 1 \pmod 8$ when $\omega(N)$ is odd and $\equiv 0, 4 \pmod 8$ when $\omega(N)$ is even. None of these congruences is an obstruction, consistently with Proposition 2.2, but they thin every hunt over the odd sector by a fixed factor.*

Formalised. *Erdos307.mod8_law* *is the law itself, resting only on* *odd_sq_mod8*; *prime_sums_congruent* *is* (b); *parity_law* *is* (a) *and is stated over* $\mathbb{N}$ *with* mod *, so it needs no* *ZMod* *API, and* *omega_even_of_even_deriva* *is the step Theorem 4.18*(i) *uses. Part* (b)*'s second half, $S(a) \equiv ab$, is* *prime_sum_eq_product*;

*part* (d) *is* ***plus_quantity_mod8*** *with the square–residue filter* ***plus_hit_residue*** *and the* $\omega$*–parity split* ***plus_hit_residue_split****, which supplies the* $\equiv 1$ *versus* $\equiv 0, 4$ *dichotomy; part* (c) *is* ***mixed_member_mod16****,* ***even_member_mod16*** *and* ***mixed_prime_sum_mod8****, the mod–*16 *statements chaining onto the law through its integer form* ***mod8_law_int****. All four parts are therefore formal* (`lean/Erdos307/Mod8.lean`)*.*

*Proof.* $q^2 \equiv 1 \pmod 8$ for odd $q$, so $m/q \equiv mq \pmod 8$; summing over $q \mid m$ gives $m' \equiv m\,S(m) \pmod 8$, and multiplying by $m$ (with $m^2 \equiv 1$) gives the law. (a) is the law modulo 2, since $S(m) \equiv \omega(m)$ for odd primes. (b) is the law at $a$ and at $b$ with $a' = b$, $b' = a$. In (c), Leibniz gives $b = (2k)' = k + 2k'$, and the law at $k$ gives $k' \equiv kS(k) \pmod 8$, whence $b \equiv k + 2kS(k) \pmod{16}$; the second congruence is the law at $b$ with $b' = 2k$, and the companion follows from $2k(k + 2k') \equiv 2k^2 + 4kk' \pmod{16}$ with $k^2 \equiv 1 \pmod 8$ and $kk' \equiv S(k) \pmod 8$. In (d), $s^2 \bmod 8 \in \{0, 1, 4\}$ and $s \equiv N' \equiv \omega(N) \pmod 2$. All parts were verified exhaustively over the 405,285 odd squarefree $m \leq 10^6$ and all odd plus–hits below $10^6$: zero exceptions (`code/mod8_plusthin.py`). □

**Proposition 5.15** (Structure of the plus–layer)**.** *Call squarefree* $N$ *with* $\omega(N) \geq 2$ *a* plus–hit *if* $N' + 2N$ *is a perfect square; the first are* 130, 145, 305, 689, 1745, 1870, 1970, 2353, … (114 *below* $2 \times 10^5$*; OEIS [38]*)*.* (i) *For a semiprime* $N = pq$, $2(N' + 2N) + 1 = (2p + 1)(2q + 1)$*, so a plus–hit forces* $(2s)^2 \equiv -2 \pmod \ell$ *for every prime* $\ell \mid (2p + 1)(2q + 1)$*, whence* $\ell \equiv 1, 3 \pmod 8$ *and* $p \equiv q \equiv 1 \pmod 4$ *(verified on all* 59 *semiprime instances; this explains the dominance of* 5 *and* 13*).* (ii) [*empirical*] $\#\{\textit{plus–hits} \leq X\} \approx 0.25\sqrt{X}$ *(ratios* 0.250, 0.253, 0.255 *at* $X = 5 \times 10^4, 10^5, 2 \times 10^5$*): the plus factor behaves like a random integer of its size, so (with Proposition 5.13) the whole integer obstruction sits in the minus factor.* (iii) *Fixing* $p = 5$, $q = (s^2 - 5)/11$ *with* $s$ *even and* $s \equiv \pm 4 \pmod{11}$ *gives* $q \equiv 1 \pmod 4$*, and each prime value yields the plus–hit* $N = 5q$ *(*68 *with* $s < 2000$*, up to* $N =$ 1,774,805*); Bunyakovsky's conjecture for this quadratic thus implies infinitely many plus–hits. The two discriminant factors separate cleanly: the plus–layer is Bunyakovsky–class, the minus–layer is the barrier.* (iv) *[Remark] The full cycle condition states that* $(d^2, N', s^2)$ *is a three–term arithmetic progression with common difference* $2N$ *whose outer terms are squares; a congruent–number–shaped configuration; the elliptic–curve machinery of that theory does not transfer because* $N'$ *is not an algebraic function of* $N$*. A* 251*–term b–file to* $N < 10^6$ *(counting ratio* 0.251*) and an OEIS submission draft accompany this note; the plus–hit sequence was not in OEIS when we searched; it is now A396915, submitted by us and cited above, so the entry is not independent corroboration.* (v) *[P, C]* General local law (QR tournament)*. For every odd prime* $\ell \mid N$ *of a plus–hit,* $N' + 2N \equiv N/\ell \pmod \ell$ *(all cofactor terms* $N/p$ *with* $p \neq \ell$ *carry a factor of* $\ell$*; only* $N/\ell$ *survives), so* $N/\ell$ *is a nonzero quadratic residue mod* $\ell$*. Verified with zero violations on all* 114 *plus–hits below* $2 \times 10^5$ *and all* 251 *terms to* $10^6$*; the semiprime* $p \equiv q \equiv 1 \pmod 4$ *theorem of* (i) *is its* $\omega = 2$ *shadow via quadratic reciprocity.*

The empirical $\frac{1}{4}\sqrt{X}$ law of part (ii) can be proved, up to logarithms, on the stratum that dominates the small–scale census.

**Proposition 5.16** (The semiprime plus–layer is provably thin)**.** *The number of semiprime plus–hits* $N = pq \leq X$ *is* $O\big(\sqrt{X}\,(\log X)^2\big)$*, by a fully elementary argument; Erdős's bound* $\sum_{s \leq Y} \tau(2s^2 + 1) \ll Y \log Y$ *[47] sharpens this to* $O(\sqrt{X} \log X)$*. Under Bunyakovsky's conjecture the* $p = 5$ *family of Proposition 5.15(iii) alone contributes* $\gg \sqrt{X}/\log X$*, so conditionally*

$$\frac{\sqrt{X}}{\log X} \ll \#\{\textit{semiprime plus–hits} \leq X\} \ll \sqrt{X}\,\log X :$$

*the* $\sqrt{X}$ *order of the empirical law is genuine on the semiprime stratum, not an artefact of the tested range.*

*Proof.* For a semiprime plus–hit $N = pq \le X$ one has $s^2 = N' + 2N = p + q + 2pq \le 3X + 1$ (as $p + q \le pq + 1$), and Proposition 5.15(i) gives $2s^2 + 1 = (2p + 1)(2q + 1)$: the map $N \mapsto \big(s, \{2p+1, 2q+1\}\big)$ into pairs (root, unordered nontrivial factorisation) is injective, so the count is at most $\sum_{s\le Y} \tau(2s^2 + 1)$ with $Y = \sqrt{3X + 1}$. The polynomial $2s^2 + 1$ has discriminant $-8$, prime to every odd modulus, so its roots modulo an odd prime are simple and lift uniquely by Hensel: $2s^2 + 1 \equiv 0 \pmod u$ has at most $2^{\omega(u)} \le \tau(u)$ solutions modulo $u$ (none for even $u$). Pairing each divisor of $2s^2 + 1$ with its cofactor and keeping the smaller member $u \le \sqrt{2s^2 + 1} \le \sqrt{2}\,Y$,

$$\sum_{s\le Y} \tau(2s^2 + 1) \;\le\; 2 \sum_{u\le\sqrt{2}Y} \#\{s \le Y : u \mid 2s^2 + 1\} + Y \;\le\; 2 \sum_{u\le\sqrt{2}Y} \tau(u)\Big(\frac{Y}{u} + 1\Big) + Y \;\ll\; Y(\log Y)^2,$$

by Dirichlet's $\sum_{u\le Z} \tau(u) \sim Z \log Z$ and $\sum_{u\le Z} \tau(u)/u \sim \frac{1}{2}(\log Z)^2$. For the conditional lower bound, the family of Proposition 5.15(iii) has $\asymp \sqrt{X}$ eligible $s \le \sqrt{2.2X + 5}$ (two residue classes modulo 22), of which Bunyakovsky for $(s^2 - 5)/11$ makes $\gg \sqrt{X}/\log X$ prime. The injection, the Hensel root count, and the censuses were verified numerically (59 semiprime hits below $2\times10^5$, all reproduced; $\sum_{s\le Y} \tau(2s^2{+}1)/(Y \log Y)$ stable near 1.05–1.09 for $Y = 500$–2000; `code/mod8_plusthin.py`). The $\omega \ge 3$ stratum remains empirical. □

**Remark 5.17** (What of the last argument is formal)**.** The proof splits into an elementary skeleton and an analytic tail, and the split is worth making explicit because the two have different status. The skeleton is formalised: `Erdos307.plus_semiprime_identity` is the factorisation $2s^2 + 1 = (2p + 1)(2q + 1)$, `plus_range` the bound $s^2 \le 3N + 1$ that confines $s$ to $Y$, `plus_recover` the injectivity, and `count_le_divisor_sum` the reduction of a count to $\sum_{s\le Y} \tau(2s^2 + 1)$ by injection into the disjoint union of the divisor sets. That reduction is stated abstractly, over any finite set carrying a root and a divisor: the notion "plus–hit" is not itself defined in Lean, so the instantiation to this layer is done in the prose above and not machine–checked. `card_large_le_card_small` and `tau_le_two_small` are the divisor–pairing step that reorganises that sum over the small member alone, and `two_roots_quadratic` is the Hensel root count at the prime level, in a form needing only $2 \ne 0$ rather than the discriminant. 0 `sorry` (`lean/Erdos307/PlusThin.lean`).

The rest of the elementary chain is formal as well. The count of $s \le Y$ in the admissible residue classes is `count_in_residue_classes`, and the divisor sum is bounded in the printed shape by `sum_tau_div_le_log_sq`, $\sum_{u\le Z} \tau(u)/u \le (1 + \log Z)^2$, from Mathlib's `harmonic_le_one_add_log`. Only Dirichlet's *asymptotics* are absent from the library, and the upper bound does not use them.

The step that was missing is now supplied. The chain needs $|R| \le \tau(u)$ for $R$ the roots of $2s^2 + 1 \equiv 0$ modulo $u$, and the half $2^{\omega(u)} \le \tau(u)$ was already `two_pow_omega_le_tau`. The other half, $|R| \le 2^{\omega(u)}$, is the multiplicativity of the root count, and is now `Erdos307.rootSet_card_le_two_pow_omega`, with the composition `rootSet_card_le_tau`. Two steps, both elementary in $\mathbb{Z}/u$: the Chinese remainder ring equivalence carries a root modulo $ab$ to a pair of roots injectively, so the count is submultiplicative on coprime factors (`rootSet_card_mul_le`); and modulo an odd prime power there are at most two roots, because $(x - y)(x + y) = 0$ while $p$ cannot divide the representatives of both factors, their sum representing the unit $2x$ (`rootSet_card_prime_pow_le_two`). Modulo a power of 2 there are none. 0 `sorry` (`lean/Erdos307/RootCount.lean`).

What remains outside Lean is the *instantiation*, not a lemma: "plus–hit" is not defined in Lean, so tying the abstract reduction to this layer is done in the prose above. Erdős's sharpening and Bunyakovsky's lower bound stay cited, and neither is used by the upper bound.

The semiprime argument generalises: the slot calculus of Proposition 7.17 counts.

**Proposition 5.18** (Slot counting: every fixed–$\omega$ stratum of the plus–layer is sparse)**.** *Fix $r \ge 2$. The number of plus–hits $N \le X$ with $\omega(N) = r$ is $O\big(X^{1-1/r+o(1)}\big)$.*

*Proof.* Write $p = P^+(N)$ for the largest prime of $N$ and $M = N/p$, so $\omega(M) = r - 1$ and, since $N < p^r$, $M \le N^{1-1/r} \le X^{1-1/r}$. The slot identity of Proposition 7.17 (base $M$, slot $p$) gives $N' + 2N = p\,E_0(M) + M$ with $E_0(M) = M' + 2M$, so a plus–hit $s^2 = N' + 2N$ forces

$$s^2 \equiv M \pmod{E_0(M)}, \qquad p = \frac{s^2 - M}{E_0(M)},$$

and the map $N \mapsto (M, s)$ is injective. For squarefree $M$ one has $\gcd(M, M') = 1$ (modulo $q \mid M$ only the cofactor $M/q$ survives in $M'$), hence $\gcd(M, E_0(M)) = 1$: at every odd prime $\ell \mid E_0(M)$ a root of $s^2 \equiv M$ has $\ell \nmid s$, is nonsingular, and lifts uniquely (Hensel), so the congruence has at most $4 \cdot 2^{\omega(E_0)} = X^{o(1)}$ roots modulo $E_0(M)$. Since $s^2 = N\big(\sigma(N) + 2\big)$ and $\sigma(N) \ll \log\log X$, each base carries at most $X^{o(1)}\big(X^{1/2}/E_0(M) + 1\big)$ admissible $s$, and with $E_0(M) \ge 2M$,

$$\#\{N\} \;\le \sum_{\substack{M \le X^{1-1/r} \\ \omega(M) = r-1}} X^{o(1)}\Big(\frac{X^{1/2}}{M} + 1\Big) \;\ll\; X^{1/2+o(1)} + X^{1-1/r+o(1)} \;=\; X^{1-1/r+o(1)}.$$

(For $r = 2$ this recovers the order of Proposition 5.16, which is sharper by logarithms.) The injection, the identity, $\gcd(M, M') = 1$ and $P^+(N) \ge N^{1/r}$ were verified on all 251 plus–hits below $10^6$: zero exceptions (`code/strata_count.py`). $\square$

**Remark 5.19** (What separates the strata from the full layer)**.** The bound is honest but far from the truth: at $X = 10^6$ the strata $\omega = 2, \ldots, 5$ hold 124, 69, 50, 8 hits (every one below $\sqrt{X}$) because the congruence $s^2 \equiv M \pmod{E_0(M)}$ is usually *insoluble*: over random odd semiprime bases the average number of roots is 0.34, the no–residue branch of the trichotomy of Proposition 7.17 dominating. Summing the proposition over $r$ does *not* give the full layer density zero, because the $X^{o(1)}$ root–count factors are not uniform in $r$; an average bound for $\tau(M' + 2M)$ (a Titchmarsh–divisor–type sub–problem) remains the route to the true ($\sqrt{X}$–order) count. Density zero itself, however, needs less: a smooth/rough dichotomy bypasses the average entirely, because roughness buys a factor $X^{-1/\log\log X}$ against which the divisor bound costs only $X^{\log 2/\log\log X}$, and $\log 2 < 1$.

**Proposition 5.20** (The plus–layer has density zero)**.** *Unconditionally, with* $\log_3 = \log\log\log$*,*

$$\#\{\textit{plus–hits } N \le X\} \;\ll\; \frac{X}{(\log X)^{(1+o(1))\log_3 X}}.$$

*In particular the plus–layer has density zero. The proof applies verbatim to* $\{N \textit{ squarefree} : N' + cN = \square\}$ *for any fixed* $c \ge 1$*.*

*Proof.* Split at $y = X^{1/u}$ with $u = \log\log X$. *Smooth part.* The number of $N \le X$ with $P^+(N) \le y$ is $\Psi(X, y) = X\,u^{-u(1+o(1))} = X(\log X)^{-(1+o(1))\log_3 X}$ (Rankin's bound; [18], III.5). *Rough part.* If $P^+(N) = p > y$, the injection of Proposition 5.18 applies with $M = N/p \le X/y = X^{1-1/u}$: the hit determines $(M, s)$ with $s^2 \equiv M \pmod{E_0(M)}$, a congruence with at most $\rho(E_0) \le 4 \cdot 2^{\omega(E_0)} \le \exp\big((\log 2 + o(1))\log X/\log\log X\big)$ roots by the maximal order of the divisor function [17]; each base therefore carries at most $\rho(E_0)\big(X^{1/2+o(1)}/E_0(M) + 1\big)$ values of $s$. Summing, with $E_0(M) \ge 2M$:

$$\#\{\text{rough hits}\} \;\ll\; X^{1/2+o(1)} + X^{1-1/u}\exp\Big((\log 2 + o(1))\frac{\log X}{\log\log X}\Big) \;=\; X^{1/2+o(1)} + X\exp\Big(-(1 - \log 2 - o(1))\frac{\log X}{\log\log}$$

and $1 - \log 2 = 0.3068\ldots > 0$: the rough part is smaller than every $X/(\log X)^A$. The smooth part dominates and gives the stated rate. (At $X = 10^6$ the split reads $251 = 225$ rough $+ 26$ smooth, the injection collision–free; `code/pluszero.py`.) $\square$

On a derivative *line* the slot is not merely constrained; it is pinned, and the counting sharpens to every line at once.

## 5.4 Slot and line anatomy

**Proposition 5.21** (The determined slot: all derivative lines are thin, uniformly). *For integers $e \geq 1$ and $c$,*

$$\#\{N \leq X \text{ squarefree}: \ N' = eN + c\} \ \ll \ \frac{X}{(\log X)^{(1+o(1))\log_3 X}},$$

*uniformly in $e$ and $c$. In particular the primary pseudoperfect line $n' = n - 1$, the Giuga quotient–one line $n' = n + 1$, every line $n' = n \pm d$ of [34], and every slope–two line $n' = 2n + c$ obey this bound. (Continuity of the limit law of Proposition 2.6 already forces each line to have density zero; the content is the rate, far beyond what concentration inequalities give, and the uniformity. Classical lines with extra divisibility structure may well admit sharper bounds; we claim none.)*

*Proof.* Split at $y = X^{1/u}$, $u = \log\log X$; smooth members number $\Psi(X, y) = X(\log X)^{-(1+o(1))\log_3 X}$ (Rankin). For a rough member $N = Mp$, $p = P^+(N) > y$, Leibniz gives $N' - eN = p\,(M' - eM) + M$, and $M' - eM \neq 0$: $\sigma(M) = e$ would make the auto–reduced fraction of Theorem 2.1 an integer, forcing $D_M = 1$, $M = 1$. Hence

$$p \ = \ \frac{c - M}{M' - eM}$$

is *determined* by $M$ (the line has no free root at all) so $N \mapsto N/P^+(N)$ is injective on rough members, whose number is at most $X^{1-1/u}$. The smooth part dominates. Two classical objects fall out of the formula: on the stratum $M' - M = -1$ of the two slope–one lines it reads $p = M + 1$ ($c = -1$) and $p = M - 1$ ($c = +1$) (the oblong chains $6 \mapsto 42 \mapsto 1806$ and $30 = 6 \cdot 5$, $1722 = 42 \cdot 41$) while off–stratum members stay slot–determined ($858 = 66 \cdot 13$ with $M' - M = -5$; $47058 = 1518 \cdot 31$ with $M' - M = -49$). Verified: the formula reproduces $P^+(N)$ on all listed members and identically on the 51,201 squarefree $N \leq 10^5$ with $\omega \geq 2$ (`code/lines_slot.py`). □

The members of a line carry far more structure than thinness: every prime factor is pinned modulo its cofactor, and the top of the factorisation is capped by the bottom.

**Proposition 5.22** (Line anatomy: pinning, coupling, caps). *Let $M$ be squarefree on the line $M' = eM + c$, let $p > q$ be its two largest primes and $R = M/(pq)$.*

- (a) (Pinning.) $M/t \equiv c \pmod t$ *for* every *prime $t \mid M$; the Giuga congruence of the line, independent of $e$. Equivalently, each prime factor lies in one determined residue class modulo its cofactor: $t \equiv c\,(M/(tr))^{-1} \pmod r$ for all primes $r \mid M/t$.*
- (b) (Top cap.) *If $M/p \neq c$ then $p \leq \sqrt{M} + |c|$: line members are $(\sqrt{M} + |c|)$–smooth.*
- (c) (Coupling.) $pq \mid R(p + q) - c$. *Here $\gcd(pq, c) = 1$ is automatic: $p \mid c$ together with $p \mid qR - c$ would give $p \mid qR$, impossible as $p$ exceeds every prime of $qR$.*
- (d) (Second cap.) *If $R(p + q) \neq c$ then $q < R + \sqrt{R^2 + |c|}$; the degenerate stratum $R(p + q) = c$ forces $c \geq 2\sqrt{RM}$, so it is empty on every line with $c < 2\sqrt{M}$; in particular on all the near–critical minus lines.*

*Proof.* (a) Cofactor survival gives $M' \equiv M/t \pmod t$ while $eM \equiv 0$; the second form is CRT. (b) $M/p - c$ is a multiple of $p$; if nonzero, $M/p \geq p - |c|$. (c) $p \mid qR - c$ and $q \mid pR - c$ give $pq \mid (qR - c)(pR - c) = pqR^2 - cR(p + q) + c^2$, so $pq \mid c\,(R(p + q) - c)$, and the displayed coprimality removes $c$. (d) If $R(p + q) \neq c$ then $pq \leq |R(p + q) - c| \leq R(p + q) + |c| < 2Rp + |c|$, whence $q^2 < 2Rq + |c|$; if $R(p + q) = c$ then $c \geq 2R\sqrt{pq} = 2\sqrt{RM}$ by AM–GM. All parts were verified on the 7,184,234 squarefree $n \leq 2 \times 10^7$ with $\omega \geq 3$, for both $e = 1, 2$ with $c := n' - en$ (zero exceptions, the second cap attained with slack 1 at $n = 70$) and on the classical members, where the coupling quotient $\big(R(p+q)-c\big)/pq$ equals 1 on the oblong–chain members and 5 on both off–stratum members 47058 and 66198 (`code/line_anatomy.py`). $\square$

The lines themselves close into a family under peeling, and the closure explains why no reorganization of the counting has broken through.

**Proposition 5.23** (The peeling groupoid)**.** *For integers $a \geq 1$, $b$, $c$ let* $L(a,b,c) = \{n \text{ squarefree} : an' = bn + c\}$.

(i) (Closure; the slope is a mass.) *If $n = qm \in L(a,b,c)$ with $q = P^+(n)$, then $m \in L(aq, bq-a, c)$, and the slope moves by $b/a \mapsto b/a - 1/q$: the full peel chain of $n$ descends the partial sums of $\sigma(n)$; the classical lines, their $\mu$–Sondow shifts, and the slope–two lines of the minus square are one orbit family.*

(ii) (Rigidity at every level.) *Along the chain of $n$ on an integer line $(a,b) = (1,e)$, the level–$j$ denominator vanishes iff $\sigma(m_j) = b_j/a_j$, which telescopes to the single $j$–independent condition $\sigma(n) = e$; impossible, since an integer mass forces $n = 1$ by Theorem 2.1. So every denominator on every genuine chain is nonzero.*

(iii) (Determined slot at every level.) *Consequently $q = (c - am)/(am' - bm)$ at each step: a line member is a tower of single determined candidates over its own core.*

*Every partial–peel reorganization of the line counts (peeling one prime, several, in any order, from either layer of the split criterion) stays inside this groupoid, where part* (iii) *reproduces the one–candidate–per–cofactor obstruction verbatim. The obstruction is therefore* invariant *under the natural changes of viewpoint: a proof must inject input from outside the groupoid; primality along determined thin sequences, or a decomposition of integers not indexed by their primes.*

*Proof.* (i) is Leibniz: $a(qm'+m) = bqm+c$ gives $aq\,m' = (bq-a)m+c$, and $(bq-a)/(aq) = b/a-1/q$. In (ii), $\sigma(m_j) = b_j/a_j = e - \sum_{i<j} 1/q_i$ says $\sigma\big(m_j \prod_{i<j} q_i\big) = \sigma(n) = e$; by the auto–reduced fraction of Theorem 2.1 an integer mass forces $D = 1$. (iii) is the display above (i). Verified: closure and slot on $10^5$ random $(n,a,b)$ (the only slot exceptions being exactly the $\sigma(m) = b/a$ locus, excluded on genuine chains, 73 of $10^5$ random triples, 0 of 310,788 genuine chain steps at $e = 1,2,3$), and the slope telescope exact on every chain (`code/peeling_groupoid.py`). $\square$

The determined slot bounds more than itself: unwound one level further it constrains the *second* prime by a mass.

### 5.5 A second derivation of the barrier constant

**Proposition 5.24** (Mass–defect bound on the second prime)**.** *Let $M$ be squarefree with $\omega(M) \geq 3$ on the line $M' = eM + c$, and write $p = P^+(M)$, $m = M/p$, $P = P^+(m)$, $n = m/P$. If $\sigma(m) < e$ then*

$$P\big(e - \sigma(n)\big) \;<\; 2 - \frac{c}{m}, \qquad \textit{equivalently} \qquad P^2\,(en - n') \;<\; 2Pn - c.$$

*For $e = 2$ the hypothesis is automatic below $\Pi_{59}$; that is, throughout the range in which the minus layer is counted. On the near–critical slope–two lines ($|c|$ small) the bound reads $P < 2/(2-\sigma(n)) + O(|c|/n)$:* the second–largest prime of $M$ is bounded by twice the reciprocal of the mass defect of the remaining cofactor.

*Proof.* $M$ squarefree gives $p > P$. The determined slot of Proposition 5.23 is $p = (c-m)/(m'-em)$, and $\sigma(m) < e$ makes the denominator negative, so $p = (m-c)/\big(m(e-\sigma(m))\big)$ and $p > P$ reads $P\,m(e-\sigma(m)) < m-c$, i.e. $P(e-\sigma(m)) < 1-c/m$. Since $m = Pn$ and $\sigma(m) = \sigma(n)+1/P$, the left side is $P(e-\sigma(n))-1$, which gives the first display; multiplying by $Pn$ clears denominators. Verified with 0 violations on every squarefree $M \le 2\times 10^6$ with $\omega \ge 3$, for $e = 1, 2, 3$; at $e = 1$ the 21,146 exceptions are exactly the $\sigma(m) \ge 1$ cases excluded by the hypothesis, and the bound is within a factor 2 of equality for 11% of members and within 100 for 59% (`code/slot_massbound.rs`). □

**Remark 5.25** (The exact form, and a window that is real but inert)**.** Proposition 5.24 has an exact ancestor. With $N = pm$, $p = P^+(N)$, $k = 2m - m'$, Leibniz gives $N' = pm' + m$, so on the minus layer

$$d^2 \;=\; N' - 2N \;=\; m - pk, \qquad \text{whence} \qquad \sigma(N) \;=\; 2 + \frac{1}{P^+(N)} - \frac{k}{m}, \qquad k \ge 1.$$

Since $k \ge 1$ this is a *two–sided* window; the barrier supplies only the lower half:

$$2 \;<\; \sigma(U) \;<\; 2 + \frac{1}{\max U}, \qquad \text{equivalently} \qquad \sigma\big(U \setminus \{\max U\}\big) < 2,$$

so a minus–layer support is *mass–minimal*: deleting any one of its primes drops the mass below 2 (the largest is the binding case). The upper half is nonetheless *inert* where it can be tested: all 49,961 admissible level–59 supports already satisfy it, the tightest margin being $1.26 \times 10^{-3}$, so it prunes nothing there. We record it because it is exact, because it is the identity behind Proposition 5.24, and because it may bind at higher levels, where the largest admissible prime grows. (The enumeration also reproduces the count 49,961 of Proposition 4.5 independently, from a different traversal; `code/masswindow.rs`.)

**Remark 5.26** (The minus layer is the anatomy of a quadratic polynomial)**.** The identity $d^2 = m - pk$ of Remark 5.25 can be read as a parametrisation rather than a constraint. On the minus layer $d^2 = N' - 2N \ge 0$ is a square, so

$$m \;=\; pk + d^2,$$

and for *fixed* $(p, k)$ the cofactor $m$ runs over the values of the quadratic polynomial $f_{p,k}(d) = d^2 + pk$. The minus layer is therefore not an arbitrary exact coincidence but a question about the *anatomy of quadratic–polynomial values*: for which $d$ does $f_{p,k}(d)$ carry mass $\sigma = 2 - k/f_{p,k}(d)$, i.e. $\omega \approx 58$ prime factors concentrated on the small primes, with $P^+(f_{p,k}(d)) < p$?

This is the first place in this circle where a *named* toolkit applies. The anatomy of polynomial values is a developed subject; Erdős–Kac for $\omega(f(n))$, Halberstam–Richert sieve bounds for almost–primes in polynomial sequences, and the divisor sums of [47] already used in Proposition 5.16; whereas "primality of a determined value" (Remark 7.15) had none. What the toolkit must supply is unusual: not the typical behaviour of $\omega(f(d))$, which is $\log\log$ by Erdős–Kac, but a *large deviation* to $\omega \approx 58$ with the prime factors forced onto the bottom of the spectrum, since only the small primes carry mass. Existing polynomial–anatomy results control typical and almost–prime behaviour, not deviations of this shape; so the reformulation names the tool without yet applying it. We record it as the most promising translation of the residual question, and note that the $p = 5$ family of Proposition 5.15(iii) is the plus–layer shadow of the same quadratic mechanism. (Identity verified on all 1,606,956 squarefree $M \le 3 \times 10^6$ with $\omega \ge 2$: 0 violations.)

**Remark 5.27** (The residue amplifier is real, and exactly paid for)**.** Sieving $f_c(d) = d^2 + c$ by its roots exposes a mechanism the union–side reading of Corollary 7.16 misses. A prime $\ell$ divides some value iff $-c$ is a quadratic residue modulo $\ell$, which removes half the primes; the $2^{-\omega}$ penalty the corollary records. But an *admissible* odd $\ell$ has *two* roots, so it divides values with density $2/\ell$ rather than $1/\ell$: on the primes that survive, the weight is doubled. Only the penalty has ever been counted, and one might expect the doubling to tilt the heuristic. It does not: the two effects cancel exactly.

Per fixed $c$ the effect is genuine. Writing $\sigma_B$ for the mass on primes $\leq B$, measured over $d \leq 2 \times 10^6$ with $B = 2 \times 10^4$: a generic $c = 1009$ gives mean $\sigma_B = 0.368$, a friendlier $c = 4966$ (admitting 35 of the 46 primes below 200) gives 0.400, against 0.452 for random integers; a *perfectly* friendly $c$, admitting every odd prime, would give $\frac{1}{4} + 2\sum_{\ell>2} \ell^{-2} = 0.654$, a factor 1.45 above random. So friendliness helps, and in the limit beats a random integer outright.

But friendliness is priced. Each odd $\ell$ is admissible for only half of the $c$, so a $c$ admitting the first 58 odd primes has density $2^{-58}$, exactly cancelling the $2^{58}$ gained from the doubled densities:

$$2^{-58} \cdot \tfrac{1}{2} \prod_{i=2}^{59} \frac{2}{\ell_i} \;=\; \prod_{i=1}^{59} \frac{1}{\ell_i},$$

the random–integer probability. Averaging over $c = 1, \ldots, 400$ confirms it numerically: the mean mass ratio is 0.9990, and the tail ratios $\Pr[\sigma_B > T]$ are 0.998, 0.996, 0.978 at $T = 0.8, 1.0, 1.2$; unity to within sampling, while individual $c$ range from 0.257 to 0.635 in mean. *The quadratic reformulation is heuristically neutral*: it neither strengthens nor weakens the YES lean of Remark 1.8, and the $2^{-59}$ filter of Corollary 7.16 is correctly priced after all. What Remark 5.26 buys is a toolkit, not a bias (`code/quad_anatomy.rs`, `code/amplifier_cost.rs`).

The bound has enough force to regenerate the barrier on its own.

**Corollary 5.28** (A second, independent derivation of the barrier)**.** *Let $N$ be a minus–layer hit with $\omega(N) \geq 3$; more generally a member of any line $N' = 2N + c$ with $c \geq 0$. Write $p = P^+(N)$, $m = N/p$, $P = P^+(m)$, $n = m/P$. Then $N > 10^{112.94}$, without using the arithmetic–geometric mean inequality.*

*Proof.* If $\sigma(m) \geq 2$ then $m$ carries at least 59 primes and $N > m \geq \Pi_{59}$ already. Otherwise Proposition 5.24 applies, and $c \geq 0$ makes its right side at most 2: $P\big(2 - \sigma(n)\big) < 2$. Since $P > P^+(n)$ this gives

$$2 - \sigma(n) \;<\; \frac{2}{P^+(n)}, \qquad \text{i.e.} \qquad \sigma(n) > 2 - \frac{2}{P^+(n)}.$$

But $\sigma(n) \leq \sum_{q \leq P^+(n)} 1/q$, so $P^+(n)$ must satisfy $\sum_{q \leq p} 1/q > 2 - 2/p$. The least such prime is $p = 269$, the 57th: at $p = 269$ the sum is $1.99505 > 1.99257$, while at $p = 263$ it is $1.99133 < 1.99240$. Hence $\omega(n) \geq 57$, $P^+(n) \geq 269$, and $n \geq \Pi_{57} > 10^{108.07}$; with $p > P > P^+(n)$ the two further primes give $N > \Pi_{57} \cdot 271 \cdot 277 > 10^{112.94}$. □

**Remark 5.29** (Two mechanisms, one constant)**.** Corollary 5.28 reaches $10^{112.94}$, and $\Pi_{59} > 8.77 \times 10^{112}$ is $10^{112.94}$: the two derivations agree to two decimal places. They share no step. Theorem 4.3 divides the defining equation by $ab$ and applies AM–GM to $a/b + b/a$; the present route never forms that quotient, using instead the determined slot of Proposition 5.23, the trivial inequality $p > P^+(N/p)$, and the additivity $\sigma(m) = \sigma(n) + 1/P$. What they share is the arithmetic input underneath (how many primes Mertens needs to reach mass 2) which is why the constants coincide rather than merely being comparable. The barrier is thus not an artefact of the mass inequality one happens to use: it is the same Mertens threshold seen through two independent structures, and the critical set $S_2 = \{n : 2 - \sigma(n) < 2/P^+(n)\}$ of the second structure is empty below $10^{108}$ (verified empty for $n \leq 5 \times 10^6$, `code/critical_set.rs`).

**Remark 5.30** (The minus side: coercive versus degenerate moduli)**.** Proposition 5.20 rests on the plus modulus being *coercive*: $E_0^+(M) = M' + 2M \geq 2M$. The minus–square obeys the same slot identity with $E_0^-(M) = M' - 2M = M(\sigma(M) - 2)$, a modulus that *vanishes exactly on the mass–2 wall*: the degeneration locus of the counting method is the barrier stratum itself; the counting incarnation of the form degeneration $N_0(2 - \sigma) \to 0$ of Proposition 7.18. The minus–layer is the union of the slope–two lines $N' = 2N + d^2$ over square heights, and three facts are unconditional. (i) A rough minus–hit ($P^+(N) = p > y$) forces its base upward: if $\sigma(M) < 2 - 1/y$ then $d^2 = p\, E_0^-(M) + M < M(1-p/y) \leq 0$, impossible; so $\sigma(M) \geq 2-1/y$, and once $y > (2-T_{58})^{-1} = 793.7\ldots$ this yields $\omega(M) \geq 59$ and $M \geq \Pi_{59}$: *the 59–prime wall propagates from the hit to its base*, and $N \geq \Pi_{59}\, y$. (ii) Bases with $\sigma(M) \geq 2 + 1/y$ have moduli $E_0^- \geq M/y$, and contribute as in Proposition 5.20. (iii) The remaining bases are near–critical, $\sigma(M) \in [2-1/y, 2+1/y)$; exactly the low slope–two lines $M' - 2M = \pm k$, $k \leq M/y$. Each is thin by Proposition 5.21, but at a degenerate modulus the $d$–freedom is $\sqrt{X}$–sized, and summing the uniform bound does not close the layer. What would: *square–root bounds on slope–two lines*, $\#\{M \leq Z : M' - 2M = \pm k\} \ll Z^{1/2+o(1)}$ uniformly in $k$. There is a second, sharper route to that bound, obtained by abandoning the prime–indexed decomposition altogether. Split $M = uv$ by *size* rather than by largest prime, $u$ the coprime divisor nearest $\sqrt{M}$; Leibniz gives the exact bilinearisation

$$M' - 2M \;=\; v\,\delta(u) + u\,\delta(v), \qquad \delta(n) = n' - n,$$

so a slope–two line becomes a bilinear equation in $(u, v)$ with both variables of comparable size and, for fixed $u$, the exact line $u\,v' - (2u - u')\,v = c$, i.e. $v \in L\big(u,\, 2u - u',\, c\big)$ in the notation of Proposition 5.23. Writing $N(u)$ for the number of $v$ on that line in the balanced window $v \asymp M/u$,

$$\#\{M \leq Z \text{ on the line}\} \;\leq\; \sum_{u \leq Z^{1/2}} N(u),$$

so the missing square–root bound follows from $N(u) \ll Z^{o(1)}$ on average; a *short–window line–multiplicity* hypothesis. This is a strictly sharper target than the one–class–per–cofactor count: the summation length drops from $Z$ cofactors to $Z^{1/2}$ balanced divisors (empirically $Z^{0.43}$, the median of $\log u/\log M$ over nearest–to–$\sqrt{M}$ coprime splits being 0.432), and the surviving object is bilinear in a short window; the shape on which dispersion and Fouvry–Iwaniec methods operate, rather than a worst–case single–class count. The measured multiplicity is exactly 1 (mean 1.00, no sampled line carrying two solutions), so the hypothesis is not merely plausible but numerically flat; proving it, however, runs back into the determined slot, since $L(u, 2u - u', c)$ is itself a groupoid object. All identities verified exhaustively on the 591,423 squarefree $M \leq 10^6$, in exact rationals for the line form (`code/bilinear_balanced.py`). By Proposition 5.22(a) that missing bound is a *one–class–per–cofactor* prime count: a line member is $M = pm$ with $p$ prime in a single determined class modulo its own cofactor $m$, so the count is $\sum_m \#\{p \leq Z/m : p \equiv \alpha_c(m) \pmod{m}\}$, and the trivial estimate $Z/m^2 + 1$ per cofactor fails exactly on its $+1$: one candidate integer per $m$, thinned only by the primality of that single determined value. Bounding worst–case single–class counts summed over all moduli lies beyond Bombieri–Vinogradov and beyond Linnik uniformly; the same species of count guards the true $\sqrt{X}$ order of the plus layer. The two layers' remaining questions are one wall. The ledger: plus layer closed by coercivity; minus layer reduced to the near–critical spectrum. (A small unconditional companion, from Proposition 5.14: an odd member of $N' = 2N + c$ has $S(N) \equiv cN + 2 \pmod 8$.)

**Remark 5.31** (The primality can be dropped: the missing bound is a divisibility count)**.** The formulation above asks for a bound on how often a determined value is *prime*, which is what places it beyond Bombieri–Vinogradov. That requirement can be removed. The slot $(c - m)/(m' - em)$ is an integer at all only when

$$(m' - em) \mid (c - m),$$

and line members inject into the set of $m$ satisfying that divisibility, so

$$\#\{m \le Z : \ (m' - em) \mid (c - m)\} \ \ll \ Z^{1/2+o(1)} \quad \text{uniformly in } (e, c)$$

already implies the square–root bound, with no prime–distribution input whatever.

The substitution costs almost nothing, because integrality is where the thinness lives. Counting both conditions separately on the two lines whose members are known, over $m \le 2 \times 10^7$ (so $\log Z = 16.8$): the line $n' = n - 1$ has 21 integral slots of which 16 are prime, and $n' = n + 1$ has 29 of which 19 are prime. Integrality alone is already $O(\log Z)$; primality removes a further factor of about 1.3 and no more. The prime–counting requirement was carrying a constant, while the divisibility was carrying the whole saving.

What remains is therefore a purely multiplicative question about the derivative, of the same shape as Proposition 5.23 but with the primality stripped out, and with roughly four orders of slack against the target at the ranges where the count can be observed. We record the substitution because it changes which literature the problem faces: divisibility counts for $m' - em$ rather than primes in determined classes.

# 6 Two closures on the existence route

Atom A10 asks for a prime term of the ternary recurrence of Remark 1.16. Before attacking it, two families of approach can be removed: the relaxations that would give the construction more freedom, and the algebraic methods that would avoid the archimedean place. Both are closed outright.

**Proposition 6.1** (No free slot)**.** *Let $P, Q$ be finite sets of primes with $\sigma(P)\sigma(Q) = 1$, and put $a = \prod P$, $b = \prod Q$. Then $b = a'$ and $a = b'$ exactly. In particular $Q$ is* determined *by $P$: it is the set of prime factors of $a'$, and the existence route carries exactly one parameter, the set $P$.*

Formalised. *`Erdos307.Q_determined`: given $a' = b$, the set $Q$ is exactly the set of prime factors of $a'$. The one–prime normal form of Proposition 5.3 is `oneprime_forced`, `oneprime_cycle`, `oneprime_converse` (which closes the equivalence; earlier editions formalised one direction only), `oneprime_mass_identity` and `oneprime_mass_bound`, the last two giving part* (b)*: the mass identity holds for* every *witness, prime or composite, so the exactness layer alone carries the barrier. The slot calculus of Proposition 7.17 is `slot_layers` and `slot_recovery`, the filter of Corollary 7.16 is `qr_filter`, resting on `cofactor_survival`, the congruence $N' + 2N \equiv N/\ell \pmod{\ell}$, which is proved rather than assumed (an earlier edition stated `qr_filter` with that arithmetic as a hypothesis on abstract residues, which left it contentless); Lemma 7.1 is `Frame.csum_eq_mul_recipSum` for its first half and `symbolfact` for the Jacobi factorisation; and the reduction mechanism behind Proposition 3.10, Rigidity and Lemma 2.1 alike is isolated once as `dvd_sum_erase_iff`* (`lean/Erdos307/NormalForm.lean`

*Proof.* $\sigma(a) = a'/a$ and $\sigma(b) = b'/b$, both in lowest terms since $\gcd(m, m') = 1$ for squarefree $m$. Then $\sigma(a)\sigma(b) = 1$ reads $a'b' = ab$. From $\gcd(a', a) = 1$ and $a' \mid ab$ we get $a' \mid b$, say $b = a'k$; symmetrically $b' \mid a$, say $a = b'j$. Substituting, $a'b' = ab = b'j \cdot a'k$, so $jk = 1$ and $j = k = 1$. □

**Corollary 6.2** (The multi-slot relaxations are vacuous)**.** *There is no version of the construction in which two or more primes of $Q$ may be chosen freely. In particular one cannot fix a base $B \subset Q$ and solve $1/q_1 + 1/q_2 = r/s$ for the remaining two primes by ranging over the $\tau(s^2)$ factorisations $(rq_1 - s)(rq_2 - s) = s^2$: by Proposition 6.1 the pair $(q_1, q_2)$ is already forced to be the corresponding part of the factorisation of $a'$. The exponential parametrisation of Remark* 1.16 *is therefore intrinsic, not an artefact of the one-free-prime ansatz, and no amount of extra slots converts it into a Bunyakovsky one. What does dispose of the exponential growth is not extra slots but Proposition 4.7, which bounds the tail prime outright and so replaces the orbit by a finite divisor search.*

The second closure is not new here, but it belongs in this list because it is what forces the whole existence route to be analytic. Theorem 3.1 of Section 3 shows that over $\mathbb{F}_q[t]$ the arithmetic derivative satisfies $\deg f' \leq \deg f - 1$, hence has no two–cycles and no nonzero fixed points, and Corollary 3.2 concludes that the function–field Erdős–Barbeau equation is insoluble outright. (An independent brute-force recheck, 8192 squarefree monics over $\mathbb{F}_2$ to degree 13 and 6561 over $\mathbb{F}_3$ to degree 8, 0 violations and 0 two–cycles: `code/ff_nocycle.rs`.) What is added here is not that statement but its consequence for *method*, recorded next.

**Remark 6.3** (Why the existence route cannot be algebraic)**.** The mechanism of Theorem 3.1 is the ultrametric inequality. The cycle equation is $\sigma(a)\sigma(b) = 1$, so one of the two masses must reach 1; over $K[t]$ the mass satisfies $|\sigma(f)| = \max_i |1/\pi_i| < 1$ by ultrametricity and can never do so, whereas over $\mathbb{Z}$ the archimedean absolute value permits $\sum_{p|n} 1/p > 1$. So #307 is possible at all only because $|\cdot|_\infty$ is not ultrametric.

This has a consequence for method. Any argument for *existence* that is purely algebraic, in the sense that it applies verbatim with $\mathbb{Z}$ replaced by $K[t]$, would prove a statement that Theorem 3.1 refutes; hence no such argument exists. Together with Proposition 5.5 (no polynomial parametrisation) and Proposition 5.11 (no place of $\mathbb{Q}$ carries the descent) this closes the algebraic and geometric toolkit on *both* routes: the negative route has no place to descend at, and the positive route has no place to construct at. What remains for A10 is archimedean and analytic, which is precisely the regime in which lower bounds for primes in sparse sequences are unavailable. That is a sharper statement of the obstruction than Remark 1.16 gives, and it is the reason the present work does not resolve the existence direction.

There is a third closure, and it removes the one concrete attack that survives the other two. The 34 immune families of Remark 1.4 are precisely the tail families the congruence campaign cannot kill, so each is a live candidate and the obvious move is to test them one at a time. That move is not available, and the reason is quantitative rather than a want of effort.

**Proposition 6.4** (The immune families cannot be tested)**.** *Let $S$ be one of the* 34 *immune bases. Deciding whether the tail family $S \cup \{q\}$ contains a Pythagorean pair is, by Remark* 1.16, *deciding whether $B_S x^2 - A_S y^2 = -4D_S^2$ has an integral solution with $q = (x^2 - D_S)/A_S$ prime. Neither of the two routes to that decision is within reach.*

(i) Global route. *Producing a solution requires the fundamental unit, equivalently the class group, of the real quadratic order of discriminant $4A_SB_S$. On the first immune base that discriminant has* 225 *digits and is a nonsquare, so the subexponential index-calculus cost $L_\Delta(1/2) = \exp\big(\sqrt{\log \Delta \, \log\log \Delta}\big)$ evaluates to $10^{24.7}$ operations.*

(ii) Parametric route. *Since $A_S$ is prime and $(D_S \mid A_S) = +1$, a square root $x_0^2 \equiv D_S \pmod{A_S}$ exists and the family is genuinely parametrised by $x = x_0 + tA_S$, with $q(t) = \big((x_0 + tA_S)^2 -$*

$D_S)/A_S$. *But a solution additionally needs* $B_S q(t)+D_S$ *to be a square, and that quantity has* 223 *digits already at* $t=0$*, so the heuristic hit rate is* $10^{-112}$ *per trial. Checked at* $t=0,1,2,3$*: not a square* (`code/a10_pell_feasibility.gp`).

*There is a third route, which is to decline the computation rather than perform it, and it is worth costing because it is the natural thing to try next and because nothing structural forbids it. The size of the first candidate is governed by the regulator:* $q_1 \approx \varepsilon^2/A_S$ *with* $R = \log \varepsilon$*, so a base whose regulator happens to be near its minimum* $R_{\min} \approx \frac{1}{2}\log\Delta$ *would give* $q_1$ *of about* $10^{113}$*, which an Atkin–Morain test settles in minutes. The obstruction is not that such bases are useless but that they are rare. Near-minimal regulator means the fundamental solution has* $v=1$*, that is* $\Delta = u^2-4$*, and the* $\Delta \le X$ *of that shape number about* $\sqrt{X}$*; so the fraction of discriminants with a testable regulator is about* $\Delta^{-1/2} = 10^{-112.5}$*, against a typical* $R \approx \sqrt{\Delta} \approx 10^{113}$ *which puts* $q_1$ *at* $\exp(10^{112})$ *and out of reach of notation, let alone computation. Since* $\Delta = 4A_SB_S$ *is determined by the base and not chosen, one may only sample: the* 34 *immune bases give an expected* $10^{-111}$ *hits and all* 49,961 *give* $10^{-108}$*, against the* $10^{112.5}$ *bases one expected hit would need. The shortfall against the immune family is* $10^{111}$*. So base selection does not circumvent the two routes above; the construction is one fortunate discriminant away from being decidable, and the fortune required is* $10^{112.5}$*. The estimate is* HEURISTIC*, being a density count for the regulator.*

One hope survives all of this on its face: a lower bound for *some term of some sequence in a family* is formally weaker than one for a single sequence, and Proposition 4.5 supplies 49,961 of them. It fails, and it fails for a reason no enlargement of the family can repair.

**Proposition 6.5** (Multiplicity cannot help)**.** *Every known lower-bound method for primes in a set requires that set to have counting function at least* $X^\theta$ *for some fixed* $\theta > 0$*. The union over all* 49,961 *bases of the terms of their sequences below* $X$ *has counting function* $C\log X$ *with* $C = 49{,}961/\log\varepsilon$*, since each sequence grows like* $\varepsilon^n$ *and so contributes only* $O(\log X)$ *terms. Hence*

$$\frac{\log(\textit{union below } X)}{\log X} = \frac{\log\bigl(C\log X\bigr)}{\log X} \longrightarrow 0,$$

*and the family size* $C$ *is powerless because it sits inside a logarithm.*

*Measurement.* On an immune base $\varepsilon$ has 224 digits, so at $X = 10^{500}$ each sequence contributes 2.2 terms and the whole family contributes $10^{5.05}$, a density exponent of 0.0101. Reaching $\theta = \frac{1}{2}$ at that $X$ would need $10^{249.7}$ bases against the $10^{4.7}$ available, a shortfall of $10^{245}$. Even the absolute ceiling on the family at *every* level combined, all $2^{139} = 10^{41.8}$ subsets of the primes below 800, falls short by $10^{207.8}$ (`code/a10_multiplicity.gp`). $\square$

Each of the four blocks was then attacked on its own terms. None fell; each is now located more precisely, and two further routes are closed.

**Proposition 6.6** (The number–field barrier transfer is FALSE)**.** *Theorem 3.1 kills the algebraic route by ultrametricity, which suggests replacing* $K[t]$ *by a number ring* $\mathcal{O}_K$*, where the absolute value is archimedean and the obstruction disappears. It does disappear. Earlier versions of this paper asserted that nothing is gained, namely that for every number field* $K$ *the least* $\prod N(\pi)$ *over prime ideals with* $\sum 1/N(\pi) \ge 2$ *is at least that of* $\mathbb{Q}$. That assertion is false. *It is withdrawn, and what replaces it is the following.*

(i) The true infimum is 16. *In the biquadratic field* $K = \mathbb{Q}(\sqrt{17},\sqrt{33})$*, of discriminant* 314,721*, the prime* 2 *splits completely: there are four prime ideals of norm* 2*. They carry mass* $4\cdot\frac{1}{2} = 2$

*exactly, and their product of norms is* $2^4 = 16$*. Conversely* 2 *is the least possible ideal norm, so mass* 2 *needs at least four ideals of norm* 2 *and any admissible family has product at least* 16*; a field can carry at most* $[K : \mathbb{Q}]$ *ideals of norm* 2*, so degree* 4 *is also minimal. Hence* 16 *is the exact infimum over all number fields, against the claimed* $\prod_{i\leq 59} p_i > 8.77 \times 10^{112}$*: the withdrawn statement was wrong by a factor* $10^{112}$*.*

(ii) Where the argument broke. *The Chebotarev computation is correct as a statement about* density*: in a degree-$d$ Galois field the completely split primes have density* $1/d$*, so they carry mass* $d \cdot (1/d) \log\log Y = \log\log Y$*, exactly as over* $\mathbb{Q}$*, while each costs* $d \log p$*, and inert primes contribute* $1/p^d$*, that is, nothing. The error was to read a statement about the tail as a statement about the minimum. The barrier consumes only* finitely many *primes, and for any finite set of small primes one may choose* $K$ *in which all of them split; Chebotarev constrains how often splitting happens, not which particular primes split.*

(iii) What survives, and what closes the route instead. *For a* fixed *field the measurement stands:* $\mathbb{Q}$ *needs* 59 *primes and* $10^{112.9}$*, whereas* $\mathbb{Q}(i)$ *needs* 226 *prime ideals and* $10^{735.1}$ (`code/a10_fieldbarrier.gp`). *What does not survive is the use of this proposition as a* block*. The route is nevertheless closed, and by a sounder argument, which is Proposition 6.7 below: the ideal–theoretic #307 is* trivially soluble*, so it carries no information about the rational problem.*

**Proposition 6.7** (The difficulty of #307 is distinctness, not mass)**.** *Allow* $P$ *and* $Q$ *to be finite* multisets *of primes rather than sets. Then the #307 equation is soluble with four primes in total:* $\left(\frac{1}{2}+\frac{1}{2}\right)\left(\frac{1}{2}+\frac{1}{2}\right) = 1$*. With* $P, Q$ *genuine sets the same equation forces* $|P \cup Q| \geq 59$ *and* $\prod > 10^{112.9}$ *by Theorem 4.3. The barrier is therefore a statement about the injectivity of* $p \mapsto 1/p$ *on the primes, not about the arithmetic of the mass condition.*

*Two consequences, and the second is the operative one.*

(i) The ideal–theoretic problem is trivial. *In* $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$ *the prime* 2 *splits completely into four pairwise coprime prime ideals of norm* 2*; taking* $P = \{\pi_1, \pi_2\}$ *and* $Q = \{\pi_3, \pi_4\}$ *gives* $\sigma(P) = \sigma(Q) = 1$ *and* $|P \cup Q| = 4$ (`code/ideal307.gp`)*. Nor does the derivative transfer:* $\pi_1 + \pi_2$ *is the unit ideal, so* $\mathfrak{a}'$ *collapses. This is the correct reason the number–field route is closed, and it explains why refuting the previous proposition reopened nothing: the transferred problem was never the same problem.*

(ii) A no–go for transfers generally. *Any setting admitting two distinct objects of the same norm* $n \geq 2$ *on each side solves the mass equation at size four, whatever the setting is. So a dictionary permitting repeated denominators* dissolves *#307 rather than relocating its difficulty, and can carry no information about it. This prunes a class of attacks in one line, and it is the test any future transfer should face first. Corollary 4.6 passes it only because pairwise–coprime integers still have distinct least prime factors, which is distinctness re-entering through the back door.*

Formalised. *`Erdos307.multiset_solution` is the four–element witness, `distinctness_carries_the_barrier` the contrast against `card_ge_59_of_recipSum_ge_two`, stated as a conjunction so the two halves cannot drift apart, and `transfer_dissolves` the general no–go* (`lean/Erdos307/Distinctness.lean`)*.*

**Remark 6.8** (The three other blocks, attacked)**.** (i) Against Proposition 6.1. The rigidity $\prod Q = a'$ would dissolve if disjointness were an assumption one could relax. It is not: $P \cap Q = \varnothing$ is *derived* in Theorem 2.3 by the Israel mechanism, $r \in P \cap Q$ forcing $v_r(st) = -2 \neq 0$. There is nothing to relax.

(ii) Against Proposition 6.4. Homogenised, $B_S x^2 - A_S y^2 = -4D_S^2$ is the ternary form $B_S x^2 - A_S y^2 + 4D_S^2 z^2$, a *conic*, and conics obey Hasse–Minkowski. Since Remark 1.1 already gives local solubility everywhere, *rational* solubility follows unconditionally, with no class group anywhere in the argument. This sharpens the block rather than removing it: what is missing is not solubility but *integrality*, and extracting integral points from the rational parametrisation returns the fundamental unit. Worse, the constructive step is itself blocked, since Simon's algorithm must factor the determinant $-4A_SB_SD_S^2$, whose cofactor after removing the known primes of $D_S$ is the 224-digit balanced semiprime $A_SB_S$: the same cryptographic wall as Remark 1.4, reached from a new direction.

(iii) Against Proposition 6.5. The template that exploits a family without density is Heath-Brown's: Artin's conjecture holds for all but at most two primes, with no means of saying which. It does not transfer. That argument needs each failure to impose a *detectable* structure, which several failures then cannot share. Here a failing family would be one all of whose $q_n$ are composite, and the only known mechanism for that is a covering system, which Proposition 1.9 proves does not exist. A failure would therefore be structureless, and structureless failures cannot be played against one another.

All of the above bounds the negative direction. The positive direction admits a quantitative shadow that had not been measured: a solution is exactly $r(a) := \sigma(a)\sigma(a') = 1$, so one may ask how close $r$ actually comes to 1. The answer is instructive, and it disposes of the usual way such heuristics are calibrated.

**Proposition 6.9** (The near-miss spectrum, and why the heuristic cannot be calibrated)**.** *Over all* 4,385,727 *integers $a \le 10^7$ with $a$ and $a'$ both squarefree, the maximum of $r(a)$ is* 0.5535*, attained at $a =$* 3,972,254*; the frontier advances by decade as* 0.333, 0.345, 0.502, 0.502, 0.536, 0.554*, and at $10^8$ it reaches* 0.586076 *at $a =$* 23,667,105 (`code/sweep1e8.rs`, which reproduces every earlier decade independently)*. This last decade was run as a test of Proposition 6.11: the bound $R(10^8) \le 0.67270$ and the range $[0.55, 0.62]$ suggested by the earlier decades were both recorded before the computation, and both hold. In particular $r \ge 1$ never occurs, the defect $|a'' - a|$ is of size $a^{0.9}$ or larger in all but three cases, and the smallest relative defect is* 0.4465*. Consequently the constant in the heuristic count of two-cycles, $f(1)\log X$ with $f$ the density of $r$ at 1, is* not measurable*, and the obstruction is two–sided.*

*By Theorem 4.3 the region $r \ge 1$ lies above $10^{112.9}$. The window immediately to the* left *of 1 is barred as well, by AM–GM rather than by the barrier hypothesis: $\sigma(a)+\sigma(a') \ge 2\sqrt{\sigma(a)\sigma(a')} = 2\sqrt{r}$, and $2\sqrt{r} > T_{58} = 1.99874\ldots$ as soon as $r > (T_{58}/2)^2 = 0.998740\ldots$, whence $|P \cup Q| \ge 59$ and the product again exceeds $\Pi_{59} > 8.77 \times 10^{112}$. So the whole neighbourhood $r \in (0.998740, \infty)$ of the value 1 lies above the barrier, not merely the half-line $r \ge 1$, and no sweep reaches any of it* (`code/a10_nearmiss.rs`, `code/a10_density1.rs`)*. Earlier editions barred only $r \ge 1$, which left the left window formally open and the unmeasurability argument one–sided.*

**Proposition 6.10** (An effective approximation theorem, with a certified witness)**.** *Bado's Theorem 10.2 [27] asserts that for every $Y$ and every $\varepsilon > 0$ there are disjoint finite sets of primes $P, Q$, all greater than $Y$, with $0 \le \sigma(P)\sigma(Q) - 1 < \varepsilon$. The proof is a greedy accumulation and is ineffective. Made effective it is expensive: to land in $[T, T+\eta)$ the accumulation must start above $\approx 1/\eta$, and Mertens gives $\sum_{Y_1 < p \le W} 1/p \approx \log\log W - \log\log Y_1 = T$, so $W = Y_1^{e^T}$; two stages with $T \approx 1$ put the largest prime and the cardinality both at $\approx \varepsilon^{-e^2} = \varepsilon^{-7.389\ldots}$.*

*A Newton ladder does the same work with a handful of primes. Take $Q = \{2, 3, 5\}$, so $\sigma(Q) =$* 31/30 *and $P$ must approach* 30/31*; from a pool, repeat $p \leftarrow$ the least prime $\ge 1/d$ and $d \leftarrow d - 1/p$, where $d$ is the running deficit. Each rung squares it, $d_{\mathrm{new}} = d - 1/p \approx t\,d^2$ with $t$ the local prime gap, so the largest prime is $\approx \varepsilon^{-1/2}$ and the number of rungs is $O(\log\log(1/\varepsilon))$. Taking the pool to be the primes $7 \le p \le 271$ and eleven rungs gives $|P| = 66$, $|Q| = 3$, largest prime of* 2,010 *digits,* *and, in exact rational arithmetic,*

$$\bigl|\sigma(P)\sigma(Q) - 1\bigr| \;=\; 4.98523\ldots \times 10^{-4016},$$

*a rational whose numerator has* 129 *digits and whose denominator has* 4,145*. Every rung prime is* proved *prime by* APR–CL*, not merely a pseudoprime* (`code/nearmiss_effective.gp`)*. The certificate is tiered, and a reader may stop where they like:*

| *rungs* | 9 | 10 | 11 |
|---|---|---|---|
| *largest prime (digits)* | 505 | 1,007 | 2,010 |
| $\bigl\lvert\sigma(P)\sigma(Q) - 1\bigr\rvert$ | $9.57 \times 10^{-1007}$ | $2 \times 10^{-2010}$ | $4.99 \times 10^{-4016}$ |
| *cumulative certification* | $< 10$ *s* | $\approx 110$ *s* | $\approx 27$ *min* |

*The statement is machine–checked at the three–rung tier, where the largest prime is* 2,348,039,453 *and the defect is* $2.1111\ldots \times 10^{-18}$*: the primality of all* 58 *primes, the disjointness of the two sides, the exact value of* $\sigma(P)$ *as a ratio of a* 126*– and a* 129*–digit integer, and the bound* $|\sigma(P)\sigma(Q) - 1| < 10^{-17}$ *are all verified in* ***`Erdos307.Witness.effapprox_witness`****, on the standard axioms and with no appeal to kernel–external evaluation. Beyond that tier the primes are certified by APR–CL, which Mathlib does not carry; the higher rungs are therefore cited to the* PARI/GP *certificate rather than formalised.* 0 ***`sorry`*** (`lean/Erdos307/Witness.lean`)*.*

The gap is not marginal. At $\varepsilon = 10^{-13}$ the greedy needs a largest prime near $10^{94}$ and some $10^{92}$ primes; the ladder needs a largest prime near $10^7$ and three rungs. The ladder is exponentially better in the size of the largest prime and doubly exponentially better in the count, and the reason is visible: greedy convergence is linear in the number of primes, Newton convergence is quadratic. Truncating gives certificates of any size — five rungs and a 34–digit largest prime already give $6.07 \times 10^{-66}$, and seven rungs give $3.22 \times 10^{-254}$. The witness above approaches 1 from below; taking the last rung's prime just under $1/d$ instead overshoots, so either side is available and the statement of Theorem 10.2 as printed is met.

What this is not: progress towards a solution. A free pair of disjoint prime sets carries none of the rigidity an exact hit requires — $\sigma(P)\sigma(Q) = 1$ forces $\prod P = (\prod Q)'$ and $\prod Q = (\prod P)'$, and nothing in the ladder produces that — and by Proposition 6.9 the size of $|\sigma(P)\sigma(Q) - 1|$ does not measure progress towards the integer identity at all. The value of the construction is that it makes an existence theorem effective and exhibits a witness anyone can check.

**Proposition 6.11** (The near–miss frontier obeys a double–logarithmic law)**.** *For* $X \geq 2$ *put* $R(X) = \max\{r(a) : a \leq X \text{ squarefree}\}$*, where* $r(a) = \sigma(a)\sigma(a')$*. Let* $\omega_{\max}(X)$ *be the largest* $k$ *with* $\Pi_k \leq X$*, let* $\sigma_{\max}(X) = T_{\omega_{\max}(X)}$*, and let* $n(X)$ *be the largest* $n$ *with* $\Pi_n \leq X^2\sigma_{\max}(X)$*. Then*

$$R(X) \;\leq\; \Bigl(\tfrac{1}{2}\, T_{n(X)}\Bigr)^2.$$

*Proof.* Rigidity gives $\gcd(a, a') = 1$, so $U = \operatorname{supp}(a) \sqcup \operatorname{supp}(a')$ has $|U| = \omega(a) + \omega(a')$ distinct primes and $\prod_{p\in U} p \mid aa'$. By AM–GM, $\sum_{p\in U} 1/p = \sigma(a) + \sigma(a') \geq 2\sqrt{\sigma(a)\sigma(a')} = 2\sqrt{r}$, while the same sum is at most $T_{|U|}$ by the extremality of the smallest primes; hence $r \leq (T_{|U|}/2)^2$. For the size, $\Pi_{|U|} \leq \prod_{p\in U} p \leq aa' = a^2\sigma(a) \leq X^2\sigma_{\max}(X)$, so $|U| \leq n(X)$, and $T$ is increasing. $\square$

The bound is unconditional, needs no squarefreeness of $a'$, and is close: against the measured frontier of Proposition 6.9 it reads

| $X$ | $10^2$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $10^7$ |
|---|---|---|---|---|---|---|
| bound | 0.4014 | 0.4920 | 0.5296 | 0.5879 | 0.6129 | 0.6342 |
| $\max r$ observed | 0.333 | 0.345 | 0.502 | 0.502 | 0.536 | 0.554 |

(`code/frontier_bound.gp`). Two consequences. First, since $T_{n(X)} = \log\log\bigl(X^2\sigma_{\max}\bigr) + M + o(1)$ by Mertens, the frontier can climb only like

$$R(X) \;\le\; \tfrac{1}{4}\bigl(\log\log X^2 + M + o(1)\bigr)^2,$$

a *doubly logarithmic* law: this is why the measured frontier crawls, and why an extrapolation of its per–decade advance is worthless. Second, the bound reaches 1 exactly when $T_n \ge 2$, that is $n \ge 59$, that is $X^2\sigma_{\max} \ge \Pi_{59}$, giving $X \ge 2.0942 \times 10^{56}$ — the constant of Theorem 4.3, recovered here from the frontier side rather than from the cycle equations. The two derivations are independent and agree to all printed digits.

**Corollary 6.12** (The cost of a near–miss; the barrier as a function of $r$)**.** *For $r > 0$ put $n(r) = \min\{n : T_n \ge 2\sqrt{r}\}$. Every squarefree $a$ with $r(a) \ge r$ has $|U| \ge n(r)$, hence $aa' \ge \Pi_{n(r)}$ and*

$$\min(a, a') \;\ge\; \sqrt{\Pi_{n(r)}/T_{n(r)}}.$$

*Proof.* The first display of the proof of Proposition 6.11 gives $2\sqrt{r} \le \sum_{p\in U} 1/p \le T_{|U|}$, so $|U| \ge n(r)$; the rest is Theorem 4.3 verbatim with 59 replaced by $n(r)$. $\square$

At $r = 1$ this returns $n = 59$ and $2.0929 \times 10^{56}$, so Theorem 4.3 is the endpoint of a one–parameter family rather than an isolated statement. The family is what makes the crawl quantitative (`code/nearmiss_cost.gp`):

| $r$ | 0.5 | 0.554 | 0.6 | 0.7 | 0.8 | 0.9 | 0.99 | 1 |
|---|---|---|---|---|---|---|---|---|
| $n(r)$ | 8 | 9 | 11 | 16 | 24 | 37 | 56 | 59 |
| $\min(a,a') \ge$ | $2.6{\cdot}10^3$ | $1.2{\cdot}10^4$ | $3.6{\cdot}10^5$ | $4.4{\cdot}10^9$ | $1.2{\cdot}10^{17}$ | $4.3{\cdot}10^{30}$ | $4.7{\cdot}10^{52}$ | $2.1{\cdot}10^{56}$ |

Each increment in $r$ is paid for exponentially in the product, which is the same double–logarithmic law seen from the other side: $r$ enters the bound only through $2\sqrt{r}$, and $T_n$ grows like $\log\log p_n$. It also explains the shape of the swept data. The record $r = 0.5535$ needs $\min(a, a') \ge 1.2 \times 10^4$ and was found at $a = 3{,}972{,}254$; the next decade of $r$ costs four decades of $a$, and by $r = 0.7$ the requirement has passed $10^9$, beyond any sweep that has been run.

The restriction $r \le 1$ is not used in the proof, and above 1 the corollary says how expensive it is for the second derivative to *exceed* its argument:

| $r$ | 1 | 1.03944 | 1.25 | 1.5 | 2 | 9/4 |
|---|---|---|---|---|---|---|
| $n(r)$ | 59 | 71 | 212 | 940 | 37,963 | 361,139 |
| $\min(a,a') \ge$ | $10^{56.3}$ | $10^{71.3}$ | $10^{274.8}$ | $10^{1579.7}$ | $10^{98275.7}$ | $10^{1127769.3}$ |

So $a'' \ge 2a$ forces a union support of 37,963 primes and $\min(a, a') > 10^{98275}$: the second derivative cannot double its argument except at wholly inaccessible size. The witness of Theorem 6.34, with $r = 1.03944$, needs $|U| \ge 71$ and $\min \ge 10^{71.3}$, and its 142 digits supply both.

**Remark 6.13** (One inequality, six barriers)**.** Corollary 6.12 and Propositions 4.20 and 4.22 are the same statement read at different constants. Every barrier in this note is an instance of

$$\sum_{p\in U} \frac{1}{p} \;\ge\; c \quad\Longrightarrow\quad |U| \;\ge\; n(c), \qquad \prod_{p\in U} p \;\ge\; \Pi_{n(c)},$$

with $n(c) = \min\{n : T_n \geq c\}$, and the arithmetic enters only in producing $c$:

| configuration | $c$ | $n(c)$ | recorded as |
|---|---|---|---|
| near–miss ratio $r$ | $2\sqrt{r}$ | varies | Corollary 6.12 |
| $r = 1$, i.e. a two–cycle | 2 | 59 | Theorem 4.3 |
| $k$–cycle | $k/\lfloor k/2 \rfloor$ | 59 ($k$ even) | Proposition 4.20 |
| three–cycle | 3 | 361,139 | Proposition 4.19 |
| all–odd, $\min U \geq 3$ | 2 on $p \geq 3$ | 1,412 | Theorem 4.18 |
| $a'' \geq 2a$ | $2\sqrt{2}$ | 37,963 | above |
| orbit of growth $G$ in $L$ steps | $\frac{L}{\lceil L/2 \rceil} G^{1/L}$ | varies | Proposition 6.21 |

The coincidence $2\sqrt{9/4} = 3 = c(3)$ is why $r = 9/4$ and the three–cycle carry the identical bound 361,139: they are the same value of $c$. What looked like six separate barriers is one mass–to–size dictionary, and the only mathematics specific to each row is the derivation of its $c$.

**Proposition 6.14** (The mass window, and the forced core)**.** *Let $(a, b)$ be a two–cycle with $|U| = n$ and let $t_n > 1$ solve $t + 1/t = T_n$. Then*

$$\frac{1}{t_n} \;\leq\; \sigma(a),\, \sigma(b) \;\leq\; t_n,$$

*and every prime $p$ with $1/p \geq T_{n+1} - 2$ lies in $U$.*

*Proof.* The masses are positive with product 1 and sum at most $T_n$, so each is between the roots of $z^2 - T_n z + 1$. If $p \notin U$ then the $n$ primes of $U$ carry at most $T_{n+1} - 1/p$, and $\sigma(U) > 2$ forces $1/p < T_{n+1} - 2$. □

Both halves are sharp and both are informative at $n = 59$. The window is $[0.952682, 1.049668]$: *each member's mass is pinned within five per cent of* 1. And the forcing puts every prime up to 167 in $U$ — 39 of the 59 — so two thirds of the support is determined before any search begins. The window also recovers Proposition 6.20: the smaller member needs $T_k \geq 1/t_n$, which gives $k \geq 3$ for $n \leq 68$, $k \geq 2$ from $n = 69$ and $k \geq 1$ at $n = 1413$, reproducing that proposition's three thresholds exactly. The forcing decays with the level:

| $n$ | 59 | 60 | 61 | 62 | 63 | 65 | 68 | 72 |
|---|---|---|---|---|---|---|---|---|
| all $p \leq$ | 167 | 103 | 73 | 61 | 47 | 37 | 23 | 19 |
| forced | 39 | 27 | 21 | 18 | 15 | 12 | 9 | 8 |

(`code/forced_core.gp`). This accounts for the size of the level–59 enumeration. The 39 forced primes carry mass $1.91162\ldots$, so the remaining 20 must supply more than 0.08838 from primes above 167; the twenty smallest such primes reach exactly that, at 277. The free part is therefore a choice of 20 primes from a narrow band, which is why the admissible count is 49,961 and not astronomical. At level 60 the forced core has fallen to 27 and the free part to 33, and by level 68 — the top of the window Proposition 6.20 says a level search must cross — only 9 primes remain forced.

**Proposition 6.15** (The free band; why exactly one level is finite)**.** *Let $U$ be an admissible support with $|U| = n$ and let $q = \max U$. If $T_{n-1} < 2$ then*

$$q \;<\; \frac{1}{2 - T_{n-1}}.$$

*For $n \leq 58$ no admissible support exists, since $T_n < 2$. For $n = 59$ the hypothesis holds, $T_{58} = 1.99874\ldots$, and $q < 793.68$, so $q \leq 787$. For $n \geq 60$ it fails, since $T_{59} = 2.00235\ldots > 2$, and no bound on $q$ is available.*

*Proof.* The $n-1$ primes of $U$ other than $q$ carry at most $T_{n-1}$, so $\sigma(U) > 2$ forces $1/q > 2-T_{n-1}$. $\square$

This is the mechanism of Proposition 4.8 in its sharpest form. Level–by–level enumeration is finite at 59 and infinite from 60, and the switch is the single fact

$$T_{58} \;<\; 2 \;<\; T_{59}.$$

Below 59 there is not enough mass to reach 2 at all; at 59 every prime is capped at 787; at 60 the first 59 primes already carry mass $2.00235 > 2$, so appending *any* prime leaves an admissible support and the count is infinite at once. There is nothing gradual about the transition, and it happens at one level.

Together with Proposition 6.14 this closes level 59 from both sides. The support contains every one of the 39 primes $\leq 167$ and is contained in the 138 primes $\leq 787$, so the free part is a choice of 20 primes from the 99 candidates in $(167, 787]$. Unconstrained that is $\binom{99}{20} = 4.288 \times 10^{20}$ choices; the mass condition $\sigma(U) > 2$ cuts it to 49,961, a reduction by a factor $8.6 \times 10^{15}$ (`code/free_band.gp`). The level–59 computation of Proposition 4.5 is therefore not a brute force over an unbounded space but an exact traversal of a set whose two ends are both pinned by the same inequality.

**Corollary 6.16** (The level–59 count, from the two pins)**.** *The admissible supports at level* 59 *are exactly the sets*

$$\{p : p \leq 167\} \cup S, \qquad S \subset \{p : 167 < p \leq 787\}, \quad |S| = 20, \quad \sum_{p\in S} \frac{1}{p} \;>\; 2-T_{39} \;=\; 0.0883759429\ldots,$$

*and there are* 49,961 *of them.*

*Proof.* Membership of the core is Proposition 6.14, the ceiling is Proposition 6.15, and the displayed inequality is $\sigma(U) > 2$ after subtracting the forced mass $T_{39} = 1.9116240570\ldots$. The count is a 20–from–99 knapsack, evaluated in `code/level59_count.rs`. $\square$

This is not a closed formula — subset–sum counts do not have one — but it is a closed *characterisation*, and it reduces the level to a search of 119,509 nodes in place of the original enumeration. That the two independent routes agree on 49,961 is the point: the pins are not merely necessary conditions that happen to hold, they cut out the level exactly, and nothing outside them is doing any work. The evaluation also reports no subset sum within $10^{-12}$ of the threshold, so the floating comparison is measured to be safe rather than assumed. For orientation, the baseline 20 smallest candidates carry $0.0907260944\ldots$, exceeding the threshold by $T_{59}-2 = 0.0023501514\ldots$, and every admissible $S$ is a perturbation of that baseline within this budget.

**Proposition 6.17** (How many rungs the ladder has)**.** *Let $(a, b)$ be a Pythagorean pair with $a < b$, $t = b/a$ and $|U| = n$. Positivity of the two masses gives $-t < k \leq 0$, so $k$ takes at most $\lceil t\rceil$ values, and $t \leq t_n$ where $t_n + 1/t_n = T_n$. Hence the number of rungs available at level $n$ is at most $\lceil t_n\rceil$, and*

$$t_n > m \iff T_n > m + \tfrac{1}{m}.$$

*In particular the ladder has exactly the two rungs $k \in \{0, -1\}$ at every level $n \leq 1412$, and the third rung $k = -2$ opens only at $n = 1413$.*

Two readings. First, the hypothesis $\sigma(ab) < 5/2$ under which Proposition 5.5 confines $k$ to $\{0, -1\}$ is not a restriction anywhere a search operates: it is equivalent to $t < 2$, hence to $n \leq 1412$, and at level 59 one has $t = 1.049668$ with $\lceil t\rceil = 2$. The confinement is unconditional throughout the range the rest of this note inhabits, and the hypothesis only becomes real past level 1412.

Second, 1413 is not a new constant. It is the level at which $T_n$ reaches $5/2$, and that is the same threshold as the all–odd case of Theorem 4.18: an all–odd cycle needs mass 2 on primes $\geq 3$, hence $T_n \geq 2 + \frac{1}{2}$. The level at which the ladder acquires a third rung and the level at which an all–odd cycle becomes possible are the same level, for the same reason (`code/level_structure.rs`). The fourth rung would need $T_n > 10/3$, beyond $\pi(4 \times 10^7)$.

**Proposition 6.18** (The admissible supports are not a tree)**.** *There are admissible supports at level* 60 *with no admissible subset at level* 59*. Explicitly,*

$$U \;=\; \{p : p \leq 271\} \;\cup\; \{809,\, 811\}$$

*has* $|U| = 60$ *and* $\sigma(U) = 2.0012091\ldots > 2$*, while every* 59*–element subset has mass below* 2*.*

*Proof.* Deleting the largest prime loses the least mass, so if that deletion is inadmissible every deletion is. Here $\sigma(U) - 1/811 = 1.9999761\ldots < 2$. □

Such supports are exactly those of the form $V \cup \{q\}$ with $|V| = 59$, $\sigma(V) \leq 2$, $q > \max V$ and $q < 1/(2 - \sigma(V))$, and they are not exceptional. Taking $V = \{p \leq 271\} \cup \{r\}$ with $r > 794$, so that $\sigma(V) < 2$, each $r$ admits every prime $q$ in $\bigl(r,\ 1/(2 - \sigma(V))\bigr)$: at $r = 797$ that interval reaches 190,414 and contains 17,061 primes, and summing over $r < 4000$ gives 60,831 such supports from this one shape alone (`code/level_structure.rs`).

The consequence is for the organisation of a level search, and it is independent of Proposition 4.8. That proposition says level 60 has infinitely many admissible supports; this one says that even the part reachable with bounded primes is not exhausted by the base–plus–tail organisation, because the admissible supports do not form a tree under adjunction. Closing every one–new–prime family at level 60, which is question (i) of Section 9, would therefore still not close level 60. The searches of Propositions 1.2 and 4.5 are stated over one–new–prime families throughout and are unaffected; what changes is the reading of what completing them would buy.

**Proposition 6.19** (The level–minimal structure at any $c$)**.** *Let $U$ satisfy $\sum_{p \in U} 1/p \geq c$ with $|U| = n(c) = \min\{n : T_n \geq c\}$. Then every prime $p$ with $1/p \geq T_{n(c)+1} - c$ lies in $U$, and $\max U < 1/\bigl(c - T_{n(c)-1}\bigr)$, the denominator being positive by minimality of $n(c)$.*

*Proof.* Both are Propositions 6.14 and 6.15 with 2 replaced by $c$; neither argument uses the value. □

The slack $T_{n(c)} - c$ is the total mass a support may waste, and it collapses as $c$ grows, so the support becomes more forced, not less (`code/level_structure.rs`):

| $c$ | 2 | 9/4 | 7/3 | 5/2 | 3 |
|---|---|---|---|---|---|
| $n(c)$ | 59 | 231 | 400 | 1,413 | 361,139 |
| $T_{n(c)} - c$ | $2.35{\cdot}10^{-3}$ | $2.33{\cdot}10^{-4}$ | $2.44{\cdot}10^{-4}$ | $2.07{\cdot}10^{-5}$ | $3.09{\cdot}10^{-8}$ |
| forced primes | 39 | 181 | 259 | 1,175 | 314,407 |
| forced fraction | 66% | 78% | 65% | 83% | 87% |
| $\max U <$ | 793.7 | 2,194 | 8,284 | 15,592 | $6.19{\cdot}10^6$ |

The row $c = 2$ is Propositions 6.14 and 6.15, recovered. The row $c = 3$ is the three–cycle of Proposition 4.19: at its minimal level a three–cycle contains *every* prime up to 4,476,608, that is 314,407 of its 361,139, and no prime beyond $6.19 \times 10^6$. Its support is almost completely determined, with a slack of $3.09 \times 10^{-8}$ to distribute among the remaining 46,732 places.

The same reading applies with the mass restricted to primes $\geq P$, which is the setting of Proposition 4.22. At $P = 3$, the all–odd case of Theorem 4.18, the minimal level is $n = 1412$ with slack $2.069 \times 10^{-5}$; every odd prime up to 9,484 is forced, 1,174 of the 1,412 or 83%, no prime exceeds 15,592, and the product is $10^{5040.1}$. At $P = 5$ the figures are $n = 40{,}259$, 24,770 forced, $\max U < 1.61 \times 10^6$ and $10^{209582}$, reproducing Proposition 4.22 from the structural side.

That the all–odd case is 83% forced does not make it a feasible search, and it is worth recording why, since the shape invites the comparison with level 59. There the free part was 20 primes from 99 candidates and the count was 49,961. Here it is 238 from 643: the forced mass is $1.97751561\ldots$, so the free primes must carry more than $0.02248438\ldots$ against a baseline of $0.02250507\ldots$, an excess of $2.069 \times 10^{-5}$. A traversal capped at $4 \times 10^8$ nodes reaches $1.977 \times 10^8$ admissible supports without exhausting the space (`code/level_structure.rs`). The slack per free place, not the forced fraction, is what decides enumerability, and here it is 87 times larger per place than at level 59. Question (ii) of this section does not close this way.

**Proposition 6.20** (The barrier stratified by the smaller support)**.** *Let $(a, b)$ be a two–cycle and put $k = \min(\omega(a), \omega(b))$. Then*

$$k = 1 \;\Rightarrow\; |U| \geq 1413, \qquad k = 2 \;\Rightarrow\; |U| \geq 69, \qquad k \geq 3 \;\Rightarrow\; |U| \geq 59,$$

*and the last is attained. Consequently every two–cycle with $|U| \leq 68$ has $\min(\omega(a), \omega(b)) \geq 3$.*

*Proof.* Write $T = T_{|U|}$. The supports are disjoint, so $\sigma(a) + \sigma(b) \leq T$, and $\sigma(a) \leq T_k$ for the member with $k$ primes. The product $x(T - x)$ is increasing for $x \leq T/2$, so $\sigma(a)\sigma(b) \leq m(T - m)$ with $m = \min(T_k, T/2)$, and $\sigma(a)\sigma(b) = 1$ forces $m(T - m) \geq 1$. For $k \geq 3$ one has $T_k \geq T_3 = 31/30 > T/2$ in the relevant range, so $m = T/2$ and the condition is $T \geq 2$, i.e. $|U| \geq 59$. For $k \leq 2$ one has $T_k < T/2$, so $m = T_k$ and the condition is $T \geq T_k + 1/T_k$: at $k = 1$, $T \geq 2.5$ and $|U| \geq 1413$; at $k = 2$, $T \geq 2.0333\ldots$ and $|U| \geq 69$. □

The stratification is sharp at each end. At $k = 1$ the member is a prime $p$, so $\sigma(b) = p \geq 2$ and $b$ must carry mass 2 on primes other than $p$, which needs 1412 of them; at $k = 2$ the best case is $a = 6$, whence $\sigma(b) \geq 6/5$ on primes $\geq 5$, which needs 67. This contains and extends Proposition 16 of [9], that no two–cycle has both members a product of two primes: the present statement rules out *either* member having two prime factors unless $|U| \geq 69$, and either having one unless $|U| \geq 1413$. Since Proposition 4.5 gives $|U| \geq 60$, the window $60 \leq |U| \leq 68$ that any level–by–level search must cross admits only splits with both $\omega(a) \geq 3$ and $\omega(b) \geq 3$ (`code/split_bound.gp`).

**Proposition 6.21** (The derivative cannot outgrow its own support)**.** *Let $a_0 \to a_1 \to \cdots \to a_L$ be an orbit segment of the arithmetic derivative with every $a_i$ squarefree, put $G = a_L/a_0$ and let $U = \bigcup_{i<L} \operatorname{supp}(a_i)$. Then*

$$\sum_{p \in U} \frac{1}{p} \;\geq\; \frac{L}{\lceil L/2 \rceil}\, G^{1/L}, \qquad \textit{equivalently} \qquad G \;\leq\; \Big( \frac{\lceil L/2 \rceil}{L}\, T_{|U|} \Big)^{L},$$

*and for even $L$ the geometric growth rate satisfies $G^{1/L} \leq T_{|U|}/2$.*

*Proof.* Squarefreeness gives $\gcd(a_i, a_i') = 1$, so consecutive members are coprime and $S_p = \{i < L : p \mid a_i\}$ has no two consecutive elements: it is independent in the path on $L$ vertices, whence $|S_p| \leq \lceil L/2 \rceil$. Then $\sum_{i<L} \sigma(a_i) = \sum_{p \in U} |S_p|/p \leq \lceil L/2 \rceil \sum_{p \in U} 1/p$, while $\prod_{i<L} \sigma(a_i) = a_L/a_0 = G$ gives $\sum_{i<L} \sigma(a_i) \geq L G^{1/L}$ by AM–GM. The second form inverts the first using $\sum_{p \in U} 1/p \leq T_{|U|}$. □

This is the general shape of which the rest of the dictionary is a specialisation: at $L = 2$ it reads $\sum_{p\in U} 1/p \geq 2\sqrt{G}$, which is Corollary 6.12 with $G = r$; at $G = 1$ it returns $c = L/\lceil L/2\rceil \to 2$, which is Proposition 4.20 on the path in place of the cycle. The statement it adds is dynamical rather than Diophantine, and it bounds the growth branch of Ufnarovski–Åhlander's Conjecture 2: an orbit may tend to infinity, but only slowly unless it accumulates an enormous support (`code/growth_bound.gp`):

| rate $g$ | 1.05 | 1.10 | 1.25 | 1.50 | 1.75 | 2 |
|---|---|---|---|---|---|---|
| $\lvert U\rvert \geq$ | 97 | 170 | 1,413 | 361,139 | $4.6 \times 10^9$ | $4.3 \times 10^{16}$ |
| largest prime $\geq$ | $5.4{\cdot}10^2$ | $1.0{\cdot}10^3$ | $1.2{\cdot}10^4$ | $5.2{\cdot}10^6$ | $1.2{\cdot}10^{11}$ | $1.8{\cdot}10^{18}$ |

A squarefree orbit that doubles at every step needs some $4.3 \times 10^{16}$ distinct primes in its supports and a prime past $1.8 \times 10^{18}$: sustained rapid growth is not merely unobserved but impossible on a small support. The hypothesis that every $a_i$ is squarefree is essential and is not generic along orbits; it is automatic on cycles, by Ufnarovski–Åhlander, which is why the cycle rows of the dictionary carry no such caveat.

**Corollary 6.22** (Sustained growth needs scale, and the scale is enormous)**.** *Let $a_0 \to \cdots \to a_L$ be a squarefree orbit segment with every member at most $X$ and geometric growth rate at least $g > 1$. Then*

$$\log_2 X \;\geq\; \sqrt{n(2g)\,\log_2 g}\,, \qquad n(c) = \min\{n : T_n \geq c\}.$$

*Proof.* Proposition 6.21 gives $\sum_{p\in U} 1/p \geq 2g$, so $|U| \geq n(2g)$. Every member is at most $X$, so $\omega(a_i) \leq \log_2 X$ and $|U| \leq \sum_{i<L} \omega(a_i) \leq L \log_2 X$; growth at rate $g$ from $a_0 \geq 2$ gives $a_L \geq g^L$, so $L \leq \log_g X = \log_2 X/\log_2 g$. Combining, $(\log_2 X)^2/\log_2 g \geq n(2g)$. ☐

The threshold is explosive (`code/growth_bound.gp`):

| $g$ | 1.10 | 1.25 | 1.50 | 1.75 | 2 |
|---|---|---|---|---|---|
| members must exceed | $10^{1.5}$ | $10^{6.4}$ | $10^{138}$ | $10^{18420}$ | $10^{6.23\times10^7}$ |

The arithmetic derivative can therefore grow slowly at any scale and quickly only at scales no computation reaches. Rate 1.25 needs members past $2.6 \times 10^6$, so it is visible inside the sweep of Proposition 6.9; rate 1.5 needs $10^{138}$; and a squarefree orbit cannot double at every step unless its members exceed $10^{6.2\times10^7}$.

Inverting the corollary bounds the rate itself. The largest $g$ a squarefree orbit can sustain up to $X$ solves $n(2g)\log_2 g = (\log_2 X)^2$, and since $n(c)$ grows like $\mathrm{Li}\big(e^{e^{c-M}}\big)$ this gives $e^{e^{2g}} \asymp (\log_2 X)^2$, that is

$$g \;\sim\; \tfrac{1}{2}\log\log\log X.$$

| $X$ | $10^8$ | $10^{20}$ | $10^{100}$ | $10^{1000}$ | $10^{10^5}$ | $10^{10^7}$ |
|---|---|---|---|---|---|---|
| max sustainable $g$ | 1.2768 | 1.3590 | 1.4808 | 1.6174 | 1.8134 | 1.9537 |

The ceiling is a triple logarithm, so it is effectively a constant: even at $X = 10^{10^7}$ the rate stays below 2, and for any range a computation can address it is below 1.5. The reading for the escape branch of Ufnarovski–Åhlander's Conjecture 2 is that an orbit tending to infinity while remaining squarefree must do so sub–geometrically in every bounded window, its rate decaying like the triple logarithm of that window. Escape is possible; escape at a fixed geometric rate is not.

The mass window transfers to the normal form as well. In Proposition 5.3 a solution is $a = Mp$, so $\sigma(a) = \sigma(M) + 1/p$, and Proposition 6.14 places $\sigma(a)$ in $[1/t_n, t_n]$. Hence

$$\sigma(M) \;\in\; \big[\,1/t_n - 1/p,\; t_n\,\big],$$

which at level 59 is $[0.9527 - 1/p,\ 1.0497]$: the base of the one–prime normal form carries mass within five per cent of 1, just as each member does, and $\omega(M) \geq 2$ by Proposition 6.20.

The region is not unreachable, only unsweepable. The witness $n_0$ of Theorem 6.34 has $n_0$ and $n_0'$ both squarefree and $r(n_0) = \sigma(n_0)\sigma(n_0') = 1.03944\ldots$, so $n_0'' > n_0$: the value 1 that a solution must hit is interior to the range of $r$, not extremal, and no inequality on $r$ alone can settle #307. The same construction approaches 1 from below. Tuning $\sigma(a)$ against the forced $\sigma(a') = \frac{31}{30} + \varepsilon$ gives

$$a = 3359 \prod_{\substack{7 \leq p \leq 269 \\ p \notin \{229,271\}}} p, \qquad a' = 2 \cdot 3 \cdot 5 \cdot 383 \cdot 113717 \cdot P_{99},$$

with $P_{99}$ prime, both $a$ and $a'$ squarefree, 108 digits, and

$$r(a) = 0.99207796\ldots, \qquad |r(a) - 1| = 7.9220 \times 10^{-3},$$

against the smallest relative defect 0.4465 over the swept range (`code/nearmiss_tune.gp`).

The same tuning with its acceptance window mirrored, $\sigma(a)$ just *above* 30/31 rather than just below, reaches 1 from the other side and closer. With

$$a = 3301 \prod_{\substack{7 \leq p \leq 293 \\ p \notin \{137,263\}}} p, \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{117},$$

where $P_{117}$ is a 117–digit prime, one has $a$ of 118 digits and, exactly,

$$r(a) = 1.001014244\ldots, \qquad |r(a) - 1| = 1.0142 \times 10^{-3}, \qquad a'' > a.$$

This is exact rather than a lower bound, since $a'$ factors completely: the cofactor after trial division is prime, so $\sigma(a')$ is known and not merely bounded below. It improves the record on both sides at once, against $7.9220 \times 10^{-3}$ from below and $3.944 \times 10^{-2}$ from above, the latter being $r(n_0)$ of Theorem 6.34 (`code/nearmiss_above.gp`, re–checked independently in `code/nearmiss_above_verify.gp`).

That record is not the natural limit, and seeing why identifies what the near–miss problem actually is. In every witness of this shape $a' = 2 \cdot 3 \cdot 5 \cdot P$ with $P$ prime, so $\sigma(a') = \frac{31}{30} + P^{-1}$ with $P^{-1}$ of size $10^{-118}$, and therefore

$$|r(a) - 1| = \tfrac{31}{30} \left|\sigma(a) - \tfrac{30}{31}\right|$$

to any precision that matters. The quantity $a'$ has dropped out. What remains is the Diophantine question of how closely a sum of *distinct prime reciprocals* approaches 30/31 from above, and the $10^{-3}$ above is only the granularity of adjusting by a single prime below 4000.

Closing the deficit in stages removes that ceiling. With $d = \frac{30}{31} - \sigma(S)$ for $S$ the primes in $[7, 271]$, take $p_1$ the least prime exceeding $1/d$, then $p_2$ the least exceeding $1/(d - p_1^{-1})$, and finally $p_3$ the greatest prime below the reciprocal of what remains; the overshoot is then of order $\mathrm{gap}(p_3)/p_3^2$. Each rung squares the precision. Sweeping the choice at the first two rungs and retaining those that satisfy the forcing congruence and have $a' = 2 \cdot 3 \cdot 5 \cdot P$ gives, with $(p_1, p_2, p_3) = (569, 1949, 15461)$,

$$a = 569 \cdot 1949 \cdot 15461 \prod_{7 \leq p \leq 271} p \quad \text{(58 primes, 120 digits)}, \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{118},$$

and, exactly,

$$r(a) = 1 + 6.278713 \times 10^{-9}, \qquad a'' > a.$$

This is $1.6 \times 10^5$ times closer to 1 than the preceding record.

That record is on the wrong side of 1, and identifying the family says why. Every witness here has $a' = 2 \cdot 3 \cdot 5 \cdot P$ with $P$ prime, which is exactly the stratum $M = 30$ of Proposition 5.3: with $M' = 31$ the equation $MN - M'N' = M^2$ reads $30N - 31N' = 900$, and the proposition's $N = M + M'p$, $N' = Mp$ become $a = 31P + 30$, $a' = 30P$. Two identities follow, both exact: $a'' = (30P)' = 31P + 30$, so $r = a''/a = (31P + 30)/a$, and a solution of the stratum is precisely $30a - 31a' = 900$. Hence

$$\sigma(a) = \frac{30P}{31P + 30} = \frac{30}{31} - \frac{900}{31a} < \frac{30}{31}$$

for every solution of this stratum, strictly. Approaches to 1 from *above*, which is what the $6.28 \times 10^{-9}$ above and the $3.76 \times 10^{-9}$ of a wider sweep are, therefore lie on a side the stratum cannot populate. They stand as records for the spectrum of $r$ and cannot converge on a solution.

Aimed below instead, where the stratum's solutions must lie, the same ladder gives

$$a = 431 \cdot 68111 \cdot 2792213 \prod_{7 \le p \le 271} p \quad (58 \text{ primes}, \ 123 \text{ digits}), \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{122},$$

and, exactly,

$$r(a) = 1 - 1.654663 \times 10^{-13}, \qquad \sigma(a) < \tfrac{30}{31}, \qquad a'' < a,$$

which is $4.8 \times 10^{10}$ times closer to 1 than the previous record from below. For this witness $30a - 31a'$ is a 112–digit number where a solution needs 900, which is the honest measure of the distance remaining: the ratio is close, the integer is not.

Neither construction is exhausted. Over the same sweeps the smallest *available* deficit was $2.043 \times 10^{-18}$ from below and $1.209 \times 10^{-14}$ from above; what limits the reported values is the rate at which $a'$ takes the form $2 \cdot 3 \cdot 5 \cdot P$, not the tuning (`code/nearmiss_ladder.gp`, `code/nearmiss_below.gp`, re–checked in `code/nearmiss_ladder_verify.gp` and `code/nearmiss_below_verify.g`

We stop here, and the reason is a measurement rather than a budget. Between the two from–below witnesses the ratio improved by three orders of magnitude while $30a - 31a'$ fell from 114 digits to 112, against the 3 a solution requires. Exhausting the available deficit would gain perhaps five more digits of the 112. The near–miss frontier measures $|r - 1|$, a ratio; membership of the stratum is the integer identity $30a - 31a' = 900$, and the two are not commensurable at this scale. Driving the ratio further is therefore not evidence about the integer, and the line is closed as a line of attack. What survives it is structural and is stated above: the family is the $M = 30$ stratum of Proposition 5.3, its solutions satisfy $\sigma(a) < 30/31$ strictly, and approaches from above cannot reach one. Those three facts are machine–checked in `lean/Erdos307/StratumM30.lean`.

The constant 30/31 is not special. In the normal form of Proposition 5.3 a solution of the $M$–stratum is $a = Mp$, $b = N = M + M'p$, so $\sigma(a) = \sigma(M) + 1/p$ and, by the identity $(\sigma(M) + 1/p)\sigma(N) = 1$ of part (b), $\sigma(b) = 1/\sigma(a) < 1/\sigma(M) = M/M'$, with defect exactly $M^2/(M'(M + M'p))$. Hence *every* stratum is one–sided: for no squarefree $M$ does the half–space $\sigma(b) \ge M/M'$ contain a solution, and $M = 30$ recovers 30/31. This is a corollary of Proposition 5.3(b) rather than a new result, but it is the constraint that governs search direction, so it is machine–checked in `lean/Erdos307/StratumGeneral.lean`. The consequence for the near–miss programme is that the mirrored tuning of the previous paragraph is not merely unlucky at $M = 30$; the corresponding tuning is empty in every stratum. Consistently with the two–sided statement above, $a$ lies past the barrier: $118 > 113$ digits, and $r$ sits inside the barred neighbourhood $(0.998740, \infty)$.

What remains unmeasurable is the density of $r$ at 1, since constructed points do not estimate a density, and that is what the heuristic constant needs. The frontier reported above is therefore superseded as a record and unchanged as an argument.

The situation is not the usual one. Heuristics of Bateman–Horn or Hardy–Littlewood type are normally checked against data, and the agreement is what justifies confidence in them. Here the prediction that solutions exist rests on a constant that cannot be estimated by sampling, because the barrier and the sampleable range are disjoint. The expected-count heuristic for #307 is therefore unverifiable in the usual way.

It is not, however, beyond computation, and the reason is that the constant is the barrier.

**Proposition 6.23** (The heuristic constant is the barrier's own probability)**.** *Model a random squarefree $a$ by $\mathbb{P}(p \mid a) = 1/(p+1)$, and $a'$ by the coprimality model that Proposition* 6.27 *validates, so that $\mathbb{P}(q \mid a') = 1/q$ for $q \nmid a$. Then each prime independently lies in $a$, in $a'$, or in neither, with probabilities $\frac{1}{p+1}, \frac{1}{p+1}, 1 - \frac{2}{p+1}$, and*

$$\sigma(a) + \sigma(a') = \sum_{p \mid aa'} \frac{1}{p}, \qquad \mathbb{P}(p \mid aa') = \frac{2}{p+1}.$$

*Since a two-cycle forces $\sigma(a)\sigma(a') = 1$ and hence $\sigma(a) + \sigma(a') \geq 2$ by* AM–GM*, the density $f$ of $r = \sigma(a)\sigma(a')$ is supported in the event $B : \sigma(a) + \sigma(a') \geq 2$, so $f(1) \leq C\,\mathbb{P}(B)$ with $C$ the conditional density of $r$ at 1 given $B$. Numerically*

$$\mathbb{P}(B) = 2.74 \times 10^{-90}.$$

The point is the order and not the digits. $\mathbb{P}(B)$ is a *tail* probability of a one-dimensional sum, not a density, which is what makes it stable: it moves from 4.09 to 3.01 to $2.74 \times 10^{-90}$ as the grid is refined from $2 \times 10^4$ to $8 \times 10^4$ bins and the prime range from $10^4$ to $10^6$, while a direct attempt at $f(1)$ on a two-dimensional grid moves by four orders of magnitude per refinement and converges to nothing. Two checks support it. The same computation with $\mathbb{P}(p \mid n) = 1/p$ returns $\mathbb{P}(\sigma(n) \geq 1) = 0.0420$, reproducing the measured density of prime-abundant integers to four places; and the single configuration in which $aa'$ absorbs exactly the first 59 primes has probability $\prod_{p \leq 277} 2/(p+1) = 10^{-96.0}$, below $\mathbb{P}(B)$ by the factor one expects from the many other prime sets of mass 2 (`code/barrier_probability.rs`).

One more thing follows, and it disarms the near-miss spectrum entirely.

**Proposition 6.24** (The near-miss infimum is zero, so closeness is not evidence)**.**

$$\inf\Big\{ \big|\sigma(P)\sigma(Q) - 1\big| \ : \ P, Q \textit{ finite disjoint prime sets} \Big\} = 0,$$

*and #307 asks exactly whether the infimum is attained.*

*Proof.* Split the primes into two infinite classes, say by parity of index. Each class has terms tending to 0 and divergent reciprocal sum, so by the classical subseries theorem each carries a subseries summing to exactly 1. This produces disjoint *infinite* $P_\infty, Q_\infty$ with $\sigma(P_\infty) = \sigma(Q_\infty) = 1$. Truncating both at height $y$ gives finite disjoint sets with $\sigma(P_y)\sigma(Q_y) \to 1$. □

The analytic content is formalised, and in a slightly stronger form than the proof above uses. For the infimum one does not need the subseries theorem, only that each mass can be placed in $(1-\delta, 1)$: `Erdos307.exists_block_sum_near` takes the first consecutive block whose sum exceeds $T - \varepsilon$ and observes that it overshoots by less than one term, and `infimum_zero` assembles two such blocks into a product within $\varepsilon$ of 1. Both carry 0 `sorry` and the three standard axioms (`lean/Erdos307/InfZero.lean`). The arithmetic input, that the primes split into two infinite classes each with reciprocal terms tending to 0 and unbounded partial sums, is cited rather than reproved, as in Theorem 6.30.

The greedy form of that argument reaches $|\sigma(P)\sigma(Q)-1| = 6.50\times10^{-9}$ with 144 primes, against 0.4465 for the swept frontier of Proposition 6.9 (`code/infimum_zero.gp`). So the analytic half of #307 is empty: the mass equation can be approximated as closely as one likes, and no near-miss, however good, is evidence that a solution exists.

What the near-misses do measure is a trade-off, and it is the honest picture of the obstruction. A solution must kill two defects at once, the mass defect $|\sigma(P)\sigma(Q) - 1|$ and the size defect $\log_{10}\bigl(\prod Q/\sigma(P)\prod P\bigr)$, the latter vanishing exactly when $\prod Q = N_P$, which is Theorem 2.3. Each is reachable alone and neither construction moves the other:

| construction | mass defect | size defect |
|---|---|---|
| greedy, $Q$ free | $6.50 \times 10^{-9}$ | $10^{341.3}$ |
| $Q = \text{primes}(a')$, `nearmiss_tune.gp` | $7.92 \times 10^{-3}$ | 0 |

Fixing $Q$ as the prime set of $a'$ makes the size defect vanish identically and leaves the mass defect at $10^{-3}$; choosing $Q$ freely drives the mass defect to $10^{-9}$ and costs 341 orders of magnitude in size. #307 is the demand that both columns read zero simultaneously, and nothing in this paper moves them together.

**Remark 6.25** (What the calibrated heuristic says, and it is not encouraging)**.** With $\mathbb{E}(X) \sim f(1)\,\delta\log X$ and $\delta = 0.3721$, Proposition 6.23 gives an expected number of two-cycles below the proved barrier of about $10^{-88}$, and $\mathbb{E}(X)$ does not reach 1 until $X$ is of order $10^{4\times10^{89}}$. So the heuristic favours YES only in the limiting sense that $\log X$ diverges; at every scale that any search or any barrier argument will reach, it predicts nothing. Two consequences, both negative and both worth stating. A proof of the negative direction would have to contradict a divergence, and a search for a solution is hopeless not merely at present but at any scale expressible in the notation of this paper. The constant is small for a reason that is not incidental: it *is* the barrier, seen as a probability rather than as a counting bound, which is the third independent appearance of the same Mertens threshold in this work.

The label is HEURISTIC and the model is doing the work. What is claimed is only that the constant, said above to be unavailable to sampling, is available to computation once the coprimality model of Proposition 6.27 is granted.

**Remark 6.26** (What is left, on both sides)**.** The two directions can now be specified rather than merely called open.

*Negative.* Theorem 6.32 closes the certificate route on both branches, so a proof must count rather than certify; and by Remark 6.33 the count it must defeat diverges. A proof of the negative direction would therefore have to contradict a divergent expected count, which is to say that the direction is not merely unproved but probably false.

*Positive.* Three routes are available in principle, and two are closed quantitatively.

(a) *Exhibit a witness.* Excluded by Remark 6.25: the first is expected near $10^{4\times10^{89}}$.

(b) *Bound the count below by analytic means.* Excluded by size. At the proved barrier the main term $f(1)\delta\log X$ lies $10^{200.6}$ below the trivial count $X$, and the saving a method would need in order to keep the main term above its own error grows like $X$ itself. Sieve and circle-method errors save a power of $X$ or a power of $\log X$. So an analytic proof here would have to be *exact*, with error identically zero, rather than sharp.

(c) *Construct exactly.* This is the only branch alive, and it exists: the Pell orbit of Remark 1.16. It is alive but narrower than it was, and the narrowing should not be mistaken for proximity. Two cautions, both quantitative. The searches of Remark 1.11 came back empty over $3.99 \times 10^{11}$ candidates, but empty was the *predicted* outcome: the expected count was $2.4 \times 10^{-56}$, so the computation confirms the heuristic model rather than deciding anything, and a confirmed model is not a decided problem. And by Proposition 4.8 those searches cannot be continued upward at all, since the admissible supports are infinite at every level past 59. The unconditional exclusions of this note stop where enumeration stops, which is now. Proposition 4.7 bounds the tail prime by $4N$, so the orbit contributes nothing past a bounded initial segment and the construction is a finite search over the divisors of $4D$ rather than a Lehmer primality question; Remark 1.11 then makes that finite search heuristically empty on all 34 immune families, by two independent estimates. What survives is the branch itself, not its expectation: the construction must now either come from a base outside the immune list or exploit the parametrisation $q = 2(N \pm k)/m^2$ directly.

The two exclusions interlock, and the interlock is the real statement. Exactness is what algebra supplies, and Theorem 3.1 excludes every argument that survives the passage to $K[t]$, where the conclusion is false. The Pell construction escapes that exclusion for one reason only: its final input is the primality of a determined term, and primality does not survive to $K[t]$. So a successful method must be *archimedean and exact at the same time*, a combination which is unusual but not unprecedented, continued fractions solving Pell being the standard instance, and Remark 1.16 is exactly the analogue here.

The labels differ across the three branches and the difference matters. The exclusions of (a) and (b) are HEURISTIC: both rest on the value of $f(1)$ from Proposition 6.23, hence on the coprimality model, and a reader who rejects that model recovers both branches. The interlock itself is not heuristic: Theorem 3.1 and Proposition 5.5 are PROVED, and so is the reduction of branch (c) to a primality question. So what follows is stated at the strength of its weakest link, which is the model.

What remains of #307 is therefore one question and not a programme: whether some base admits a Pythagorean tail prime, together with the sign selection $k = 0$, which Remark 1.16 shows is a choice between $\{0, -1\}$ rather than a coincidence of measure zero. By Proposition 5.5 the orbit parametrising the Pythagorean locus is forced to be exponential rather than polynomial. That was previously read as placing the question in the Lehmer class rather than the Schinzel class, hence open for every non-degenerate sequence; but by Proposition 4.7 a prime tail satisfies $q \leq 4N$, so the exponential orbit supplies no candidate past a bounded initial segment and the question is neither Lehmer nor Schinzel. It is a finite search over the divisors $m \mid 4D$ with $AB + Dm^2$ a square, which Remark 1.11 makes heuristically empty on all 34 immune families. The residue is thus smaller and better located than it was, and still undecided: the bound is proved, the emptiness is not, and no base outside the immune list has been examined at all. That is why nothing in this paper decides the problem.

The measurement does expose a sharp structural fact, which turns out to be exactly the barrier seen statistically.

**Proposition 6.27** (The barrier is statistically complete)**.** $\mathbb{E}[\sigma(a')]$ *falls by an order of magnitude as* $\sigma(a)$ *grows, from* 0.2035 *unconditionally to* 0.0207 *on* $\sigma(a) \geq 1.3$. *The whole of this suppression is coprimality. Writing the prediction for an integer chosen at random subject to* $\gcd(n, a) = 1$, *namely* $\sum_{p \nmid a} p^{-2}$, *the ratio of measured to predicted is*

$$0.65, \quad 0.82, \quad 0.91, \quad 0.92, \quad 0.95, \quad 1.01$$

*at $\sigma(a) \geq 0$, 0.5, 0.9, 1.0, 1.2, 1.3: it converges to* 1 *precisely in the regime a solution would inhabit* (`code/a10_anticorr.rs`)*.*

The mean is not the moment that $f(1)$ depends on. Since $f(1)$ is an integral of the joint density of $(\sigma(a), \sigma(a'))$ along $\sigma(a)\sigma(a') = 1$, a model could match the conditional mean and still misstate the density; the next moment is therefore the sharper test. The model makes a prediction for it too, and a per–$a$ one: given $\mathrm{supp}(a)$, each $q \nmid a$ lands in $a'$ independently with probability $1/q$, so

$$\mathbb{E}[\sigma(a')] = \sum_{q \nmid a} \frac{1}{q^2}, \qquad \mathrm{Var}[\sigma(a')] = \sum_{q \nmid a} \frac{1}{q^3}\Big(1 - \frac{1}{q}\Big).$$

Measured against these over all $a \leq 10^7$ with $a, a'$ squarefree (`code/f1_variance.rs`), the ratio of measured to predicted variance is

$$1.111,\ 0.968,\ 0.894,\ 0.956,\ 0.938,\ 0.860,\ 0.948,\ 1.030$$

at $\sigma(a) \geq 0$, 0.5, 0.7, 0.9, 1.0, 1.1, 1.2, 1.3, against $0.648, \ldots, 1.010$ for the mean. The variance is predicted as well as the mean, it shows no systematic drift, and the two converge together. The top band rests on 438 values and is correspondingly noisy, so we claim only the absence of a trend, not the precise value. What the test was designed to detect — a model matching the first moment while its tail degrades, which would inflate $f(1)$ and could in principle send the true value to 0 — does not occur.

So on the sampled range $a'$ is mass-poor for one reason only, that it is coprime to $a$ and $a$ has absorbed the small primes, and there is no further effect to exploit. This is an independent confirmation of Proposition 7.25: the barrier is not merely near-optimal as a bound, it already contains the entire statistical structure of the obstruction, and no refinement by correlation is available.

The range matters, and the statement is about averages. Every $a \leq 4 \times 10^8$ with $\sigma(a) > 1$ is even, so the measurement never sees an $a$ that omits the small primes. Such $a$ exist, since the prime reciprocals diverge, and for them $a'$ is free to absorb 2, 3 and 5 and is not mass-poor at all: Theorem 6.34 exhibits one with $\sigma(a') = \frac{31}{30}$. Proposition 6.27 bounds an expectation on a range, and nothing here bounds a supremum off it.

## A Lyapunov candidate for the negative direction

Theorem 3.1 rules out cycles over $\mathbb{F}_q[t]$ by exhibiting a function that provably decreases, namely the degree. Proposition 5.11 shows no *absolute value* on $\mathbb{Q}$ plays that role. It does not exclude functions that are not absolute values, and the following is the outcome of a search among such functions. It is a conjecture, not a theorem, and it is recorded because it is falsifiable and because the mechanism protecting it is one of the paper's own results.

**Definition 6.28.** For $\lambda > 0$ and squarefree $n$ set $\delta_\lambda(n) = \log n + \lambda\, \sigma(n)$.

**Proposition 6.29** (A decreasing $\delta_\lambda$ would settle #307 negatively)**.** *If for some $\lambda > 0$ one has $\delta_\lambda(n') < \delta_\lambda(n)$ for every squarefree $n$ with $n'$ squarefree, then the arithmetic derivative has no cycle of any length among such $n$, and in particular #307 has answer* NO.

*Proof.* A cycle would give $\delta_\lambda(n) < \delta_\lambda(n)$. □

Since $n' = n\sigma(n)$, the requirement is the pointwise inequality

$$\log \sigma(n) \ < \ \lambda\big(\sigma(n) - \sigma(n')\big). \tag{$*$}$$

Around a two-cycle the two instances of $(*)$ have increments summing to $\log(\sigma(a)\sigma(b)) = 0$, so they cannot both hold: the admissible range of $\lambda$ closes exactly at a cycle. The content is therefore whether that range is non-empty.

Below the barrier the criterion is not conjectural: it holds with the explicit value $\lambda = \frac{1}{2}$, by an elementary inequality.

**Theorem 6.30** (The half-mass Lyapunov function)**.** *Put $\delta(n) = \log n + \frac{1}{2}\sigma(n)$. Let $n$ be squarefree with $n'$ squarefree. If $\sigma(n\,n') < 2$ then $\delta(n') < \delta(n)$.*

*Proof.* By Theorem 2.3 $n$ and $n'$ are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n')$. The hypothesis therefore gives $\sigma(n') < 2 - \sigma(n)$, hence

$$\tfrac{1}{2}\big(\sigma(n) - \sigma(n')\big) \ > \ \sigma(n) - 1 \ \geq \ \log \sigma(n),$$

the last step by $\log x \leq x - 1$. If $\sigma(n) = 1$ the two ends read 0 and the first inequality is still strict, since $\sigma(n) + \sigma(n') < 2$ forces $\sigma(n') < 1$. Now $n' = n\sigma(n)$ gives $\delta(n') - \delta(n) = \log \sigma(n) - \frac{1}{2}(\sigma(n) - \sigma(n')) < 0$. □

Verified over all 7,480,605 squarefree $n \leq 2 \times 10^7$ with $n'$ squarefree lying in the region: 0 violations, least slack 0.242 (`code/invent_lyapunov.rs`). The analytic content is formalised: `Erdos307.log_lt_half_sub` proves $\log s < (s - t)/2$ for $s > 0$ and $s + t < 2$, `delta_decreasing` gives the increment form, and `two_le_add_of_mul_eq_one` is the AM–GM step of Corollary 6.31; all three carry 0 `sorry` and only the three standard axioms (`lean/Erdos307/HalfLyap.lean`). The arithmetic inputs, that $n$ and $n'$ are coprime and that $n' = n\sigma(n)$, are Theorem 2.3 and the definition, so the statement is VERIFIED modulo those.

**Corollary 6.31** (A Lyapunov proof of the barrier)**.** *The arithmetic derivative has no cycle, of any length, all of whose consecutive pairs satisfy $\sigma(n\,n') < 2$. In particular a two-cycle $a' = b$, $b' = a$ with $a \neq b$ has $\sigma(ab) = \sigma(a) + \sigma(b) \geq 2\sqrt{\sigma(a)\sigma(b)} = 2$, with equality only if $\sigma(a) = \sigma(b) = 1$ and hence $a = b$; so $\sigma(ab) > 2$ and therefore $ab \geq \Pi_{59}$.*

*Proof.* A cycle would give $\delta(n) < \delta(n)$ by Theorem 6.30. The second statement is AM–GM together with the mass estimate of Theorem 4.3. □

There are exactly two ways to exploit the fact that increments cancel around a cycle, and the second is foreclosed. Around a two-cycle the two increments of any function sum to zero, so an obstruction must pin down either their *sign*, which is the Lyapunov route above, or their *value in a finite group*: if some $\varphi$ satisfied $\varphi(n') - \varphi(n) \equiv c \pmod{m}$ for all $n$ with $2c \not\equiv 0$, a two-cycle would give $0 \equiv 2c$. No such $\varphi$ exists, and not for want of searching: Proposition 1.9 proves the cycle system soluble modulo every $m$, so any obstruction expressible as a congruence is empty a priori. Measured, the increments of $\omega(n)$, of $\#\{p \mid n : p \equiv 3\ (4)\}$, of $n$, and of $n + \omega(n)$ each hit every residue class modulo 2, 3 and 4 (`code/invent_lyapunov.rs`). The sign route is therefore the only one of the two available, which is another form of the thesis that the obstruction is ordered–archimedean rather than congruential (Remark 3.4).

The sign route is available but, as it turns out, not weaker than the problem. Both branches of the dichotomy can now be closed at once.

**Theorem 6.32** (Both branches are closed)**.** *Let $S$ be the set of squarefree $n \geq 2$ with $n'$ squarefree.*

(1) (Finite-group branch.) Conjecture, *downgraded. The claim that there is no $\varphi$ on $S$ and modulus $m$ with $\varphi(n') - \varphi(n) \equiv c \pmod{m}$ for all $n$ and $2c \not\equiv 0 \pmod{m}$ is* not *established. Proposition 1.9 settles only the moduli it covers, $m \leq 210$, and the general statement is equivalent to the target rather than independent of it: on a functional graph with no periodic orbit one may build a constant-increment $\varphi$ for any $(m, c)$ by assigning $\varphi$ along depth, so nonexistence for all $m$ is equivalent to the existence of a periodic orbit. Branch (1) therefore has exactly the status of branch (2), which Proposition 6.38 already proves for the sign side. What is proved is the mechanism recorded after the proof.*

(2) (Sign branch, not weaker than the target.) *A $\delta$ on $S$ with $\delta(n') < \delta(n)$ throughout exists if and only if $n \mapsto n'$ has no periodic orbit in $S$; that condition implies a negative answer to #307 and is not implied by it.*

*Proof.* (1) is not proved; see the statement. Proposition 1.9 makes the cycle system soluble modulo every $m \leq 210$, which does not reach the quantifier the branch needs. (2) is Proposition 6.38; a two-cycle is a periodic orbit, and periodic orbits of length other than two are not excluded by a negative answer to #307. □

The mechanism of branch (1) is formalised. `Erdos307.two_nsmul_eq_zero_of_period_two` proves that a constant increment $c$ in any abelian group forces $2c = 0$ at a point of period two, `const_increment_iterate` gives the general $\varphi(D^{[k]}n) = \varphi(n) + kc$, `no_period_two_of_two_nsmul_ne_zero` is the certificate such a $\varphi$ would supply, and `zmod_two_mul_eq_zero_of_period_two` states it over $\mathbb{Z}/m$; all with 0 `sorry` and on `propext` alone (`lean/Erdos307/Congruential.lean`). What is cited rather than formalised is the *emptiness* of the class, that no such $\varphi$ with $2c \neq 0$ exists for the arithmetic derivative, which is Proposition 1.9 and is arithmetic. Branch (1) is therefore VERIFIED in its mechanism and PROVED in its emptiness, and the two should not be conflated.

The taxonomy that makes this a closure rather than two isolated facts is the displayed thesis above, that an invariant whose increments cancel around a cycle can be exploited only through their sign or their value in a finite group. That thesis is not a theorem, and an argument outside it is not excluded by anything here.

This is a statement about methods, not about #307, and it is the natural endpoint of the structural programme: Proposition 5.11 removes the absolute values, Theorem 3.1 removes the arguments that survive to $K[t]$, Proposition 1.9 removes the congruences, and Theorem 6.32 removes what is left of the pointwise route. A proof of the negative direction, if there is one, must be global: it must count, and not certify.

**Remark 6.33** (What a counting proof would have to defeat)**.** The count it would have to defeat diverges. A two-cycle is exactly $r(a) = 1$ for $r(a) = \sigma(a)\sigma(a') = a''/a$, so if $r$ has a limiting density $f$ on the class of squarefree $a$ with $a'$ squarefree, then $\mathbb{P}(a'' = a) = \mathbb{P}\big(r \in [1, 1 + \frac{1}{a})\big) \sim f(1)/a$ and

$$\mathbb{E}\big(\#\{a \leq X : a'' = a\}\big) \;\sim\; f(1)\,\delta \log X, \qquad \delta = 0.3721\ldots,$$

$\delta$ being the measured density of the admissible class (`code/region_shape.rs`, 3,721,148 admissible $a \leq 10^7$). This diverges, so the heuristic predicts infinitely many two-cycles, and any proof of the negative direction must account for a divergent expected count producing no solutions at all.

The same formula locates them, and shows why the location is not knowable. Setting the expectation to 1 gives a first solution near $\exp\big(1/(\delta f(1))\big)$, which is exponential in $1/f(1)$: at $f(1) = 10^{-2}$ it is $10^{116.7}$, at $f(1) = 5 \times 10^{-3}$ it is $10^{233.4}$, at $f(1) = 10^{-3}$ it is $10^{1167}$. Halving the constant squares the answer. Consistency with Theorem 4.3 needs $f(1) < 0.01034$, which is evidence of the softest kind, since an expectation near 1 with no solution observed is an ordinary event and not a contradiction. What Proposition 6.9 shows is that $f(1)$ is the one quantity here that cannot be sampled (`code/heuristic_location.gp`). That $f(1) > 0$ is not established either; the constructed witnesses of Theorem 6.34 and of the near-miss frontier show only that $r$ takes values on both sides of 1 and within $8 \times 10^{-3}$ of it.

Corollary 6.31 recovers the barrier by a route independent of the minimisation used in Theorem 4.3: no extremality argument, no enumeration, only $\log x \le x - 1$ and the coprimality of Theorem 2.3. It is also stronger in one direction, since it excludes cycles of every length rather than two-cycles alone. What it does not do is decide #307: the region it covers is exactly the region where the barrier already says nothing can happen, and above $\Pi_{59}$ the hypothesis $\sigma(n\,n') < 2$ fails, which is where any solution must live. The general conjecture is what remains.

Over all 153,397,754 such $n \le 4 \times 10^8$, $(*)$ holds for every $\lambda \in (0.275438, 1.250828)$ and fails for no step (`code/lyap_battery.rs`). The interval narrows with the range, of width 1.654, 1.267, 1.237, 1.116, 1.092, 1.004, 0.975 at cutoffs $10^4$ through $4 \times 10^8$; the binding lower constraint is at $n = \Pi_9 = 223{,}092{,}870$. Six candidate functions were tested and five died, among them the control $\delta = \log n$, which fails first at $n = 30$ where $\sigma = 1.0333$; a battery that did not kill it would not be testing anything (`code/invent_lyapunov.rs`).

That interval is an artefact of the range swept. No $\lambda$ works.

**Theorem 6.34** (The criterion fails, for every $\lambda$)**.** *There is no real $\lambda$ for which $(*)$ holds at every squarefree $n$ with $n'$ squarefree. Two witnesses suffice. Let*

$$n_0 = \prod_{\substack{7 \le p \le 373 \\ p \notin \{307, 317, 359\}}} p,$$

*a squarefree integer with 68 prime factors and 142 decimal digits. Then $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ with $C$ a prime of 141 digits, so $n_0'$ is squarefree and $\sigma(n_0') = \frac{31}{30}$ exactly, while $\sigma(n_0) = 1.0059067\ldots$. Since $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$,*

$$\log \sigma(n_0) \; > \; 0 \; > \; \lambda\big(\sigma(n_0) - \sigma(n_0')\big) \qquad (\lambda > 0),$$

*so $(*)$ fails at $n_0$ for every positive $\lambda$. At $n = 30$, with $30' = 31$ prime, $\sigma(30) = \frac{31}{30} > 1$ and $\sigma(30) - \sigma(31) = \frac{931}{930} > 0$, so $\log \sigma(30) > 0 \ge \lambda(\sigma(30) - \sigma(31))$ for every $\lambda \le 0$. Together the two exclude every real $\lambda$.*

*Proof.* Both statements are finite verifications in exact rational arithmetic. For $n_0$: $\sigma(n_0)$ and $\sigma(n_0')$ are computed as rationals, $\gcd(n_0, n_0') = 1$ and $\sigma(n_0) = n_0'/n_0$ confirm Theorem 2.3, the factorisation $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ multiplies back to $n_0'$ and has all exponents 1, and $C$ carries an Atkin–Morain certificate, generated and validated (`certs/lyap_refute_cofactor_ecpp.txt`). The comparisons $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$ are exact, and $\sigma(n_0') = \sigma(n_0)$ is impossible in any case by Lemma 7.2. The case $n = 30$ is arithmetic (`code/lyap_refute_verify.gp`). □

The argument is formalised. `Erdos307.criterion_fails_of_pos` and `criterion_fails_of_nonpos` give the two sign clashes, `no_lambda_works` combines them, and `lyapunov_criterion_false` instantiates the combination at $\sigma(n_0)$, carried as the exact quotient $n_0'/n_0$, and at the pair $(\frac{31}{30}, \frac{1}{31})$ of $n = 30$; the two numeric comparisons $1 < \sigma(n_0)$ and $\sigma(n_0) < \frac{31}{30}$ are checked on the 142-digit numerals themselves. All six declarations carry 0 `sorry` and only `propext`, `Classical.choice`, `Quot.sound`, with no `native_decide` (`lean/Erdos307/LyapFalse.lean`). What Lean does not see is the factorisation $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ and the primality of $C$, which is the Atkin–Morain certificate;

the status is therefore VERIFIED modulo that certificate, in the same sense that Theorem 6.30 is VERIFIED modulo Theorem 2.3.

The witness is built, not found. For squarefree $n$ and a prime $w \nmid n$ one has $n' \equiv n \sum_{p|n} p^{-1} \pmod{w}$, hence

$$w \mid n' \iff \sum_{p|n} p^{-1} \equiv 0 \pmod{w}. \tag{F}$$

Imposing (F) for $w \in \{2, 3, 5\}$ forces $\sigma(n') \geq \frac{31}{30}$, and building $n$ from primes $\geq 7$ with $\sigma(n)$ just above 1 gives $\sigma(n') > \sigma(n)$. Reaching mass 1 without $2, 3, 5$ needs primes only to about 360, which is what keeps $n_0'$ small enough to factor completely; the squarefreeness of $n_0'$ is therefore verified rather than assumed. Three of the larger pool primes are dropped to satisfy (F), at negligible cost in mass (`code/lyap_refute.gp`).

The sweep could not have seen this. Every $n \leq 4 \times 10^8$ with $\sigma(n) > 1$ is even, so the battery measured only $n$ containing 2, whereas the failure needs $n$ to omit 2, 3 and 5 so that $n'$ may absorb them. By Theorem 6.35 below, any failure must have $n\,n' \geq \Pi_{59}$, and $n_0 n_0'$ has 284 digits: the region was out of reach of any sweep, and only a construction enters it.

One route to the criterion had looked provable rather than measured, and it fails for the same reason. Write $\rho = \sup\{\sigma(n')/\sigma(n) : \sigma(n) > 1\}$, the supremum being over the same class of $n$. If $\rho \leq c < 1$ then $\sigma(n) - \sigma(n') \geq (1-c)\sigma(n)$, and since $\log x/x \leq 1/e$,

$$L := \sup_{\sigma(n)>1} \frac{\log \sigma(n)}{\sigma(n) - \sigma(n')} \leq \frac{1}{e(1-c)}.$$

Measured over $n \leq 4 \times 10^8$, $\rho = 0.519746$, which would give $L \leq 0.766$ against a measured upper demand of 1.274; a proof of $\rho < 1 - 1/(eU)$, for $U$ a lower bound on the upper demand, would then have given $(*)$. The measurement is not the truth. Already

$$a_0 = \prod_{5 \leq p \leq 163} p, \qquad a_0' = 2 \cdot 3 \cdot 173 \cdot 227 \cdot 1973 \cdot 82890547 \cdot 68126695363 \cdot P_{36},$$

with $P_{36}$ prime, is squarefree with squarefree derivative, $\sigma(a_0) = 1.0723027\ldots > 1$ and $\sigma(a_0') = 0.8440258\ldots$, so $\rho \geq 0.7871152\ldots$; the bound the route would yield is $1/(e(1-\rho)) \geq 1.7280$, already above the measured upper demand, so it constrains nothing. The witness $n_0$ of Theorem 6.34 gives $\rho \geq 1.02726$, so $\rho < 1$ is false outright (`code/rho_verify.gp`).

The mechanism advanced for $\rho < 1$ was that $\sigma(n) > 1$ forces $n$ to absorb the small primes while $n'$ stays coprime to $n$. It does not: the reciprocals of the primes diverge, so $\sigma(n) > 1$ is reachable from primes $\geq 7$ alone, and then $n'$ is free to take 2, 3 and 5. Both $a_0$ and $n_0$ are of that shape. The earlier probe of this regime evaluated $\sigma(n')/\sigma(n)$ at the least odd $n$ with $\sigma(n) > 1$, obtaining 0.0312, and read the regime off that single point; the quantity to be bounded is a supremum, and the least member of a family does not report it. This is the error recorded in Remark 7.26, in a second instance.

What survives is the location of the failure, and it is exactly where the barrier puts it.

**Theorem 6.35** ($\rho < 1$ below the barrier)**.** *Let $n$ be squarefree with $n'$ squarefree and $\sigma(n) > 1$. If $\sigma(n') \geq \sigma(n)$ then $n\,n' \geq \Pi_{59} > 8.77 \times 10^{112}$. Consequently $\rho < 1$ for every $n$ with $n\,n' < \Pi_{59}$, and since $\sigma(n) > 1$ gives $n' > n$, for every $n < \sqrt{\Pi_{59}}$, that is for every $n < 10^{56.5}$.*

*Proof.* $\sigma(n') \geq \sigma(n) > 1$ gives $\sigma(n) + \sigma(n') > 2$. By Theorem 2.3 $n$ and $n'$ are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n') > 2$. A squarefree integer of mass at least 2 has at least 59 prime factors and is therefore at least $\Pi_{59}$, since $\sigma(\Pi_{58}) = 1.998740 < 2 \leq 2.002350 = \sigma(\Pi_{59})$;

this is the mass estimate underlying Theorem 4.3. Finally $\sigma(n) > 1$ gives $n' = n\sigma(n) > n$, so $n^2 < n\,n'$. ☐

The counting step is formalised. `Erdos307.card_ge_59_of_recipSum_ge_two` isolates the mass estimate, that a set of primes of reciprocal sum at least 2 has at least 59 elements, and `Erdos307.card_union_ge_59_` applies it to two disjoint sets each of mass exceeding 1; both carry 0 `sorry` (`lean/Erdos307/RhoBarrier.lean`). Their axiom lists are those of Theorem 4.3 itself, the three standard axioms together with the `native_decide` evaluation of the first 58 primes. The same core proves `card_ge_59` of the barrier, which assumes $\sigma(P)\sigma(Q) = 1$ instead; the two hypotheses differ and the shared step has been extracted rather than duplicated.

Theorem 6.35 therefore locates the failure before exhibiting it: $\rho \geq 1$ forces $n\,n' \geq \Pi_{59}$, so no sweep can meet a counterexample, and any counterexample must have $n$ of at least 57 digits. The witness $n_0$ has 142, and $n_0 n_0'$ has 284. The regime in which $\rho \geq 1$ is possible is the regime in which a two-cycle is possible, since both need two coprime squarefree integers of total mass at least 2; what Theorem 6.34 shows is that the first of those is genuinely non-empty, which says nothing about the second.

**Corollary 6.36** (The half-mass hypothesis cannot be relaxed)**.** *For every $\lambda > 0$ and every $c > 2.03925$, the implication "$\sigma(n\,n') < c$ implies $\delta_\lambda(n') < \delta_\lambda(n)$" is false. Indeed $\sigma(n_0\,n_0') = 2.039240\ldots$ while $\delta_\lambda(n_0') - \delta_\lambda(n_0) = \log\sigma(n_0) + \lambda(\sigma(n_0') - \sigma(n_0)) > 0$ for every $\lambda > 0$.*

Two is also the exact threshold for the argument of Theorem 6.30, in the following sense, which concerns the inequality alone and not the arithmetic of $\sigma$.

**Proposition 6.37** (The mass-sum relaxation stops at 2)**.** *Let $c > 0$. There is $f : (0,\infty) \to \mathbb{R}$ with $f(x) - f(y) > \log x$ for all $x, y > 0$ satisfying $x + y < c$ if and only if $c \leq 2$. For $c = 2$ the linear solutions are exactly $f(x) = \frac{1}{2}x + const.$*

*Proof.* The region $x + y < c$ is symmetric, so a solution gives both $f(x) - f(y) > \log x$ and $f(y) - f(x) > \log y$; adding, $0 > \log(xy)$. Taking $x = y \uparrow c/2$ forces $c^2/4 \leq 1$, that is $c \leq 2$. For $c \leq 2$ take $f(x) = \frac{1}{2}x$: if $x + y < 2$ then $\frac{1}{2}(x - y) > x - 1 \geq \log x$. If $f = \lambda x$ then letting $y \uparrow 2 - x$ gives $2\lambda(x - 1) \geq \log x$ for all $x \in (0, 2)$, and dividing by $x - 1$ and letting $x \to 1$ from each side gives $2\lambda \geq 1$ and $2\lambda \leq 1$. ☐

Both halves are formalised: `Erdos307.half_works` for $c \leq 2$ and `no_f_above_two` for $c > 2$, the latter by the diagonal point $x = y$ with $1 < x < c/2$, where the right side vanishes and the left is positive; `mass_sum_threshold` states the two together. The uniqueness of the constant is formalised as well: `lambda_eq_half` shows that a linear solution at $c = 2$ forces $\lambda = \frac{1}{2}$, by the local-minimum route rather than a two-sided limit, since $g(x) = 2\lambda(x - 1) - \log x$ is nonnegative on $(0, 2)$ and vanishes at 1, so its derivative $2\lambda - 1$ vanishes there (`lean/Erdos307/NoInvariant.lean`).

Any strengthening must therefore use an input about $n$ beyond the pair $(\sigma(n), \sigma(n'))$, and by the next proposition there is no room in the choice of function either.

**Proposition 6.38** (The ansatz is universal, so the search was the problem)**.** *Let $S$ be the set of squarefree $n \geq 2$ with $n'$ squarefree. The following are equivalent:*

(1) *$n \mapsto n'$ has no periodic orbit inside $S$;*

(2) *there is $\delta : S \to \mathbb{R}$ with $\delta(n') < \delta(n)$ whenever $n, n' \in S$;*

(3) *there is $f$ on $\sigma(S)$ with $f(\sigma(n)) - f(\sigma(n')) > \log\sigma(n)$ for all such $n$.*

*Proof.* (2) implies (1), since a periodic orbit would give $\delta(n) < \delta(n)$. For (1) implies (2), the transitive closure of $n \mapsto n'$ is irreflexive by (1) and transitive by construction, hence a strict partial order on the countable set $S$; it has a linear extension, and a countable linear order embeds order-preservingly in $\mathbb{Q}$, which supplies $\delta$. For (2) and (3), $\sigma$ is injective on squarefree integers by Lemma 7.2, so every $\delta$ on $S$ is $\log n + f(\sigma(n))$ for exactly one $f$, namely $f(\sigma(n)) = \delta(n) - \log n$; and $n' = n\sigma(n)$ turns $\delta(n') < \delta(n)$ into the inequality of (3). □

The collapse of the ansatz is formalised. `Erdos307.exists_f_of_injective` proves that injectivity of $\sigma$ forces *every* $\delta$ into the shape $\log n + f(\sigma(n))$, `increment_iff` converts $\delta(n') < \delta(n)$ into $(*)$ verbatim, and `no_two_cycle_of_decreasing` is the implication of Proposition 6.29; all with 0 `sorry` and the three standard axioms (`lean/Erdos307/NoInvariant.lean`).

So Definition 6.28 restricts nothing. The tournament that produced $\delta_\lambda$ was searching the space of all functions on $S$, and by Proposition 6.38 the existence of any member of that space is equivalent to the absence of cycles, which is strictly stronger than a negative answer to #307. The criterion was never a weakening of the problem, and Theorem 6.34 settles the one form of it that had been left open. What remains true, and is the residue of the whole construction, is Theorem 6.30 together with Corollary 6.31: an elementary Lyapunov proof of the barrier, valid exactly up to mass 2 and, by Corollary 6.36, not one step further.

**Remark 6.39** (Four independent blocks on A10)**.** The existence route is therefore obstructed at four separate levels, and no two of them are the same obstruction. Proposition 6.1 removes the freedom: the construction carries one parameter and cannot be relaxed into a denser family. Theorem 3.1 removes the method: no argument that survives the passage to $K[t]$ can prove existence, because there the statement is false. Proposition 6.4 removes the computation: even the 34 most reduced objects in the whole problem, the ones every sieve has failed to kill, cannot be examined. Proposition 6.5 removes the last refuge, the hope that a large family of sequences is easier than one: the union stays logarithmically thin however many sequences it has. What is left is a lower bound for primes in a sparse exponential sequence, which exists for no non-degenerate linear recurrence at all. We state plainly that the present work does not decide the existence direction and does not contain a route to deciding it.

## 7 The square sieve at the mass-two wall

Remark 7.12 left the residue inside $\{\sigma(m) \geq 2 - \varepsilon\}$. We now attack that set directly, by a change of question. The minus layer is $m' - 2m = d^2$, so $\sigma(m) = 2 + d^2/m > 2$: it lives *above* the wall, and the layer is exactly the set on which the arithmetic function $m \mapsto m' - 2m$ takes square values. Atom A9 is therefore an instance of *how often is an arithmetic function a perfect square*, and for that question there is a tool: Heath-Brown's square sieve. It has not previously been applied to the arithmetic derivative, because $m'$ is not a polynomial and no Weil bound is available. The obstruction dissolves because the relevant symbol factorises.

**Lemma 7.1** (Factorisation of the symbol)**.** *Let $r$ be odd and let $m$ be squarefree with* $\gcd(m, r) = 1$ (*equivalently, no prime factor of $m$ divides $r$*). *Put $\sigma_r(m) = \sum_{p|m} p^{-1} \in \mathbb{Z}/r$. Then $m' \equiv m\,\sigma_r(m)$* $\pmod r$ *and*

$$\left(\frac{m' - 2m}{r}\right) = \left(\frac{m}{r}\right)\left(\frac{\sigma_r(m) - 2}{r}\right).$$

*Proof.* $m' = \sum_{p|m} m/p$ and $m/p \equiv m\,p^{-1} \pmod r$ for each $p \mid m$, since $p$ is invertible modulo $r$. Summing gives $m' \equiv m\sigma_r(m)$, hence $m' - 2m \equiv m(\sigma_r(m) - 2)$, and the Jacobi symbol is completely multiplicative in its numerator. □

Verified exhaustively for $r = 101, 1009, 10007$ over all admissible squarefree $m \leq 3 \times 10^7$: 54,514,790 checks, 0 violations (`code/sqsieve_lemmas.rs`).

The point of Lemma 7.1 is that $\sigma_r$ is *additive* over the distinct prime divisors, so after a Gauss-sum expansion the second factor becomes a product over $p \mid m$ and the whole sum turns multiplicative. That is what puts it inside Halász's theorem.

### 7.1 The square sieve and Theorem A.10

The argument recorded in this section is a square sieve at $P \asymp (\log N)^{1/4}$ whose analytic input does not close in the form attempted there, and why Remark 6.26 keeps this material off the critical path; the rate is nonetheless proved, by a different route, in Theorem A.13. The density statement itself, Theorem A.10, is proved, by a different sieve at a far smaller range (Proposition A.8). The section occupied eight pages in the main line. It is moved verbatim to Appendix A; nothing is omitted and every label is unchanged, so the cross-references from §7.2 still resolve. What the argument establishes, what it assumes, and exactly where the gap sits are stated there.

### 7.2 Pair averages and the mass–two wall

**Lemma 7.2** ($\sigma$ is injective on squarefree integers)**.** *For squarefree $d, d'$, $\sigma(d) = \sigma(d')$ forces $d = d'$. Consequently $F(d, d') := D(d)\, d' - D(d')\, d = dd'(\sigma(d) - \sigma(d'))$ vanishes only on the diagonal.*

*Proof.* By Theorem 2.3, $\sigma(d) = D(d)/d$ is already in lowest terms, since $\gcd(D(d), d) = 1$ for squarefree $d$; so $d$ is recoverable from $\sigma(d)$ as its denominator. $\square$

Verified over all 182,377 squarefree $d \leq 3 \times 10^5$: 0 collisions (`code/sqsieve_pairavg.rs`). The lemma is formalised: in the notation of Section 7, `Erdos307.dprod_eq_of_mass_eq` and `Erdos307.csum_eq_of_mass` derive it from `rigidity_coprime`, with 0 `sorry` and axioms `propext, Classical.choice, Quot.sound` (`lean/Erdos307/Injective.lean`). Its label is therefore VERIFIED.

**Theorem 7.3** (The average pair bound)**.** *Let $\mathcal{R} = \{pq : p \neq q \in \mathcal{P}\}$ and $P(D, r) = \#\{(d, d') \sim D : \sigma_r(d) \equiv \sigma_r(d')\}$. Then*

$$\sum_{r \in \mathcal{R}} P(D, r) \ \ll\ |\mathcal{R}|\, D + D^{2+o(1)}.$$

*Proof.* As $\gcd(d, r) = \gcd(d', r) = 1$, the condition $\sigma_r(d) \equiv \sigma_r(d')$ is exactly $r \mid F(d, d')$, so $\sum_{r\in\mathcal{R}} P(D, r) = \sum_{(d,d')} \#\{r \in \mathcal{R} : r \mid F(d, d')\}$. On the diagonal $F = 0$ is divisible by every $r$, contributing $|\mathcal{R}| \cdot \#\{d \leq D : \mu^2(d) = 1\}$. Off it $F \neq 0$ by Lemma 7.2, and $|F| \leq 2D^2\sigma_{\max}$, so $\#\{r \in \mathcal{R} : r \mid F\} \leq \tau(F) \ll D^{o(1)}$, contributing $D^{2+o(1)}$. $\square$

Measured against $D^2/r + D$ the average has ratio 1.011 at $|\mathcal{R}| = 78$ (`code/sqsieve_pairavg.rs`). Note the proof uses no equidistribution of $\sigma_r$ and no uniformity in $r$: only the divisor bound and Lemma 7.2, the latter being the same rigidity that makes #307 a two-cycle problem at all. The counting skeleton is formalised as `Erdos307.pairavg_bound`, with the divisor step `card_filter_dvd_le` and the double count `sum_filter_comm`; 0 `sorry`, standard axioms (`lean/Erdos307/PairAvg.lean`). The arithmetic input enters only through the hypothesis that $F$ vanishes exactly on the diagonal, which is Lemma 7.2, itself VERIFIED and transferred from prime sets to squarefree integers by `Erdos307.eq_of_cross`, so that the hypothesis consumed by the skeleton is exactly the statement the paper uses.

The decomposition demanded by Remark A.22 can be bypassed entirely. Rather than splitting $\sum_{m\leq N}$ into bilinear pieces, one may reverse the roles of the two arguments of the symbol and appeal to the quadratic large sieve, which averages over the modulus. This is the more promising route and we record it, together with the one hypothesis it still needs.

**Lemma 7.4** (Reversal)**.** *Write $a_m = m' - 2m$ and $a_m = \pm s_m^2 k_m$ with $k_m$ squarefree. For odd squarefree $r$,*

$$T_r(N) = \sum_k b_k^+ \Big(\frac{k}{r}\Big) + \Big(\frac{-1}{r}\Big) \sum_k b_k^- \Big(\frac{k}{r}\Big) \; + \; O\big(\#\{m \le N :\ p^2 \mid a_m \text{ for some } p \mid r\}\big),$$

*where $b_k^\pm = \#\{m \le N : \mu^2(m) = 1,\ k_m = k,\ \pm a_m > 0\}$ depend on $N$ only. For $r = pq$ with $p, q \asymp N^{1/2}$ one would want the error term to be $O(1)$, and an earlier revision cited Lemma* A.9 *in the form $\#\{m \le N : p^2 \mid a_m\} \ll N/p^2$. That form is the one that lemma explicitly disowns, since its implied constant depends on $p$ while $p$ here grows with $N$; the citation is withdrawn and the error term is not controlled at this range.*

Regime caveat, and its retraction. *That $O(1)$ holds only at $p \asymp N^{1/2}$; at $P \asymp (\log N)^{1/4}$ the same estimate gives only $N/p^2 \asymp N/(\log N)^{1/2}$. Two earlier revisions read that as a second, independent reason for the* CONJECTURE *label then carried by Theorem A.13. It is not, for two reasons. The comparison there was made against $N$, whereas the benchmark is that statement's own target, now $N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$ (Theorem A.13), against which $N(\log N)^{-1/2}$ is* smaller *— an error term below the claimed rate cannot obstruct it. And Theorem A.13 does not travel this route at all: re–derived as in Proposition A.8 it needs no kernel reversal, and Proposition 7.6 shows the reversal is no reduction in any case. The operative blocker is recorded at the corollary itself.*

*Proof.* If $\gcd(a_m, r) = 1$ then $(a_m/r) = (\pm 1/r)(k_m/r)(s_m/r)^2 = (\pm 1/r)(k_m/r)$. The two sides can differ only when some $p \mid r$ divides $s_m$ but not $k_m$, that is when $p^2 \mid a_m$, which is the error term. The coefficients $b_k^\pm$ carry no dependence on $r$. □

Measured at $N = 5 \times 10^6$: 30.0% of the $a_m$ are non-squarefree, but only 0.035% have $s_m > 10^3$, so for $p \asymp N^{1/2}$ the error term is empty in practice (`code/sqsieve_holes.rs`). The kernels $k_m$ are even 22.2% of the time, so the moduli are split into the four classes mod 8 before the sieve is applied, at a cost of a factor 4.

**Hypothesis 7.5** (Kernel collisions)**.** $\sum_k \big(|b_k^+|^2 + |b_k^-|^2\big) \ll N^{1+o(1)}$, equivalently $\#\{(m, m')$ squarefree, $\le N :\ a_m a_{m'}$ is a perfect square$\} \ll N^{1+o(1)}$.

**Proposition 7.6** (The route is circular)**.** *Hypothesis* 7.5 *implies the bound it would be used to prove, so it cannot serve as a reduction. Indeed $b_1^+ = \#\{m \le N :\ \mu^2(m) = 1,\ a_m \text{ is a positive perfect square}\}$ is precisely the minus layer, and $(b_1^+)^2 \le \sum_k |b_k^+|^2$, so Hypothesis* 7.5 *forces $b_1^+ \ll N^{1/2+o(1)}$, which is the conclusion.*

**Remark 7.7** (Why computation cannot detect this)**.** The circularity is invisible to measurement. Below the barrier $\sigma(m) > 2$ is impossible, so $a_m > 0$ never occurs and $b_1^+ = 0$ identically: at $N = 5 \times 10^6$ one finds $b_1^+ = 0$ while $b_1^- = 1213$, the latter contributing 12.5% of $\sum_k |b_k|^2$ (`code/sqsieve_kernel.rs`). Every computed check of Hypothesis 7.5 therefore tests a statement from which the load-bearing term has been deleted. This is the same blindness recorded in Proposition 6.9, appearing here in a form that would otherwise have passed as supporting evidence.

Accordingly, the large sieve does apply and Lemma 7.4 is correct, but the resulting statement is a *reformulation*: CRUX-A9 is equivalent to a kernel-collision bound, with the equivalence carried by the $k = 1$ term. The new attack surface is genuine, since the reformulated statement is unsigned and about products, but it is not a reduction and is not counted as progress on the crux. The unconditional record is Proposition A.8.

**Remark 7.8** (The hypothesis measures, and its constant identifies itself)**.** $P(D,r)$ against $D^2/r+D$ has ratio 0.3688, 0.3688, 0.3689 at $r = 1009$; 0.3705, 0.3696, 0.3695 at $r = 10007$; and 0.3867, 0.3727, 0.3702 at $r = 100003$, for $D = 10^6, 5 \times 10^6, 2 \times 10^7$ (`code/sqsieve_bilinear.rs`). The ratio is not merely bounded but constant, and the constant is $(6/\pi^2)^2 = 0.3696$, the square of the squarefree density, which is exactly what equidistribution of $\sigma_r$ predicts once both $d$ and $d'$ are restricted to squarefree integers.

Compare Hypothesis A.16. That one asks for signed square-root cancellation, and Proposition A.19 showed the only available route to it fails. Theorem 7.3 instead bounds an *unsigned* second moment, and does so unconditionally; what it does not do is bound $T_r$, since the passage from bilinear forms to $T_r$ is exactly the gap recorded in Remark A.22.

**Remark 7.9** (What made this work)**.** Every previous attack on A9 counted members of the minus layer directly, and each ran into the same obstruction: the layer is a union of lines $m' = 2m - k$, each thin by Proposition 5.21 but not summably so, and the peeling reorganisations that would combine them are all conjugate under the groupoid of Proposition 5.23. The square sieve never counts a member. It asks only whether $m' - 2m$ is a square, and answers by averaging a Jacobi symbol, so the groupoid has nothing to act on. The one obstacle, that $m'$ is not a polynomial, is removed by Lemma 7.1: the symbol splits into a character in $m$ times a character in $\sigma_r(m)$, and $\sigma_r$ is additive, which is exactly the hypothesis Halász needs. The analytic inputs are all standard: Halász's theorem, the Gauss sum evaluation $|\tau(\chi)| = \sqrt{r}$, the classical zero–free region, the $\log L$ bound for non–principal characters, and Siegel–Walfisz. No Salié or Weil bound is used: the sum in question is a Gauss sum, as Lemma A.1 records, and an earlier revision listed them here in error. Note also that the theorem does not decide #307, and cannot: it is a statement about density, and a single two-cycle is a density-zero event.

**Proposition 7.10** (Slot stratification of the divisibility set)**.** *Fix $c$ and let $D(c) = \{m \text{ squarefree} : (2m - m') \mid (m - c),\ m \neq c\}$. For $m \in D(c)$ put $p = (m - c)/(2m - m')$, the* slot*. Then*

$$p\,m' \;=\; (2p-1)\,m + c, \qquad \textit{equivalently} \qquad \sigma(m) = 2 - \frac{m-c}{p\,m},$$

*so $D(c) = \bigsqcup_{p \neq 0} L(p, 2p-1, c)$ is a disjoint union of generalized lines in the sense of Proposition 5.23, the stratum $p = 1$ being the slope-*one *line $m' = m + c$. The identity was verified with 0 violations on all 2,294 members with $m \le 2 \times 10^7$, $|c| \le 400$ (`code/slot_stratify.rs`).*

*Proof.* $m - c = p(2m - m')$ rearranges to $pm' = (2p-1)m + c$; dividing by $pm$ gives the mass form. Distinct $p$ give disjoint strata since $p$ is a function of $m$. $\square$

**Theorem 7.11** (The residual difficulty sits entirely at the mass-two wall)**.** *For every $\varepsilon > 0$ and every $c$,*

$$\#\big(D(c) \cap [1, Z]\big) \;\le\; \big(\delta(2-\varepsilon) + o_\varepsilon(1)\big)\, Z + O(|c|),$$

*where $\delta(x)$ is the density of $\{m : \sigma(m) \ge x\}$. Moreover*

$$\delta(x) \;\le\; \inf_{t>0}\; e^{-tx} \prod_p \Big(1 + \frac{e^{t/p} - 1}{p}\Big),$$

*which evaluates to $\delta(1) \le 10^{-0.8}$, $\delta(1.5) \le 10^{-7.2}$ and $\delta(2) \le 10^{-105.6}$ (`code/chernoff_mass.rs`).*

*Proof.* Split $D(c)$ at $|p| \le P$ and $|p| > P$. The first part is a union of at most $2P$ generalized lines, each of size $\ll Z/(\log Z)^{(1+o(1))\log_3 Z}$ by Proposition 5.21 (*caveat*: Proposition 5.21 is stated for *integer* slope $e \geq 1$, while the strata $L(p, 2p-1, c)$ of Proposition 7.10 have slope $(2p-1)/p$, which is non–integral for $p \neq 1$. The bound is therefore invoked outside the range in which it is proved, and the first part of this proof is CONJECTURE pending a version of Proposition 5.21 uniform over rational slopes); taking $P = P(Z) \to \infty$ slowly makes the total $o(Z)$. For the second, $|p| > P$ forces $|2 - \sigma(m)| < (1 + |c|/m)/P$ by the mass form, so for $m > |c|$ we get $\sigma(m) > 2 - 2/P$, and the count is at most $\delta(2-2/P)Z(1+o(1))$. For the Chernoff bound, $f(m) = \prod_{p|m} e^{t/p}$ is multiplicative with $f(p) \geq 1$, so $\sum_{m \leq Z} f(m) = \sum_{d \leq Z} \mu^2(d) \prod_{p|d}(f(p)-1)\lfloor Z/d \rfloor \leq Z \prod_p (1 + (f(p)-1)/p)$, and $\#\{\sigma \geq x\} \leq e^{-tx} \sum_{m \leq Z} f(m)$. The product converges because $(e^{t/p} - 1)/p \sim t/p^2$. □

**Remark 7.12** (What this settles and what it does not)**.** Theorem 7.11 replaces the constant $\delta = 0.0420$ of Proposition 7.13 (ii) by one below $10^{-105}$, an unconditional gain of more than a hundred orders of magnitude, and it does so by *using* the divisibility rather than merely the mass condition it forces. It also localises the residue exactly: away from the mass-two wall the set is a bounded union of thin lines, and *all* the remaining difficulty lies in $\{\sigma(m) \geq 2 - \varepsilon\}$. It does not prove A9. The set $\{\sigma \geq 2\}$ has *positive* density, at least $1/\Pi_{59} = 10^{-112.9}$ since every multiple of $\Pi_{59}$ lies in it, so the method cannot reach $o(Z)$: the limiting constant is positive, not zero. The bound $10^{-105.6}$ sits within seven orders of that truth, so little is lost in the Chernoff step, and the obstruction is genuine rather than an artefact of the estimate. The residual question is therefore sharp: is $D(c)$ thin *within* the mass-two wall, the one place where the barrier (Theorem 4.3) also lives?

**Proposition 7.13** (The divisibility set: slope one is dense, slope two is not)**.** *Let $e \geq 1$ and $c$ be integers and let $m$ be squarefree with $(em - m') \mid (m - c)$ and $m \neq c$. Then*

$$\sigma(m) \;\geq\; e - 1 - \frac{|c|}{m}.$$

*Consequently the two slopes behave oppositely.*

- (i) *For $e = 1$ the condition is vacuous and the set has positive density: every prime $m$ satisfies it when $c = 1$, since then $m' = 1$ and $em - m' = m - 1 = m - c$. Over $m \leq 2 \times 10^7$ this accounts for* 1,270,651 *of the members, exactly* $\pi(2 \times 10^7)$.
- (ii) *For $e = 2$ the condition reads $\sigma(m) \geq 1 - |c|/m$, so by Proposition 2.6*

$$\#\{m \leq Z :\; (2m - m') \mid (m - c),\; m \neq c\} \;\leq\; (\delta + o(1))\, Z + O(|c|),$$

*with $\delta = 0.0420\ldots$ the abundance density, certified in Proposition 2.6 to lie in* $[0.04202, 0.04223]$.

*Proof.* If $|em - m'| > |m - c|$ the divisibility forces $m = c$. So $|em - m'| \leq |m - c| \leq m + |c|$; below the barrier $\sigma(m) < e$ for $e \geq 2$, so $em - m' = m(e - \sigma(m)) > 0$ and $m(e - \sigma(m)) \leq m + |c|$, which is the display. Part (i) is the computation just given, and (ii) is Proposition 2.6 applied to $\{\sigma \geq 1 - \varepsilon\}$. The mass condition was checked with 0 violations over all squarefree $m \leq 2 \times 10^7$ and all $|c| \leq 400$ at $e = 1, 2, 3$, where the nontrivial member counts are 2,197,230, 28 and 8 respectively (`code/a9prime_falsify.rs`). □

**Remark 7.14** (Consequence for the residual question)**.** Proposition 7.13 settles the shape of the residual bound. A uniform statement over all slopes is false, and false for a trivial reason: at $e = 1$ the primes sit on the set by an identity. The correct target is confined to $e \geq 2$, where a genuine mass condition appears, and there Proposition 7.13(ii) gives the first unconditional bound, $\delta Z$ with $\delta < 0.043$. The gap to the $Z^{1/2}$ needed for Remark 5.30 is therefore now a gap between two *proved* quantities rather than between a proof and a wish, and the observed counts at $e = 2$ (28 over the whole range $|c| \leq 400$, against $Z^{1/2} = 4472$) sit far below both.

**Remark 7.15** (The two open routes are one question, asked in opposite directions)**.** The existence route of Remark 1.16 and the counting route of Remark 5.30 look unrelated (one is a search for a single solution, the other a density estimate) but they are the same question about the same kind of object. In each case an index determines a single integer, and everything turns on whether that integer is prime: in Remark 1.16 the Pell index $n$ determines $q_n = (x_n^2 - D_S)/A_S$, a term of a ternary linear recurrence, and #307 needs it prime *once*; in Remark 5.30 the cofactor $m$ determines the slot $(c - am)/(m' - em)$ of Proposition 5.23, and the density needs it prime *rarely*. Existence asks for a lower bound on the number of prime values, density for an upper bound, on determined values along a thin deterministic sequence, and neither estimate is available: no unconditional infinitude of prime terms is known for any non–degenerate linear recurrence of order $\geq 2$, and no worst–case upper bound is known for single–class prime counts summed over all moduli. #307 sits exactly between two missing estimates on one object, which is why the heuristics can be read either way and why progress on either side would be informative about the other. The remaining walls are, in this sense, one wall seen from two directions.

### 7.3 Slot structure

**Corollary 7.16** (Quadratic–residue filter for #307 solutions)**.** *Let $(P, Q)$ solve Erdős #307 and set $a = D_P$, $b = D_Q$, $N = ab$. Since $N' = a^2 + b^2$ and hence $N' + 2N = (a + b)^2$ automatically, every prime $\ell \in P \cup Q$ satisfies $(N/\ell \mid \ell) = +1$: the cofactor $N/\ell$ is a nonzero quadratic residue modulo $\ell$. Indeed $\ell$ divides exactly one of $a, b$ (by $P \cap Q = \varnothing$), so $\ell \nmid a + b$ and $(a + b)^2 \not\equiv 0 \pmod{\ell}$; combined with the identity $N' + 2N \equiv N/\ell \pmod{\ell}$ of* (v)*, this forces $N/\ell$ to be a nonzero square mod $\ell$. As a search filter the condition prunes candidate unions by a factor $\leq 2^{-|P\cup Q|} \leq 2^{-59}$ (heuristically each of the $\geq 59$ primes imposes an independent QR constraint); for genuine cycles it is automatic and does not alter the existence heuristics. We note that Corollary 7.16 is* not *independent of Bado: it is the squared form of his congruence $D_Q \equiv D_P/\ell \pmod{\ell}$ (itself $N_P$ reduced mod $\ell$ by cofactor survival), giving $N/\ell \equiv D_Q^2 \pmod{\ell}$. Squaring is lossy ($r$ and $-r$ share a square) so Bado's linear congruence is strictly stronger, pinning $D_Q \bmod \ell$ exactly where the residue filter determines it only up to sign; the filter's one added value is being* split–free *(the residue $N/\ell \bmod \ell$ is computable from the candidate union alone, with no knowledge of the $P \mid Q$ partition). The general law of Proposition 5.15(v), by contrast, holds for* any *squarefree plus–hit with no $P \mid Q$ split and is not subsumed by Bado.*

The plus– and minus–hits both organise into one–prime "slot" families, which makes the structure of Proposition 5.13 explicit at every $\omega$.

**Proposition 7.17** (Slot calculus for the discriminant layers)**.** *Let $N_0$ be squarefree, $r \nmid N_0$ a prime, $N = N_0 r$, and $E_0 := N_0' + 2N_0$. By Leibniz ($N' = N_0' r + N_0$),*

$$N' + 2N = rE_0 + N_0, \qquad N' - 2N = r(N_0' - 2N_0) + N_0,$$

*so $N$ is a plus–hit iff $r = (s^2 - N_0)/E_0$ for some integer $s$ (equivalently $s^2 \equiv N_0 \pmod{E_0}$).*

(a) Every $\omega$–stratum. *A plus–hit $N$ ($\omega \geq 2$) factors, through any prime divisor $r$, as $N = N_0 r$ with $N_0 = N/r$ squarefree, and lies in the quadratic family $\{N_0(s^2 - N_0)/E_0\}_s$ of its base; generalising the $p = 5$ family of Proposition 5.15(iii) beyond semiprimes (each step adds one prime, $\omega(N) = \omega(N_0) + 1$). The congruence $s^2 \equiv N_0 \pmod{E_0}$ is solvable for* 669 *of the* 3041 *squarefree $N_0 \in [2, 5000]$ (22.0%); solvability is necessary but not sufficient for the family to contain a hit.*

(b) Recursive towers: a trichotomy. *When the base is itself a plus–hit, $E_0 = s_0^2$ is a square; below $2 \times 10^5$ the* 114 *plus–hit bases split into three kinds.* No residue *(*84*): $s^2 \equiv N_0 \pmod{E_0}$ is unsolvable and the family is empty; a quadratic–residue obstruction, occurring for both even $E_0$ ($N_0 = 305$, with 305 a non–residue mod 13) and odd $E_0$ ($N_0 = 1870$, $E_0 = 73^2$).* Fixed divisor *(*13*, each with fixed divisor* 2*): the family is nonempty but every slot is even, so only $r = 2$ could serve and never gives a square; sterile (e.g. $N_0 = 145$, $E_0 = 18^2$).* Live *(*17*): fixed divisor* 1 *and a prime slot is exhibited (e.g. $N_0 = 130$, first prime slot $r = 83399$, giving the plus–hit $N = 10{,}841{,}870$ with $N' + 2N = 5487^2$).*

(c) Minus layer; the barrier in slot form. *For $\sigma(N_0) < 2$ the minus identity gives $N' - 2N \geq 0$ iff $r \leq 1/(2 - \sigma(N_0))$, i.e. iff $\sigma(N) \geq 2$: this is the slot–variable form of Corollary 10.3, not an independent proof of it. The bound $1/(2 - \sigma(N_0))$ is unbounded as $\sigma(N_0) \to 2^-$; over squarefree $N_0 < 10^5$ its maximum is only* 1.524 *(at $N_0 = 30030$), so no such base admits a prime minus–slot, yet for the* 58*–prime primorial the bound exceeds* 793 *and the next prime* 277 *fits, producing the* 59*–prime primorial, the smallest minus–hit base. A prime minus–slot thus first becomes feasible* exactly *at the* 59*–prime barrier; the minus–empty–below–barrier statement rests on the primorial growth of Corollary 10.3, not on the figure* 1.524.

(The two identities are checked symbolically and on $2 \times 10^4$ random pairs; the base trichotomy, slot lists, and the 1.524 maximum by exhaustive enumeration.)

The two square conditions over a base combine into a single binary–form representation, which places the obstruction inside Gauss's genus theory.

**Proposition 7.18** (The Pythagorean condition as a binary–form representation)**.** *For squarefree $N_0$ and prime $r \nmid N_0$, $N = N_0 r$, the identity*

$$(2N_0 - N_0')(N' + 2N) + (2N_0 + N_0')(N' - 2N) = 4N_0^2$$

*holds unconditionally (Leibniz; checked on $2 \times 10^4$ random pairs). Hence if $N$ is a Pythagorean pair product over $N_0$ (both $N' + 2N = s^2$ and $N' - 2N = d^2$ squares) then $(s, d)$ represents $4N_0^2$ by the binary form $Q_{N_0}(s, d) = A\,s^2 + B\,d^2$ with $A = 2N_0 - N_0' = N_0(2 - \sigma(N_0))$, $B = 2N_0 + N_0' = N_0(2 + \sigma(N_0))$, of discriminant $-4AB = 4(N_0'^2 - 4N_0^2)$; conversely such a representation yields a Pythagorean pair over $N_0$ exactly when the recovered slot $r = (s^2 - N_0)/E_0$ is a positive integer, prime, and coprime to $N_0$ (the integrality $s^2 \equiv N_0 \pmod{E_0}$ holds on every representation we found, $N_0 < 2 \times 10^4$). For $\sigma(N_0) < 2$ the form is positive–definite ($A > 0$), and below the barrier representations are rare and all "ghosts": the only squarefree $N_0 \leq 3000$ representing $4N_0^2$ are $N_0 = 30$ (with $(s, d) = (11, 1)$) and $N_0 = 1722$ ($(83, 1)$), each carrying the non–admissible slot $r = 1$, and no live prime slot occurs. As $\sigma(N_0) \to 2^-$ the leading coefficient $A = N_0(2 - \sigma(N_0)) \to 0$ and the ellipse $A\,s^2 + B\,d^2 = 4N_0^2$ degenerates to an arbitrarily thin sliver: the geometric face of the mass–2 barrier. Consequently #307 is the existence of a base with $\sigma(N_0) \geq 2 - 1/r$ whose form $Q_{N_0}$ represents $4N_0^2$ with an admissible, integral,* prime *slot and deviation $k = 0$.*

(We make no class–number claim: each base yields a form of a *different* discriminant, and the question is single–value representation, not a class count; only the degeneration $A \to 0$ is used.)

**Proposition 7.19** (An explicit Pythagorean family, and why it stops at rung $-1$)**.** *Let $N_0$ be squarefree with $N_0' = 2N_0 - 1$ (the slope–two line $c = -1$, i.e. $k_0 = 1$) and suppose $N_0 - 1$ is prime. Then $(a, b) = (N_0 - 1,\ N_0)$ is a Pythagorean pair:*

$$a^2 + b^2 \;=\; 2N_0^2 - 2N_0 + 1 \;=\; (ab)'.$$

*It arises from the form equation of Proposition 7.18 at its degenerate end: with $k_0 = 1$, $E_0 = 4N_0 - 1$, the pair $(s, d) = (2N_0 - 1,\, 1)$ represents $4N_0^2$ exactly, and the slot it yields is $r = (s^2 - N_0)/E_0 = N_0 - 1$. Every member of the family sits on rung $k = -1$, and no member can sit on rung 0: $a$ is prime, so $a' = 1$, and $a' = b$ would force $N_0 = 1$.*

*Proof.* $(ab)' = N_0'(N_0 - 1) + N_0(N_0 - 1)' = (2N_0 - 1)(N_0 - 1) + N_0 = 2N_0^2 - 2N_0 + 1 = (N_0 - 1)^2 + N_0^2$, using $(N_0 - 1)' = 1$ for $N_0 - 1$ prime. The form identity is $(2N_0 - 1)^2 + (4N_0 - 1) = 4N_0^2$, and $(s^2 - N_0)/E_0 = (4N_0^2 - 5N_0 + 1)/(4N_0 - 1) = N_0 - 1$. The rung is $k = (a' - b)/a = (1 - N_0)/(N_0 - 1) = -1$. □

**Remark 7.20** (What the family shows)**.** This is the first explicit construction of Pythagorean pairs in this note: everything earlier established that they are rare (Corollary 10.3: none below $10^{112}$) without exhibiting a mechanism. The mechanism is a clean reduction (the pair exists as soon as the line $n' = 2n - 1$ carries a member whose predecessor is prime) and it is Bunyakovsky–shaped, matching the observation in Remark 1.16 that a single square condition is of that class. It is also conditional in a way the barrier controls: $\sigma(N_0) = 2 - 1/N_0$ forces $\omega(N_0) \geq 58$, so the family is empty below the scale where the critical set of Remark 5.30 switches on.

Two further consequences. First, combining with Corollary 5.5: for odd $N_0$ the pair has mixed parity with odd member $N_0$, so $k \equiv \omega(N_0) \pmod 2$; since $k = -1$ here, *any odd member of the line $n' = 2n - 1$ whose predecessor is prime has $\omega(N_0)$ odd*; a necessary condition on that line, derived rather than observed. Second, and this is the honest limitation, the construction is *permanently* off the cycle rung: primality of $a$ is exactly what makes it work and exactly what forces $a' = 1 \neq b$. A $k = 0$ analogue would need $a$ composite with $a' = b$, which is the original problem. So the family exhibits the Pythagorean locus without approaching #307; useful as the first worked mechanism, not as a route.

**Proposition 7.21** (The composite mechanism: $|P| = 2$ is one line and one primality)**.** *Write $a = 2m$ with $m$ odd squarefree, so that $b = a' = m + 2m'$ (Theorem 4.18). The two–cycle condition is then the single equation*

$$b' = 2m, \qquad b = m + 2m',$$

*in the one unknown $m$. For $m = q$ prime it reads $b = q + 2$ and*

$$\boxed{b' = 2b - 4} \qquad \textit{with } b - 2 \textit{ prime},$$

*and any such $b$ is a solution: $a = 2q$, $b = q + 2$ satisfy $a' = b$, $b' = a$ and $\sigma(a)\sigma(b) = \left(\frac{1}{2} + \frac{1}{q}\right)\left(2 - \frac{4}{q+2}\right) = 1$ identically. Hence*

$$\#307 \textit{ with } |P| = 2 \iff \exists\, b \textit{ squarefree}: \; b' = 2b - 4 \textit{ and } b - 2 \textit{ prime}.$$

*Proof.* $(2m)' = m + 2m'$ by Leibniz, so $a' = b$ by definition of $b$, and the cycle needs only $b' = a = 2m$. For $m = q$ prime, $b = q + 2$ and $b' = 2q = 2(b - 2) = 2b - 4$; the mass identity is the displayed product. □

**Remark 7.22** (Two lines, two rungs)**.** Comparing with Proposition 7.19 isolates exactly what separates the cycle from its neighbour. Both are "a member of a slope–two line whose shifted predecessor is prime":

$$\begin{array}{lll} \text{line } n' = 2n - 1, \quad N_0 - 1 \text{ prime} & \Longrightarrow & \text{Pythagorean pair, rung } k = -1; \\ \text{line } n' = 2n - 4, \quad b - 2 \text{ prime} & \Longrightarrow & \text{genuine two–cycle, rung } k = 0. \end{array}$$

The difference between the open problem and its solved neighbour is the constant in the line. And the obstruction changes character with it: Proposition 7.19 is barred from rung 0 *structurally* (its $a$ is prime, so $a' = 1 \neq b$) whereas Proposition 7.21 is barred only by *scale*, since $\sigma(b) = 2 - 4/b$ forces $\omega(b) \geq 58$ and $b \geq \Pi_{58}$, putting the whole family above $10^{108}$ and the resulting $N = 2qb$ at the barrier. A direct search of the reduction over odd squarefree $m \leq 2 \times 10^6$ (810,590 admissible $m$, of which 111,464 prime) finds none, with nearest miss $|b' - 2m| = 5$ at $m = 3$ (`code/composite_a.rs`); exactly as Corollary 10.3 demands. This is the composite mechanism the rung–0 case requires; what it does not do is evade the barrier, which reappears as the mass of the line.

**Proposition 7.23** (The barrier is a function of the split)**.** *For a solution with $|P| = j$ write $a = \prod P$, $b = \prod Q$. Since $\sigma(a)\sigma(b) = 1$ and $P \cap Q = \varnothing$, fixing $j$ forces $\sigma(b) = 1/\sigma(a)$ and leaves $Q$ only the odd primes outside $P$, which bounds $N$ from below split by split. Taking $a = 2 \cdot$ (the $j-1$ smallest odd primes), the most favourable configuration at that size:*

| $\lvert P\rvert$ | $\sigma(a)$ | $\sigma(b)$ | $\lvert Q\rvert \geq$ | $\log_{10} N \geq$ |
|---|---|---|---|---|
| 2 | 0.8333 | 1.2000 | 67 | 137.9 |
| 3 | 1.0333 | 0.9677 | 56 | 112.9 |
| 4 | 1.1762 | 0.8502 | 63 | 132.8 |
| 5 | 1.2671 | 0.7892 | 73 | 161.0 |
| 6 | 1.3440 | 0.7440 | 85 | 195.3 |
| 12 | 1.5927 | 0.6279 | 181 | 491.1 |

*The minimum over splits is $10^{112.9}$ at $|P| = 3$, recovering the constant of Theorem 4.3 by an independent route, and every other split is strictly dearer.*

*Proof.* $\sigma(b) = 1/\sigma(a)$ is the cross–match. $Q$ is disjoint from $P$, so its mass must be assembled from odd primes outside $P$; the least number of them reaching $\sigma(b)$, and the product of that many, are finite computations (`code/sector_barrier.rs`). Enlarging any prime of $a$ lowers $\sigma(a)$, raises $\sigma(b)$ and hence $|Q|$, so the displayed configuration is extremal for each $j$. □

**Remark 7.24** (Why the composite mechanism is out of reach)**.** Proposition 7.23 settles the standing of the clean reduction of Proposition 7.21. There $|P| = 2$, and the mechanism is forced through the dearest small sector: $b$ is odd, so $Q$ must assemble mass $\sigma(b) = 2q/(q+2)$ from odd primes alone, and the sector cannot begin below $10^{137.9}$; some twenty–five orders of magnitude *above* the general barrier $10^{112.9}$. The line $n' = 2n - 4$ is therefore a correct and complete description of the $|P| = 2$ case and simultaneously the most expensive place to look.

This also explains the balance the barrier strikes. The cheapest configuration is the near–balanced one, $\sigma(a) \approx \sigma(b) \approx 1$ at $|P| = 3$; pushing the split either way (towards a small $P$ with $\sigma(b) \to 2$, or towards a large $P$ with $\sigma(b) \to 0$) costs orders rapidly, because mass near 2 needs the odd primes to $\approx 10^4$ and mass near 0 needs many large ones. The 59–prime barrier is thus not merely a statement about $|P \cup Q|$ but the *minimum of a convex profile over splits*, and Theorem 4.3's constant is the value at that minimum. (Rows with $|P| + |Q| < 60$ are further excluded by Proposition 4.5, which sharpens the $|P| = 3$ entry without moving the minimum's location.)

**Proposition 7.25** (The cycle condition does not improve the barrier)**.** *A two–cycle satisfies more than the mass inequality: it forces $b = a'$, that is $b = a\,\sigma(a)$ exactly. Since $\sigma(b) = 1/\sigma(a)$ and $P \cap Q = \varnothing$, writing $S$ for the small primes of $a$ and $\Pi_{\min}(S)$ for the least product of primes outside $S$ with reciprocal sum $\geq 1/\sigma(S)$, one gets*

$$a \;\geq\; \max\Big(\prod S,\ \Pi_{\min}(S)/\sigma(S)\Big), \qquad N \;=\; a^2\sigma(a) \;\geq\; \max\Big(\prod S,\ \Pi_{\min}(S)/\sigma(S)\Big)^2 \sigma(S).$$

*Minimised over $S$, this yields $\log_{10} N \geq 113.2$, attained at $S = \{odd\ p \leq 151\} \setminus \{3, 97\}$ with $\sigma(S) = 1.0495$ and $|Q| \geq 26$. The barrier of Theorem 4.3 is $\log_{10} N \geq 112.9$:* the cycle condition buys essentially nothing beyond the mass condition*, and Theorem 4.3 is already close to optimal for arguments of this class.*

Formalised. *The deductive content is **`Erdos307.cycle_bound_max`**: two independent lower bounds on $a$ combine through the maximum, and $N = a^2\sigma(a)$ is monotone in both arguments, so the constraints give a bound on $N$. The conclusion drawn — that the gain over Theorem 4.3 is under one order — is an* upper *bound on a minimum, and an upper bound on a minimum needs one witness, not the search. Since $\Pi_{\min}(S)$ is itself a minimum, bounding the expression at $S$ from above needs a witness for it too: any $T$ disjoint from $S$ with $\sigma(T) \geq 1/\sigma(S)$ has $\Pi_{\min}(S) \leq \prod T$. Taking $S$ to be the odd primes to 151 less $\{3, 97\}$ and $T$ its greedy completion, 26 primes ending at 277, one checks $\prod T \leq \prod S \cdot \sigma(S)$, so the maximum is $\prod S$, and*

$$\left(\left(\prod S\right)^2 \sigma(S)\right)^2 \;<\; 10^{227},$$

*squared so the exponent is an integer: the bound on $\log_{10} N$ at this $S$ is at most 113.5, and the gain over 112.9 is under 0.6 of one order. Note $\left(\prod S\right)^2 \sigma(S)$ is an integer, $\sigma(S)$ having denominator exactly $\prod S$, so the whole check is exact integer arithmetic. 0* ***`sorry`*** (`lean/Erdos307/NoGain.lean`, `lean/Erdos307/NoGainWitness.lean`)*. The exhaustive minimisation certifies the sharper claim that* 113.2 *is exactly the minimum; that is not needed here, and it was the only reason this proposition was carried as a formalisation refusal.*

**Remark 7.26** (Why restricted families overstated the gain)**.** Compare the conclusion of Proposition 7.25 with what restricted families suggest, because the discrepancy is large and instructive. Confining $S$ to an initial segment $\{2, 3, 5, \dots\}$ gives a minimum of $10^{222.9}$; enumerating all subsets of the first 14, 16, 18 primes gives $10^{195.9}$, $10^{188.9}$, $10^{181.7}$; still falling, hence not converged. Each of these looks like a large improvement of the barrier and none of them is one. The optimum sits outside all of them: it hands 2 to $Q$ (half of $Q$'s mass budget in a single prime), keeps the odd primes to 151 in $S$, and drops two of them, a configuration no initial segment and no small subset can express. Only after searching that structure does the bound settle at 113.2.

The lesson is about the shape of the search rather than the arithmetic. A bound minimised over a family that cannot express the optimum reports the family's best, not the problem's, and here that error was worth eighty orders of magnitude. We record the negative result as the substantive one: imposing $b = a'$, the strongest structural constraint available beyond the masses, does not move the barrier, which is evidence that $10^{112.9}$ is not an artefact of the AM–GM route but the true cost of the mass condition (`code/full_minimisation.rs`, `code/minimise_structured.rs`).

**Remark 7.27** (Decoupling the two squares: what it buys, and what it costs)**.** Remark 1.16 locates the exponential difficulty in the *conjunction* of the two squares, so it is worth asking for a family on which one of them is free. The slot calculus supplies one: on

$$\mathcal{F}(N_0) \;=\; \left\{\, N_0 r \;:\; r = (s^2 - N_0)/E_0 \,\right\}$$

the plus–square $N' + 2N = s^2$ holds *identically*, by construction. The minus–square then costs exactly the form equation of Proposition 7.18, $k_0 s^2 + E_0 d^2 = 4N_0^2$, and the class of the residual question genuinely changes. On a coupled tail family the solutions run along a Pell orbit and the candidate slots grow like $\varepsilon^n$; here, for $\sigma(N_0) < 2$ the form is positive definite, so $4N_0^2$ has only *finitely* many representations and each base carries finitely many candidate slots. Decoupling therefore trades a Lehmer sequence for a finite per–base check.

What it does not do is help. Over the 30,397 squarefree bases $N_0 \le 5 \times 10^4$ with $\sigma(N_0) < 2$ the form represents $4N_0^2$ exactly *twice* (at most once per base, largest $s = 83$) and both representations are the ghost slot $r = 1$; there is no live prime slot at all (`code/plus_automatic.rs`, reproducing the sub–barrier census of Proposition 7.18 from an independent enumeration). The reason is the degeneration already noted: the barrier forces $\sigma(N_0) \uparrow 2$, so $k_0 = N_0(2-\sigma)$ collapses and the form becomes ever more lopsided, while the target stays at $4N_0^2$. So freeing one square does not free the problem; it relocates the difficulty from the primality of an exponential sequence to the solubility of a degenerating quadratic form. The pattern is the circularity of Section 5 once more, but the relocation is informative, since a finite, degenerating representation problem is a different kind of object from a Lehmer sequence, and no no–go here forbids attacking it.

The emptiness of that census is structural, and the reason fixes the range in which the question lives.

**Proposition 7.28** (The live–slot threshold)**.** *Let $N_0$ be squarefree and suppose $Q_{N_0}(s,d) = A\,s^2 + B\,d^2$ represents $4N_0^2$ with a slot $r = (s^2 - N_0)/B$ satisfying $r \ge 2$. Then*

$$\sigma(N_0) \;\ge\; \tfrac{3}{2},$$

*hence $\omega(N_0) \ge 10$ and $N_0 \ge \Pi_{10} = 6{,}469{,}693{,}230$.*

*Proof.* Assume first $\sigma = \sigma(N_0) < 2$, so $A = N_0(2-\sigma) > 0$ and the form is positive definite. From $r \ge 2$ and $B = N_0(2+\sigma)$,

$$s^2 \;\ge\; N_0 + 2B \;=\; N_0(5 + 2\sigma),$$

while $A\,s^2 \;\le\; A\,s^2 + B\,d^2 \;=\; 4N_0^2$ gives $s^2 \;\le\; 4N_0^2/A \;=\; 4N_0/(2-\sigma)$. The two are compatible only if $(5+2\sigma)(2-\sigma) \le 4$, that is $2\sigma^2 + \sigma - 6 \ge 0$, whose positive root is exactly $\frac{3}{2}$. Since among $k$–element prime sets the reciprocal sum is maximised by the first $k$ primes (Lemma 4.2) and $\sigma(\Pi_9) = 1.498956\ldots < \frac{3}{2} \le 1.533439\ldots = \sigma(\Pi_{10})$, the mass bound forces $\omega(N_0) \ge 10$ and $N_0 \ge \Pi_{10}$. If instead $\sigma(N_0) \ge 2$ then $N_0 \ge \Pi_{59} > \Pi_{10}$ by Theorem 4.3, so the conclusion holds a fortiori. □

The threshold is formalised. `Erdos307.live_slot_threshold` proves the analytic core, that a representation of $4N_0^2$ with $s^2 \ge N_0 + 2B$ forces $\sigma \ge \frac{3}{2}$, by dividing out $N_0^2$ and factoring $4 - (2-\sigma)(5+2\sigma) = (2\sigma - 3)(\sigma + 2)$ against $\sigma + 2 > 0$; and `threshold_bracket` checks the two numerals $\sigma(\Pi_9) < \frac{3}{2} \le \sigma(\Pi_{10})$ that place the entry point at ten primes. Both carry 0 `sorry` and the three standard axioms (`lean/Erdos307/LiveSlot.lean`). The passage from the mass bound to $\omega(N_0) \ge 10$ is the step already formalised as `card_ge_59_of_recipSum_ge_two` for the barrier, and is cited rather than repeated.

The census reported above ran to $N_0 \le 5 \times 10^4$, which is below $\Pi_{10}$ by a factor $1.3 \times 10^5$: it could not have met a live slot, and its emptiness is therefore not evidence about the family. The two representations it did find, at $N_0 = 30$ and $N_0 = 1722$, have $\sigma = 1.0333$ and $\sigma = 1.0006$, both far below $\frac{3}{2}$, which is why both are ghosts.

Three mass thresholds now organise the problem, and each is a Mertens count: $\sigma = 1$ at three primes, $\sigma = \frac{3}{2}$ at ten, and $\sigma = 2$ at fifty–nine. The first bounds the trivial regime, the third is Theorem 4.3, and the second is the entry point of the decoupled family. The entry region has now been swept: over the 105,104,561 squarefree $N_0 \le 10^{18}$ with $\sigma(N_0) \ge \frac{3}{2}$, the form $Q_{N_0}$ represents $4N_0^2$ not once, so there is no live slot and no ghost either (`code/live_slot_census2.rs`). Two checks accompany it. The same program without the mass gate returns the two representations of Proposition 7.18 exactly, at $N_0 = 30$ with $(s,d) = (11,1)$ and $N_0 = 1722$ with $(83,1)$, both carrying

$r = 1$; and the primorial bases $\Pi_{10}, \ldots, \Pi_{13}$ were re-checked by an independent implementation, each admitting no representation. The sweep uses the reduction $A \mid N_0'(N_0' - 2d^2)$, which follows from $2N_0 \equiv N_0' \pmod{A}$ and lets a single quadratic non-residue retire a base without any search; it retires 45% of them. The same program run without the mass gate returns the two representations of Proposition 7.18 exactly, at $N_0 = 30$ with $(s, d) = (11, 1)$ and $N_0 = 1722$ with $(83, 1)$, both with $r = 1$, which is the control that the search sees what it should.

### 7.4 Classical layers: genus, Giuga, parity

**Proposition 7.29** (A principal–genus necessary condition, and a cascade on solutions)**.** *As* $\gcd(N_0, N_0') = 1$*, the value* $4N_0^2$ *is a square coprime to the odd part of* $\operatorname{disc} Q_{N_0}$*, so every genus character is* $+1$ *on it and a primitive form represents it only from the principal genus [35]. Reading the odd assigned characters off* $Q_{N_0}(1,1) = 4N_0$ *gives the computable necessary condition*

$$(N_0 \mid \ell) = +1 \quad \textit{for every odd prime } \ell \mid (N_0'^2 - 4N_0^2) \textit{ to odd multiplicity}$$

*(a 2–adic condition accompanies it; for odd* $N_0$ *of even* $\omega$ *the form is imprimitive and must be primitivised first; so we state the criterion on the odd part, with a 2–adic caveat). Both ghost bases* $30, 1722$ *satisfy it, and about* 16% *of squarefree bases pass in the ranges tested (an* upper bound, *using only the odd characters, and mildly range–dependent).* Cascade. *In a solution of #307 with* $N = D_P D_Q$*, each* $\ell \in P \cup Q$ *gives a cofactor* $N/\ell$ *realising the Pythagorean structure of* $N$ *over base* $N/\ell$*; every cofactor with* $\sigma(N/\ell) < 2$ *(all of them for a minimal first–*59*–primes solution) must then satisfy the principal–genus criterion, a cascade of* $\geq 59$ *computable genus conditions complementing the quadratic–residue filter of Corollary 7.16.* (The further class–group and slot–admissibility stages of the fertility gap we observe only numerically at toy scale.)

Below the barrier the form has exactly one source of solutions, and it is entirely classical.

**Proposition 7.30** (The sub–barrier ghosts are pronic Giuga numbers)**.** *Every representation of* $4N_0^2$ *by* $Q_{N_0}$ *over squarefree* $N_0 \leq 3 \times 10^4$ *is a ghost* $(s, d) = (\sqrt{4N_0 + 1}, 1)$*, occurring exactly when* $N_0' = N_0 + 1$ *with* $N_0$ *pronic; in range only* $30 = 5 \cdot 6$ *and* $1722 = 41 \cdot 42$*. The two lines through such bases are classical: since* $n' = n\,\sigma(n)$ *for squarefree* $n$*,*

$$\{n : n' = n-1\} = \textit{primary pseudoperfect numbers } \left(\textstyle\sum_{p|n} \frac{1}{p} + \frac{1}{n} = 1\right), \qquad \{n : n' = n+1\} = \textit{Giuga numbers (quotien}$$

*The derivative characterisations are due to Lava (OEIS [36, 37], 2009; the Giuga form* $n' = n + 1$ *being Lava's* conjecture, *with the generalisation* $n' = an + 1$, $a \in \mathbb{N}$*, the theorem of Grau–Oller-Marcén [32]); the underlying number classes are Butske–Jaje–Mayernik [31] (exactly one primary pseudoperfect number per* $\omega \leq 8$*, eight known) and Giuga [28, 29], and the oblong chain (*$N$ *with* $N + 1$ *prime breeds the next term* $N(N + 1)$*) is classical in both directions (Forgues and Sondow, OEIS [36]). To* $2 \times 10^6$ *the lines are* $\{n' = n - 1\} = \{2, 6, 42, 1806, 47058\}$ *and* $\{n' = n + 1\} = \{30, 858, 1722, 66198\}$*, and the chain correctly predicts the next terms:* 47059 *is prime, so* $47058 \cdot 47059 = 2{,}214{,}502{,}422$ *is the sixth primary pseudoperfect number, and* 47057 *is prime, so the pronic* $47057 \cdot 47058 = 2{,}214{,}408{,}306$ *is the fifth Giuga number; both already known, recovered here by structure. Our contribution is the reframing: the chain step is the one–line Leibniz identity* $N' = N - 1 \Rightarrow (N(N+1))' = N'(N+1) + N^2 = N(N+1) - 1$*, exhibiting the chain as orbits of the Artin–Schreier map* $x \mapsto x(x+1) = x^2 + x$*; the same map, with the same halt (*$2 \to 6 \to 42 \to 1806$*, stopping at* $1807 = 13 \cdot 139$*), as the* $\mathbb{F}_2[t]$ *tower of Proposition 5.8: characteristic* 2 *and* $\mathbb{Z}$ *run identical breeders on the two lines.*

**Remark 7.31** (Converse, and the bridge)**.** Every pronic solution of $n' = n+1$ is $m(m+1)$ with $m$ prime and $m+1$ primary pseudoperfect: Leibniz gives $m'(m+1) + m(m+1)' = m^2 + m + 1$, and writing $m' = km + 1$ forces $(m+1)' = (1-k)m - k$, which is $\geq 0$ only for $k = 0$, whence $m' = 1$ ($m$ prime) and $(m+1)' = m$. (The congruence $m' \equiv 1 \pmod{m}$ alone does *not* force $m$ prime; its composite solutions are precisely the Giuga numbers $30, 858, 1722, 66198, \ldots$, so primality comes from the sign elimination $k = 0$.) Thus the only sub–barrier life in the form equation is the classical Giuga/primary–pseudoperfect world, known to sit in a web with Sylvester's sequence, Znám's problem and the Erdős–Moser equation [33] and with the arithmetic derivative [32, 34]; we claim none of this web, only that #307 meets it through $n' = n\,\sigma(n)$, and that its $a \geq 2$ (equivalently $\sigma > 2$) branch lands on the same $\geq 59$–prime barrier [30] that bars genuine cycles (Corollary 10.3).

Writing both members of a cycle as members of these lines stratifies #307 by the gap.

**Proposition 7.32** (Gap stratification: #307 as adjacency of $\mu$–Sondow lines)**.** *For a squarefree two–cycle $a' = b$, $b' = a$ with $a > b$ and gap $d = a - b$, the smaller member lies on the line $n' = n + d$ and the larger on $n' = n - d$ (as $b' - b = d$ and $a' - a = -d$), with* $\gcd(d, ab) = 1$ *(a prime dividing $d$ and $a$ would divide $b = a - d$); conversely $x' = x + d$ with $(x+d)' = x$ is a two–cycle. These opposite–sign lines are the quotient–one $\mu$–Sondow numbers of parameter $\mp d$ [34]. Hence #307 holds iff for some $d \geq 1$ the lines $n' = n + d$ and $n' = n - d$ carry members exactly $d$ apart; for $d = 1$ this is* a Giuga number adjacent to a primary pseudoperfect number*. By the parity dichotomy (Theorem 4.18) the mixed–parity case forces $d$ odd and the both–odd case ($d$ even) carries the $10^{2519}$ bound, so every cycle below that bound has odd gap and $d = 1$ is the lead stratum. A $d = 1$ pair $(k, k+1)$ has exactly one odd member, hence requires either an odd Giuga number (known to have at least* 9 *prime factors [29, 30]) or an odd primary pseudoperfect number (none is known): the lead stratum inherits two classical open problems. Tabulating cycle–eligible members (squarefree, coprime to $d$) to $2 \times 10^6$ for all $d \leq 50$ gives* 21 *on the plus lines (none of even gap, consistent with the parity remark) and* 81 *on the minus lines, with* no *adjacent pair at any $d \leq 50$; the barrier in gap–stratified form.* (*For coprime $M$ on $n' = n - \alpha$ and $N$ on $n' = n + \beta$, Leibniz gives $(MN)' - 2MN = \beta M - \alpha N$, so $\beta M - \alpha N = \square$ would manufacture a minus–side square in the sense of Proposition 5.13; the case $(\alpha, \beta) = (1, 1)$, $M - N = 1$ is the $d = 1$ cycle itself, and the known finite Giuga/primary–pseudoperfect lists contain no coprime pair, all members being even.*) *To our knowledge this gap–stratified equivalence and the pairing of opposite–sign lines at fixed additive distance are new; the lines themselves are [34].*

The line parameter $\delta(n) = n' - n$ obeys an affine recursion that organises the classical members into a genealogy.

**Proposition 7.33** (The $\delta$–recursion and the genealogy of the classical lines)**.** *With $\delta(n) = n' - n$, for squarefree $M$ and a prime $p \nmid M$,*

$$\delta(Mp) = p\,\delta(M) + M \qquad (\textit{Leibniz; checked on } 10^3 \textit{ random pairs}).$$

*Solving $p\,\delta(M) + M = \pm d$ gives the child rule $p = (M \mp d)/e$ from a parent $M$ on the minus line $\delta(M) = -e$. Every known Giuga number arises this way from a small–parameter parent: $30 = 6 \cdot 5$ ($e = 1$), $858 = 66 \cdot 13$ ($e = 5$), $1722 = 42 \cdot 41$ ($e = 1$), $66198 = 1122 \cdot 59$ ($e = 19$), $2{,}214{,}408{,}306 = 47058 \cdot 47057$ ($e = 1$); the seed $47058$ itself resolves as $6 \to 66 \to 1518 \to 47058$ (with $\delta = -1, -5, -49, -1$), so the previously unexplained members $858$ and $66198$ acquire genealogies. Two members of a cycle can never share a parent: siblings $M(M \mp d)/e$ differ by $2Md/e > d$, since $e = M - M' < 2M$. The oblong case $e = 1$ is the classical chain of Proposition 7.30 (its $\mu$–Sondow* *form is [34]); the recursion, the general–e genealogy, and the resulting small–$|\delta|$ tree do not appear there.*

**Corollary 7.34** (Nonemptiness families for the $\mu$–Sondow conjecture)**.** *In the notation of [34] (their $S_\mu$; the quotient–one slice is the line $n' = n - \mu$), the recursion produces explicit members exceeding $|\mu|$ in three infinite parameter families: $2p \in S_{p-2}$ for every odd prime $p$; $Mp \in S_{p-M}$ for every primary pseudoperfect number $M$ and prime $p \nmid M$, and $Gp \in S_{-(p+G)}$ for every Giuga number $G$ and prime $p \nmid G$; in each case $\mu/n + \sum_{q|n} 1/q = 1$ exactly (checked symbolically and on 151 instances). This settles their nonemptiness conjecture ($S_\mu \cap (|\mu|, \infty) \neq \varnothing$, their Conjecture 1(ii)) for infinitely many parameters, including infinitely many outside their verified range $-145 < \mu < 673$; these families do not appear among their constructions, which scale by $\mathrm{rad}(\mu)$ rather than extend by a new prime. Their two open instances resist precisely because every quotient–one parent route hits a composite: for $\mu = -145$, $M - 145$ is composite for all eight known primary pseudoperfect numbers ($1661 = 11 \cdot 151$, $46913 = 43 \cdot 1091$, and the three larger cases, including the recently found eighth member) while $p + G = 145$ has no prime solution; for $\mu = 673$, $M + 673$ is composite for all eight ($679 = 7 \cdot 97$, $2479 = 37 \cdot 67$, $47731 = 59 \cdot 809$, the rest ending in 5) and $p = 675$ fails. A prime in either family at the next generation would settle a published open instance; an exhaustive search found none. Concretely: no quotient–one member of $S_{-145}$ or $S_{673}$ exists with* any *prime–factor cofactor $\leq 10^{11}$ (every squarefree parent $M \leq 10^{11}$ was scanned against the child rules, a region whose members reach $\sim 10^{22}$, together with a depth–two descent), the quotient–zero slice of $S_{-145}$ is empty outright ($n' = 145$ forces $n \leq 5256$ by $n' \geq 2\sqrt{n}$, and the finite check is empty), and quotients $\geq 2$ force $\sigma \gtrsim 2$, hence at least 59 prime factors. Both instances therefore remain open, with the quotient–one frontier moved from the $10^{10}$ of [34] to cofactor scale $10^{11}$.*

**Remark 7.35** (The look–back)**.** Along small–$|\delta|$ paths the forced prime is $p \approx M/e$, so members grow doubly exponentially and the $10^{56}$–$10^{113}$ wall zone lies only a few generations from the seeds: a sparse, recursively enumerable tree (a child needs $(M - j)/e$ prime for small $j$). The adjacency of two such trees at a fixed additive distance places the final condition in the Pillai/$S$–unit genre (two multiplicatively generated sequences meeting at a fixed additive distance) where fixed–support finiteness is classical but growing–support statements are conjectural; a lens, not a transfer. Re–expressed in $\delta$–coordinates the obstruction is once more $\sigma > 1$ (positivity of $\delta$) plus integer reachability $\delta = \pm d$ under forced primes: the same additive–multiplicative exactness reappears in every coordinate tried (mass, the deviation $k$, the form $Q_{N_0}$, genus theory, $\mu$–Sondow lines, $\delta$–dynamics). The reframings buy structure, not a reduction.

**Proposition 7.36** (Even–gap plus lines are empty below $3.23 \times 10^9$)**.** *An even squarefree $x = 2u$ has $\delta(x) = 2u' - u$ odd, so every line member of even $\delta$ is odd (with $\omega$ odd). An odd member of a plus line has $\sigma > 1$ over odd primes, and the least odd squarefree integer of mass exceeding 1 is $3 \cdot 5 \cdots 29 = 3{,}234{,}846{,}615$ (the 8–prime product $3 \cdots 23 = 111{,}546{,}435$ has mass $0.99896$). Hence every plus line of even gap $d$ is empty below $3{,}234{,}846{,}615$, proving the census observation (no even–gap plus member to $2 \times 10^7$) and placing an explicit floor under the smaller member of any both–odd configuration.*

The lines also admit parity, infinitude, and congruence refinements.

**Proposition 7.37** (Parity floor for odd line members)**.** *For odd squarefree $n$ every cofactor $n/p$ is odd, so $n' \equiv \omega(n) \pmod 2$. Hence an odd primary pseudoperfect number ($n' = n - 1$, even) and an odd Giuga number ($n' = n + 1$, even) both have $\omega(n)$ even; combined with the classical bound $\omega \geq 9$ [29, 30] this forces $\omega \geq 10$, a parity sharpening (for the primary pseudoperfect side, the eight smallest odd primes give $\sum 1/p = 0.99896 < 1$, so $\omega \geq 9$ directly). The floor cascades when* *small primes are avoided: for an odd squarefree member of any plus line, or of a quotient–one line (where $\sigma(n) \geq 1 - 1/n$), exact greedy/exchange computation gives $\omega \geq 27$ if $3 \nmid n$; $\omega \geq 66$ if $3, 5 \nmid n$; $\omega \geq 144$ if $3, 5, 7 \nmid n$; and $\omega \geq 16$ if $3 \mid n$, $5 \nmid n$; each avoided small prime multiplies the required complexity, and for odd Giuga or primary pseudoperfect numbers the parity constraint rounds odd floors up (e.g. $3 \nmid n$ forces $\omega \geq 28$). (The restriction to plus or quotient–one lines is essential: on a general minus line $\sigma$ can be small (every prime $p$ trivially satisfies $p' = p - (p - 1)$) so no such floor exists there.)*

**Proposition 7.38** (Infinitely many gaps carry a nonempty plus line). *By the recursion of Proposition 7.33, $\delta(30p) = p\,\delta(30) + 30 = p + 30$ for every prime $p > 5$ (as $\delta(30) = 1$), so the line $n' = n + (p + 30)$ is nonempty; by Euclid infinitely many distinct gaps $d$ carry a nonempty plus line, and the same runs from any Giuga number $G$ ($\delta(Gp) = p + G$). This fixes the census tail: $330, 390, 510, 570 = 30{\cdot}\{11, 13, 17, 19\}$ lie on the gaps $d = 41, 43, 47, 49$. (Per–gap infinitude remains open, of Bunyakovsky type.)*

**Proposition 7.39** (A difference–36 congruence modulo 72). *Let $n = 6m$ be a Giuga or primary pseudoperfect number with $\gcd(m, 6) = 1$, $n' = n + \varepsilon$ ($\varepsilon = +1$ Giuga, $-1$ primary pseudoperfect). Then $6m' = m + \varepsilon$, and with $m' \equiv \omega(m) \pmod 2$ this yields $n \equiv 36\,\omega(n) - 78 \pmod{72}$ on the Giuga side and $n \equiv 36\,\omega(n) - 66 \pmod{72}$ on the primary pseudoperfect side; in particular equal $\omega$ forces equal residue. The congruence is verified on all twelve Giuga numbers and all seven 6–divisible primary pseudoperfect numbers listed in the OEIS (we recomputed the factorisations of the five largest Giuga members; all are squarefree, with $\omega = 7, 7, 8, 8, 8$), and gives, at the modulus 72, the difference–36 arithmetic progression observed by Sondow–MacMillan [33]; their finer modulus–288 pattern remains empirical.*

The modulus–72 argument is the $D = 6$ case of a general transport of the line equation across a fixed divisor.

**Proposition 7.40** (Divisor transport). *Let $n = Dm$ with $D, m$ coprime squarefree and $n' = n + \varepsilon$ ($\varepsilon = \pm 1$) a quotient–one member. Leibniz transports the line equation to*

$$D\,m' = (D - D')\,m + \varepsilon.$$

(a) *If $\sigma(D) > 1$ (so $D' > D$; least case $D = 30$), the right side is negative for $m > 1$ while the left is nonnegative; the only escape is $m = 1$, $\varepsilon = +1$. Hence* no primary pseudoperfect number is divisible by 30, and the only Giuga number divisible by 30 is 30 itself *(elementary; we have not located it in the literature), consistent with the full known corpus, of which only* 30 *carries the factor* 5.

(b) *At a primary pseudoperfect block $D$ (where $D - D' = 1$) the slice $m' = 1$ reads $m = D - \varepsilon$ prime: the two classical chain rules ($D(D + 1)$ primary pseudoperfect when $D + 1$ is prime, $D(D - 1)$ Giuga when $D - 1$ is prime [34, 33]) are the two signs of one transported equation ($1806 = 42 \cdot 43$ and $1722 = 42 \cdot 41$ sit on the same slice $D = 42$).*

(c) *For any gap $d$ the same transport stratifies the line $n' = n + d$ by divisor blocks $D$ into the steeper lines $D\,m' = (D - D')m + d$, a further enumeration coordinate for the genealogy of Proposition 7.33.*

Finally, the form obstruction can be averaged: it eliminates almost every integer, though not the thin stratum where a solution could live.

**Proposition 7.41** (Average rarity of form–barrier survivors)**.** *Call a squarefree $N$* blocked *at an odd prime $\ell$ if $\ell \nmid N$, $N' \equiv \pm 2N \pmod{\ell}$, and $(N \mid \ell) = -1$, and a* survivor *if it is blocked at no odd prime. A blocked $N$ admits no representation $E\,s^2 + D\,d^2 = 4N^2$ (reducing mod $\ell$ forces $N$ to be a quadratic residue, against $(N \mid \ell) = -1$; this is the genus fragment of Proposition 7.29), so every union of a #307 solution is a survivor. Now $N' \equiv N\,S_\ell(N) \pmod{\ell}$ with $S_\ell(N) = \sum_{p \mid N} p^{-1}$ an* additive *function, so blocking at $\ell$ is an additive–function event ($S_\ell \equiv \pm 2$) twisted by the quadratic character, of limiting density $1/(\ell+1)$* [HEURISTIC*: this constant is asserted, not derived. Equidistribution of $S_\ell$ over $\mathbb{Z}/\ell$ together with density $\frac{1}{2}$ for $(N \mid \ell) = -1$ would give $2/\ell \cdot \frac{1}{2} = 1/\ell$ instead. The two agree to leading order and $\sum_\ell 1/\ell$ diverges as well, so the shape of the conclusion is unaffected; the exact constant is not established here*]*. As $\sum_\ell 1/(\ell+1)$ diverges (Mertens), the Selberg–Delange/Halász theory of additive functions in residue classes [18] gives that the survivors have natural density zero among the squarefree integers (unconditional), with $\#\{\text{survivors} \le X\} = O\big(X/\log\log\log X\big)$ under a standard, but here unproved, uniformity estimate. Thus the single–prime form obstruction removes* almost every *squarefree integer. We stress the scope: survivors remain* infinite*, and a #307 solution; one integer beyond* $2.09 \times 10^{56}$*, in the density–zero stratum $\sigma(N) \ge 2$; is untouched by such an average statement, which speaks only to the typical integer, not to existence.*

# A The minus–layer density theorem, and the square sieve that does not reach its rate

Relegated from §7.1. This appendix is the historical route to Theorem A.10: a square sieve at $P \asymp (\log N)^{1/4}$ whose analytic input was never supplied, since the uniformity it needs is exactly what Remark A.3 forecloses at that range. The theorem is now proved instead by Proposition A.8, at the much smaller $P = \log\log N/(\log\log\log N)^{1+\varepsilon}$, where the same uniformity *is* available (Lemma A.4) — at the cost of a far weaker rate. The sieve is kept in full because its algebraic layer is correct and reusable, because the located gap is what identified the workable range, and because Theorem A.13 — now proved, via Lemma A.5 — is what this route gives once its input is supplied.

**Lemma A.1** (Cancellation in the character sums)**.** *Fix an odd squarefree $r > 1$. Then*

$$T_r(N) \;:=\; \sum_{\substack{m \le N,\ \mu^2(m)=1 \\ \gcd(m,r)=1}} \Big(\frac{m' - 2m}{r}\Big) \;=\; o_r(N) \qquad (N \to \infty).$$

*The statement is qualitative and per fixed $r$: the implied constant depends on $r$, and no uniformity is claimed. Remark A.3 records why the uniform version is not available and what was previously claimed here in error; Remark A.2 records what the constant is and is not.*

*Proof.* Five standard facts are used, all at fixed $r$: Halász's theorem; $|\tau(\chi)| = \sqrt{r}$ for a primitive character; the classical zero–free region; $|\log L(s,\psi)| \ll_r \log\log(3 + |\tau|)$ for $\psi$ non–principal at $s = 1 + 1/\log N + i\tau$; and Siegel–Walfisz. None is formalised.

*The expansion.* Write $\chi = (\cdot \mid r)$. For $r$ odd squarefree the Jacobi symbol is a primitive real character modulo $r$, so $\chi(x) = \tau(\chi)^{-1} \sum_{t \bmod r} \chi(t) e_r(tx)$ holds for *every* $x$, including $\gcd(x,r) > 1$, where both sides vanish. That case occurs: at $r = 15$ already $m = 2$ gives $\sigma_r(m) - 2 = 6$. By Lemma 7.1, $(m' - 2m \mid r) = \chi(m)\,\chi(\sigma_r(m) - 2)$, and expanding the second factor,

$$T_r(N) = \tau(\chi)^{-1} \sum_{t \bmod r} \chi(t)\, e_r(-2t)\; S_t(N), \qquad S_t(N) = \sum_{\substack{m \le N,\ \mu^2(m)=1 \\ \gcd(m,r)=1}} h_t(m),$$

where $h_t(m) = \chi(m)\prod_{p|m} e_r(tp^{-1})$, using that $\sigma_r$ is a sum over $p \mid m$ and $e_r$ carries sums to products. Extend $h_t$ to $\mathbb{N}$ by $h_t(p^k) = 0$ for $k \geq 2$ and $h_t(p) = 0$ for $p \mid r$, the only extension preserving $S_t$. It is multiplicative with $|h_t| \leq 1$, which is Halász's hypothesis; $|h_t| = 1$ on the support only.

*The distance.* Halász needs $M = \min_{|\tau|\leq \log N} \sum_{p\leq N} \big(1 - \operatorname{Re} h_t(p)p^{-i\tau}\big)/p \to \infty$. Now $h_t$ is not a function of $m \bmod r$, being built from the factorisation: at $r = 7$, $h_1(3) = 0.2225 + 0.9749i$ while $h_1(10) = -1$, though $3 \equiv 10$. Its restriction to primes is, which is all that is needed. Put $g_t(a) = \chi(a)e_r(ta^{-1})$ on $(\mathbb{Z}/r)^\times$, so $h_t(p) = g_t(p \bmod r)$ for $p \nmid r$, and expand $g_t = \sum_\psi c_\psi \psi$. Reindexing by the involution $a \mapsto ta^{-1}$ of $(\mathbb{Z}/r)^\times$ and using $\chi(a^{-1}) = \chi(a)$,

$$c_{\psi_0} = \frac{1}{\varphi(r)} \sum_{a\in(\mathbb{Z}/r)^\times} \chi(a)e_r(ta^{-1}) = \frac{\chi(t)\,\tau(\chi)}{\varphi(r)}, \qquad |c_{\psi_0}| = \frac{\sqrt{r}}{\varphi(r)}.$$

This identity uses no primitivity: it holds for any quadratic character on any finite abelian group, which is what the formalisation records. Primitivity enters twice elsewhere, for $|\tau(\chi)| = \sqrt{r}$ and for the inversion at $\gcd(x, r) > 1$ above.

Splitting the prime sum by $\psi$,

$$\sum_{p\leq N} h_t(p)p^{-1-i\tau} = c_{\psi_0} \sum_{p\leq N,\, p\nmid r} p^{-1-i\tau} + \sum_{\psi\neq\psi_0} c_\psi \sum_{p\leq N} \psi(p)p^{-1-i\tau}.$$

The first sum is at most $\sum_{p\leq N} 1/p = \log\log N + O(1)$ in modulus, trivially and uniformly in $\tau$. For the coefficients, $|g_t| = 1$ gives $\sum_\psi |c_\psi|^2 = 1$ by Parseval, hence $\sum_{\psi\neq\psi_0} |c_\psi| \leq \sqrt{\varphi(r)}$ by Cauchy–Schwarz; they are *not* small individually. Indeed $c_\psi = \varphi(r)^{-1} \sum_b (\chi\psi)(b)e_r(tb)$, so $|c_\psi| = \sqrt{\operatorname{cond}(\chi\psi)}/\varphi(r)$, and the maximum $\sqrt{r}/\varphi(r)$ is attained exactly on the $\prod_{p|r}(p-2)$ characters with $\chi\psi$ primitive: three of the eight at $r = 15$, five of the twelve at $r = 21$. Their contribution is small because the $L$–functions have no pole, and reaching that needs care: the cited bound governs $\log L$, an Euler product over all $p$, while what is required is the truncation. Both routes go by partial summation against Siegel–Walfisz and both shed the same factor $1 + |\tau|$; what differs is the *range*, and it is the range and not any shift that does the work. On $[2, N]$ one is left with $\int_{\log 2}^{\log N} u^{-A} du = O_A(1)$, so $(1 + |\tau|) \cdot O(1)$ is $\ll \log N$, worse than trivial. On $[N, \infty)$, writing $A(v) = \sum_{p\leq v} \psi(p) \ll_{A,r} v(\log v)^{-A}$,

$$\sum_{p>N} \psi(p)p^{-s} \;=\; -A(N)N^{-s} + s\int_N^\infty A(v)v^{-s-1}\,dv \;\ll\; (\log N)^{-A} + (1+|\tau|)\frac{(\log N)^{1-A}}{A-1} \;\ll\; (\log N)^{2-A},$$

which is $o(1)$ for $A > 2$ and $|\tau| \leq \log N$. Note what is *not* used: the damping $p^{-1/\log N}$ attached to the shifted point $s = 1 + 1/\log N + i\tau$ is bounded by 1 and is simply discarded above. It contributes nothing to the tail, and indeed the shift alone leaves the tail of size $\int_1^\infty e^{-w}w^{-1}dw = E_1(1) = 0.2194$. The shift costs $O(1)$ by Mertens, since $\sum_{p\leq N} \log p/(p \log N) \ll 1$, and it is convenient for keeping the Euler logarithm absolutely convergent, but it is not what makes the tail small. The zero–free region then gives $|\log L(s, \psi)| \ll_r \log\log(3 + |\tau|)$, the argument of the double logarithm being taken above $e$ so that the bound is positive at $\tau = 0$; the non–principal terms are $O_r(\log\log(3 + |\tau|)) = O_r(\log\log\log N)$ uniformly for $|\tau| \leq \log N$, since at the top of that range the cited input gives $\log\log(3 + \log N) \asymp \log\log\log N$ and not $O(1)$. Hence

$$M \;\geq\; c_r \log\log N - O_r(\log\log\log N), \qquad c_r := 1 - \frac{\sqrt{r}}{\varphi(r)} \;\geq\; 1 - \frac{\sqrt{3}}{2} \;=\; 0.1339\ldots,$$

the extremum being at $r=3$: each local factor of $\sqrt{r}/\varphi(r)$ is $\sqrt{p}/(p-1)$, and $(p-1)/\sqrt{p}\ge 2/\sqrt{3}$ for every odd prime, so the supremum is the single factor at $r=3$ (`code/charcancel_check.gp`).

*Conclusion.* With $T=\log N$, Halász gives $S_t(N)=\sum_{m\le N}h_t(m)\ll N\big((1+M)e^{-M}+(\log N)^{-1/2}\big)$. Note $M\to\infty$ alone would not suffice: at fixed $T$ the bound is only $\ll NT^{-1/2}$. Both limits are in play, and $S_t(N)\ll_r N(\log N)^{-\min(c_r,1/2)+o(1)}=o(N)$, the $\frac{1}{2}$ being the floor the $(\log N)^{-1/2}$ term imposes however large $M$ becomes. Only $r=3$ and $r=5$ have $c_r\le\frac{1}{2}$, so for every other modulus, and in particular for every $r=pq$ arising in Theorem A.10, the floor is the binding constraint. Only the $\varphi(r)$ values of $t$ coprime to $r$ contribute, since $\chi(t)=0$ otherwise, so

$$|T_r(N)|\;\le\;r^{-1/2}\sum_{\gcd(t,r)=1}|S_t(N)|\;\le\;\frac{\varphi(r)}{\sqrt{r}}\max_t|S_t(N)|\;\le\;\sqrt{r}\,\max_t|S_t(N)|\;=\;o_r(N).$$

The net factor is a *loss* of $\sqrt{r}$: the $\varphi(r)$ summands outweigh $|\tau(\chi)|^{-1}$. At fixed $r$ that is a constant.
□

**Remark A.2** (What the constant $c_r$ is and is not)**.** Three things about $c_r=1-\sqrt{r}/\varphi(r)$ are worth recording, because an earlier revision overstated all three.

*Sharpness is conditional.* The bound $M\ge c_r\log\log N-O_r(1)$ is attained only when the main term $\chi(t)\tau(\chi)$ is real and positive. Since $\overline{\tau(\chi)}=\chi(-1)\tau(\chi)$ with $\chi(-1)=(-1)^{(r-1)/2}$, the Gauss sum is purely imaginary for $r\equiv 3\pmod 4$, and there the true $M$ has constant 1 rather than $c_r$. For $\chi(t)=-1$ the main term has the wrong sign, and no admissible $\tau$ rotates it back, since $|\sum_{p\le X}p^{-1-i\tau}|$ collapses to $O(\log\log(3+|\tau|))$ away from $\tau=0$; there the constant is $1+\sqrt{r}/\varphi(r)$. Measured at $r=5$: the growth rate $\Delta M/\Delta\sum_p 1/p$ tends to $0.440887\to 1-\sqrt{5}/4=0.440983$ for $t\equiv\pm1$, and to $1.558921\to 1+\sqrt{5}/4=1.559017$ for the other two residues, whose $M(10^7)$ is 4.378 against the leading term 1.341. So the constant is attained for $r\equiv 1\pmod 4$ and $\chi(t)=+1$, and is not sharp otherwise.

*The finite-X deficit is a constant, though the proved error term is not.* The bound carries $-O_r(\log\log\log N)$, which is what the $L$–bound gives at the top of the $\tau$–range; what is *measured* at a single modulus is smaller and flat. At $r=5$, $t=1$ the deficit $c_r\sum_{p\le X}1/p-M$ is $0.1399, 0.1401, 0.1402, 0.1402$ at $X=10^4,\dots,10^7$, flat to three places and still rising in the fourth, while $\log\log\log X$ rises by 28% over the same range. Both sums here run over *all* $p\le X$: $h_t$ is extended by $h_t(p)=0$ at $p\mid r$, so those primes contribute $1/p$ to $M$, and omitting them shifts every number in this remark (`code/deficit_check.gp`). The $\log\log\log$ enters only through uniformity in $\tau$.

*The rate that follows, and the one that does not.* Halász gives $T_r(N)\ll_r N(\log N)^{-c_r+o(1)}$. Since $c_3=0.134$ and $c_5=0.441$ are below $\frac{1}{2}$, this does *not* yield $T_r(N)\ll\sqrt{r}\,N(\log N)^{-1/2}$ at those moduli, and an earlier revision asserted that it did. Quantitatively the lemma is asymptotic in the strong sense: at $r=5$, with the measured constants and an implied Halász constant of 1, the bound first beats the trivial $|T_r(N)|\le N$ at $\log N=143.0$, that is $N=10^{62.1}$. Measured against the honest trivial bound, which is $0.5066N$ because $T_r$ has only that many terms, the crossing moves to $\log N=1013.7$, that is $N=10^{440}$.

**Remark A.3** (Erratum: the $\tau$ bound above is false near the threshold)**.** The step "if $|\tau|\ge 1/\log N$ then $\left|\sum_{p\le N}p^{-1-i\tau}\right|\ll\log\log(3+|\tau|)$" is *false* in the range that matters. Near the threshold the sum has size about $\log(1/|\tau|)$, which is of order $\log\log N$, not $\log\log(3+|\tau|)$, which is of order 1. Measured at $N=2\times10^7$ with $\tau=2/\log N=0.11897$: the left side is 2.6797 against a claimed bound of 0.12883, a violation by a factor 20.8; at $\tau=1/\log N=0.05948$ the left side is 2.9773 against 0.11176, a factor 26.6 (`code/charcancel_check.gp`). A second defect is that the inequality bounds the untwisted sum $\sum p^{-1-i\tau}$, whereas the Halász distance $M$ requires the $h_t$–twisted sum.

Consequences, stated exactly. Neither defect touches the lemma as now proved, because that proof does not use the offending step: it bounds the *twisted* sum through the character expansion of $h_t$, where the principal coefficient contributes at most $\sqrt{r}/\varphi(r) \leq 0.8661$ of $\sum_{p\leq N} 1/p$ and the non–principal ones contribute $O_r(\log\log\log N)$, both uniformly in $\tau$. The trivial bound $\left|\sum_{p\leq N} p^{-1-i\tau}\right| \leq \sum_{p\leq N} 1/p$ is all that is needed of the untwisted sum, so the false inequality is not merely repaired but unnecessary.

Two claims must be distinguished, and an earlier revision ran them together. What is shown false above is the *r–uniform form of one input*, the $\log L$ bound, whose lower half needs $L(1,\psi)$ away from 0 and has only $L(1,\psi) \gg r^{-1/2}$ effectively or $\gg_\varepsilon r^{-\varepsilon}$ by Siegel, ineffectively. That is *not* the same as the uniform form of the lemma being false, and the distinction turned out to be the whole opening: the obstruction is *range–dependent.* At $r \leq (\log\log N)^2$ one has $\log r \leq 2\log\log\log N$, so even a Siegel–worst–case $|\log L(1,\psi)|$ costs $\eta\log r + O_\eta(1)$, which is negligible against the main term $\log\log N$; that is Lemma A.4, and with Lemma A.7 it proves Theorem A.10 with a rate. One caveat on how much this explains: at the far larger moduli of Theorem A.13 the $L$–function bound does also fail, but it is not what binds there. Even an $O(1)$ per–character bound would leave the requirement $\sqrt{\varphi(r)} = o(\log\log N)$, from the *coefficient mass* rather than from any $L$–function — a limitation of the character expansion, not of the arithmetic; see the blocker recorded at that corollary and the question posed in Remark A.12. What the erratum forecloses is uniformity at the far larger $r$ that Theorem A.13 needs, where $\log r$ is comparable to the main term — and that foreclosure binds only the route that expands in characters. The rate is proved in Theorem A.13 by a route that does not expand (Lemma A.5); two further blockers recorded in earlier revisions, this one and the regime mismatch after Lemma 7.4, were retracted before that.

**Lemma A.4** (Cancellation, uniform at polyloglog moduli)**.** (Label discipline, as for Theorem A.10: this statement went through six rounds of adversarial review, each by a session that did not write the text under review. Rounds one to five each found defects — twice in a repair made by the round before — which were repaired; round six, auditing statement and proof line by line and recomputing every printed constant, found nothing, which is what the label records. Proposition A.8 passed clean at round three.) *Fix $\varepsilon > 0$ and write $L = \log\log N$. For every odd squarefree $r$ with $7 \leq r \leq L^2/(\log L)^{2+\varepsilon}$,*

$$T_r(N) \ \ll\ N\,(\log N)^{-2/5},$$

*with an implied constant that depends only on $\varepsilon$ and is ineffective; the same bound, by the same proof with $e_r(+2t)$ in place of $e_r(-2t)$, holds for the sum with $(m'+2m \mid r)$ in place of $(m'-2m \mid r)$. The moduli $r = 3, 5$ are excluded: there $c_r < 1/2$, and they are covered instead at fixed $r$ by Lemma A.1 with the exponents of Remark A.2.*

*Proof.* The proof of Lemma A.1 is followed step by step; what is new is that every implied constant is tracked in $r$, and each is absolute in the stated range. Here $r \leq L^2/(\log L)^{2+\varepsilon} \leq L^2$, so $\log r \leq 2\log L$. The range is exactly the one the argument supports: the binding requirement below is $\sqrt{r}\,\log L = o(L)$, that is $r = o\big((L/\log L)^2\big)$. Every earlier version of this lemma stated the strictly smaller $r \leq L^{2-\delta}$ — weaker than its own proof gives.

*Expansion.* Unchanged: $|T_r(N)| \leq r^{-1/2}\sum_t |S_t(N)| \leq r^{1/2}\max_t |S_t(N)|$, the sum over the $\varphi(r)$ values of $t$ with $\gcd(t,r) = 1$, since $\chi(t) = 0$ otherwise; the loss $r^{1/2} \leq L$ is absorbed below.

*The distance.* As before, $\sum_{p\leq N} h_t(p)p^{-1-i\tau} = c_{\psi_0}\sum_{p\leq N,\, p\nmid r} p^{-1-i\tau} + \sum_{\psi\neq\psi_0} c_\psi S_\psi(\tau)$ with $S_\psi(\tau) = \sum_{p\leq N}\psi(p)p^{-1-i\tau}$ and $|c_{\psi_0}| = \sqrt{r}/\varphi(r)$. The principal part is at most $(\sqrt{r}/\varphi(r))\log\log N$

in modulus, trivially and uniformly in $\tau$; dropping the primes $p \mid r$ from the untwisted sum costs $\sum_{p\mid r} 1/p \le \omega(r)/3 + O(1) = O(\log L)$, absorbed into the error below.

*The non–principal sums, uniformly.* Fix $\psi \ne \psi_0$ mod $r$ and $|\tau| \le \log N$, and put $s = 1 + 1/\log N + i\tau$. For composite $r$ the character $\psi$ need not be primitive; let $\psi^*$ of conductor $d \mid r$, $d > 1$, induce it. Then $S_\psi$ and $S_{\psi^*}$ differ only at the primes $p \mid r$, by at most $\sum_{p\mid r} 1/p = O(\log L)$, absorbed into the bound below, and Siegel–Walfisz, the zero–free region and Siegel's theorem are applied to $\psi^*$. Partial summation against Siegel–Walfisz, exactly as in Lemma A.1 with $A = 3$, replaces the truncated sum by $\log L(s, \psi) + O(1)$ at a cost $O((\log N)^{-1})$; Siegel–Walfisz is uniform, with an absolute ineffective constant, for conductors up to $(\log N)^3$, and $r \le L^2 \le (\log N)^3$ holds for all $N \ge 16$. For $\log L(s, \psi)$ two cases arise. If $L(\cdot, \psi)$ has no exceptional zero, the classical zero–free region gives $|\log L(s, \psi)| \ll \log\log\bigl(r(3 + |\tau|)\bigr) \ll \log L$ with an absolute constant. If $\psi$ is real with an exceptional zero $\beta$, split the $\tau$–range. For $|\tau| \le 1$ the local decomposition $\log L(s, \psi) = \log(s - \beta) + O\bigl(\log\log(r(3 + |\tau|))\bigr)$ applies, and

$$|s - \beta| \;\ge\; \mathrm{Re}(s) - \beta \;>\; 1 - \beta \;\ge\; c(\eta)\, r^{-\eta}$$

by Siegel's theorem in zero–location form, applied with any fixed $\eta > 0$ — a parameter unrelated to the $\varepsilon$ of the range — so $|\log(s-\beta)| \le \log\bigl(1/(1-\beta)\bigr) + O(1) \le \eta \log r + O_\eta(1) \ll \eta \log L + O_\eta(1)$; the decomposition is *not* asserted beyond $|\tau| \le 1$, where $|s - \beta|$ can be as large as $\log N$ and $\log|s - \beta|$ as large as $L$. For $|\tau| > 1$ the zero $\beta$ lies at distance greater than 1 from $s$, outside the local zero–density disc, and the unconditional bound $|\log L(s, \psi)| \ll \log\log(r(3 + |\tau|))$ applies with no exceptional case. In both regimes $|S_\psi(\tau)| \ll \log L$, absolutely and ineffectively, uniformly in $|\tau| \le \log N$.

*Assembly.* By Cauchy–Schwarz on the coefficients, $\sum_{\psi\ne\psi_0} |c_\psi| \le \sqrt{\varphi(r)} \le \sqrt{r} \le L/(\log L)^{1+\varepsilon/2}$, so the non–principal contribution to the distance is $\ll L(\log L)^{-\varepsilon/2} = o(L)$, and

$$M \;\ge\; \Bigl(1 - \frac{\sqrt{r}}{\varphi(r)}\Bigr)L \;-\; O\bigl(L(\log L)^{-\varepsilon/2}\bigr) \;\ge\; \Bigl(1 - \frac{\sqrt{r}}{\varphi(r)} - o(1)\Bigr)L.$$

The constant $1 - \sqrt{r}/\varphi(r)$ does *not* tend to 1 over the whole range: it is $0.133\ldots$ at $r = 3$ and $0.441\ldots$ at $r = 5$, and exceeds $1/2$ for every odd squarefree $r \ge 7$, with minimum $1 - \sqrt{15}/8 = 0.515\ldots$ at $r = 15$. Indeed $\sqrt{r}/\varphi(r) = \prod_{p\mid r} \sqrt{p}/(p-1)$, every factor is $< 1$, and $p \mapsto \sqrt{p}/(p-1)$ is decreasing on $p \ge 3$; so adjoining a prime only shrinks the product, and over $\omega(r) \ge 2$ the supremum is attained at $r = 15$, giving $\sqrt{r}/\varphi(r) \le \sqrt{15}/8 = 0.4841\ldots$, while for $\omega(r) = 1$ with $p \ge 7$ it is at most $\sqrt{7}/6 = 0.4409\ldots$. Only $r = 3, 5$ exceed $1/2$ (values confirmed for $r \le 20001$ in `code/uniform_min_sweep.gp`). Halász (Montgomery's form; the constants are absolute and the function enters only through $|h_t| \le 1$) then gives $\max_t |S_t(N)| \ll N(1+M)e^{-M} + N(\log N)^{-1/2} \ll N(\log N)^{-1/2+o(1)}$ for $r \ge 7$, and the expansion loss $r^{1/2} \le L$ leaves $T_r(N) \ll N(\log N)^{-1/2+o(1)} \ll N(\log N)^{-2/5}$.

Two honesty notes, as in Remark A.2. The $o(1)$ terms above are vacuous at every computable $N$: the crux inequality $\sqrt{r}\log L \ll L$ first bites at the top of the range around $\log\log N \approx e^{36}$, and the widening over the earlier $r \le L^{2-\delta}$ is asymptotic only, $L^\delta$ overtaking $(\log L)^{2+\varepsilon}$ far beyond even that. And the constant is ineffective twice over, through Siegel–Walfisz and through Siegel's theorem; no effective form is claimed. □

**Lemma A.5** (The distance, without expanding: uniform to power-of-log moduli)**.** *Fix $A > 0$ and write $L = \log\log N$. Let $r > 1$, let $g : (\mathbb{Z}/r)^\times \to \mathbb{C}$ satisfy $|g| \le 1$, and let $h$ be the multiplicative function with $h(\ell) = g(\ell \bmod r)$ for primes $\ell \nmid r$, $h(\ell) = 0$ for $\ell \mid r$, and $h(\ell^k) = 0$ for $k \ge 2$. Put*

$\widehat{g}_0 = \varphi(r)^{-1}\sum_a g(a)$, *the mean of* $g$, *and let*

$$M \;=\; \min_{|\tau|\le \log N}\ \sum_{\ell\le N}\frac{1-\operatorname{Re} h(\ell)\ell^{-i\tau}}{\ell}$$

*be Halász's distance for* $h$. *Then*

$$M \;\ge\; \bigl(1-|\widehat{g}_0|\bigr)\log\log N \;-\; o(\log\log N),$$

*uniformly for* $r \le (\log N)^A$ *and for all such* $g$, *the* $o(\cdot)$ *being ineffective and of unspecified rate.*

*Proof.* Only the values of $h$ at primes are used. Fix $\varepsilon\in(0,1)$ and put $T_0 = \exp\bigl((\log N)^{\varepsilon}\bigr)$; $\varepsilon$ is an auxiliary of the proof and is removed at the end. Since $\sum_{\ell\le N}1/\ell = L + O(1)$, it suffices to bound $\bigl|\sum_{\ell\le N} h(\ell)\ell^{-1-i\tau}\bigr|$ uniformly in $|\tau|\le\log N$.

*Small primes.* Trivially $\bigl|\sum_{\ell\le T_0} h(\ell)\ell^{-1-i\tau}\bigr| \le \sum_{\ell\le T_0}1/\ell = \log\log T_0 + O(1) = \varepsilon L + O(1)$, with no $\tau$–dependence.

*Large primes.* Every $\ell > T_0$ is coprime to $r\le(\log N)^A$, so $h(\ell) = g(\ell \bmod r)$ there. For $a\in(\mathbb{Z}/r)^{\times}$ write $E_a(x) = \pi(x;r,a) - \operatorname{li} x/\varphi(r)$ and $E(x) = \sum_a g(a)E_a(x)$, and set

$$S(x) = \sum_{T_0<\ell\le x} h(\ell) = \widehat{g}_0\bigl(\operatorname{li} x - \operatorname{li} T_0\bigr) + R(x), \qquad R(x) = E(x) - E(T_0),$$

using $\sum_a g(a) = \varphi(r)\widehat{g}_0$; in particular $R(T_0) = 0$ identically. Siegel–Walfisz with the exponent $B = A/\varepsilon$ gives $|E_a(x)| \ll_B x\exp(-c_B\sqrt{\log x})$ uniformly in $a$, legitimately for every $x\ge T_0$, since then $(\log x)^{A/\varepsilon}\ge(\log T_0)^{A/\varepsilon} = (\log N)^A \ge r$. Summing over the $\varphi(r)\le r$ classes with $|g|\le 1$,

$$|R(x)| \;\ll\; r\Bigl(x\,e^{-c\sqrt{\log x}} + T_0\,e^{-c\sqrt{\log T_0}}\Bigr) \;\ll\; r\,x\,e^{-c\sqrt{\log x}},$$

the second term being dominated because $y\mapsto y\,e^{-c\sqrt{\log y}}$ increases once $\log y > c^2/4$. The factor $r$ comes from that class sum, not from Siegel–Walfisz, whose error carries no modulus.

Now integrate. Writing the large–prime sum as a Stieltjes integral against $dS$,

$$\sum_{T_0<\ell\le N} h(\ell)\ell^{-1-i\tau} = \widehat{g}_0\int_{T_0}^{N}\frac{x^{-i\tau}}{x\log x}\,dx + \Bigl[x^{-1-i\tau}R(x)\Bigr]_{T_0}^{N} + (1+i\tau)\int_{T_0}^{N} R(x)\,x^{-2-i\tau}dx.$$

The first term has modulus at most $|\widehat{g}_0|\int_{T_0}^{N}dx/(x\log x) = |\widehat{g}_0|(1-\varepsilon)L$. The bracket vanishes at $T_0$ and is $\ll r\,e^{-c\sqrt{\log N}} = o(1)$ at $N$. For the last, substituting $u=\log x$ and using $\int_U^\infty e^{-c\sqrt{u}}\,du = 2c^{-2}(1+c\sqrt{U})e^{-c\sqrt{U}}$,

$$(1+|\tau|)\int_{T_0}^{N}|R(x)|x^{-2}dx \;\ll\; (1+\log N)\,r\int_{(\log N)^{\varepsilon}}^{\log N} e^{-c\sqrt{u}}du \;\ll\; (1+\log N)(\log N)^A(\log N)^{\varepsilon/2}e^{-c(\log N)^{\varepsilon/2}},$$

which is $o(1)$ uniformly in $|\tau|\le\log N$, since $c(\log N)^{\varepsilon/2}$ eventually exceeds $(A+1+\frac{\varepsilon}{2})\log\log N$. Collecting,

$$\Bigl|\sum_{\ell\le N} h(\ell)\ell^{-1-i\tau}\Bigr| \;\le\; \varepsilon L + |\widehat{g}_0|(1-\varepsilon)L + O(1)$$

uniformly in $|\tau|\le\log N$, whence $M \ge L - \varepsilon L - |\widehat{g}_0|(1-\varepsilon)L - O(1) = (1-\varepsilon)\bigl(1-|\widehat{g}_0|\bigr)L - O(1)$.

*Removing $\varepsilon$.* For each $j \geq 1$ take $\varepsilon_j = 1/(2j)$ and $B_j = 2Aj$, both *fixed*, so that $c_{B_j}$ and the implied constant $C_j$ of Siegel–Walfisz are constants (ineffective, but constants). The display above then holds beyond a threshold $N_j$ depending only on $(A, j)$, and

$$\Big(1 - \tfrac{1}{2j}\Big)\big(1 - |\widehat{g}_0|\big)L - C_j \;\geq\; \big(1 - |\widehat{g}_0|\big)L - \tfrac{1}{j}L \iff \tfrac{L}{2j}\big(1 + |\widehat{g}_0|\big) \geq C_j,$$

which holds once $L \geq 2jC_j$; enlarge $N_j$ so that it does, and so that $N_j$ increases. Setting $j(N) = \max\{j : N \geq N_j\} \to \infty$ gives $M \geq (1-|\widehat{g}_0|)L - L/j(N) = (1-|\widehat{g}_0|)L - o(L)$, uniformly in $r$ and $g$ as claimed. The ineffectivity is exactly Siegel's, entering through Siegel–Walfisz; no exceptional–zero case split is needed, because no $L$–function is bounded anywhere above. $\square$

**Remark A.6** (What this changes, and what it does not)**.** The constant is *not* an improvement: at $g_t(a) = \chi(a)e_r(ta^{-1})$ one has $\widehat{g}_0 = \chi(t)\tau(\chi)/\varphi(r)$, so $|\widehat{g}_0| = \sqrt{r}/\varphi(r)$ and Lemma A.5 returns the same $1 - \sqrt{r}/\varphi(r)$ that Lemma A.4 already gives; and in the error term it is worse, an $o(L)$ of unspecified rate against that lemma's explicit $O\big(L(\log L)^{-\varepsilon/2}\big)$. Both are ineffective, so ineffectivity is not the difference; the rate is.

What changes is the *range*, and by an exponential. Lemma A.4 is capped at $r \leq L^2/(\log L)^{2+\varepsilon}$ because expanding $g$ in multiplicative characters pays $\sum_{\psi \neq \psi_0} |c_\psi| \asymp \sqrt{r}$ against a per–character bound $O(\log L)$. The proof above never expands, so never pays it; the price is instead a factor $r$ against $\exp(-c\sqrt{\log x})$, which is free at $r \leq (\log N)^A$. Remark A.12 posed exactly this substitution as its open question, and the measurement recorded there — that $\max_s |\sum_{\ell \leq 10^6} e_p(s\ell^{-1})/\ell|$ stays near 1.3 while $\sqrt{p}$ grows from 3.6 to 44.8 — is the numerical shadow of the same fact.

Two honesty notes. The $o(L)$ is ineffective and of unspecified rate, so nothing here is quantitative at any particular $N$; and the error estimate is vacuous at every computable $N$, the bound $(1 + \log N)(\log N)^A(\log N)^{\varepsilon/2}e^{-c(\log N)^{\varepsilon/2}}$, at $A = 1, \varepsilon = 0.2, c = \frac{1}{2}$, *peaking* at $\log N = 10^{16.23}$ with value $9.29 \times 10^{24}$, and falling below 1 only past $\log N = 10^{23.58}$ (`code/swdirect_check.gp`). An earlier revision of this sentence said "turning over near $10^{18}$", which its own certificate contradicts in both directions. The uniform Mertens estimate underlying the untwisted case is classical — Norton, *Illinois J. Math.* **20** (1976), Lemma 6.3, gives $\sum_{p \leq x,\, p \equiv \ell\,(k)} 1/p = \log\log x/\varphi(k) + \delta_\ell/\ell + O(\log 3k/\varphi(k))$ with an absolute but ineffective constant — but we are not aware of a reference carrying the $\tau$–uniformity used here, and deduce it from Siegel–Walfisz (Montgomery–Vaughan, *Multiplicative Number Theory I*, §11.3) by the partial summation above.

**Lemma A.7** (Two uniform deficit bounds)**.** (Label discipline, as for Lemma A.4: three rounds of adversarial review, each by a session that did not write the text. The first two found five defects — including a principal–coefficient value false in one of three cases, and a numerical certificate that tested the wrong congruence with a silently inoperative coprimality filter — all repaired; the third found nothing, which is what the label records.) *Let $c \in \{2, -2\}$, $L = \log\log N$, and $P = L/(\log L\, w(L))$ with $w(L) \to \infty$. Uniformly for primes $p \neq q$ in $(\frac{1}{2}P, P]$, with implied constants that depend only on $w$ and are ineffective,*

$$\#\{m \leq N : \; \mu^2(m) = 1, \; \gcd(m, p) = 1, \; \sigma_p(m) \equiv c \pmod{p}\} \;\ll\; N/p,$$

$$\#\{m \leq N : \; \mu^2(m) = 1, \; \gcd(m, pq) = 1, \; \sigma_p(m) \equiv c \pmod{p} \textit{ and } \sigma_q(m) \equiv c \pmod{q}\} \;\ll\; N/(pq).$$

*Proof.* Here $p, q \in (P/2, P]$ and $pq \leq P^2 = L^2/(\log L\, w(L))^2$, inside Lemma A.4's range. Recall $\sigma_p(m) = \sum_{\ell | m} \ell^{-1} \bmod p$, so that $e_p\big(t\sigma_p(m)\big) = \prod_{\ell | m} e_p(t\ell^{-1})$ is multiplicative in squarefree $m$.

*The second bound; the first is the same argument with one prime.* Detect both congruences additively:

$$\#\{\cdots\} \;=\; \frac{1}{pq}\sum_{s \bmod p}\sum_{t \bmod q} e_p(-sc)\, e_q(-tc)\, S_{s,t}, \qquad S_{s,t} = \sum_{\substack{m\le N,\ \mu^2(m)=1\\ \gcd(m,pq)=1}} h_{s,t}(m),$$

where $h_{s,t}(m) = \prod_{\ell\mid m} e_p(s\ell^{-1})e_q(t\ell^{-1})$, extended by $h_{s,t}(\ell^k) = 0$ for $k \ge 2$ and $h_{s,t}(\ell) = 0$ for $\ell \mid pq$. It is multiplicative with $|h_{s,t}| \le 1$, which is Halász's hypothesis. The term $(s,t) = (0,0)$ contributes $(pq)^{-1}\#\{m \le N : \mu^2(m) = 1, \gcd(m,pq) = 1\} \le N/(pq)$, which is the asserted bound; the phases $e_p(-sc)e_q(-tc)$ are unimodular and play no further role, so the two values of $c$ are on the same footing. It therefore suffices to prove

$$\max_{(s,t)\ne(0,0)} |S_{s,t}(N)| \;\ll\; N/(pq), \tag{$*$}$$

since the $pq-1$ remaining terms then contribute $\ll N/(pq)$ in total.

*The distance.* Put $g_{s,t}(a) = e_p(sa^{-1})e_q(ta^{-1})$ on $(\mathbb{Z}/pq)^\times$, so $h_{s,t}(\ell) = g_{s,t}(\ell \bmod pq)$ for $\ell \nmid pq$, and expand $g_{s,t} = \sum_\psi c_\psi \psi$ over the characters mod $pq$. By the Chinese remainder theorem $g_{s,t}$ factors and so do its coefficients, $c_\psi = c^{(p)}_{\psi_1} c^{(q)}_{\psi_2}$ with $\psi = \psi_1\psi_2$. At a single prime $p$ and $s \not\equiv 0$, reindexing by $b = sa^{-1}$,

$$c^{(p)}_{\psi_1} = \frac{1}{p-1}\sum_a e_p(sa^{-1})\overline{\psi_1}(a) = \frac{\overline{\psi_1}(s)}{p-1}\sum_{b\ne 0} e_p(b)\psi_1(b),$$

so $c^{(p)}_{\psi_{1,0}} = -1/(p-1)$ by the Ramanujan sum $\sum_{b\ne0} e_p(b) = -1$, while $|c^{(p)}_{\psi_1}| = \sqrt{p}/(p-1)$ for $\psi_1 \ne \psi_{1,0}$ by $|\tau(\psi_1)| = \sqrt{p}$. For $s \equiv 0$ the factor $g$ is identically 1 and $c^{(p)}_{\psi_{1,0}} = 1$, the other coefficients vanishing. Hence for $(s,t) \ne (0,0)$ at least one factor is a Ramanujan sum: the three cases $s,t \ne 0$; $s \ne 0, t = 0$; $s = 0, t \ne 0$ give $|c_{\psi_0}|$ equal to $1/((p-1)(q-1))$, $1/(p-1)$ and $1/(q-1)$ respectively, so in every case, both $p$ and $q$ exceeding $P/2$,

$$|c_{\psi_0}| \;\le\; \max\Bigl(\frac{1}{p-1}, \frac{1}{q-1}\Bigr) \;\le\; \frac{2}{P-2} \;=\; o(1),$$

whereas Parseval $\sum_\psi |c_\psi|^2 = 1$ gives, by Cauchy–Schwarz, $\sum_{\psi\ne\psi_0} |c_\psi| \le \sqrt{\varphi(pq)} \le \sqrt{pq} \le L/(\log L\, w(L))$.

*The non–principal sums.* This is verbatim the corresponding step of Lemma A.4, at modulus $pq \le L^2/(\log L\, w(L))^2$: for $\psi \ne \psi_0$ and $|\tau| \le \log N$, partial summation against Siegel–Walfisz (legitimate since $pq \le L^2 \le (\log N)^3$) replaces the truncated sum by $\log L(s,\psi^*) + O(1)$ at the inducing character, and the classical zero–free region gives $|\log L(s,\psi^*)| \ll \log\log(pq(3+|\tau|)) \ll \log L$; in the exceptional case the same $\tau$–split applies, the branch $|\tau| \le 1$ using $|s-\beta| > 1-\beta \ge c(\eta)(pq)^{-\eta}$ for any fixed $\eta > 0$, unrelated to the $w$ of the range, and hence $|\log(s-\beta)| \le \eta\log(pq) + O_\eta(1) \ll \eta \log L + O_\eta(1)$, and the branch $|\tau| > 1$ needing no exceptional case. So $|S_\psi(\tau)| \ll \log L$ uniformly.

*Assembly and Halász.* Therefore

$$M \;=\; \min_{|\tau|\le\log N} \sum_{\ell\le N} \frac{1 - \operatorname{Re} h_{s,t}(\ell)\ell^{-i\tau}}{\ell} \;\ge\; \bigl(1-o(1)\bigr)L \;-\; O\bigl(L/w(L)\bigr) \;=\; \bigl(1-o(1)\bigr)L,$$

the principal part costing at most $|c_{\psi_0}|L = o(L)$ and the non–principal part $O\big(L/w(L)\big) = o(L)$. Halász's theorem (Montgomery's form; absolute constants, and $h_{s,t}$ enters only through $|h_{s,t}| \le 1$) with $T = \log N$ gives

$$|S_{s,t}(N)| \;\ll\; N(1+M)e^{-M} + N(\log N)^{-1/2} \;\ll\; N(\log N)^{-1/2},$$

the floor dominating since $(1+M)e^{-M} \ll (\log N)^{-1+o(1)}$. Finally $pq \le L^2$, so $pq = o\big((\log N)^{1/2}\big)$ and hence $N(\log N)^{-1/2} = o\big(N/(pq)\big)$, which is $(*)$. The one–prime bound is the same computation with the single detector $p^{-1}\sum_s e_p(-sc)S_s$, principal coefficient $-1/(p-1)$, $\sum_{\psi\ne\psi_0}|c_\psi| \le \sqrt{p} \le \sqrt{P}$, and the requirement $\max_{s\ne 0}|S_s| \ll N/p$, which is weaker than $(*)$.

Two notes, as for Lemma A.4. The constants are ineffective, through Siegel–Walfisz and Siegel. And the distance constant here is $1 - 1/(p-1) \to 1$, larger than the $1 - \sqrt{r}/\varphi(r)$ of Lemma A.4: the deficit bounds have more room than the cancellation bound they are used beside, which is why no analogue of the $r \ge 7$ restriction is needed. Numerically, at $N = 2\times 10^6$ and $c = 2$, the ratio of the counted set to $N_{\mathrm{sf}}(p)/p$, where $N_{\mathrm{sf}}(p)$ counts the squarefree $m \le N$ coprime to $p$, lies in $[0.988, 1.023]$ for $3 \le p \le 31$, and the two–prime ratio against $N_{\mathrm{sf}}(pq)/(pq)$ lies in $[0.997, 1.057]$ for pairs up to $(23, 29)$: the densities are not merely bounded but asymptotically exact, with no growth in $p$ (`code/deficit_numeric.gp`). The condition measured is $\sigma_p(m) \equiv 2$, not $m' \equiv 2$; these differ by the unit $m$, and an earlier version of that script tested the latter and silently dropped its coprimality filter. $\square$

**Proposition A.8** (A rate on the minus layer). *Write $L = \log\log N$. For every function $w$ with $w(L) \to \infty$, however slowly,*

$$\#\{m \le N:\ \mu^2(m) = 1,\ m' - 2m \textit{ a positive perfect square}\} \ll_w N\,\frac{(\log\log\log N)^2\, w(L)}{\log\log N},$$

*which is the ceiling of the method (Remark A.12); in particular the bound $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N$ holds for every fixed $\varepsilon > 0$, and the same for $m' + 2m$. In particular Theorem A.10 follows with a rate, by an argument with no diagonalisation over $r$.*

*Proof.* Let $P = L/(\log L\, w(L))$, $\mathcal{P}$ the primes in $(P/2, P]$, $\Pi = |\mathcal{P}| \gg P/\log P$, and $a_m = m' - 2m$. First, $a_m \ne 0$ for every squarefree $m$: $a_m = 0$ forces $\sigma(m) = 2$ exactly, impossible for $m > 1$ since $\gcd(D(m), m) = 1$ (Theorem 2.3) and $a_1 = -2$. If $a_m$ is a positive square then $(a_m \mid p) \ne -1$ for every $p$, and $(a_m \mid p) = 0$ exactly when $p \mid a_m$, so $\sum_{p\in\mathcal{P}}(a_m \mid p) \;\ge\; \Pi - \#\{p \in \mathcal{P} : p \mid a_m\}$. Hence $\Pi^2\,\#\{\text{squares}\} \;\le\; \sum_m\big(\sum_p (a_m \mid p) + \#\{p \in \mathcal{P} : p \mid a_m\}\big)^2$, the sum over all squarefree $m \le N$; expanding the square, the cross term is bounded through $\big|\sum_p (a_m \mid p)\big| \le \Pi$, which gives

$$\Pi^2\,\#\{\text{squares}\} \;\le\; \sum_{p\ne q}\big(T_{pq}(N) + E_{pq}\big) + \Pi N + 2\Pi\sum_p \#\{m : p \mid a_m\} + \sum_{p\ne q}\#\{m:\ p \mid a_m,\ q \mid a_m\} + \sum_p \#\{m : p \mid$$

where $|E_{pq}| \;\le\; N/p + N/q \;\le\; 4N/P$ collects the $m$ with $\gcd(m, pq) > 1$, dropped from $T_{pq}$'s summation range exactly as in the argument for Theorem A.10. By Lemma A.4 at $r = pq \le L^2/(\log L\, w(L))^2$, inside its range once $w(L) \ge (\log L)^{\varepsilon/2}$ or by enlarging $w$ (each $r$ odd squarefree, $r \ge 7$), $\sum_{p\ne q}|T_{pq}| \ll \Pi^2 N(\log N)^{-2/5}$. By Lemma A.7 — applicable because $p \mid m$ forces $a_m \equiv m' \equiv m/p \not\equiv 0 \pmod p$ for squarefree $m$, so the $m$ counted here are automatically coprime to $p$, and likewise to $pq$ in the two–prime count — $\sum_p \#\{m : p \mid a_m\} \ll N\Pi/P \ll N/\log P$ and the two–prime sum is $\ll \Pi^2 N/P^2 \ll N/(\log P)^2$. Dividing by $\Pi^2$: the diagonal $\Pi N$ gives the main term $N/\Pi \ll N\log P/P$, and every other term is smaller. Since $\log P \asymp \log\log\log N$, the bound follows: $\log P \asymp \log L$, so $N \log P/P \asymp N(\log L)^2 w(L)/L$. The fixed–$\varepsilon$ form is the case $w = (\log L)^\varepsilon$. Note $\mu^2(m)$ and the coprimality conditions are carried inside every count; no Möbius expansion is performed. $\square$

Measured at $r = 1009$, $N = 3 \times 10^7$: $|\sum_m h_t(m)|/N$ equals $3.6 \times 10^{-6}$, $1.3 \times 10^{-3}$, $2.1 \times 10^{-3}$, $5.7 \times 10^{-4}$, $1.5 \times 10^{-3}$ at $t = 0, 1, 2, 17, 500$, all of size $\sqrt{N}$ rather than merely $o(N)$. The sums $T_r(N)$ themselves were measured at $N = 5 \times 10^7$ for $r = 101, 1009, 10007, 100003$ and $r = pq = 101 \cdot 1009$, $1009 \cdot 10007$, giving $|T_r|/\sqrt{N} = 8.1$, 0.1, 1.2, 1.0, 1.0, 0.2 (`code/sqsieve_test.rs`).

**Lemma A.9** (Local densities at $p^2$)**.** *For each odd prime $p$, $\limsup_{N\to\infty} N^{-1}\#\{m \le N : \mu^2(m) = 1,\ p^2 \mid (m' - 2m)\} \le p^{-2}$, with absolute constant.* (*The density form is the one Theorem A.10 needs, because it sends $P \to \infty$; the form "$\ll N/p^2$ for each fixed $p$" advertises a $p$–dependent implied constant and would not license that limit.*)

*Proof.* For $p \nmid m$ the condition is $\sigma_{p^2}(m) \equiv 2 \pmod{p^2}$ by Lemma 7.1 applied with modulus $p^2$. Detecting it by additive characters, the non-trivial frequencies give multiplicative sums $\prod_{\ell \mid m} e_{p^2}(t/\ell)$ whose values at primes depend only on $\ell \bmod p^2$ and which are not Dirichlet characters, because $u \mapsto e_{p^2}(t/u)$ is not multiplicative; the Halász argument of Lemma A.1 applies, but *not* verbatim, and the difference is worth naming. That lemma uses primitivity of $\chi = (\cdot \mid r)$ twice, for $|\tau(\chi)| = \sqrt{r}$ and for the principal coefficient $\chi(t)\tau(\chi)/\varphi(r)$; here $(\cdot \mid p^2)$ is trivial on $(\mathbb{Z}/p^2)^\times$, so there is no character twist at all. The principal coefficient is instead a Ramanujan sum $c_{p^2}(t)/\varphi(p^2)$, which vanishes for $\gcd(t, p^2) = 1$ and equals $-1/(p-1)$ when $p \parallel t$; the distance bound follows more easily than in Lemma A.1, not by the same computation. Hence the density is $\frac{6}{\pi^2} \cdot \frac{p}{p+1} \cdot p^{-2} + o(1) \le p^{-2}$. The two factors in front were omitted in an earlier revision: the count runs over *squarefree $m$* coprime to $p$, so it carries the squarefree density $6/\pi^2$ and the coprimality factor $p/(p+1)$. The error was in the safe direction, the stated $p^{-2}$ being an over-estimate, so the inequality the lemma asserts survives with room to spare. The case $p \mid m$ does not arise at all: writing $m = pn$ with $p \nmid n$ gives $m' = pn' + n$, so $m' - 2m = n + p(n' - 2n) \equiv n \not\equiv 0 \pmod{p}$, and $p$ does not even divide $m' - 2m$, let alone $p^2$. $\square$

Measured at $N = 3 \times 10^7$ for $p = 5, 7, 11, 13, 101$, the count divided by $\frac{6}{\pi^2}\frac{p}{p+1} N/p^2$ is 0.842, 0.887, 0.930, 0.925, 0.993, approaching 1 as the corrected density predicts. Against the bare $N/p^2$ the same data read 0.512, 0.539, 0.566, 0.562, 0.604, approaching $6/\pi^2 = 0.6079$; an earlier revision printed the first list while describing it as the second.

**Theorem A.10** (The minus layer has density zero on the squarefree stratum)**.** Label: PROVED. Two independent routes reach this statement, and the label rests on the second. The first is the square sieve at fixed $r$ recorded in Remark A.11 below: it needs no uniformity, since it sends $P \to \infty$ only after $N$, and it is why the statement was believed through several releases — but it yields no rate, and its *text* failed three consecutive reviews, a load–bearing display silently shedding a hypothesis each time, so it has never had a clean round and the label does not rest on it. The second is Proposition A.8, which proves the statement *with a rate* from Lemma A.4 and Lemma A.7; each of those, and the proposition itself, earned its label under this project's standard — a review by a session that did not write it, finding nothing.

$\#\{m \le N :\ \mu^2(m) = 1,\ m' - 2m$ *is a positive perfect square*$\} = o(N)$*; indeed, by Proposition A.8, the count is $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N$ for every $\varepsilon > 0$. Equivalently, atom* A9 *holds on the squarefree stratum: the squarefree minus layer has density zero. The same argument with $m' + 2m$ gives the squarefree plus–analogue, so the squarefree Pythagorean layer has density zero unconditionally. That is* not *a reproof of Proposition* 5.16*, which is a sharper $O(\sqrt{X}(\log X)^2)$*

*on the narrower semiprime stratum; the two are incomparable, and that proposition rests on its own elementary divisor–sum proof.*

*The squarefree restriction is not cosmetic and earlier versions of this paper omitted it. Every tool in the proof, Rigidity, the cofactor identity, and Lemma A.1, is stated for squarefree m; the non-squarefree stratum, of density* $1 - 6/\pi^2 \approx 0.392$*, is untreated. Since a #307 solution is squarefree by Rigidity, the restriction costs nothing for the application, but the unrestricted statement is not proved here.*

*Proof.* Immediate from Proposition A.8, which gives the minus–layer count $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N = o(N)$ on the squarefree stratum for any fixed $\varepsilon > 0$, and the plus–layer count by the same statement. Remark A.11 records an independent route to the qualitative form. □

**Remark A.11** (The original route, and what it does and does not give)**.** This is the argument that carried Theorem A.10 before Proposition A.8 existed. It is recorded because it is independent of that proposition, because it needs no uniformity in $r$ whatsoever, and because the errata inside it are the ones that cost the most to find. It is *not* what the label rests on: its text failed three consecutive reviews and has never had a clean one, and it yields no rate, since $P \to \infty$ only after $N$.

Write $a_m = m' - 2m$ and fix $P$, letting $\mathcal{P}$ be the primes in $(P, 2P]$ and $\Pi = |\mathcal{P}| \sim P/\log P$. If $a_m$ is a perfect square coprime to every $p \in \mathcal{P}$ then $(a_m/p) = 1$ for all such $p$, so $\sum_{p\in\mathcal{P}}(a_m/p) = \Pi$. Hence

$$\Pi^2\,\#\{m \le N:\ \mu^2(m) = 1,\ a_m = \Box,\ \gcd(a_m, \textstyle\prod\mathcal{P}) = 1\} \le \sum_{\substack{m\le N\\ \mu^2(m)=1}} \Big(\sum_{p\in\mathcal{P}}\Big(\frac{a_m}{p}\Big)\Big)^2 = \sum_{p,q\in\mathcal{P}}\ \sum_{\substack{m\le N\\ \mu^2(m)=1}}\Big(\frac{a_m}{pq}\Big).$$

Both $m$–sums carry $\mu^2(m) = 1$ throughout, and so does the counting set on the left. This is not a convenience: every tool below is squarefree–only, and an earlier revision wrote these sums unrestricted, which made the identity with $T_{pq} + E_{pq}$ false by the non–squarefree terms. Those are not negligible: measured at $N = 3 \times 10^7$ the omitted sum is 119,473 against a $T_{pq}$ of 2,699 at $r = 101 \cdot 1009$, and 169,949 against 314 at $r = 1009 \cdot 10007$, exceeding the term the lemma controls by a factor 44 to 540. The paper bounds it by nothing better than the trivial $(1 - 6/\pi^2)N$. The diagonal $p = q$ contributes $(a_m/p^2) = (a_m/p)^2 \in \{0, 1\}$, hence at most $\Pi N$. The off-diagonal is $\sum_{p\ne q}\big(T_{pq}(N) + E_{pq}(N)\big)$, where $T_{pq}$ carries the coprimality condition $\gcd(m, pq) = 1$ of Lemma A.1 and $E_{pq}(N) = \sum_{m\le N,\ \mu^2(m)=1,\ \gcd(m,pq)>1}(a_m/pq)$ collects the terms it omits. That correction is harmless but is not empty and must be carried: $|E_{pq}(N)| \le \#\{m \le N : p \mid m \text{ or } q \mid m\} \le N(1/p + 1/q) \le 2N/P$ — the primes here lie in $(P, 2P]$, whereas Proposition A.8 takes them in $(P/2, P]$ and so carries $4N/P$ — so $\Pi^{-2}\sum_{p\ne q}|E_{pq}(N)| \le 2N/P$, which is dominated by the diagonal's own $N/\Pi \asymp N\log P/P$ and changes nothing below. Each $T_{pq}(N)$ is $o(N)$, for each of the finitely many pairs, by Lemma A.1. A square divisible by $p$ is divisible by $p^2$, so the excluded $m$ are counted by Lemma A.9 in its *density* form: $\limsup_N N^{-1}\#\{m \le N : \mu^2(m) = 1,\ p^2 \mid a_m\} \le p^{-2}$, with absolute constant 1. That form is what the endgame needs, since sending $P \to \infty$ against a $p$–dependent implied constant would not be legitimate; the statement "$\ll N/p^2$ for each fixed $p$" advertises exactly such a constant and was cited here in an earlier revision. Summing, $\sum_{p\in\mathcal{P}} p^{-2} \le \Pi/P^2 \asymp 1/(P\log P)$. Here $p \mid m$ is impossible, since $m = pn$ with $p \nmid n$ gives $a_m = n + p(n' - 2n) \equiv n \not\equiv 0$, a computation that itself needs $\mu^2(m) = 1$. The case $a_m = 0$ does not arise: the theorem counts $a_m = \Box > 0$, and in any case $m' = 2m$ has *no* squarefree solution, since $\sigma(m) = 2 \in \mathbb{Z}$ forces $D_m = 1$ by rigidity. Altogether

$$\limsup_{N\to\infty}\frac{1}{N}\#\{m \le N:\ \mu^2(m) = 1,\ a_m = \Box > 0\} \ll \frac{1}{\Pi} + \frac{1}{P\log P}.$$

The left side does not depend on $P$; letting $P \to \infty$ gives 0. The restriction $\mu^2(m) = 1$ must be carried here as everywhere above: without it the display would assert density zero for the *full* minus layer, which is exactly the statement this theorem disclaims, and three successive revisions of this proof dropped it from one display or another.

The proof of Theorem A.10 sends $P \to \infty$ only after $N$, which costs all quantitative information. Keeping $P$ a function of $N$ would turn it into an explicit rate, and what that requires is uniformity in $r$ as $P \to \infty$.

*That uniformity is exactly what is not available*, and earlier editions of this paragraph asserted the opposite. Lemma A.1 gives $M \geq \left(1 - \sqrt{r}/\varphi(r)\right) \log\log N - O_r(\log\log\log N)$, with the error depending on $r$ through the non–principal $L$–functions, and its statement and the erratum following it both say so. The constant is $\sqrt{r}/\varphi(r)$, not the $3/\sqrt{r}$ printed here previously; the two are not interchangeable, since $\varphi(r) \geq r/3$ fails for odd squarefree $r$, first at $r = 3 \cdot 5 \cdots 23 = 111{,}546{,}435$ with $\varphi(r)/r = 0.3272$. The statement below was therefore stated as a CONJECTURE when this remark was written, and its argument here is a sketch rather than a proof. The uniform input it wanted is Lemma A.5, and the rate itself is Theorem A.13, proved by a different route.

**Remark A.12** (The ceiling of the method, and the one input that would break it)**.** Two things are worth recording about how far Proposition A.8 can be pushed, since the same accounting error was made twice in this paper's history.

*The ceiling.* The binding constraint there is again the coefficient mass: $\sqrt{pq} \cdot O(\log\log\log N) = o(\log\log N)$, that is $P = o(L/\log L)$ with $L = \log\log N$. The proposition takes $P = L/(\log L)^{1+\varepsilon}$, which is short of that ceiling by $(\log L)^{\varepsilon}$ — exactly as the earlier edition of Lemma A.4 was short of *its* ceiling by $L^{\delta}$. Taking $P = L/(\log L\, w(L))$ for any $w \to \infty$ gives

$$\#\{m \leq N : \ \mu^2(m) = 1, \ m' - 2m = \square > 0\} \ \ll \ N \, \frac{(\log\log\log N)^2 \, w(\log\log N)}{\log\log N},$$

so the true ceiling of the method is $N(\log\log\log N)^{2+o(1)}/\log\log N$ with the $o(1)$ positive, unremovable, and arbitrarily slow. We state the proposition with a fixed $\varepsilon$ because it is the usable form; the sharpening is asymptotic and worth nothing numerically. What matters is the shape: no choice of $P$ puts a power of $\log N$ in the denominator *so long as the distance is estimated through the character expansion.* That proviso is not decoration — see the next paragraph, where it is exactly what is in question.

*Where the loss actually is, and what it is not.* Both walls — this one and the corollary's — are the same $\sqrt{\varphi(r)}$ paid when the phase $\ell \mapsto e_r(t\ell^{-1})$ is expanded in multiplicative characters. It is tempting to ask for a Weil–type bound for that phase over primes, and an earlier draft of this remark did. That framing is *wrong*, and the error is worth recording. The phase depends only on $\ell$ mod $p$, so

$$\sum_{\ell \leq X} e_p(s\ell^{-1})\ell^{-1-i\tau} \ = \ \sum_{a \not\equiv 0} e_p(sa^{-1}) \sum_{\substack{\ell \leq X \\ \ell \equiv a\,(p)}} \ell^{-1-i\tau},$$

an exact reorganisation with no loss; its discrete Fourier transform is a Gauss sum, so a "direct" estimate is the character expansion restated rather than an independent route. The Kloosterman literature over primes — Fouvry–Michel, Fouvry–Kowalski–Michel, Garaev, Korolev, Irving — is *not* the relevant technology: every bound there is nontrivial only for modulus at least a fixed power of $X$, and ours is at most a power of $\log X$. Note also that the sum is not $o(\log\log X)$ at fixed $p$: since $\sum_{a \not\equiv 0} e_p(sa^{-1}) = -1$, Mertens in progressions gives $-\log\log X/(p-1) + O(1)$, measured as $-0.2502$, $-0.1667$, $-0.0830$, $-0.0105$ at $p = 5, 7, 13, 101$ against $-1/(p-1)$ (`code/inverse_phase_check.gp`). That failure is confined to bounded $p$ and is harmless, $1/(p-1)$ being itself $o(1)$.

What the same computation does show is that the *accounting* is lossy: measured, $\max_s |\sum_{\ell \le 10^6} e_p(s\ell^{-1})/\ell|$ is $1.04, 1.17, 1.35, 1.32, 1.33$ at $p = 13, 101, 503, 1009, 2003$ — flat, while $\sqrt{p}$ grows from 3.6 to 44.8. So the triangle inequality over characters discards almost everything, and the wall above is an artefact of that step rather than an arithmetic obstruction. It can be avoided, and that is Lemma A.5: applying Siegel–Walfisz to the residue classes directly pays a factor $r$ against an error $\exp(-c\sqrt{\log x})$ rather than $\sqrt{r}$ against a bound on $\log L$, which is free at $r \le (\log N)^A$. The ceiling computed in this remark is therefore the ceiling of the *character–expansion* route only; Theorem A.13 passes it by a quarter–power of $\log N$. The measurement above — $\max_s |\cdot|$ flat near 1.3 while $\sqrt{p}$ grows to 44.8 — was the numerical shadow of exactly this.

**Theorem A.13** (A rate for A9)**.** (Label: PROVED. This statement stood at CONJECTURE through many releases, with two blockers recorded for it that were both later retracted — Lemma A.9's non–uniformity, an artefact of a superseded route that needs a $p^2$ count where Proposition A.8 needs only first–power divisibility; and the regime mismatch after Lemma 7.4, which was measured against $N$ rather than against this statement's own target, and is smaller than it. The operative blocker was the pretentious distance, and Lemma A.5 supplies it.) *For every $\delta > 0$,*

$$\#\{m \le N : \ \mu^2(m) = 1, \ m' - 2m \textit{ is a positive perfect square}\} \ll_\delta N \, \frac{(\log\log N)^{1/2+\delta}}{(\log N)^{1/4}},$$

*and the same for $m' + 2m$. The squarefree restriction is inherited from Theorem A.10 and from Lemma A.1; an earlier revision stated this unrestricted. The implied constant is ineffective, through Siegel.*

*Proof.* Run the sieve of Proposition A.8 at $P = (\log N)^{1/4}(\log\log N)^{1/2}$, with $\mathcal{P}$ the primes in $(P/2, P]$ and $\Pi = |\mathcal{P}| \gg P/\log P$; here $\log P \asymp \log\log N$. The expanded square and its five terms are exactly as there, and we bound each after dividing by $\Pi^2$.

*The distance, at these moduli.* Every modulus arising is $r = pq \le P^2 \ll (\log N)^{1/2} \log\log N$, so Lemma A.5 applies with $A = 1$. For the character sums it gives, at $g_t(a) = \chi(a)e_r(ta^{-1})$ and hence $|\widehat{g}_0| = \sqrt{r}/\varphi(r) = o(1)$, $M \ge (1 - o(1)) \log\log N$; for the deficit sums, at $g_{s,t}(a) = e_p(sa^{-1})e_q(ta^{-1})$ and hence $|\widehat{g}_0| \le \max\left(\frac{1}{p-1}, \frac{1}{q-1}\right) \le 2/(P-2) = o(1)$, the same. In both cases Halász (Montgomery's form, $T = \log N$) gives $N(1 + M)e^{-M} + N(\log N)^{-1/2} \ll N(\log N)^{-1+o(1)} + N(\log N)^{-1/2} \ll N(\log N)^{-1/2}$: the floor binds.

*The five terms.* The diagonal contributes the main term $N/\Pi \asymp N \log P/P \asymp N(\log\log N)^{1/2}(\log N)^{-1/4}$, which is the stated bound. The character term is $\Pi^{-2} \sum_{p \ne q} |T_{pq}| \le \max_{p \ne q} \sqrt{pq} \cdot \max_t |S_t| \ll 2P \cdot N(\log N)^{-1/2} \asymp N(\log\log N)^{1/2}(\log N)^{-1/4}$, the same order — this is what fixes $P$, the balance $N \log P/P \asymp PN(\log N)^{-1/2}$ giving $P \asymp (\log N)^{1/4}(\log\log N)^{1/2}$. The correction $|E_{pq}| \le 4N/P$ contributes $\ll N(\log N)^{-1/4}(\log\log N)^{-1/2}$, smaller by $\log\log N$. For the two divisibility terms, note first that no $p^2$–count is needed and Lemma A.9 does not enter: what the sieve requires is $\#\{m : p \mid a_m\} \ll N/\Pi$, weaker than $N/p$ by the factor $\log P$, and the two–prime count needs nothing at all, since $\#\{m : p \mid a_m, \ q \mid a_m\} \le \#\{m : p \mid a_m\}$ makes $\sum_{p \ne q} \#\{\cdots\} \le \Pi \sum_p \#\{\cdots\}$. By the detector identity of Lemma A.7 — whose $1/(pq)$ prefactor cancels the $pq - 1$ non–principal terms exactly, with no $\sqrt{r}$ amplification — and the distance bound above, $\#\{m : p \mid a_m\} \le N/p + \max_{s \ne 0} |S_s| \ll N/p + N(\log N)^{-1/2} \ll N/\Pi$, as required. Every non–diagonal term is therefore at most the main term, and dividing by $\Pi^2$ gives the bound. The proof in fact delivers the sharper exponent $\frac{1}{2}$ on $\log\log N$; the $\delta$ in the statement is a deliberate weakening, so that no reader has to track which constants are absolute. The plus case is identical with $c = -2$. $\square$

**Remark A.14** (Why this is stronger than the rate it replaces)**.** Earlier revisions stated this corollary as $\ll N\log\log N(\log N)^{-1/4}$, balancing at $P \asymp (\log N)^{1/4}$. Carrying the balance properly against the Halász floor gives $P \asymp (\log N)^{1/4}(\log\log N)^{1/2}$ and a bound better by $(\log\log N)^{1/2}$. Compared with Proposition A.8, which remains the statement proved without Lemma A.5, the gain is a full quarter–power of $\log N$: that proposition's ceiling is $N(\log\log\log N)^{2+o(1)}/\log\log N$, and no choice of $P$ improves it while the distance is estimated through the character expansion.

*Sketch, conditional on a uniform-in-r form of Lemma A.1.* Take $\mathcal{P}$ to be the primes in $(P, 2P]$ with $P = P(N)$ to be chosen, so $\Pi \sim P/\log P$. The three terms of Theorem A.10 are then

$$\frac{N}{\Pi} \ll \frac{N\log P}{P}, \qquad \frac{1}{\Pi^2}\sum_{p\neq q}|T_{pq}(N)| \le \max_{p\neq q}|T_{pq}(N)|, \qquad \sum_{p\in\mathcal{P}} O\left(\frac{N}{p^2}\right) \ll \frac{N}{P\log P}.$$

Granting a uniform-in-$r$ distance bound, which Lemma A.1 does *not* supply, Halász would give $T_r(N) \ll \sqrt{r}\,N(\log N)^{-1/2}$, and $r = pq \le 4P^2$ gives $\sqrt{r} \le 2P$, so the middle term is $\ll PN(\log N)^{-1/2}$. Balancing it against the first, $N\log P/P = PN(\log N)^{-1/2}$, gives $P \asymp (\log N)^{1/4}$, at which both equal $N\log\log N(\log N)^{-1/4}$ up to constants; the third term is then $O\big(N(\log N)^{-1/4}/\log\log N\big)$ and is absorbed. ☐

**Remark A.15** (Where the exponent comes from)**.** The exponent $\frac{1}{4}$ is not intrinsic to the arithmetic. It is what remains after balancing against the floor $N(\log N)^{-1/2}$ in the Montgomery form of Halász's theorem, which holds however large the pretentious distance $M$ becomes. Any improvement of that floor transfers here without change: a floor of $N(\log N)^{-\alpha}$ yields the exponent $\alpha/2$. The measured character sums are of size $\sqrt{N}$ rather than $N(\log N)^{-1/2}$ (Section 7), so the truth is far stronger than what the general theorem can deliver, and the gap is entirely in the analytic input rather than in the sieve.

The exponent $\frac{1}{4}$ is an artefact of the general theorem, not of the arithmetic, and the gap can be closed off exactly. Everything in the sieve except the character-sum input is elementary, so the reachable exponent is a function of one measurable quantity: how $T_r(N)$ grows with $r$.

**Hypothesis A.16** (Square-root cancellation, uniform in the modulus)**.** There is an absolute $A \ge 0$ with $|T_r(N)| \ll r^A N^{1/2+o(1)}$ for every odd squarefree $r$.

**Theorem A.17** (Conditional power saving, and the crux exactly)**.** *Assume Hypothesis* A.16*. Then*

$$\#\{m \le N:\ \mu^2(m) = 1,\ m' - 2m \textit{ is a positive perfect square}\} \ll N^{1-\delta}, \qquad \delta = \frac{1}{2(2A+1)}.$$

*In particular $A = 0$ gives $N^{1/2+o(1)}$, which is exactly the bound* CRUX-A9 *asks for.*

*Proof.* Run the sieve of Theorem A.10 with $\mathcal{P}$ the primes in $(P, 2P]$, $\Pi \sim P/\log P$, and dispose of the $m$ with $p \mid a_m$ by the divisor bound rather than Lemma A.9: a perfect square $a_m$ has at most $\log(2N)/\log P$ prime factors in $\mathcal{P}$, so $\sum_{p\in\mathcal{P}}(a_m/p) \ge \Pi/2$ once $\Pi \ge 2\log(2N)/\log P$. Then

$$\#\{\,\cdot\,\} \ll \frac{N}{\Pi} + \frac{1}{\Pi^2}\sum_{p\neq q}|T_{pq}(N)| \ll \frac{N\log P}{P} + P^{2A}N^{1/2+o(1)},$$

since $pq \le 4P^2$. Balancing the two terms gives $P = N^{1/(2(2A+1))}$ and the stated exponent; the side condition $\Pi \ge 2\log(2N)/\log P$ holds comfortably at that $P$. For $A = 0$ take $P = N^{1/2}$, whence both terms are $N^{1/2+o(1)}$. ☐

**Remark A.18** (The hypothesis is measured, and $A$ reads as zero)**.** Hypothesis A.16 is not a wish. At $N = 4 \times 10^7$, so $\sqrt{N} = 6.325 \times 10^3$, the ratio $|T_r(N)|/\sqrt{N}$ across twelve moduli spanning ten orders of magnitude is

| | | | | | | |
|---|---|---|---|---|---|---|
| $r$ prime : | 101 | 1009 | 10007 | 100003 | $10^6{+}3$ | $10^7{+}19$ |
| $\lvert T_r\rvert/\sqrt{N}$ : | 7.38 | 0.17 | 0.84 | 1.07 | 0.05 | 0.15 |
| $r = pq$ : | 101909 | 10097063 | 1000730021 | 100003300009 | | |
| $\lvert T_r\rvert/\sqrt{N}$ : | 0.95 | 0.21 | 1.30 | 0.64 | | |

together with 2.34 and 1.00 at the squarefree 1155 and 1113121. The implied exponent $A = \log(|T_r|/\sqrt{N})/\log r$ is $+0.006$, $-0.004$, $+0.013$, $-0.018$ and $-0.000$ at the five largest moduli; the outlying $+0.433$ occurs at $r = 101$, where $\log r$ is too small for the ratio to be meaningful. Note that the last two moduli *exceed* $N$, and the sums are still of size $\sqrt{N}$ (`code/sqsieve_uniform.rs`).

So the sharp form of A9 is not blocked by the sieve, which is elementary, nor by the arithmetic, which cooperates: it is blocked by the absence of a Weil-type bound for a character sum whose argument is not a polynomial. That is a single, isolated, and precisely stated analytic deficit, and Lemma 7.1 has already reduced it to a sum over a multiplicative function of $\sigma_r$.

The obvious route to Hypothesis A.16 is the one already used in Lemma A.1: expand by Gauss sums and bound each multiplicative sum $\sum_{m \le N} h_t(m)$ separately, where square-root cancellation would follow from a Weil-type input or, in the analytic form, from GRH applied to $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$. That route is closed. The reason is not the one a first look suggests.

**Proposition A.19** (The decomposition destroys the cancellation)**.** *Write $\alpha(u) = (u/r)e_r(tu^{-1})$, so that $h_t(p) = \alpha(p \bmod r)$. The principal Fourier coefficient $\widehat{\alpha}(\chi_0) = \varphi(r)^{-1}\sum_u \alpha(u)$ is a Salié sum; measured, it has modulus exactly $\sqrt{r}$ before normalisation, at $r = 1009$ and $r = 10007$ giving* 31.8 *and* 100.0 *against $\sqrt{r} = 31.76$ and* 100.03*. It is therefore nonzero, and $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$ retains a $\zeta$-type branch point at $s = 1$. This does* not *by itself decide the matter, since the measured sums fall* 50 *to* 350 *times below what that branch point predicts. What decides it is the growth in $N$: at $r = 10007$, $t = 1$,*

| | | | | | |
|---|---|---|---|---|---|
| $N$ : | $10^6$ | $3 \times 10^6$ | $10^7$ | $3 \times 10^7$ | $6 \times 10^7$ |
| $\lvert\sum h_t\rvert/\sqrt{N}$ : | 0.67 | 0.48 | 0.69 | 2.08 | 2.29 |

*a fitted exponent of* 0.800 *over that range, with $|\sum h_t|/(N/\log N)$ flat near* 0.005*. The individual multiplicative sums thus grow strictly faster than $\sqrt{N}$* (`code/sqsieve_route.rs`).

**Remark A.20** (Where the cancellation actually lives)**.** Since $T_r(N)$ itself measures at $\sqrt{N}$ across ten orders of magnitude in $r$ (Remark A.18) while each $\sum_m h_t(m)$ measured here grows like $N^{0.8}$, the cancellation in $T_r$ is not present frequency by frequency: it lies in the sum over $t$. The magnitude is not marginal. At $r = 1009$, $N = 4 \times 10^7$, bounding each frequency separately and summing trivially gives $\sqrt{r}\,N/\log N \approx 7 \times 10^7$, against a measured $|T_r| = 1050$: a factor of about $7 \times 10^4$ is lost the moment the Gauss expansion is taken.

The deficit behind Hypothesis A.16 is therefore not a missing Weil bound for a single character sum, which is how Remark A.18 first framed it. It is that the only available decomposition of $T_r$ scatters the cancellation across frequencies, and no current technique grips cancellation in that position. Stating it this way is more useful than the Weil framing, because it says what would have to be new: an estimate for $T_r$ that never passes through the multiplicative decomposition at all.

Remark A.20 asked for an estimate for $T_r$ that never passes through the multiplicative decomposition. The following is the mechanism such an estimate would use, isolated and proved; the reduction of $T_r$ to it is *not* established here, and we say precisely what is missing.

**Proposition A.21** (Kernel norm)**.** *Since $\sigma_r$ is additive, for coprime $d, e$ Lemma 7.1 gives*

$$\Big(\frac{(de)' - 2de}{r}\Big) = \Big(\frac{d}{r}\Big)\Big(\frac{e}{r}\Big) K\big(\sigma_r(d), \sigma_r(e)\big), \qquad K(u,v) = \Big(\frac{u+v-2}{r}\Big).$$

*The matrix $K$ is Toeplitz in $u+v$, so $K(u,v) = \sum_\xi \widehat{f}(\xi) e_r(\xi u) e_r(\xi v)$ diagonalises it against the orthogonal system $(e_r(\xi u))_u$ of norm $\sqrt{r}$. As $\widehat{f}(\xi)$ is a normalised Gauss sum, $|\widehat{f}(\xi)| = r^{-1/2}$ for $\xi \neq 0$ and $\widehat{f}(0) = 0$, so every singular value of $K$ equals $\sqrt{r}$ and*

$$\|K\|_{\mathrm{op}} = \sqrt{r}.$$

*Consequently* any *bilinear form $\sum_{d\sim D} \sum_{e\sim E} \alpha_d \beta_e K(\sigma_r(d), \sigma_r(e))$ is at most $\sqrt{r}\,\|A\|_2 \|B\|_2$, where $A(u) = \sum_{d\sim D,\ \sigma_r(d)\equiv u} \alpha_d$ and $B$ is defined likewise. No multiplicative decomposition is used; the only input is the modulus of a Gauss sum.*

**Remark A.22** (What is missing, stated exactly)**.** Proposition A.21 bounds bilinear forms, not $T_r$. Passing from one to the other requires a decomposition of $\sum_{m\le N}$ into bilinear pieces, and two obstacles are real rather than cosmetic.

(i) Summing over coprime pairs $(d, e)$ with $de = m$ weights each $m$ by $2^{\omega(m)}$: at $N = 2 \times 10^6$ the coprime-pair count is 10,014,190 against 1,215,876 squarefree $m$, a factor 8.24. Since the summand is *signed*, no inequality relates $\sum_m F(m)$ to $\sum_m 2^{\omega(m)} F(m)$ in either direction.

(ii) A single dyadic block $d \sim D$ sees only a minority of $m$: for $D = \sqrt{N}$ at $N = 2 \times 10^6$, 24.7% (`code/sqsieve_bilinear.rs`).

The natural repair is the largest-prime-factor split $m = pn$ with $p = P^+(m)$, which is unique and so avoids (i); the $A$-side is then supported on primes, where $\|A\|_2^2 \ll \pi(D)^2/\varphi(r) + \pi(D)$ follows from Brun–Titchmarsh uniformly in $r$, and the $B$-side is governed by Theorem 7.3. What remains is the coupling $P^+(n) < p$ and the dyadic bookkeeping, a Type I/II analysis that we have not carried out. Until it is, the power saving below is **CONJECTURAL**. The best unconditional rate proved here is Theorem A.13's $N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$; Proposition A.8 is what the character–expansion route gives without Lemma A.5.

**Conjecture A.23** (Power saving, pending the decomposition)**.** *If the decomposition of Remark* A.22 *is carried out with the stated bounds, then combining Proposition A.21 with Theorem 7.3 and Theorem A.17 at $P = N^{1/4}$ gives*

$$\#\{m \le N:\ \mu^2(m) = 1,\ m' - 2m \textit{ is a positive perfect square}\} \ \ll\ N^{3/4+o(1)}.$$

The two ingredients that would feed it are themselves unconditional, and are recorded separately because they do not depend on the missing step.

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